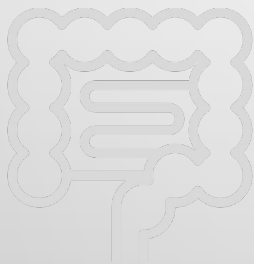
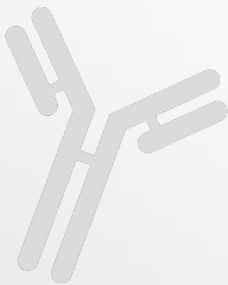
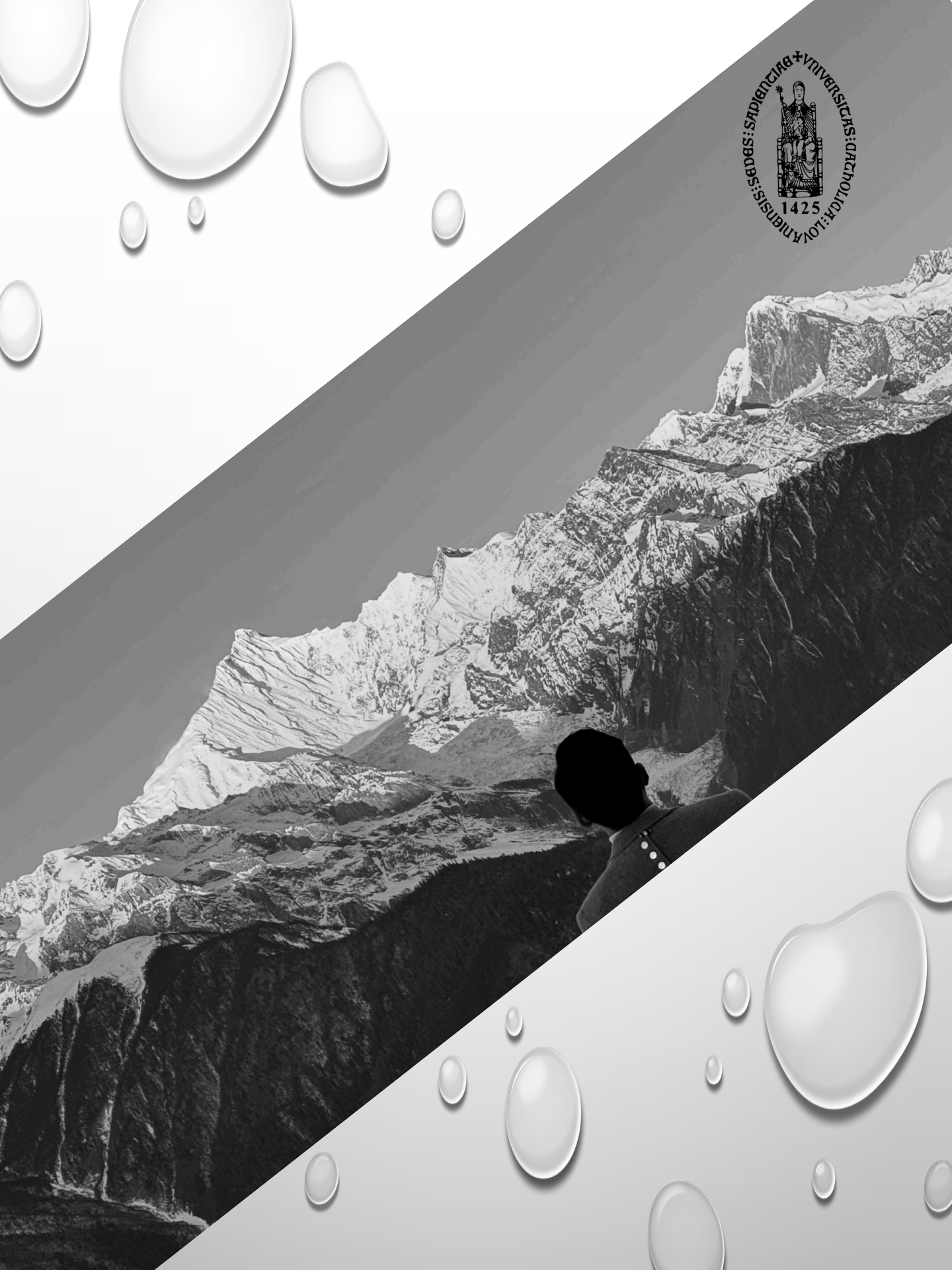


A Pharmacometrics Approach to Individualizing Dosing of Biologicals in Patients with Inflammatory Bowel Diseases



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Dissertation presented in partial
fulfilment of the requirements
for the degree of Doctor in
Pharmaceutical Sciences

Leuven, 18 November 2024

Monday 18 November 2024, 17h00

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Words of thanks

Four years ago, I made the decision to pursue a PhD in pharmacometrics. Reflecting on this journey, I feel both confident and fortunate that it was the right path.

First, I want to extend my heartfelt thanks to Prof. Dr. Erwin Dreesen, Prof. Dr. Isabel Spriet, and Prof. Dr. Paul Declerck for interviewing me and offering me this PhD position four years ago. I am forever grateful for the trust you placed in me from the very beginning.

My deepest appreciation goes to my promoter, Prof. Dr. Erwin Dreesen. Thank you for your dedicated supervision and unwavering support throughout my PhD. I have learned tremendously from you, and this thesis would not have been possible without your guidance. Your kindness, inspiration, and consistent support have been invaluable. I always knew I could rely on you for insightful advice and feedback. It has been a true honor and pleasure to learn from you and work alongside you.

I also express my profound gratitude to my promoter and co-promoter, Prof. Dr. Paul Declerck. Thank you for accepting and funding me as your PhD student four years ago. I thoroughly enjoyed our research update meetings and the enriching discussions we shared throughout my PhD. Your guidance and support have been instrumental to my progress and success.

This work would not have been possible without the support of my co-promoter, Prof. Dr. Séverine Vermeire. Thank you for providing the clinical data and context that grounded my research, allowing me to address critical, clinically relevant questions and contribute toward improving healthcare for patients.

I also wish to extend my sincere thanks to my internal jury members, Prof. Dr. Pieter Annaert, Prof. Dr. Isabel Spriet, and Prof. Dr. Karen van Hoeve, for challenging me and providing constructive feedback on my research over the past four years. Additionally, I am grateful to Prof. Dr. Peter Verwilt for coordinating the evaluation of my PhD thesis and chairing my doctoral defense.

A heartfelt thanks goes to my external jury members, Prof. Dr. Elisabet Nielsen and Prof. Dr. Jurij Hanžel. Thank you for reviewing my PhD thesis and for traveling a considerable

distance to attend my public defense in Leuven. Your presence and input are genuinely valued.

I am grateful to my colleagues in the Leuven Pharmacometrics Research Group: Dr. Wannee Kantasiripitak, Dr. Omar Elkayal, Nada Dia, Elma Edwina, Yannick Hoffert, Zoë Vander Elst, My Luong Vuong, Juma Ayubu Mohamedi, Wei Zhang, Emmanuel Niyigena, and Vesna Krapež. Thank you all for the camaraderie, the countless funny and memorable moments, and for your moral and research support during these past four years.

My sincere appreciation extends to my collaborators at UZ Leuven, including Prof. Dr. Marc Ferrante, Prof. Dr. Bram Verstockt, and Prof. Dr. João Sabino; colleagues from the MODIFI project including Ine De Greef, Sophie Tops, Melissa Nigro, An Outtier, Dr. Debby Thomas, Griet Compernelle, and the lab technicians at Translational Research in Gastrointestinal Disorders Unit; as well as collaborators from Ghent University, including Prof. Dr. Jo Lambert, Dr. Rani Soenen, Dr. Lisa Schots, and all my co-authors. Your collaborative spirit and expertise made this thesis possible.

A special thanks goes to Celltrion for generously providing the clinical data essential to my research and for sponsoring my PhD defense. I am also thankful to R-Biopharm AG for their support in the IGNITE trial.

To my PhD and postdoctoral colleagues, lab technicians, and everyone with whom I shared office space in the Laboratory of Therapeutic and Diagnostic Antibodies, PharmAbs, and the Clinical Pharmacology and Pharmacotherapy Unit, thank you for creating a welcoming and enjoyable work environment.

Finally, I extend my deepest gratitude to my family for their unwavering support, love, and belief in me—not only throughout this journey but in every step of my life.

Lastly, I want to thank myself for the dedication and hard work over the past four years, leading to the completion of this doctoral thesis. “*Omnia mirari etiam tritissima*” – Find wonder in everything, even the most commonplace. This motto has guided me through this journey and will continue to do so in the future.

Abstract

Background: The dictum of “one dose fits all” has been shown problematic in patients with inflammatory bowel diseases (IBD) starting monoclonal antibody (mAb) therapies. Over a decade ago, personalized medicine was introduced to infliximab treatment, offering tailored therapeutic recommendations through therapeutic drug monitoring (TDM). TDM-guided dosing of therapeutic mAbs typically relies on flowcharts or decision trees, which are intuitive and straightforward for clinical use, but have been criticized for being imprecise and inefficient for dose optimization. Model-informed precision dosing (MIPD) typically uses population pharmacokinetics (popPK) models, patient-specific monitoring data, and a custom-made MIPD software tool to predict the optimal dose or dosing interval for attaining a desired drug concentration in a patient. However, whether MIPD can improve treatment outcomes in patients with IBD remains unclear.

Aims: The aim of this doctoral research project was to develop popPK-pharmacodynamics (popPK-PD) models to characterize dose-exposure-(biomarker)-response relationships of mAb therapies and investigate the potential benefits of MIPD of mAbs therapies in patients with Crohn’s disease (CD) and ulcerative colitis (UC).

Methods: PopPK-PD modeling and simulation were performed based on data from patients with IBD receiving ustekinumab therapy (*chapter I*) and subcutaneous (SC) infliximab therapy (*chapter III*). We also performed popPK simulations based on published popPK models of SC infliximab and vedolizumab to explore the exposure profiles following various label and off-label dosing scenarios of intravenous (IV)-to-SC transitions (*chapter II*). Bayesian forecasting based on measured infliximab and ustekinumab concentrations, covariates, and popPK models was performed to investigate the benefits of clearance monitoring of infliximab and ustekinumab in patients with CD (*chapter IV*). Lastly, we conducted the monocentric, open-label, historically controlled Phase 4 MODIFI trial comparing infliximab interval extension based on TDM with interval extension guided by MIPD in patients with IBD who sustained treatment response following an earlier interval shortening (*chapter V*).

Results: For patients with CD receiving ustekinumab, a popPK-PD model consisting of three sequential models was developed: a two-compartment popPK model linking ustekinumab dose to exposure, a popPK-PD model describing the effect of ustekinumab exposure on fecal calprotectin (FC), and a logistic regression outcome model linking FC to endoscopic outcomes.

Results from popPK simulations of SC infliximab indicated that patients on IV infliximab every six weeks (Q6W) or Q8W can transition to the standard 120 mg SC infliximab injections Q2W without a significant reduction in trough concentrations at steady state. Additionally, we developed a clinical support tool for TDM of infliximab during the IV-to-SC switch. Simulations with our popPK-PD model of SC infliximab demonstrated that standard dosing of SC infliximab provides an enhanced FC response compared to both standard and escalated IV dosing up to Q6W. Patients on Q4W IV infliximab require higher SC dosing to avoid an FC increase.

We also demonstrated that *Bayesian* forecasted clearance of infliximab and ustekinumab is a better predictor than trough concentrations for endoscopic outcomes in patients with CD, with a discriminatory ability statistically similar to the current gold-standard biomarker FC. Additionally, we developed an open-access software tool for clearance monitoring of infliximab and ustekinumab.

Lastly, the MODIFI study enrolled 21 adult patients in the TDM arm (11 CD, 10 UC) and 31 in the MIPD arm (19 CD, 12 UC). MIPD achieved a numerically higher rate of sustained steroid-free clinical and biological remission (52% vs. 29%, $p=0.17$) and a significantly higher rate of sustained steroid-free clinical remission (81% vs. 38%, $p<0.01$) over one year compared to TDM.

Conclusions: The aims of this doctoral research were successfully achieved. Key accomplishments include developing popPK-PD models of infliximab and ustekinumab for MIPD, identifying PK and biomarker targets that correlate with therapeutic outcomes, translating scientific findings into user-friendly MIPD software tools, validating MIPD through prospective clinical trials, and establishing guidelines for effective implementation in clinical practice. These contributions significantly enhance the integration of MIPD into the management of IBD, facilitating improved patient care and treatment outcomes.

List of abbreviations

5-ASA	5-aminosalicylates
ADA	Anti-drug antibody
AE	Adverse events
ADME	Absorption, distribution, metabolism, excretion
AGA	American Gastroenterological Association
AI	Artificial intelligence
AIC	Akaike information criterion
anti-TNF α	Anti-tumor necrosis factor-alpha
ASCA	Anti- <i>Saccharomyces cerevisiae</i>
ATI	Antibodies towards infliximab
AUC	Area under the concentration–time curve
AUROC	Area under the receiver operating characteristic curve
BMI	Body mass index
BSA	Body surface area
C_{avg}	Average concentration
CD	Crohn’s disease
CDAI	Crohn’s Disease Activity Index
CDEIS	Crohn's Disease Endoscopic Index of Severity
CI	Confidence interval
CL	Clearance
C_{peak}	Peak concentration
C_{min}	Minimum concentration
CONC	Concentrations
COVID-19	Coronavirus disease 2019
CRP	C-reactive protein
C_{trough}	Trough concentration
$C_{trough,ss}$	Trough concentration at steady-state
CV	Coefficient of variation
CWRES	Conditional weighted residuals
DBS	Dried blood sampling
df	Degree of freedom
EBEs	Empirical Bayes estimates
EC	Ethics Committee
EHR	Electronic health record
ELISA	Enzyme-linked immunosorbent assay
EMA	European Medicines Agency
EPAR	European Public Assessment Report
ESR	Erythrocyte sedimentation rate
F	Bioavailability
FC	Fecal calprotectin
Fc γ R	Fc gamma receptor
FcRn	neonatal Fc receptor
FREM	Full random effects model
GDPR	General Data Protection Regulation
GI	Gastrointestinal

GP	General practitioner
HBI	Harvey-Bradshaw Index
IBD	Inflammatory bowel disease
IBDQ-32	32-Item Inflammatory Bowel Disease Questionnaire
Ig	Immunoglobulins
IIV	Interindividual variability
IOV	Inter-occasion variability
iPCQ	The Institute for Medical Technology Assessment Productivity Cost Questionnaire
IPDMA	Individual patient data meta-analysis
IPRED	Individual predictions
IQR	Interquartile range
IV	Intravenous
JAK	Janus kinase
K _a	Absorption rate constant
LBW	Lean bodyweight
LLOQ	Lower limit of quantification
LRT	Likelihood ratio test
MAA	model averaging algorithm
mAb	Monoclonal antibody
MAdCAM-1	Mucosal addressin cell adhesion molecule-1
MIPD	Model-informed precision dosing
ML	Machine learning
MSA	model selection algorithm
NAb	Neutralizing antibodies
OFV	Objective function value
P _{max,01}	Maximum probabilities of achieving endoscopic remission
P _{max,10}	Maximum probabilities of losing endoscopic remission
pANCA	perinuclear anti-neutrophil cytoplasmic antibody
pcVPC	prediction-corrected visual predictive check
PD	Pharmacodynamics
PI	Prediction interval
PK	Pharmacokinetics
POCT	Point-of-care testing
PopPD	Population pharmacodynamics
PopPK	Population pharmacokinetics
PRED	Population prediction
PRO	Patient-Reported Outcome
PRO2	Two-item Patient-Reported Outcome
QxW	Every x weeks
rBias	Relative bias
RCT	Randomized controlled trial
ROC	Receiver operating characteristics
rRMSE	Relative mean squared error
RSE	Relative standard error
RUV	Residual unexplained variability
SAE	Serious adverse event
SC	Subcutaneous

SCM	Stepwise covariate modeling
SES-CD	Simple Endoscopic Score for Crohn's disease
SIR	Sampling importance resampling
STRIDE-II	The updated Selecting Therapeutic Targets in Inflammatory Bowel Disease
SS	Steady state
T0	Baseline
TCI	Target concentration intervention
TDM	Therapeutic drug monitoring
TMDD	Target-mediated drug disposition
UC	Ulcerative colitis
ULOQ	Upper limit of quantification
V	Volume of distribution
V_c	Volume of distribution in the central compartment
VAMS	Volumetric absorptive microsampling
V_p	Volume of distribution in the peripheral compartment
VPC	Visual predictive check

General introduction

Parts of the introduction are based on

Wang Z, Dreesen E. Therapeutic drug monitoring of anti-tumor necrosis factor agents: lessons learned and remaining issues. *Curr Opin Pharmacol*. 2020;55:53–9.¹

Kantasiripitak W*, **Wang Z***, Spriet I, Ferrante M, Dreesen E. Recent advances in clearance monitoring of monoclonal antibodies in patients with inflammatory bowel diseases. *Expert Rev Clin Pharmacol*. 2021;14(12):1455-1466.²

Alsoud D*, Moes D*, **Wang Z***, Soenen R, Layegh Z, Barclay M, Mizuno T, Minichmayr I, Keizer R, Wicha S, Wolbink G, Lambert J, Vermeire S, De Vries A, Papamichael K, Padullés-Zamora N, Dreesen E. Best Practice for Therapeutic Drug Monitoring of Infliximab: Position Statement from the International Association of Therapeutic Drug Monitoring and Clinical Toxicology. *Ther Drug Monit*. 2024;46(3):291-308.³

Minichmayr I, Dreesen E, Centanni M, **Wang Z**, Hoffert Y, Friberg L, Wicha S. Model-Informed Precision Dosing: State of the Art and Future Perspectives. *Adv Drug Delivery Reviews*. 2024. Epub ahead of print.⁴

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* Designated shared co-first authorship

1. Inflammatory Bowel Diseases

Inflammatory bowel diseases (IBD), the two main subtypes being Crohn's disease (CD) and ulcerative colitis (UC), are lifelong debilitating diseases that cause chronic inflammation in the gastrointestinal tract.^{6,7} The estimated prevalence of IBD is around 0.3% in Western countries, with a high burden of IBD in North America, Oceania, and Europe.⁸ Although the incidence of these diseases has plateaued in the Western world, newly industrialized countries are experiencing a rise in incidence, similar to the Western world in the late 20th century. The onset of IBD ranges from early childhood to after the age of 60.

In CD, inflammation can develop anywhere along the gastrointestinal tract, most commonly in the terminal ileum and colon (**Figure 1**). It typically appears as patchy, transmural inflammation. In contrast, UC is confined to the colon and rectum, presenting as a continuous area of inflammation beginning at the anal margin and affecting only the innermost lining of the bowel (**Figure 1**). UC often manifests as bloody diarrhea and is diagnosed through colonoscopy and histological analysis, whereas the common symptoms of CD are abdominal pain, chronic diarrhea, weight loss, and fatigue.^{6,7}

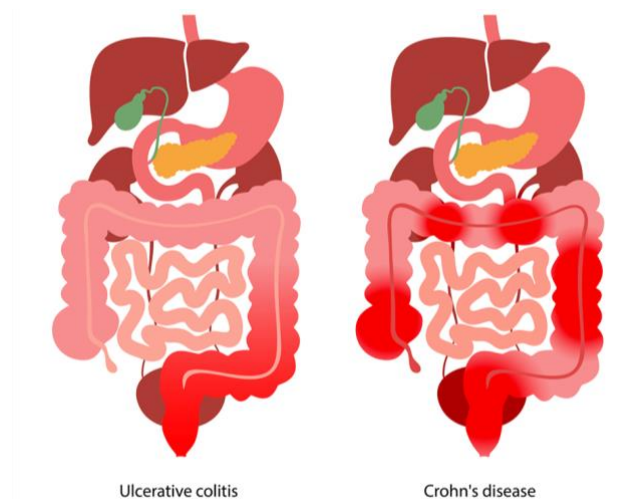


Figure 1. Gastrointestinal areas affected by Crohn's disease (left) and ulcerative colitis (right). Possible affected regions are indicated in red. Figure adapted with permission from <https://www.hopkinsmedicine.org/health/conditions-and-diseases/inflammatory-bowel-disease>.

The diagnosis of IBD involves a comprehensive assessment that includes medical history, physical examination, laboratory tests, and endoscopic findings (**Table 1**).⁹

Table 1. Comparison of diagnosis for ulcerative colitis and Crohn's disease

Characteristics	Crohn's disease	Ulcerative colitis
<i>History</i>		
Location	Anywhere in GI tract	Starts in the rectum and extends to proximal segments of the colon
Abdominal pain	Common	Uncommon
Rectal bleeding	+	++
Non-gastrointestinal symptoms	Common	Uncommon
Fatigue	+++	++
Weight loss	+++	+
<i>Physical examination</i>		
Fever	+++	+
Abdominal tenderness	+++	+
<i>Laboratory</i>		
Anemia	+++	++
Elevated CRP and/or ESR	+++	+
ASCA	++	-
pANCA	+	+++
<i>Endoscopy findings</i>		
Rectal involvement	+/-	+++
Continuous mucosal involvement	+	+++
Granulomas	+++	+

ASCA, anti-Saccharomyces cerevisiae; CRP, C-reactive protein; ESR, erythrocyte sedimentation rate; GI, gastrointestinal; pANCA: perinuclear anti-neutrophil cytoplasmic antibody; +, degree of prevalence; - not prevalent. Table reproduced with permission from Sairenji et al.⁹

1.1. Pathogenesis

IBD is an immune-mediated disease that arises due to a dysregulation of the immune response. The exact cause of IBD remains unknown. It is suspected to be caused by the interplay between heredity, intestinal microflora, and environmental factors, resulting in a malfunctioning mucosal immune response and impaired epithelial barrier function.¹⁰⁻¹² Despite the many similarities in the pathogenesis of CD and UC, essential differences exist. For instance, dysbiosis is more pronounced in patients with CD compared to those with UC.¹³ Genome-wide association studies have identified over 240 genetic loci that increase susceptibility to CD or UC.¹⁴ Besides, the gut microbiome is crucial for maintaining gut homeostasis. Dysbiosis, or alterations in gut microbial composition, is consistently observed in IBD patients.¹⁵ Compared to healthy controls, these patients show reduced microbiome diversity and changes in specific bacterial species.¹⁶ In addition, many environmental factors linked to IBD, such as smoking, diet, and antibiotic use, also impact the microbiome.

1.2. Treatment and management

IBD is incurable. Therefore, pharmacological therapies aim to alleviate symptoms, improve quality of life, and minimize long-term complications such as short bowel syndrome and colorectal cancer.⁹ Conventionally, the clinical management of IBD employs a step-up approach based on disease activity, progressing through a therapeutic pyramid that includes several medication tiers: 5-aminosalicylates (5-ASA), steroids, immunomodulators, biologicals, and, more recently, Janus kinase (JAK) inhibitors. An alternative "top-down" approach using immunosuppressives and/or biologicals as first-line treatments may be used to achieve remission more rapidly, thereby reducing complications and the need for surgery.¹⁷ While CD and UC share diagnostic considerations, their treatment strategies vary significantly.

1.2.1. *Treatment and management of Crohn's disease*

The management of CD aims for deep and long-lasting remission to prevent complications such as surgery and to halt disease progression.⁶ Assessing disease activity is crucial for guiding therapeutic decisions.⁶ To evaluate clinical disease activity, the Crohn's Disease Activity Index (CDAI) is used, categorizing the disease into remission, mild activity, moderate activity, or severe activity (**Table 2**).¹⁸ Alternatively, the Harvey-Bradshaw Index (HBI), a simplified and less time-consuming version of the CDAI, is more practical for everyday clinical use.¹⁹ In addition to clinical scores, biochemical parameters such as C-reactive protein (CRP) and fecal calprotectin (FC) are frequently measured as objective markers of disease activity. Recently, a two-item patient-reported outcome (PRO2) based on stool frequency and abdominal pain has been introduced.²⁰

Clinical disease activity alone may not correlate with objective endoscopic and biological disease activity. To evaluate endoscopic disease activity, the Crohn's Disease Endoscopic Index of Severity (CDEIS) can be used, though it is complex and time-consuming.²¹ The Simple Endoscopic Score for Crohn's Disease (SES-CD), a simplified version of the CDEIS, is more practical for clinical use (**Table 3**).²² Endoscopic remission, defined by the CDEIS, SES-CD, or the absence of ulcers, is a major treatment goal due to its correlation with reduced relapse rates and the need for surgery. Combined clinical and endoscopic remission, known as "deep remission", is currently the treatment goal for patients with IBD.²³

Table 2. Crohn's disease activity index for evaluation of clinical disease activity in patients with Crohn's disease

Item	Weighting factor
Number of liquid or soft stools, each day for 7 days (integer, maximum 2 digits)	x2
Abdominal pain, sum of seven daily ratings (0 = none, 1 = mild, 2 = moderate, 3 = severe)	x5
General well-being, sum of seven daily ratings (0 = generally well, 1 = slightly under par, 2 = poor, 3 = very poor, 4 = terrible)	x7
Number of listed complications (arthritis or arthralgia, iritis or uveitis, erythema nodosum or pyoderma gangrenosum or aphthous stomatitis, anal fissure or fistula or abscess, other fistulae, fever >37.8°C)	x20
Use of diphenoxylate or loperamide for diarrhoea (0 = no, 1 = yes)	x30
Abdominal mass (0 = no, 2 = questionable, 5 = definite)	x10
Hematocrit (male, 47-Hct [%], females 42-Hct [%])	x6
Body weight (1 - weight/standard weight) x 100 (add or subtract according to sign)	x1

Hct: hematocrit. Each item is multiplied by a weighting factor and total scores for all eight items are summed up. Crohn's disease activity index (CDAI) <150: remission; CDAI 150-219: mildly active disease; CDAI 220-450 moderately active disease; CDAI >450 severely active disease. Table reproduced with permission from Best et al.¹⁸

Table 3. Simple Endoscopic Score for Crohn's Disease (SES-CD) for the evaluation of endoscopic disease activity in Crohn's disease

Parameter	Ileum	Right colon	Transverse colon	Left colon	Rectum
Presence and size of ulcers (0 = no, 1 = aphthous (0.1-0.5 cm), 2 = large (0.5-2 cm), 3 = very large (>2 cm))	0-3	0-3	0-3	0-3	0-3
Extent of ulcerated surface (0 = 0%, 1 = <10%, 2 = 10-30%, 3 = >30%)	0-3	0-3	0-3	0-3	0-3
Extent of affected surface (0 = 0%, 1 = <50%, 2 = 50-75%, 3 = >75%)	0-3	0-3	0-3	0-3	0-3
Presence and type of narrowing (0 = no, 1 = single, can be passed, 2 = multiple, can be passed, 3 = cannot be passed)	0-3	0-3	0-3	0-3	0-3

The total scores for all four items are summed up to obtain the Simple Endoscopic Subscore for Crohn's Disease (SES-CD). SES-CD 0-2: endoscopic remission, SES-CD 3-6: mild endoscopic activity, SES-CD: 7-15 moderate endoscopic activity, SES-CD > 15: severe endoscopic activity. Table reproduced with permission from Daperno et al.²²

A rapid step-up approach is recommended for the treatment of patients with CD based on endoscopic disease activity assessment and the evaluation of risk factors, such as severe upper gastrointestinal disease, rectal disease, perianal lesions, structuring or penetrating disease, and smoking.^{6,24} However, this approach has not significantly altered the disease course, as evidenced by the high rate of surgery.⁶ Therefore, a top-down approach may be considered for patients with poor prognostic factors, severe disease, or complicated disease.²⁵ Additionally, a 'treat-to-target' treatment algorithm that combines the rapid step-up and tight monitoring approaches was proposed (Figure 2).^{6,26}

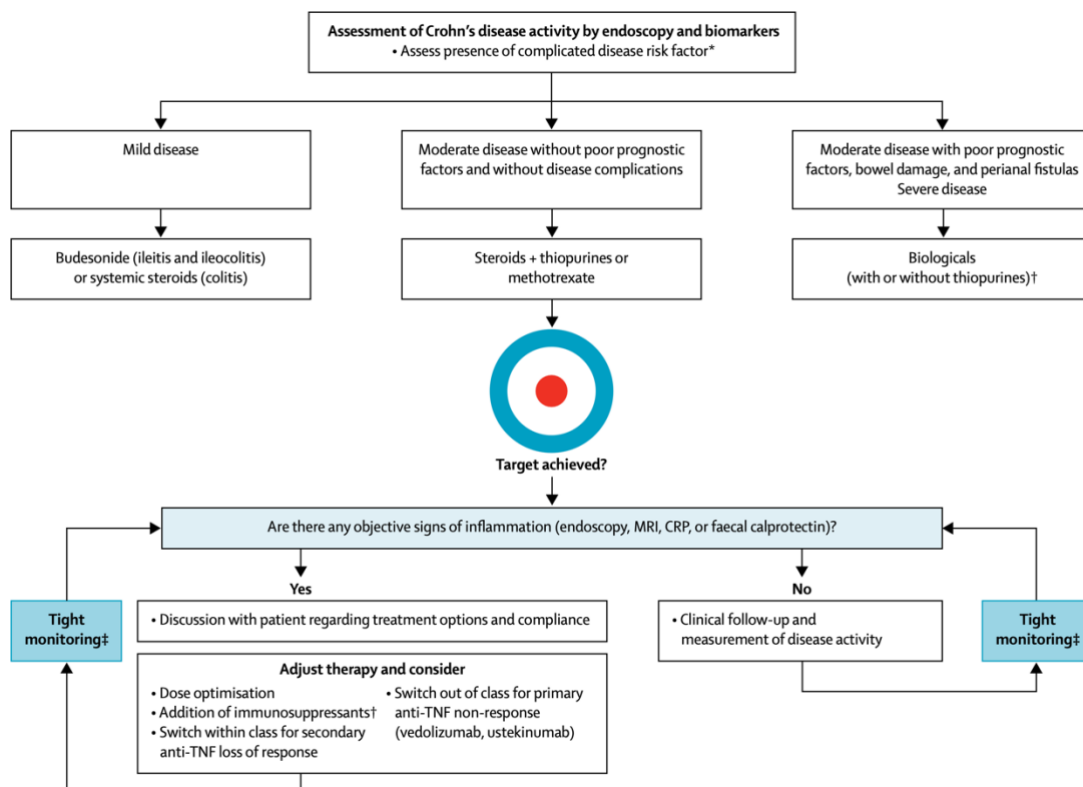


Figure 2. Proposed algorithm for the treatment of patients with Crohn's disease based on a rapid step-up approach and tight monitoring. * Poor prognostic factors include extensive small bowel disease, severe upper gastrointestinal disease, rectal disease, perianal lesions, early stricturing or penetrating disease, smoking and young age at diagnosis, and severe endoscopic lesions. † Consider anti-TNF α monotherapy in patients at high risk of adverse events, including patients older than 65 years, with a history of malignant disease, or male and younger than 35 years. ‡ Suggested interval assessment for tight monitoring: clinical assessment every 3 months for patients with active disease (every 6-12 months for patients in remission), ileocolonoscopy (or cross-sectional imaging in patients who cannot be adequately assessed with ileocolonoscopy) every 6-9 months for patients with active disease, and biomarkers (CRP/faecal calprotectin) every 3-6 months. CRP, C-reactive protein; MRI, magnetic resonance imaging; TNF α , tumor necrosis factor alpha. Figure reproduced with permission from Torres et al.⁶

1.2.2. Treatment and management of ulcerative colitis

The primary aim of UC management is to induce and maintain remission, defined as the resolution of symptoms and endoscopic healing.⁷ Long-term goals include preventing disability, colectomy, and colorectal cancer.⁷ Treatment choices are based on the activity of the disease (mild, moderate, or severe) and the extent of colonic involvement.⁹ Various indices measure UC activity, with the Mayo score being a commonly used tool (**Table 4**).²⁷ When endoscopic evaluation is not available, a partial Mayo score can be calculated using the three non-invasive parameters.²⁸ PROs are increasingly used to monitor disease activity. Many PROs have been developed and evaluated for their ability to determine disease activity, but their accuracy and applicability in clinical trials and routine practice need further validation.²⁹ Treatment options for UC include 5-ASA, corticosteroids, immunomodulators, biological drugs, and JAK inhibitors. A rapid step-up approach is recommended, tailored to disease activity and treatment response.⁷ Early use of biological drugs is advised for patients hospitalized with acute severe UC or those with steroid-refractory UC. Recently, there have been efforts to improve treatment outcomes of UC through combinational biological therapies.^{30,31} Once remission is induced, medications can be continued or combined to maintain remission.

Table 4. Mayo score for evaluation of the ulcerative colitis clinical disease activity

Item	Score
<i>Stool frequency</i>	
Normal for patients	0
1-2: 3-4: 5 or more times more than normal	1:2:3
<i>Rectal bleeding</i>	
No blood	0
Streaks of blood in stool less than half of the time	1
Obvious blood in stool most of the time	2
Blood alone passed	3
<i>Endoscopy findings *</i>	
Normal or inactive disease	0
Mild (erythema, decreased vascular pattern, mild friability)	1
Moderate (marked erythema, absent vascular pattern, friability, erosions)	2
Severe (spontaneous bleeding, ulceration)	3
<i>Physician's global assessment</i>	
Normal : Mild : Moderate: Severe disease	0 : 1 : 2 : 3

*The score of each item is summed up to obtain the Mayo score. * Also commonly referred as Mayo endoscopic subscore. Table reproduced with permission from Dignass et al.²⁷ Regulatory agencies now recommend the Mayo score, without Physician's global assessment.*

1.3. Therapeutic target

Over the past decade, IBD management has evolved from focusing solely on symptom control to adopting a "treat-to-target" paradigm that emphasizes endoscopic healing and patient-centric approaches.^{32–34} The updated Selecting Therapeutic Targets in Inflammatory Bowel Disease (STRIDE-II) guidelines have introduced both short-term and long-term treatment targets for IBD.³⁵ For the first time, quality of life and the avoidance of disability have been identified as formal targets in IBD management. The STRIDE-II consensus established a timeline for IBD management targets, progressing from disease control to disease modification (**Figure 3**).³⁵ The timeline aims for achieving clinical remission and CRP normalization in the short term to enable patients to quickly return to normal life.³⁵ In the medium to long term, the goal should be to achieve endoscopic remission and prevent disease complications, surgery, and hospitalizations.³⁵

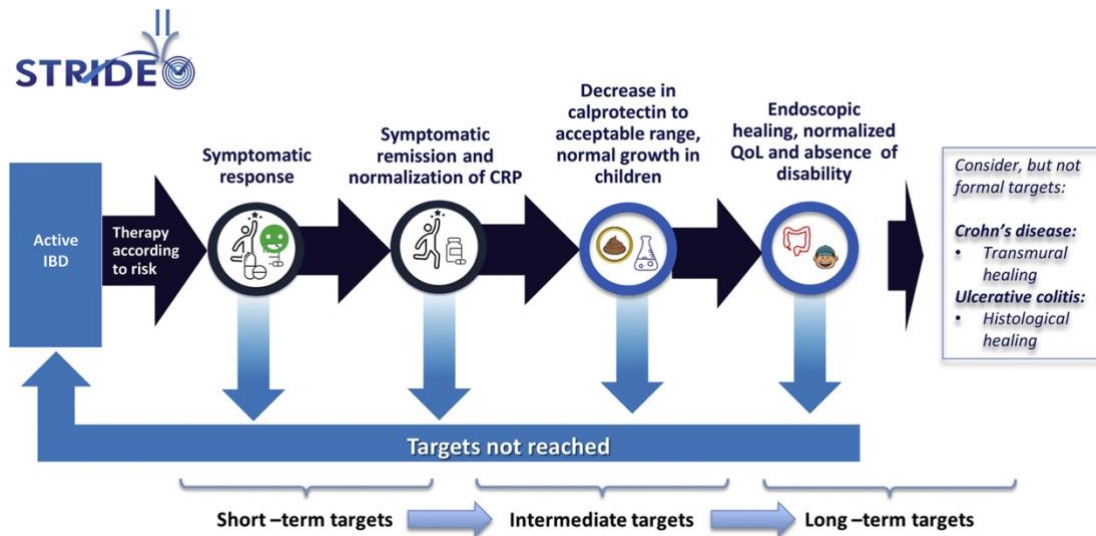


Figure 3. Summary of evolving short-term and long-term treatment targets in inflammatory bowel diseases (IBD). CRP, C-reactive protein; QoL, quality of life. Figure reproduced with permission from Turner et al.³⁵

2. Monoclonal antibodies as treatment for inflammatory bowel diseases

2.1. Antibody structure and types

Therapeutic monoclonal antibodies (mAbs), also known as immunoglobulins (Ig), are large Y-shaped glycoproteins secreted by B-cells that can recognize and bind to specific antigens. They are composed of two light chains and two heavy chains connected by disulfide bonds, forming a constant (Fc) region and an antigen-binding domain (Fab) (**Figure 4 Left**). Each Fab domain includes three hypervariable complementarity determining regions that are essential for specific antigen binding. The Fc domain of the heavy chain mediates antibody effector functions and significantly influences the antibody's *in vivo* half-life.³⁶ Human antibodies are categorized into five isotypes: IgG, IgM, IgA, IgE, and IgD, based on the heavy chain. IgG molecules feature γ -chains, IgM molecules contain μ -chains, IgA molecules are characterized by α -chains, IgE molecules have ϵ -chains, and IgD molecules include δ -chains. According to generation methodology, mAbs can be classified into murine (suffix -omab), chimeric (suffix -ximab), humanized (suffix -zumab), or fully human (suffix -umab) (**Figure 4 Right**).

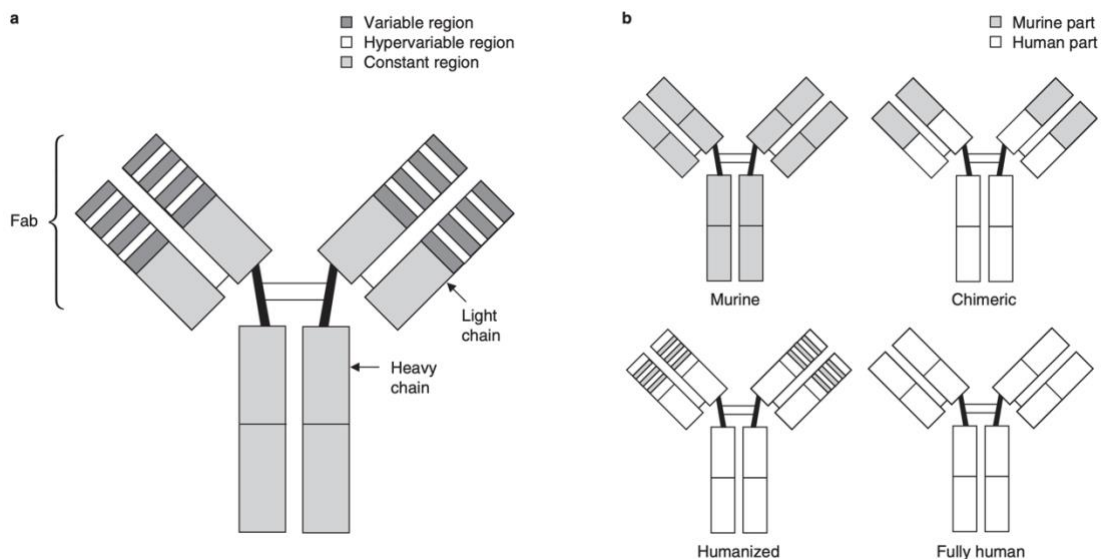


Figure 4. Schematic structure of immunoglobulins (left); overview of different origin of monoclonal antibodies (right). Figure reproduced with permission from Keizer et al.³⁷

2.2. Pharmacology of monoclonal antibodies

2.2.1. Mechanism of action of monoclonal antibodies

Anti-tumor necrosis factor-alpha (anti-TNF α) mAbs used in the treatment of IBD include Infliximab, adalimumab, golimumab, and certolizumab pegol. Anti-TNF α mAbs achieve their effects by binding to both the soluble and transmembrane forms of TNF α .³⁸ This binding disrupts the interactions between TNF α and its cell surface receptors, TNFR1 (p55) and TNFR2 (p75), thereby neutralizing the pro-inflammatory effects of TNF α on target cells.³⁸ Consequently, immune-mediated inflammation is reduced, which can lead to mucosal healing and potentially alter the natural course of IBD.³⁹

Anti- $\alpha_4\beta_7$ -integrin mAbs used in the treatment of IBD include vedolizumab and natalizumab. Anti-integrin mAbs achieve their effects by selectively blocking the trafficking of memory T cells to inflamed gut tissue.⁴⁰ This is accomplished by inhibiting the interaction between $\alpha_4\beta_7$ -integrin and mucosal addressin cell adhesion molecule-1 (MAdCAM-1).⁴⁰ The $\alpha_4\beta_7$ -MAdCAM-1 interaction mediates lymphocyte trafficking to the gut and plays a central role in the pathology of IBD.⁴¹ Natalizumab binds to the α_4 integrin, affecting both $\alpha_4\beta_7$ and $\alpha_4\beta_1$ integrins, while vedolizumab specifically targets the $\alpha_4\beta_7$ integrin.

Anti-interleukin (IL)-12/23 mAbs used in the treatment of IBD include ustekinumab. Ustekinumab works by reducing the signaling pathways of the pro-inflammatory cytokines IL-23 and IL-12, and further downregulates the gene expression of inflammatory cytokines and chemokines such as monocyte chemoattractant protein-1, TNF α , IP-10, and IL-8.⁴² IL-12 and IL-23 are heterodimeric cytokines that trigger immune and inflammatory responses, including natural killer cell activation and CD4+ T-cell differentiation and activation.⁴² The roles of IL-12 and IL-23 have been implicated in a variety of chronic inflammatory conditions, such as IBD and psoriasis.⁴²

IL-23 mAbs used in the treatment of IBD include risankizumab and mirikizumab. These agents work by reducing the signaling pathways of the pro-inflammatory cytokines IL-23.

2.2.3. Pharmacokinetics of monoclonal antibodies

Administration route and absorption

Monoclonal antibodies used in the treatment of IBD are administered parenterally, either via the intravenous (IV) route or via the subcutaneous (SC) route. Intravenous administration allows for the infusion of large volumes, resulting in an immediate peak concentration and complete systemic bioavailability ($F=100\%$). In contrast, SC injection requires smaller volumes (maximum 1 mL) and leads to slower and incomplete systemic bioavailability. The bioavailability, as indicated in the prescribing information for currently approved SC mAbs in IBD, includes adalimumab at 64%, certolizumab pegol at 80%, golimumab at 53%, infliximab at 58%, and ustekinumab at 57%.^{37,43,44} The primary pathway for systemic absorption from the subcutaneous compartment is convective transport through lymphatic vessels, although diffusion into blood vessels also plays a role.⁴⁵

Distribution

Due to their size, mAbs have a very limited ability to diffuse from the blood compartment into peripheral tissues. Extravasation likely occurs mainly through convection (engulfment in the bloodstream) or via endocytosis or pinocytosis into endothelial cells.^{46,47} The rate of convective transport into tissue interstitial fluids depends on the net rate of fluid movement and the nature of the paracellular sieving pores.⁴⁸ Antibodies have a volume of distribution of 30-80 mL/kg body weight.⁴⁹

Metabolism

The primary route of mAb elimination is through intracellular enzymatic degradation following receptor-mediated, target-mediated, or fluid-phase endocytosis, and anti-drug antibody (ADA)-mediated metabolism.

Receptor-mediated endocytosis: The Fc part of the antibody binds to Fc gamma receptor (FcγR) I, II, and III on reticuloendothelial cells, leading to internalization and subsequent degradation by lysosomes. This nonspecific elimination route is not easily saturable by mAbs, thereby exhibiting linear elimination kinetics.⁴⁸

Target-mediated endocytosis: this elimination route, *i.e.* target-mediated drug disposition (TMDD), involves degradation within the target cell after the target antigen binding to the Fab domain of mAbs.⁵⁰ This binding leads to internalization and lysosomal degradation. TMDD is saturable, resulting in non-linear elimination.

Fluid-phase pinocytosis: Elimination can occur through intracellular catabolism by lysosomal degradation to amino acids following fluid-phase endocytosis or pinocytosis (cell drinking).⁵⁰ This nonspecific fluid uptake is not preceded by specific binding to plasma membrane receptors. Upon pinocytosis, the endosome acidifies, increasing the affinity of IgG for the neonatal Fc receptor (FcRn) or Brambell receptor, followed by recycling of the FcRn-IgG complex to the cell surface where IgG is released.

ADA-mediated metabolism: mAbs may elicit a humoral immune response that can alter pharmacokinetics (PK) and thus result in a therapeutic response. Although therapeutic mAbs are designed to reduce their immunogenic potential, complete elimination of immunogenicity is not possible. ADAs can lead to enhanced uptake in the reticuloendothelial system, resulting in increased drug elimination.⁵¹ Conversely, they might act as a storage depot, decreasing drug elimination and leading to more sustained drug concentrations.⁵² ADAs can greatly impact mAbs' activity by binding to the antigen-binding site of the drug (neutralizing antibodies) or by affecting the clearance of the drug (both neutralizing and non-neutralizing antibodies).

Excretion

Hepatic and renal elimination pathways are not significant for IgG mAbs. High IBD disease activity, such as in patients with acute severe UC, can impair intestinal barrier function and consequently leading to fecal loss of proteins (*i.e.*, protein-losing enteropathy or leaky gut).⁵³ This can result in increased elimination of mAbs, lower drug exposure, and consequently, an insufficient clinical response.⁵³

2.3. Monoclonal antibodies used in the treatment of inflammatory bowel diseases

In 1999, the European Medicines Agency (EMA) approved infliximab for the treatment of severe, active CD or fistulizing CD in patients who have not responded to a full and adequate course of conventional treatment, such as corticosteroids and/or immunosuppressants.⁵⁴ Since then, emerging anti-TNFs (adalimumab, golimumab, and certolizumab pegol), anti-integrins (vedolizumab and natalizumab), and anti-interleukins (ustekinumab, risankizumab and mirikizumab) have made endoscopic remission a feasible treatment goal.³⁵ The various biologics approved for the treatment of IBD are summarized in **Table 5**. In this thesis, the focus will be on infliximab, vedolizumab and ustekinumab. The registered dosing regimens of infliximab, vedolizumab, and ustekinumab are represented in **Figure 5**.

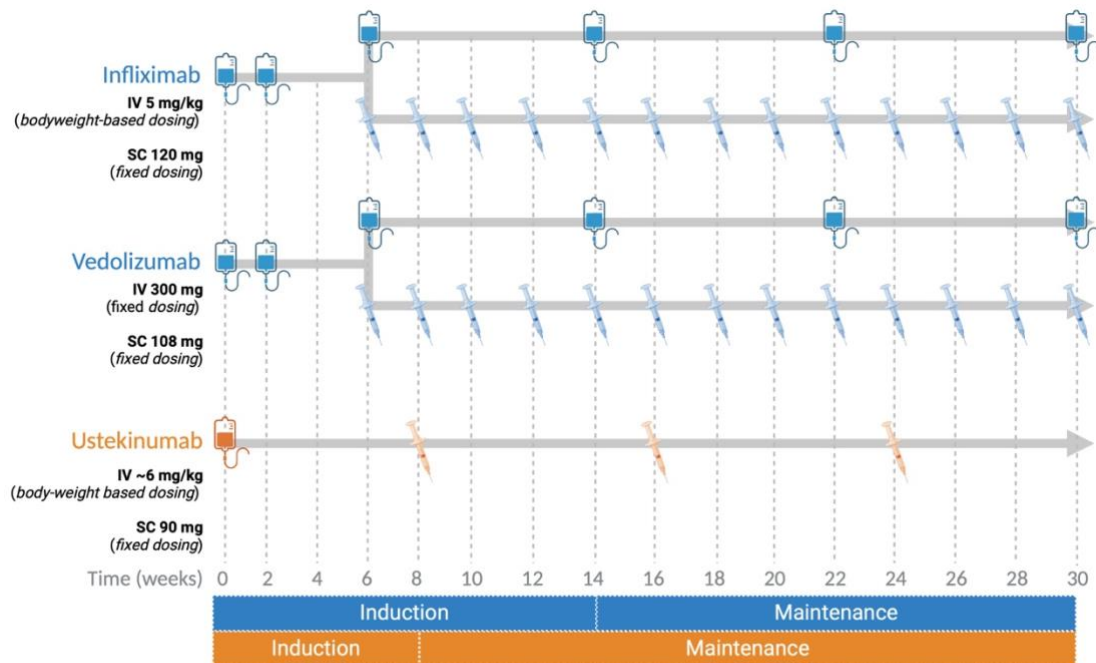


Figure 5. Registered dosing regimens of infliximab, vedolizumab and ustekinumab. IV, intravenous; mg/kg, milligram drug per kilogram bodyweight; SC, subcutaneous.

Table 5. Overview of the approved biologicals in the treatment of inflammatory bowel diseases

INN	Brand name RMP	Target	Indication	Route of administration	Date of MA
Anti-TNF					
Infliximab	Remicade	TNF	CD/UC	IV or SC or IV + SC	1999/2006 2020
Adalimumab	Humira	TNF	CD/UC	SC	2007/2012
Golimumab	Simponi	TNF	UC	SC	2013
Certolizumab pegol	Cimzia	TNF	CD in Switzerland and the USA	SC	2007/2008
Anti-integrin					
Vedolizumab	Entyvio	$\alpha_4\beta_7$	CD/UC	IV or IV + SC	2014 (IV), 2020 (SC)
Natalizumab	Tysabri	$\alpha_4\beta_1$ and $\alpha_4\beta_7$	CD in the USA	IV	2008
Anti-IL-12/23					
Ustekinumab	Stelara	p40 of IL-12/23	CD/UC	IV + SC	2016
Risankizumab	Skyrizi	p19 of IL-23	CD/UC	SC	2022
Mirikizumab	Omvoh	p19 of IL-23	UC	IV + SC	2023

CD, Crohn's disease; INN, international non-proprietary name; RMP, reference medicinal product; MA, marketing authorization; TNF, tumor necrosis factor; IL, interleukin; IV, intravenous; SC, subcutaneous; UC, ulcerative colitis; USA, United States of America.

3. Therapeutic drug monitoring in inflammatory bowel diseases

The introduction of biological therapies has revolutionized the treatment of IBD and led to the evolution of the treatment target from symptomatic disease control to endoscopic remission.³⁵ Nevertheless, the effectiveness of mAb therapies is highly variable. Approximately 10-30% of patients show no response (primary non-response), and up to 50% experience secondary loss of response or immunogenicity-induced reactions, seemingly facing a so-called therapeutic ceiling.⁵⁵⁻⁵⁸ Both primary non-response and loss of response can be attributed to either PK issues (*e.g.*, increased clearance due to immunogenicity) or pharmacodynamic (PD) issues (*e.g.*, mechanistic failure, which requires switching out of class to a drug with another molecular target).^{59,60} The unsatisfactory primary non-response and loss of response rates highlight the need for dose optimization.

In the evolving landscape of clinical pharmacology, personalized dosing strategies have gained prominence, aiming to optimize therapeutic outcomes while minimizing adverse effects. These dosing regimens are founded on PK principles (characterizing the concentration-time course of a drug in the human body), as well as on PD criteria (describing the effect following a specific drug exposure). In general, three strategies are pursued: flat dosing presents a one-size-fits-all approach, simplifying administration but overlooking potentially relevant patient-specific variables. Nonetheless, this approach is often successfully applied for drugs with a wide therapeutic window (*i.e.*, the exposure range within which a drug is efficacious and safe) or small PK/PD variability (*i.e.*, patients achieve similar drug exposure and response following the same dose). Conversely, drugs with narrower therapeutic windows and/or high PK/PD variability require dose adjustment strategies. Covariate-based dosing, considering individual patient characteristics such as bodyweight, aims for a more personalized therapy. With the goal of attaining even more accurate exposure or response, dosing with feedback through therapeutic drug monitoring (TDM) is often performed.

TDM refers to the clinical practice of measuring the concentration of a drug or its metabolites to guide individualized dosing, aiming at a concentration target that is associated with a desired treatment outcome.¹ This approach is based on the clinical pharmacology principle of the exposure-response relationship, thereby supporting the use of PK as a surrogate for PD.

3.1. Pharmacokinetic monitoring

TDM based on PK monitoring implies the measurement of serum drug concentrations. Numerous studies on exposure-response relationships have shown that higher mAb concentrations are associated with improved therapeutic outcomes.⁶¹ However, optimal cut-offs for mAb concentrations remain uncertain and are dependent on the treatment goals.⁶² Stricter outcomes like endoscopic remission typically require higher mAbs concentrations.⁶³ Despite risks of *e.g.*, infection and immunogenicity, mAbs have a good safety profile.⁶⁴ TDM of biologicals therefore usually aims for a minimally effective concentration without an upper exposure bound for toxicity. Yet, 'financial toxicity' may be considered, especially for the more recently approved mAbs, for which no biosimilars are currently available.

The American Gastroenterological Association (AGA) Institute Guideline recommends trough concentration (C_{trough}) targets of 5.0 mg/L, 7.5 mg/L, and 20.0 mg/L for infliximab, adalimumab, and certolizumab pegol, respectively, for maintaining disease remission.⁶⁵ Nevertheless, the C_{trough} targets recommended by AGA were based on low evidence and optimal C_{trough} targets during induction therapy are still unclear. For more recently approved mAbs including vedolizumab and ustekinumab, limited but emerging data support the presence of exposure-response relationships.^{66–71} Vedolizumab dosing was designed to maintain a C_{trough} above 10.0 mg/L to avoid TMDD at low concentrations.^{41,72} For ustekinumab, a C_{trough} target of 2.0–5.0 mg/L during maintenance therapy was associated with improved endoscopic response.^{73,74} Interestingly, a study reported a 46% endoscopic remission rate at week 24 with ustekinumab peak concentration (C_{max}) over 105 mg/mL, versus 7% with C_{max} under 88 mg/mL, suggesting benefits of measuring C_{max} during induction.⁷⁵

By far, most of the exposure-response analyses in IBD associate C_{trough} with treatment outcomes. However, there is a growing discussion around which exposure metric (*e.g.*, C_{trough} , average concentration, accumulative area under the concentration-time curve [AUC]) primarily drives the treatment response in IBD.⁷⁶ Infliximab AUC has been associated with risk of infection and predicted endoscopic remission.^{77,78} Besides, the drug development assessment report for SC infliximab indicates that dosing is aimed not only to exceed a C_{trough} of 5.0 mg/L but also to establish an AUC value closely aligning with that of standard 8-weekly 5 mg/kg IV dosing, hinting at the importance of an AUC target.⁴⁴

3.2. Pharmacodynamic monitoring

PD monitoring involves measuring biomarkers. Inflammatory markers have been extensively studied as potential biomarkers for assessing therapeutic response. CRP in blood and FC in feces are widely used surrogate biomarkers to monitor and assess the therapeutic effect of mAb therapies in patients with IBD.^{79–81}

CRP is a non-IBD-specific acute-phase protein produced by hepatocytes that rapidly increases in response to inflammation. Under normal physiological conditions, plasma CRP levels are low (<1 mg/L), but they can rise 100-fold during inflammation. Although elevated CRP is not specific to intestinal inflammation, it better reflects disease activity in patients with CD compared to those with UC, likely due to the transmural inflammation characteristic of CD. A CRP level below 5 mg/L is typically used to define biological remission.

FC is a protein detectable in stool that correlates with increased neutrophil infiltration in the gut. Unlike CRP, FC is a specific marker for intestinal inflammation. In CD patients, FC levels correlate well with the CDEIS and the SES-CD, making it useful for monitoring response to therapy.⁸² Thresholds for endoscopic healing are generally set at 50–250 mg/kg.^{82,83}

Serum albumin, another acute-phase protein, has been commonly used to assess IBD disease activity and is often identified as an influential covariate on mAb clearance in population PK (popPK) models.⁸⁴

3.3. State of the art and evidence from prospective clinical trials

Evidence from prospective TDM trials is mainly focused on anti-TNFs, while no prospective evidence is available for anti-integrins and anti-interleukins. Given the chronic nature of IBD, TDM can either be proactive or reactive, depending on whether it is performed pre-emptively regardless of disease activity, or reactively in response to the presence of disease flares.¹

Anti-TNFs

Seven randomized controlled trials (RCTs) have so far investigated TDM for anti-TNF therapies in patients with IBD, comprising five studies on infliximab and two studies on adalimumab (**Table 6**).^{85–91} The first RCT by Steenholdt *et al.* showed that reactive TDM during infliximab maintenance therapy reduced treatment cost, albeit without improving clinical efficacy.⁸⁵ However, sequent trials such as the TAXIT (“Trough level adapted infliximab treatment”; 2011-002061-38) and TAILORIX (“a randomized controlled trial investigating tailored treatment with infliximab for active luminal Crohn’s disease”; 2011-003038-14) trial failed to show a clear benefit of proactive TDM during infliximab maintenance therapy.^{86,87} Evidence for proactive TDM of infliximab during maintenance therapy accumulated late, supported by the Norwegian Drug Monitoring (NOR-DRUM; NCT03074656) B trial.⁸⁹ In the NOR-DRUM B trial, proactive TDM based on infliximab C_{trough} was superior to treatment without TDM for maintaining disease control in patients with chronic immune-mediated inflammatory diseases, including IBD (74% vs. 56%; $p < 0.001$).⁸⁹ Nevertheless, evidence for TDM during infliximab induction therapy is still lacking. A SC formulation of infliximab was recently added to the armamentarium for maintenance treatment in patients with chronic inflammatory disease.⁹² Although future prospective evidence remains wanted, both *in silico* simulations and real-world evidence support the use of TDM in this new clinical setting.^{76,93}

For adalimumab, the Pediatric Crohn's Disease Adalimumab Level-based Optimization Treatment (PAILOT; NCT02256462) trial in pediatric CD patients showed higher clinical remission rates with proactive TDM compared to reactive TDM after adalimumab induction therapy (82% vs. 48%; $p = 0.002$).⁹⁰ Although the AGA guidelines recommended reactive TDM in 2017, no recommendation was made for proactive TDM due to uncertainty regarding its

benefits and financial concerns.⁶⁵ The impact of recent proactive TDM trials (NOR-DRUM B, PAILOT) on the clinical application of proactive TDM approaches remains to be determined.

Anti-integrins

To date, there has not been any prospective evidence supporting TDM of anti-integrins (vedolizumab and natalizumab). Most studies on exposure-efficacy of anti-integrins focus on vedolizumab.⁹⁴ A prospective observational cohort study indicated that a vedolizumab $C_{trough} > 18.0$ mg/L at week 6 predicted endoscopic remission within one year.⁶⁸ The LOVE-CD trial ("Study to evaluate efficacy of early versus late use of vedolizumab in Crohn's disease"; NCT02646683) showed that a vedolizumab $C_{trough} > 10.0$ mg/L at week 22 correlated with endoscopic remission at week 26.⁶⁹ Despite numerous retrospective studies linking C_{trough} with endoscopic outcomes, a consensus on an optimal vedolizumab concentration target remains elusive.^{70,71}

Anti-interleukins

No prospective evidence has supported TDM of anti-integrins (ustekinumab, risankizumab and mirikizumab) yet. Emerging data on ustekinumab show an exposure-response relationship, while information on the recently approved risankizumab and mirikizumab is still limited. Posthoc analysis of clinical trials suggested an ustekinumab C_{trough} of 3.3 mg/mL at week 8 and 0.8-1.4 mg/mL during maintenance therapy for clinical remission.⁹⁵ For endoscopic response, a higher C_{trough} of 2.0-5.0 mg/L was needed.^{73,74} In the recent STARDUST RCT ("Study of treat to target versus routine care maintenance strategies in Crohn's disease patients treated with Ustekinumab"; NCT03107793), the treat-to-target method based on a week-16 endoscopic evaluation did not prove superior in enhancing clinical or endoscopic outcomes.⁹⁶

In summary, limited prospective studies explore TDM's effectiveness for optimizing biologic therapies in IBD, primarily focusing on anti-TNFs. Traditional TDM, while straightforward, suffers from issues like delayed intervention, the limited surrogate value of single drug concentrations, and imprecise TDM algorithms for dose adjustments (*i.e.*, analog flowcharts or decision trees used in the TDM trials). These shortcomings have led to the adoption of more elaborated strategies such as target concentration intervention (TCI), which employs

specific target concentrations, applies pharmacological principles for dose computation, and guides individualized interventions in a patient's therapy.⁹⁷

Table 6. Examples of previously published prospective clinical TDM and MIPD studies monitoring monoclonal antibodies in inflammatory bowel diseases

Drug Reference (trial name)	Population	Study design	PK/PD target (TDM approach)	Outcome parameter	Intervention	Control
TDM studies						
Infliximab Steenholdt <i>et al</i> 2014 ⁸⁵ [PMID: 23878167]	Adult CD patients with secondary failure during maintenance therapy (n=69)	Prospective, randomized, controlled, single-blind, multicenter study	C _{trough} ≥ 0.5 mg/L (TDM)	Patients responding after 12 weeks [†] Mean accumulated cost per patient related to treatment after 12 weeks ^{**}	Reactive TDM: 57.6% Reactive TDM: €6038	No TDM: 52.8% No TDM: €9178
Infliximab Vande Casteele <i>et al</i> 2015 ⁸⁶ (TAXIT) [PMID: 25724455]	Adult IBD patients in clinical response during maintenance therapy (n=263)	Prospective, randomized, controlled, monocenter study	C _{trough} 3.0—7.0 mg/L (TDM)	Patients with sustained clinical and biochemical remission for 1 year [†]	Proactive TDM: 68.8%	No TDM: 65.9%
Infliximab D'Haens <i>et al</i> 2018 ⁸⁷ (TAILORIX). [PMID: 29317275]	Adult CD patients initiating maintenance therapy (n=122)	Prospective, randomized, controlled, double-blind, multicenter study	C _{trough} > 3.0 mg/L (and CDAI < 150 and CRP ≤ 5 mg/L and/or FC ≤ 250 mg/kg) (TDM)	Patients in sustained CS-free clinical remission in w22-54 with absence of ulcers at w54 [†]	Proactive TDM: 30.5%	No TDM: 40.0%
Infliximab Syversen <i>et al</i> 2021 ⁸⁸ (NOR-DRUM A) [PMID: 33944876]	Adult IMID patients initiating induction therapy (n=398)	Prospective, randomized, controlled, multicenter study	C _{trough} ≥ 20.0 mg/L (2 nd infusion) C _{trough} ≥ 15.0 mg/L (3 rd infusion) C _{trough} ≥ 3.0 mg/L (≥ 4 th infusion) (and absence of ATI) (TDM)	Patients in clinical remission at w30 [†]	Proactive TDM: 50.5%	No TDM: 53.0%
Infliximab Syversen <i>et al</i> 2021 ⁸⁹ (NOR-DRUM B) [PMID: 34932077]	Adult IMID patients initiating maintenance therapy (n=454)	Prospective, randomized, controlled, multicenter study	C _{trough} 3.0—8.0 mg/L (TDM)	Patients in sustained disease control without disease worsening ^{**†}	Proactive TDM: 73.6%	No TDM: 55.9% 73.6%

Drug Reference (trial name)	Population	Study design	PK/PD target (TDM approach)	Outcome parameter	Intervention	Control
Adalimumab Assa <i>et al</i> 2019 ⁹⁰ (<i>PAILOT</i>) [PMID: 31194979]	Pediatric CD patients initiating maintenance therapy (n=78)	Prospective, randomized, controlled, multicenter study	C _{trough} ≥5.0 mg/L (TDM)	Patients in sustained CS-free clinical remission in w8-w72**†	Proactive TDM: 81.6%	Reactive TDM: 47.5%
Adalimumab D'Haens <i>et al</i> 2022 ⁹¹ (<i>SERENE CD</i>) [PMID: 35122766]	Adult CD patients initiating maintenance therapy (n=184)	Prospective, randomized, controlled, double-blind, multicenter study	C _{trough} 5.0-20.0 mg/L (and CDAI<220, CRP<10 mg/L) (TDM)	Patients in clinical remission at weeks 56 patients in endoscopic response at weeks 56 patients in endoscopic remission at weeks 56	Proactive TDM: 55.6% 25.0% 19.4%	No TDM: 53.8% 34.6% 23.1%
MIPD studies						
Infliximab Strik <i>et al</i> 2021 ⁹⁸ (<i>PRECISION</i>) [PMID: 33290108]	Adult IBD patients in clinical remission during maintenance therapy (n=80)	Prospective, randomized, controlled, multicenter study	C _{trough} ≥3.0 mg/L (MIPD)	Patients in sustained clinical remission after 1 year**†	MIPD: 87.5%	No TDM: 64.1%
Infliximab Santacana Juncosa <i>et al</i> 2021 ⁹⁹ [PMID: 32495380]	Adult IBD patients during maintenance therapy (n=108)	Prospective, monocenter study (control group: retrospective)	C _{trough} ≥3.0 mg/L (dose de-escalation possible when >7.0 mg/L) (MIPD)	Patients in clinical remission before and at least 14 weeks after last dose adjustments**†	After MIPD: 80.4% (all patients) 61.8% (dose-intensified patients)	Before MIPD: 65.7% (all patients) 35.3% (dose-intensified patients)

Explanations and abbreviations: *Statistical significance (if reported); †primary endpoint; ATI, antibodies towards infliximab; CD, Crohn's disease; C_{trough}, trough concentration; CDAI, Crohn's Disease Activity Index; CRP, C-reactive protein; CS, corticosteroid; FC, fecal calprotectin; IBD, inflammatory bowel diseases; IMID, immune-mediated inflammatory diseases; MIPD, model-informed precision dosing, PD, pharmacodynamic; PK, pharmacokinetic; TDM, therapeutic drug monitoring; w, week. Table reproduced with permission from Minichmayr *et al.*⁴

4. Pharmacometrics

Pharmacometrics integrates mathematical and statistical methods to characterize, understand, and predict a drug's PK and PD behavior, quantify uncertainties, and facilitate data-driven decision-making in drug development and pharmacotherapy.¹⁰⁰ Nowadays, it is an essential tool in drug discovery and development, widely recognized by both regulatory authorities and pharmaceutical companies.¹⁰¹

4.1. Nonlinear mixed effects modeling

The backbone of pharmacometrics is nonlinear mixed-effects models. The term "nonlinear" refers to the nonlinear relationship between the dependent variable (*e.g.*, serum drug concentration) and both the model parameters and independent variables.^{102(p2)} "Mixed-effects" denotes the modeling approach where fixed-effect parameters represent common traits across individuals, while random-effect parameters address individual variations.^{102(p2)} The goal is to find optimal fixed-effect parameters for the population while considering individual variations through random-effect parameters.¹⁰³ Parameter estimation in this context involves using maximum likelihood estimation to minimize the objective function value (OFV). The OFV is calculated as $-2\ln(\text{likelihood})$, serving as a goodness-of-fit statistic; a lower OFV indicates a better fit.¹⁰²

4.2. Population modeling

Population modeling, initially introduced by Dr. Lewis B. Sheiner and Dr. Stuart L. Beal, is a powerful technique in pharmacometrics that applies non-linear mixed effect models to quantify the interactions between physiological characteristics of the subject and the observed drug exposure or effect.¹⁰⁴ Initially developed to define dose-exposure relationship (*i.e.*, popPK), it swiftly expanded to characterize the exposure-response relationships (*i.e.*, population PD; popPD).^{105,106} The forthcoming section will detail the fundamental principles of population modeling, illustrated through a popPK example.

The human body is modeled as a system with multiple compartments for mass transfer, exemplified by the common two-compartment model with linear absorption and elimination for mAbs (**Figure 6**). This model accommodates both SC and IV administrations, where SC-

administered drugs absorb into the central compartment at a constant rate, and IV-administered drugs directly enter it. The drug then distributes between the serum and tissues, with elimination from the central compartment occurring at a constant rate. At low concentrations, dose-dependent non-linear elimination can occur if the target is not saturated.¹⁰⁷

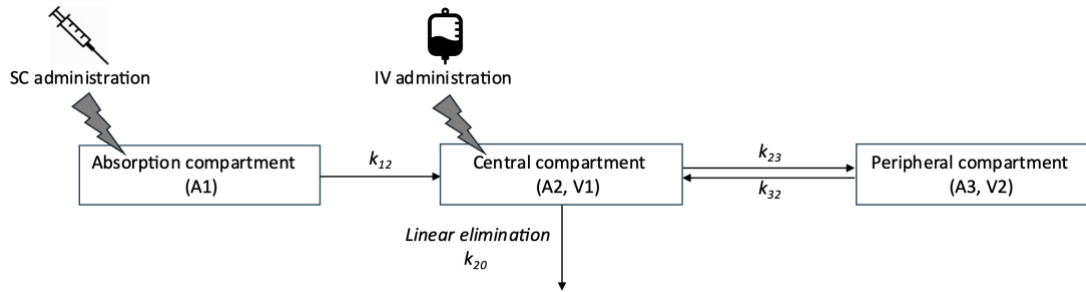


Figure 6. Diagrammatic representation of a two-compartment structural model with linear absorption and elimination. A1, A2 and A3 refers to the absorption compartment, central compartment and peripheral compartment, respectively. A, amount; IV, intravenous; k, rate constant; SC, subcutaneous; V, volume of distribution.

The drug mass transfer between compartments can be expressed as a system of ordinary differential equations:

$$\frac{\partial A_1}{\partial t} = -k_{12} \times A_1 \quad (1)$$

$$\frac{\partial A_2}{\partial t} = k_{12} \times A_1 - k_{20} \times A_2 + k_{32} \times A_3 - k_{23} \times A_2 \quad (2)$$

$$\frac{\partial A_3}{\partial t} = -k_{32} \times A_3 + k_{23} \times A_2 \quad (3)$$

where t represents time, A1, A2 and A3 are the amounts in the absorption, central and peripheral compartment, respectively, and k represents linear rate constants of mass transfer.

4.3. Population model development

A popPK model generally consists of three components: structural model, statistical model, and covariate model.¹⁰⁸ The structural and covariate models represent fixed effects, and the statistical model represents random effects.

Structural models

Structural models are mathematical functions, represented by either algebraic or differential equations, that characterize the typical drug exposure-time profile within a population.¹⁰⁸ These models define key elements such as the number of compartments and the type of absorption and elimination processes (zero-order or first-order). Additionally, the structural model determines typical PK parameters for a population, such as clearance (CL) and volume of distribution(V):

$$CL = \theta_1 \quad \text{(Equation 4)}$$

$$V = \theta_2 \quad \text{(Equation 5)}$$

Where θ_1 and θ_2 denote population values of CL and V, respectively.

Statistical models

Statistical models account for variabilities around the structural model at two levels.¹⁰⁸ Level 1 variability includes inter-individual variability (IIV) or inter-occasion variability (IOV), while Level 2 variability encompasses residual unexplained variability (RUV).¹⁰⁸ Level 2 variability may result from assay variability, errors in sample timing, or model misspecification.¹⁰⁸ The choice of statistical model functions depends on the data type being evaluated.

For IIV, each subject has a unique vector containing individual parameter estimates, known as the eta (η) vector ($\eta_{i,n}$; i for individual, n for the element index). By combining eta vectors from all subjects, we can obtain the standard deviation and variance (ω^2) of these values. Given that PK parameter distributions are positively skewed, log-normal distributions are commonly assumed, expressed by equations in below:

$$CL = \theta_1 \times e^{\eta_{i,1}} \quad (6)$$

$$V = \theta_2 \times e^{\eta_{i,2}} \quad (7)$$

Where θ_1 and θ_2 denote population values of CL and V, respectively, and $\eta_{i,1}$ and $\eta_{i,2}$ represent deviations from these values for individual i . These variances and covariances are compiled into an omega matrix (Ω) as described below:

$$\Omega = \begin{bmatrix} \text{variance of CL } (\omega_{1,1}^2) & \text{covariance of CL and } V \\ \text{covariance of CL and } V & \text{variance of } V (\omega_{2,2}^2) \end{bmatrix} \quad (8)$$

Notably, variance terms should be incorporated where covariates are expected to influence parameters; omitting these terms leads to complete shrinkage towards population typical values, limiting covariate evaluation primarily to allometric or maturation models.¹⁰⁸

IOV account for variability in PK parameters across different treatment phases (*e.g.*, induction and maintenance). IOV can be incorporated along with IIV as demonstrated below:

$$CL = \theta_1 \times e^{\eta_{i,1} + IOV} \quad (9)$$

RUV quantifies unexplained differences between observed and predicted values of the dependent variable (*e.g.*, serum drug concentration). The standard deviation and variance of these differences are denoted as ε and σ^2 , respectively, and are integrated into a sigma matrix (Σ). RUV (*e.g.*, additive error) can be integrated into the model as demonstrated below:

$$Y_{i,j} = f(\vec{\theta}, \vec{\eta}_i, t_{i,j}) + \varepsilon_{i,j} \quad (10)$$

where Y denotes the dependent variable (*e.g.*, drug concentration) for individual i at time j , modeled as a nonlinear function of the independent variables of the vector of fixed-effects parameters $\vec{\theta}$, the vector of random effect parameters specific to individual i ($\vec{\eta}_i$), and $\varepsilon_{i,j}$ represents the RUV for individual i at time point j .

Figure 7 gives a representation of a population model (structural model + statistical model).

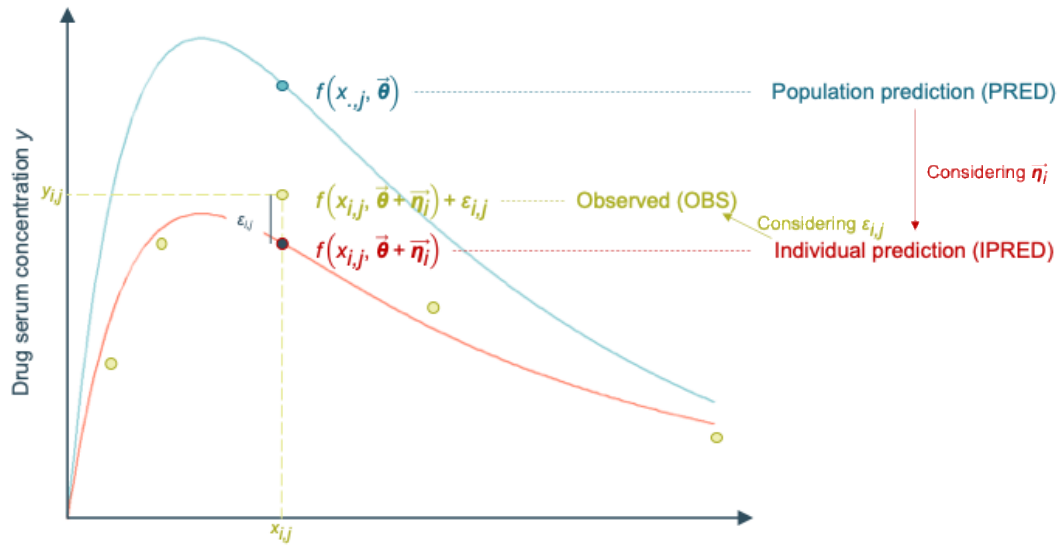


Figure 7. Summary representation of a population pharmacokinetic model.

Covariate models

Covariate models detail how factors like patient's demographics or disease characteristics (e.g., bodyweight and disease activity markers) influence the observed drug exposure or effect over time.¹⁰⁸ Identifying covariates that can explain IIV in PK parameters is crucial during popPK model development. Several methods have been developed for building covariate models, including generalized additive models, step-wise covariate models, and full random effect models.^{109–111} Parameterization of covariate models depends on the type of covariate data, such as continuous (e.g., bodyweight) or categorical (e.g., IBD type).

Model evaluation

In addition to OFV, a variety of tools are employed for model evaluation. These include graphical assessments such as goodness-of-fit plots, simulation-based methods like the visual predictive check (VPC), and the accuracy of parameter estimates assessed through techniques like bootstrapping.^{112–114} The best model was chosen based on the precision and plausibility of parameter estimates, graphical diagnostics of model fit, and hypothesis testing using the likelihood ratio test (LRT).

4.4. From therapeutic drug monitoring to model-informed precision dosing

Apart from the NOR-DRUM B trial⁸⁹, results from the major prospective RCTs failed to show a benefit of TDM of infliximab with respect to better clinical outcomes. One reason for the negative results can be inefficient TDM algorithms (*i.e.*, analog flowcharts or decision trees) used in these trials, making TDM an imprecise and trial-and-error dose optimization process, hence underrating the potential of TDM.¹ To improve the precision and accuracy of individualized dosing, model-informed precision dosing (MIPD, or also sometimes called by clinicians dashboard-driven dosing) can be used.^{1,115}

The concept of MIPD was first introduced in clinical practice over five decades ago by pioneers Lewis Sheiner and Roger Jelliffe.^{116,117} MIPD employs mathematical models to decipher TDM samples and anticipate personalized dosing strategies. Unlike TCI, MIPD can be seen as a more inclusive term, as the inputs and outputs of the model are not limited to concentrations but can encompass any discrete or continuous variable of the underlying model. Nonetheless, TCI and MIPD share the underlying concept and the terminology is still evolving. A schematic representation of the sequential steps in TDM and the enhanced capabilities of MIPD is illustrated in **Figure 8**.

MIPD uses popPK models, patient-specific monitoring data, and a custom-made MIPD software tool to predict the optimal dose or dosing interval for attaining a desired drug concentration in a patient. Today, dozens of commercially and freely available MIPD software tools make it possible to perform MIPD in clinical practice.¹¹⁸ The software tool generally uses typical PK estimates in a popPK model as prior information. With only measurements of patient covariates (*e.g.*, bodyweight, serum albumin) available, the model can provide rough estimates of the individual's PK parameters (*i.e.*, *a priori* prediction). When further considering measurements of drug concentrations in addition to the covariate values of a patient, the popPK model can be 'updated' to a model with individualized PK parameters (*e.g.*, clearance), allowing patient-specific simulations to predict the next dose required for achieving a predefined concentration target (*i.e.*, *a posteriori* prediction or *Bayesian forecasting*).

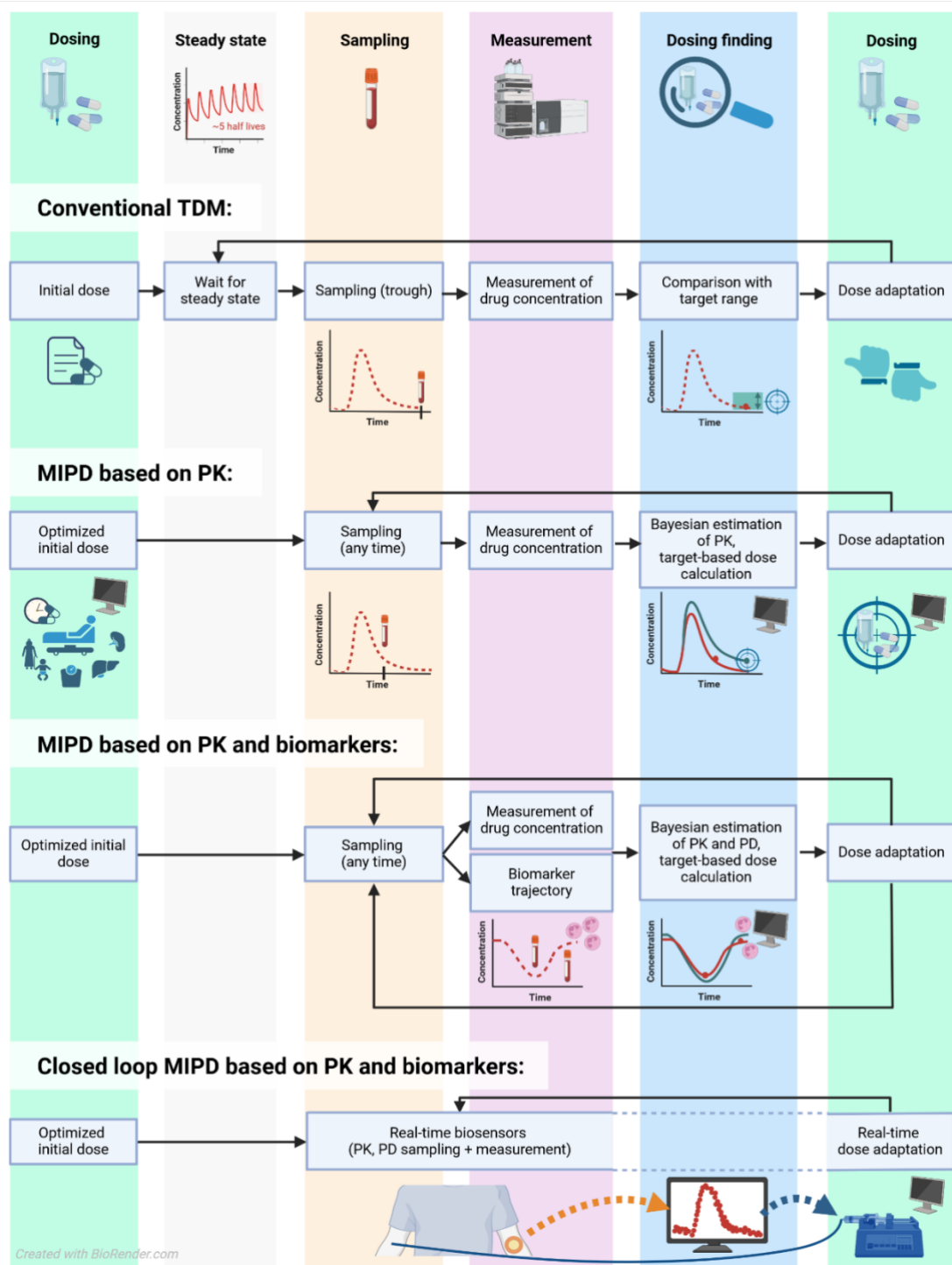


Figure 8. Different approaches of dosing individualization involving PK or biomarker measurements. MIPD, model-informed precision dosing; PD, pharmacodynamics; PK, pharmacokinetics; TDM, therapeutic drug monitoring. Figure reproduced with permission from Minichmayr et al.⁴

While the concept of MIPD was validated in *in silico* simulations¹¹⁹ and retrospective studies^{120,121}, the PRECISION trial by Strik *et al.* was the first prospective RCT to evaluate the benefit of MIPD of infliximab in patients with IBD.⁹⁸ Results from the PRECISION trial showed that MIPD of infliximab was superior to standard dosing for maintaining remission during maintenance therapy. MIPD also reduced immunogenicity and enhanced infliximab durability during induction, and improved short-term efficacy in adult IBD patients.^{122,123} In pediatric IBD patients, MIPD improved target concentration attainment and reduced immunogenicity.¹²⁴ Ongoing prospective trials like TITRATE (“Induction For Acute Ulcerative Colitis”; NCT03937609), REMODEL-CD (“Precise Infliximab Exposure and Pharmacodynamic Control”; NCT05660746), and OPTIMIZE (“Proactive Infliximab Optimization Using a PK Dashboard *versus* Standard of Care in Patients With CD”; NCT04835506) are expected to further validate the benefits of MIPD in infliximab therapy.³ Nevertheless, whether MIPD results in a better outcome than TDM (either reactive or proactive) remains undetermined.^{125,126}

There are emerging attempts to improve the predictive performance of MIPD and create further evidence supporting the implementation of MIPD in clinical practice. First, the predictive performance is substantially impacted by the type of patient information considered. MIPD of infliximab based solely on covariates (*a priori* prediction) has been shown to be biased and imprecise.^{84,119} By considering one or more measured drug concentrations, the precision and accuracy of MIPD can be greatly improved.^{84,119,127} This is expected since covariates generally only explain a small part of the total IIV (up to 6% for clearance), whereas *Bayesian forecasting* based on drug concentrations can identify the remaining, often high, “unexplained” IIV (median of 32.7%, interquartile range IQR 28.0–36.0% on clearance).^{128,129} Second, great efforts are being made to improve the methodological components of MIPD, such as methods for the selection of models^{127,130,131} and methods for estimation of PK parameters.^{132–134} For instance, dozens of popPK models of infliximab have been developed and externally evaluated in patients with inflammatory diseases.^{135–137} However, the selection and validation of the model with the best forecasting performance in often heterogeneous real-world populations remains challenging and costly. To overcome this hurdle, Uster *et al.* evaluated multi-model approaches such as a model averaging algorithm (MAA) and model selection algorithm (MSA) using vancomycin as a case

study, which showed more reliable *Bayesian forecasting* when compared with using a single-model approach.¹²⁷ In a recent retrospective study by Kantasiripitak *et al.*, the predictive performance of MIPD of infliximab based on the MAA/MSA algorithm was evaluated in adult patients with IBD who underwent infliximab dosage de-escalation.⁸⁴ Their results showed that an MAA resulted in the most accurate and precise *a posteriori* prediction as compared to the MSA and a single-model approach.⁸⁴ Besides, the predictive performance of both single- and multi-model approaches was robust to the lack of covariate data, as long as one single most recent C_{trough} of infliximab was provided.⁸⁴ However, more research is still needed to identify optimal sampling regimens that result in the most accurate *Bayesian forecasting*. Apart from multi-model approaches, novel methods like a continuous learning approach for MIPD^{130,131}, Bayesian data assimilation^{132,133}, flattened priors models¹³⁴ have also been suggested but are to be validated. Moreover, non-popPK approaches like machine learning (ML) and artificial intelligence (AI) algorithms^{138,139} are also under development, but it is still unclear if they provide any benefit over popPK-based approaches. Finally, the predictive ability of MIPD is usually described by criteria such as classification accuracy (*e.g.*, ability to predict the next C_{trough} to be higher or lower than a specific threshold) or the difference between predicted and measured values (*e.g.*, root mean square error or relative bias)^{84,127,131}. However, there is a lack of consensus on how to evaluate the success of MIPD and further investigation of the impact of different evaluation criteria on choosing a suboptimal model is needed.

There are still regulatory concerns around MIPD. The European Union's General Data Protection Regulation (GDPR) allows a citizen to know the reasons behind a clinical decision produced by ML/AI and to reject such a decision under the right of "meaningful human review".¹⁴⁰ However, it is still unclear how applicable, enforceable, and beneficial these articles are for MIPD, since MIPD only provides dose recommendations, possibly checked and authorized by a clinical pharmacist, and the final decision on adjusting the dosing is generally left at the discretion of the treating physician.¹⁴¹ Besides, with the effect of the new European Union Medical Device Regulation, MIPD software is considered a class II medical device and is subject to strict risk compliance.¹⁴²

Objectives

The PRECISION trial recently demonstrated the usefulness of MIPD of infliximab during maintenance therapy for reducing the incidence of loss of response in IBD patients in clinical remission.⁹⁸ In patients with active IBD, the benefit of MIPD of infliximab remains uninvestigated. While the popPK of infliximab has repeatedly been described, little or no data are available about the popPK of other mAbs. Also, integrated population analyses of both PK and PD are mostly lacking.

We hypothesize that population modeling and MIPD of mAbs therapies may improve therapeutic outcomes in patients with IBD.

The main objective of this doctoral research project is to develop popPK-PD models to characterize the dose-exposure-(biomarker)-response relationship of mAbs therapies and investigate the potential benefit of MIPD of mAbs therapies in patients with IBD.

In **Chapter I**, we performed popPK-PD modeling and simulation of ustekinumab in patients with CD.

In **Chapter II**, we performed popPK simulations of SC infliximab and vedolizumab in patients with IBD.

In **Chapter III**, we performed popPK-PD modeling and simulation of SC infliximab in patients with IBD.

In **Chapter IV**, we investigated mAb clearance as a better predictor than C_{trough} for endoscopic outcomes in patients with CD.

In **Chapter V**, we studied model-informed dosage de-escalation of infliximab in patients with IBD.

Research chapters

Chapter I. PopPK-PD modeling of ustekinumab

This chapter is based on

Wang Z, Verstockt B, Sabino J, Vermeire S, Ferrante F, Declerck P, Dreesen E. Population pharmacokinetic-pharmacodynamic model-based exploration of alternative ustekinumab dosage regimens for patients with Crohn's disease. *Br J Clin Pharmacol*. 2022; 88(1): 323-335.

Abstract

Background: In the UNITI endoscopy sub-study, only 17.4% of patients with CD on ustekinumab achieved endoscopic response and 10.9% of patients achieved endoscopic remission at week (w)44. We aimed to evaluate the impact of alternative ustekinumab dosage regimens on endoscopic outcomes based on popPK-PD modeling and simulation analysis.

Methods: Real-world data were obtained from 83 patients with moderate-to-severe CD (95% biological-refractory) enrolled in a prospective cohort study receiving intravenous ustekinumab (~6mg/kg) followed by Q8W subcutaneous maintenance therapy (90 mg). Three sequential models were developed: a two-compartment popPK model linking ustekinumab dose to ustekinumab exposure, an indirect response popPK-PD model describing the effect of ustekinumab exposure on FC, and a logistic regression outcome model linking FC to endoscopic outcomes.

Results: Ustekinumab clearance increased with decreasing serum albumin and increasing bodyweight. FC decreased with increasing ustekinumab exposure. The probability of endoscopic response at w24 increased from 10.0% to 17.9% with FC at w8 decreasing from 1800 $\mu\text{g/g}$ to 694 $\mu\text{g/g}$. The probability of endoscopic remission at w24 increased from 2.1% to 10.0% with FC at w8 decreasing from 1800 $\mu\text{g/g}$ to 214 $\mu\text{g/g}$. Simulation-based comparison of Q8W and Q4W maintenance dosing regimens predicted 16.7% and 22.2% endoscopic response rates, respectively. Endoscopic remission rates were estimated to be 4.2% on Q8W dosing and 6.7% on Q4W dosing.

Conclusions: The developed models can guide clinical trial design and support model-informed dose optimization (stratified or individualized dosing) to improve endoscopic outcomes.

1. Introduction

CD is a chronic IBD that can cause inflammation of any segment of the gastrointestinal tract, from the mouth to the anus. The treatment of CD mainly consists of the stepwise use of corticosteroids, immunomodulators (thiopurines, methotrexate), and biological therapies (anti-TNF agents, anti-integrin therapies, IL-12 and IL-23 therapies).

Ustekinumab is a fully human IgG1 monoclonal antibody targeting the shared p40 subunit of IL-12 and IL-23. It was initially used for the treatment of patients with moderate-to-severe plaque psoriasis and active psoriatic arthritis, where after it was studied in patients with CD and UC.^{143–148} The Phase 3 UNITI trial program demonstrated the efficacy of ustekinumab to induce and maintain clinical remission in patients with moderate-to-severe CD.^{148–150} The approved dosing of ustekinumab for patients with CD is a one-time weight-based IV infusion (260 mg for bodyweight ≤ 55 kg, 390 mg for body weight 55–85 kg, and 520 mg for body weight >85 kg) followed by maintenance therapy of 90 mg SC Q8W or Q12W.

While regulatory approval of ustekinumab was based on the assessment of clinical response and remission, therapeutic goals have evolved, now aiming for endoscopic remission.³⁵ There is a lack of consistency in defining thresholds for endoscopic response and remission. From the STRIDE-II consensus program, the following definitions prevailed: for endoscopic response a $>50\%$ decrease in the SES-CD or CDEIS and for endoscopic remission SES-CD ≤ 2 points or CDEIS <3 and lack of ulcerations (i.e., any ulcerations, including aphthous ulcers).³⁵

Endoscopic remission correlates with sustained clinical remission, steroid-free remission, and reduced rates of hospitalization and surgery.^{151–154} Although good clinical remission rates have been proven in the UNITI study, results regarding endoscopic outcomes are less promising.^{149,150,155} A UNITI post-hoc subgroup analysis reported endoscopic response ($>50\%$ decrease in the SES-CD) and remission rates (SES-CD ≤ 2) of 17.4% and 10.9%, respectively, at w44 of maintenance therapy.¹⁵⁶ Similar endoscopic response (20.5%) and endoscopic remission rates (7.1%) at w24 of maintenance therapy were reported in a prospective study in patients with moderate-to-severe CD.¹⁵⁷ The relatively low endoscopic response and remission rates imply there is a potential benefit of dose optimization.

It has been shown that normalization of FC after induction therapy (w8) is associated with attainment of endoscopic response at w16, which indicates a potential value of this less invasive FC monitoring.¹⁵⁸ Besides, Verstockt *et al.* found that a significant increase in FC after the switch from IV induction to SC maintenance therapy (beyond w8) was associated with disappointing endoscopic outcomes at w24.¹⁵⁷ However, the relationship between ustekinumab exposure, FC, and endoscopic outcomes in patients with CD has not yet been established. A popPK-PD model may provide insight in the factors that affect ustekinumab exposure and (biomarker) response.

In this study, we aimed to evaluate the impact of alternative ustekinumab dosage regimens on endoscopic outcomes. Therefore, we performed a popPK-PD modeling and simulation analysis to characterize the relationship between ustekinumab dose, ustekinumab exposure, FC concentrations, and endoscopic outcomes under standard Q8W and intensified Q4W maintenance dosing.

2. Methods

2.1. *Patients and study design*

Data from 86 patients with CD were obtained from a prospective, real-life cohort study at the University Hospitals Leuven (Leuven, Belgium). The design and conduct of the study have been previously reported.¹⁵⁷ All included patients received an ustekinumab IV loading dose of approximately 6 mg per kilogram bodyweight (260 mg for ≤ 65 kg bodyweight, 390 mg for 65–85 kg, and 520 mg for >85 kg) at w0, followed by fixed SC maintenance doses of 90 mg Q8W. The study was conducted in accordance with the principles of good clinical practice and the Declaration of Helsinki. All patients provided written informed consent prior to participation in the Institutional Review Board-approved IBD Biobank [B322201213950/S53684], whereby patients' characteristics and samples were collected prospectively on a series of predefined time points.

2.2. *PK and PD data*

Serum samples were collected at w4 (mid-dose) and w8, w16, and w24 (trough). Serum ustekinumab concentrations were quantified using a validated enzyme-linked immunosorbent assay (ELISA, 9% coefficient variation) with a lower limit of quantification (LLOQ) of 0.25 $\mu\text{g/mL}$.¹⁵⁹ Anti-ustekinumab antibodies were quantified in serum samples with undetectable ustekinumab by using a drug-sensitive bridging ELISA (cut-off of 0.005 $\mu\text{g/mL}$ MA-UST37F12 equivalents).¹⁵⁷ A drug-tolerant affinity capture elution assay was used to quantify anti-ustekinumab antibodies in serum samples with ustekinumab below 1.0 $\mu\text{g/mL}$.¹⁶⁰

Fresh stool samples were collected at baseline, w4, w8, w16, and w24. FC concentrations were quantified using the FC ELISA kit (Bühlmann, Schönenbuch, Switzerland) with a LLOQ of 30 $\mu\text{g/g}$ and an upper limit of quantification (ULOQ) of 1,800 $\mu\text{g/g}$.

Endoscopy was performed at baseline and w24. Endoscopic remission was defined as SES-CD ≤ 2 . Endoscopic response was defined as a drop in SES-CD from baseline by at least 50%.

2.3. *Model development*

2.3.1. *PopPK model*

Ustekinumab concentration data were ln-transformed and a ln-transform both sides approach was used for developing a popPK model.¹⁶¹ We used first-order conditional estimation with interaction Laplacian for parameter estimation and differential equation solver ADVAN6. Ustekinumab concentrations below the LLOQ were accounted for using the M3 method.¹⁶² The popPK base model was selected following comparisons between structural models with different numbers of compartments (one-, two- or three-compartments) and elimination processes (linear or nonlinear). Covariate model building was performed to yield the final popPK model (See 2.3.4 Covariate analysis).

2.3.2. *PopPK-PD model*

Empirical Bayes estimates (EBEs) of individual PK parameters from the final popPK model were used to predict individual ustekinumab concentrations (Individual PK Parameters [IPP] approach).¹⁶³ These were then linked to the individual observed FC concentration-time courses in an indirect response model in which ustekinumab concentrations either inhibit FC production or promote FC degradation. FC concentrations below the LLOQ and above the ULOQ were accounted for using the M3 method.¹⁶² Covariate model building was performed to yield the final popPK-PD model (See 2.3.4 Covariate Analysis).

2.3.3. *Population outcome model*

EBEs of individual parameters from the indirect response model were subsequently used to predict individual FC concentrations. Both individual predicted and observed FC were tested to predict the observed endoscopic outcomes at w24 using logistic regression models. In other words, FC concentrations were used to predict transition probabilities from active endoscopic disease at baseline to endoscopic response or remission at w24. The Laplacian method was used to approximate the marginal likelihood.

2.3.4. *Covariate analysis*

The continuous, time-varying covariates included age, bodyweight, body mass index, lean body mass¹⁶⁴, CRP, FC, serum albumin, hemoglobin, and PRO-2 at w0, w4, w8, w16, and w24,

and the SES-CD score at w0 and w24. Baseline covariates included sex, smoking status, disease activity (active/non-active endoscopic and/or Magnetic Resonance Imaging disease activity), disease location (ileal/colonic/ileocolonic), disease behavior (inflammatory/stricturing/penetrating), previous exposure to anti-TNFs (yes/no), total number of biologicals previously exposed to, concomitant corticosteroids (yes/no), and concomitant immunomodulators (yes/no).

Covariates were tested in the base model with stepwise forward addition followed by backward deletion. In the forward step, a covariate effect was accepted if it decreased the OFV by at least 6.63 points ($\alpha=0.01$, $df=1$). In the backward deletion step, an increase of more than 10.83 points in OFV ($\alpha=0.001$, $df=1$) was required to keep a covariate effect that was identified by forward addition. The resulting model after backward deletion is regarded as the final model. All covariates were tested on the base popPK, base popPK/PD and base population outcome model. For continuous covariate data, both power and exponential parameterization were tested. For categorical covariate data, linear parameterization was tested.

2.4. Model evaluation

The most parsimonious model was selected based on comparisons using standard procedures, including the precision and plausibility of parameter estimates, the OFV, goodness-of-fit plots and prediction-corrected visual predictive check (pcVPC) plots ($n=1,000$ simulated replicates of the original data set). Bootstrapping was performed to obtain nonparametric estimates of uncertainty in parameter estimates ($n=2,000$ bootstrap runs).

2.5. Simulations

The models were used to investigate if improved endoscopic outcomes could be achieved by interval shortening to Q4W maintenance dosing as compared with standard Q8W dosing. A total of 3,000 virtual patients were equally randomized into the two dosing arms (1,500 patients each). Individual albumin was sampled following a truncated normal distribution with lower and upper boundaries equal to the maximal and minimum value of albumin in our data set. In each dosing arm, patients were equally assigned into three bodyweight groups (45-65 kg, 65-85 kg, and 85-105 kg). Individual bodyweight was sampled

following a truncated normal distribution with lower and upper boundaries equal to the maximal and minimum value in the pre-assigned bodyweight group. The induction doses were given based on the pre-assigned bodyweight group (260 mg for 45-65 kg bodyweight, 390 mg for 65-85 kg, and 520 mg for 85-105 kg). The developed popPK, popPK-PD and population outcome model, were used to predict serum ustekinumab concentrations, FC concentrations, and the probability of endoscopic response and remission following the two dosing regimens.

2.6. *Software*

Data were imported in R (version 4.0.3; R Foundation for Statistical Computing, R Core Team, Vienna, Austria) for data wrangling (including R packages `data.table`, `dplyr`, `reshape2`) and visualization (`ggplot2`, `vpc`, `xpose4`) in the RStudio integrated development environment (version 1.3.1093; RStudio, Inc., Boston, MA, USA). Sample size calculation was performed using `stats` package in R, considering a 5% significance level and an 80% power of a two-sided test. All nonlinear mixed-effects modeling and simulation were performed using NONMEM (version 7.4.4; Icon Development Solutions, Gaithersburg, Maryland, USA) with a GNU Fortran 95 compiler and the Perl-speaks-NONMEM (PsN; version 4.9.0) toolkit on the interface software Pirana (version 2.9.9; Certara, Inc., Princeton, NJ, USA). The NONMEM control streams generated during the current study are available in **Appendix A**.

2.7. *Nomenclature of targets and ligands*

Key protein targets and ligands in this article are hyperlinked to corresponding entries in <http://www.guidetopharmacology.org>, the common portal for data from the IUPHAR/BPS Guide to PHARMACOLOGY,¹⁶⁵ and are permanently archived in the Concise Guide to PHARMACOLOGY 2017/2018.¹⁶⁶

3. Results

3.1 Study population

Three out of 86 patients were excluded because of immunogenicity towards ustekinumab (n=1) and the lack of serum samples (n=2). The anti-ustekinumab antibodies were detected by both the drug tolerant and drug sensitive assay. Baseline demographic and clinical characteristics of the included 83 patients are summarized in **Table 7**. The vast majority of the patients had previous exposure to at least one anti-TNF agent (94%), vedolizumab (67.5%) or both (66.3%). Seventeen patients (20.5%) achieved endoscopic response at w24. Six patients (7.2%) achieved endoscopic remission at w24.

3.2 PopPK analysis

In total, the 83 patients contributed 311 ustekinumab serum concentrations to the popPK analysis (no concentration data excluded). Ustekinumab was unquantifiable in five serum samples from two patients, but anti-ustekinumab antibodies were not detectable.

Base PK model. A two-compartment model with first-order absorption and first-order elimination was selected as the base model. Since we were not able to estimate the ustekinumab absorption rate constant (k_a), it was fixed to 0.142 day^{-1} , as estimated in a ustekinumab popPK model in adults with moderate-to-severe UC.¹⁶⁷ A schematic representation of the base model is shown in **Figure 9**. IIV was assumed to be ln-normally distributed and was estimated on F and CL. The η -shrinkage of the estimated IIV on F and CL were 18% and 2%, respectively. Allometric scaling was added on central volume of distribution V_c and peripheral volume of distribution V_p (exponents 1), and CL and intercompartmental clearance Q (exponents 0.75). An error model with a combination of a proportional and an additive error was selected.

Table 7. Patients characteristics at baseline

Parameter	Value
Number of patients	83
<i>Demographics</i>	
Sex, women, <i>n</i> (%)	56.0 (67.5)
Age, years, median [IQR]	37.5 [28.3-49.1]
Weight, kg, median [IQR]	64.5 [55.9-80.7]
Body mass index, kg/m ² , median [IQR]	23.4 [19.7-26.6]
<i>Serology</i>	
C-reactive protein, mg/L, median [IQR]	12.4 [4.5-20.0]
C-reactive protein >5 mg/L, <i>n</i> (%)	57 (68.7)
Fecal calprotectin, µg/g, median [IQR]	1398.0 [624.2-1800.0]
Serum albumin, g/L, median [IQR]	40.1 [37.9-42.5]
Hemoglobin, g/dL, median [IQR]	12.6 [11.9-13.3]
<i>Disease activity</i>	
disease activity, <i>n</i> [%]	82 (98.8)
Simple Endoscopic Score for Crohn's disease, median [IQR]	12 [8-18]
Weighted two-item patient-reported outcome, median [IQR]	16.0 [10.5-21.0]
Disease location, <i>n</i> (%)	
Ileal : Colonic : Ileocolonic	25 (30.1) : 10 (12.0) : 48 (57.9)
Disease behavior, <i>n</i> (%)	
Inflammatory : Stricturing : Penetrating	34 (41.0) : 35 (42.2) : 14 (16.9)
<i>Previous and concomitant medication</i>	
Previous exposure to anti-TNF, <i>n</i> (%)	78 (94.0)
Previous exposure to anti-TNF & vedolizumab, <i>n</i> (%)	55 (66.3)
Total number of biologicals previously exposed to, <i>n</i> (%)	
0:1:2:3	4 (4.8) : 15 (18.0) : 22 (26.5) : 42
Concomitant corticosteroids, <i>n</i> (%)	
Topical : Systemic	17 (20.5) : 12 (14.5)
Concomitant immunomodulators, <i>n</i> (%)	2 (2.4)
Smoking status, <i>n</i> (%)	
Active : Previous : Never	19 (22.9) : 22 (26.5) : 42 (50.6)

Abbreviations: anti-TNF, anti-tumor necrosis agent; IQR, interquartile range; LLOQ, lower limit of quantification; *n*, number of patients; ULOQ, upper limit of quantification. Table reproduced with permission from Wang et al.¹⁶⁸

Final PK model. Results from the univariate covariate analysis are shown in **Appendix A**. The final PK model consisted of the effect of serum albumin on CL. The inclusion of an albumin effect resulted in a drop in OFV by 38.2 points, along with a decrease in IIV on CL and F by 5.4% and 7.1%, respectively. **Equation 11** shows how ustekinumab clearance of patient i at time j ($CL_{i,j}$) increases with higher bodyweight and lower albumin:

$$CL_{i,j} = \theta_{CL} \times \left(\frac{\text{albumin}_{i,j}}{42.65}\right)^{-1.21} \times \left(\frac{\text{bodyweight}_{i,j}}{65.27}\right)^{0.75} \times e^{\eta_i} \quad (11)$$

θ_{CL} represents the typical population clearance and η_i represents the deviation of patient i from the typical clearance. The pcVPC plot shows a good agreement between model simulations and observed data (**Figure 10a**). The PK parameter estimates of the base and final model as well as results obtained from bootstrap are listed in **Table 8**. The model estimated an elimination half-life of approximately 20.4 days and a distribution half-life of 9.0 days.

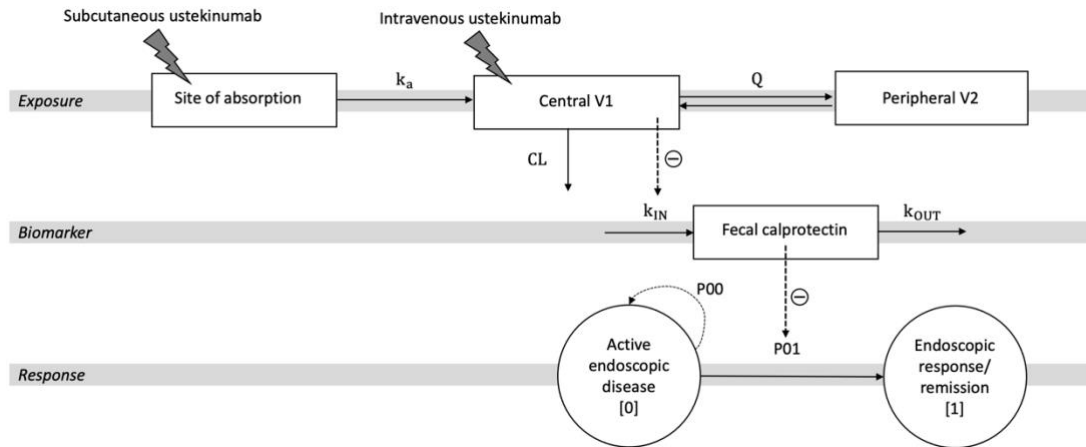


Figure 9. Schematic of the pharmacokinetic-pharmacodynamic model of ustekinumab in patients with Crohn's disease. The schematic includes the two-compartment population PK model, the indirect response model for fecal calprotectin reduction, and the logistic regression model with transitions between active endoscopic disease [0] and endoscopic response/remission [1]. CL: clearance; F, absolute bioavailability; k_a , absorption rate constant; k_{IN} , production rate constant; k_{OUT} , degradation rate constant; P00: probability of not achieving endoscopic response/remission; P01: probability of achieving endoscopic response/remission; Q: intercompartmental clearance; V1: volume of distribution in the central compartment; V2: volume of distribution in the peripheral compartment. Figure reproduced with permission from Wang et al.¹⁶⁸

Table 8. Parameter estimates of the developed models

Parameter	Base model (OFV=-305.8)		Final model (OFV=-344.7)		Bootstrap	
	Mean	%RSE [%Shrinkage]	Mean	%RSE [%Shrinkage]	Median	95% CI
Population PK model						
First-order absorption rate constant (day ⁻¹)	0.142 FIX	-	0.142 FIX	-	-	-
Absolute bioavailability (%)	59.4	8.8	62.4	7.7	62.0	(52.3, 70.7)
V in the central compartment (L)	5.25	9.8	5.05	10.6	4.98	(3.52, 6.02)
• Bodyweight on V central	1 FIX	-	1 FIX	-	-	-
V in the peripheral compartment (L)	0.713	23.8	0.748	23.7	0.790	(0.390, 1.776)
• Bodyweight on V peripheral	1 FIX	-	1 FIX	-	-	-
Inter-compartmental clearance (L/day)	0.0353	46.5	0.0426	45.3	0.0427	(0.0196, 0.283)
• Bodyweight on Inter-compartmental clearance	0.75 FIX	-	0.75 FIX	-	-	-
Clearance (L/day)	0.238	5.8	0.235	5.4	0.233	(0.212, 0.258)
• Albumin on clearance	-	-	-1.21	17.4	-1.20	(-1.58, -0.82)
• Bodyweight on clearance	0.75 FIX	-	0.75 FIX	-	-	-
IIV of F (%)	44.4	13.0 [18]	37.3	14.7 [23]	37.1	(18.7, 54.9)
IIV of CL (%)	36.6	7.7 [2]	31.2	8.0 [2]	30.8	(25.6, 37.3)
Additive residual error (mg/L)	0.247	23.0 [23.5]	0.272	23.6 [22.6]	0.269	(0.115, 0.440)
Proportional residual error (%)	14.1	8.1 [41.4]	13.6	8.2 [40.1]	13.4	(10.6, 15.8)
Population PK/PD model						
BASE (µg/g)	-	-	1550	15.9	1548	(1190, 1765)
Box-Cox transformed shape parameter on BASE	-	-	0.133	136.8	0.133	(-0.072, 0.313)
k _{OUT} (day ⁻¹)	-	-	0.0416	12.7	0.0416	(0.036, 0.052)
I _{max} (%)	-	-	1 FIX	-	-	-
IC ₅₀ (µg/mL)	-	-	2.71	11.5	2.71	(1.47, 6.38)
IIV of BASE (%)	-	-	160.5	12.5 [12]	160.5	(117.8, 214.0)
IIV of IC ₅₀ (%)	-	-	107.9	28.0 [41]	107.9	(42.6, 269.5)
IIV of k _{OUT} (%)	-	-	31.3	56.8 [56]	31.3	(9.8, 52.5)
Proportional residual error (%)	-	-	55.1	7.0	55.1	(48.2, 59.4)
Population outcome model for endoscopic response						
P01 _{max} (%)	21.2	21.6	35.8	30.2	36.7	(19.3, 72.8)
I _{max} (%)	100 FIX	-	100 FIX	-	-	-
IC ₅₀ (µg/g)	-	-	694	76.5	698	(120, 4072)
Population outcome model for endoscopic remission						
P01 _{max} (%)	7.49	39.3	19.9	52.3	20.4	(5.5, 61.2)
I _{max} (%)	100 FIX	-	100 FIX	-	-	-
IC ₅₀ (µg/g)	-	-	214	79.4	217	(30, 852)

Abbreviations: BASE, baseline fecal calprotectin (FC); IC₅₀, ustekinumab or FC concentration for achieving half-maximal effect; I_{max}, maximal inhibitory effect; IIV, interindividual variability; k_{OUT}, degradation rate of FC; P01_{max}, maximal probability for achieving endoscopic response/remission at week 24; PD, pharmacodynamic; PK, pharmacokinetic; V, volume of distribution; %CV, percentage coefficient of variation calculated as $\sqrt{e^{\text{variance}^2} - 1} \times 100\%$; %RSE, percentage relative standard error. Table reproduced with permission from Wang et al.¹⁶⁸

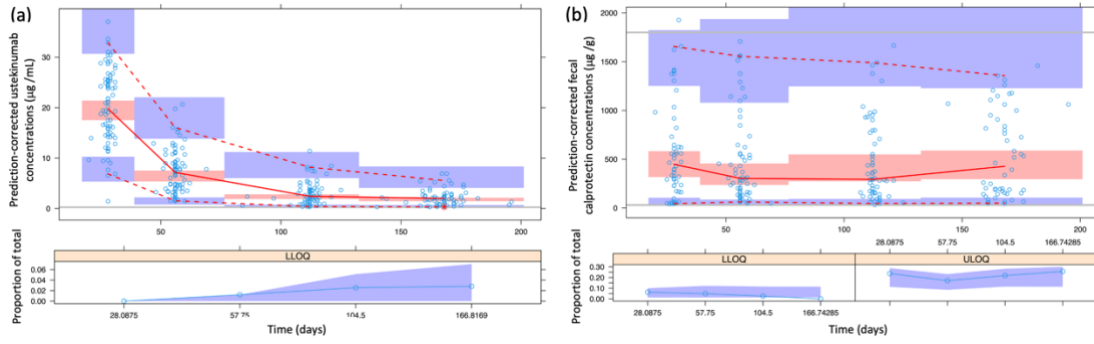


Figure 10. The prediction-corrected visual predictive check of the final population pharmacokinetic model (a) and the final population pharmacokinetic-pharmacodynamic model (b). (upper panel) Blue open circles are the prediction-corrected observed concentrations. The red solid and dash lines represent median and 2.5th and 97.5th percentile of the prediction-corrected observations, respectively. Simulation-based 95% confidence intervals around the median and 2.5th and 97.5th percentiles of the observed data are shown in the form of red and blue shaded areas, respectively. (lower panel) Simulation based 95% confidence interval (shaded blue area) around the median (solid blue line) for the fraction of observations below the lower limit of quantification (LLOQ) and above the upper limit of quantification (ULOQ). Figure reproduced with permission from Wang et al.¹⁶⁸

3.3 PopPK-PD analysis

A total of 310 FC concentrations were included for the popPK-PD analysis. Eleven fecal samples from eight patients had an FC concentration below the LLOQ of 30 µg/g. Sixty-eight fecal samples from 29 patients had an FC concentration above the ULOQ of 1,800 µg/g. The indirect response popPK-PD model, consisting of a baseline model and a maximum effect (Emax) drug model, described the data well. Baseline FC was Box-Cox transformed and modelled assuming an ln-normal distribution of random effects (**Equation 12**).

$$BASE_i = \theta_{BASE} \times e^{\frac{\eta_i^{SHP} - 1}{SHP}} \quad (12)$$

BASE represents the baseline FC concentration, θ_{BASE} represents the typical population baseline FC concentration, η_i represents the random effect around baseline FC concentration for patient i and SHP represents the shape parameter of the Box-Cox transformation.

Equation 13 shows the differential equation of the drug effect model:

$$\frac{dFC(t)}{dt} = k_{IN} \times \left[1 - I_{max} \times \frac{CONC(t)}{IC_{50} + CONC(t)} \right] - k_{OUT} \times FC(t) \quad (13)$$

The drug effect model was constructed to inhibit the production rate constant (k_{IN}) of the FC through an inhibitory Emax model equation driven by ustekinumab concentrations (CONC) (**Figure 9**). The drug effect is governed by three drug-specific parameters: I_{max} , IC_{50} and k_{OUT} .

$I_{\max}$ is the maximal drug effect. IC_{50} is the ustekinumab serum concentration required to achieve a half-maximal drug effect. k_{OUT} is the degradation rate of FC and is equal to k_{IN} divided by BASE. **Equation 14** shows the initial condition of the drug model:

$$FC(0) = BASE = \frac{k_{IN}}{k_{OUT}} \quad (14)$$

The fixed effects of BASE, IC_{50} and k_{OUT} were assumed to be ln-normally distributed. No covariate relationships were found on BASE. No covariate relationships were tested on IC_{50} and k_{OUT} due to the high η -shrinkage (>20%).

The pcVPC plot showed a good agreement between observed and model-predicted FC serum concentrations (**Figure 10b**). The parameter estimates of the final popPK-PD model as well as the results obtained from bootstrap are presented in **Table 8**.

3.4 Population outcome analysis

Logistic regression models were used to predict the transition probability from active endoscopic disease at baseline to endoscopic response and endoscopic remission at w24. The individually observed FC concentrations at w8 adequately predicted the individual probabilities of endoscopic response and endoscopic remission at w24, resulting in a significant drop in OFV of 4.5 points and 4.0 points, respectively, from the base model (no FC effect). The probability of attaining endoscopic response and endoscopic remission increased with decreasing FC at w8 (**Figure 11**). The parameter estimates of the final population outcome model as well as the results obtained from bootstrap are listed in **Table 8**. **Equation 15** shows the effect of FC on the transition probability:

$$P_{\text{endoscopic response/remission}} = P01_{\max} - P01_{\max} * I_{\max} * \left(\frac{FC}{IC_{50} + FC} \right) \quad (15)$$

$P01_{\max}$ represents the maximum probability of attaining endoscopic response/remission at w24. $I_{\max}$ stands for the maximal inhibitory effect of FC on $P01_{\max}$. IC_{50} represents the FC concentration required to achieve a half-maximal inhibitory effect. The maximum probability of attaining endoscopic response and endoscopic remission at w24 was estimated to be 35.8% and 19.9%, respectively. The probability of endoscopic response at w24 increased from 10.0% to 17.9% with FC at w8 decreasing from 1800 $\mu\text{g/g}$ to 694 $\mu\text{g/g}$. The probability of Endoscopic remission at w24 increased from 2.1% to 10.0% with FC at w8 decreasing from

1800 $\mu\text{g/g}$ to 214 $\mu\text{g/g}$. The binary predictive check plot of the outcome model demonstrates that patients with an FC concentration above 620 $\mu\text{g/g}$ at w8 have no chance of achieving endoscopic remission at w24 (**Appendix A Figure S1**).

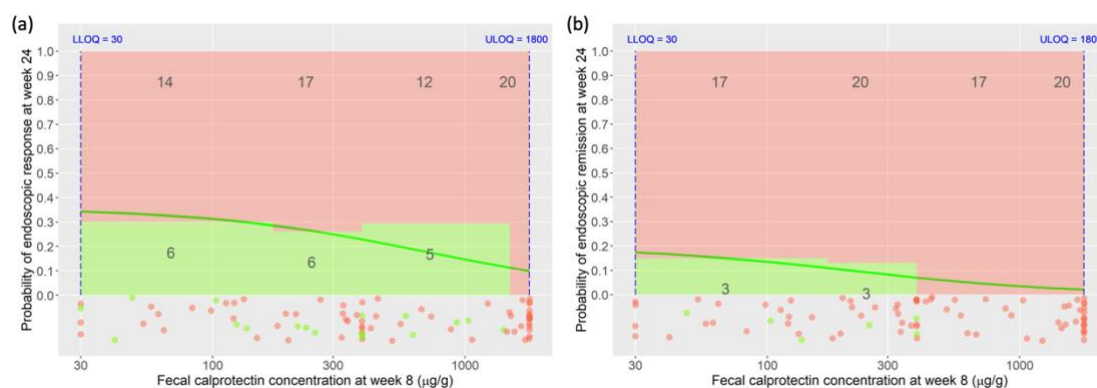


Figure 11. Goodness-of-fit plot of the population outcome model. Observed (green tiles) and predicted (line) fractions of patients achieving endoscopic response (a) or endoscopic remission (b) at week 24 are presented. Red tiles represent the observed fractions of patients not attaining endoscopic outcomes. Grey numbers indicate the number of patients in each quartile of fecal calprotectin concentration. Patients achieving endoscopic outcomes (green dots) or not (red dots) at week 24 are plotted. LLOQ: lower limit of quantification, ULOQ: upper limit of quantification. Figure reproduced with permission from Wang et al.¹⁶⁸

3.5 Simulations

Simulation-based comparison of Q8W and Q4W maintenance dosing predicted median C_{trough} at steady state of 1.3 $\mu\text{g/mL}$ and 5.3 $\mu\text{g/mL}$ ustekinumab, and of 796 $\mu\text{g/g}$ and 424 $\mu\text{g/g}$ FC, respectively (**Figure 12**). Associated endoscopic response rates were estimated to be 16.7% on Q8W dosing and 22.2% on Q4W dosing. Associated endoscopic remission rates were estimated to be 4.2% on Q8W dosing and 6.7% on Q4W dosing. A summary of the simulated results at steady state is shown in **Table 9**. Study sample sizes of 812 and 1,294 patients per dosing arm would be required for identifying statistically significant differences in the rates of endoscopic response and endoscopic remission, respectively, between both study arms (1:1 randomization, $\alpha=5\%$, power=80%). Dose doubling (180 mg Q8W), as opposed to interval halving (90 mg Q4W), was predicted to result in ustekinumab C_{trough} of 2.7 $\mu\text{g/mL}$ instead of 5.3 $\mu\text{g/mL}$ (**Appendix A Figure S2**).

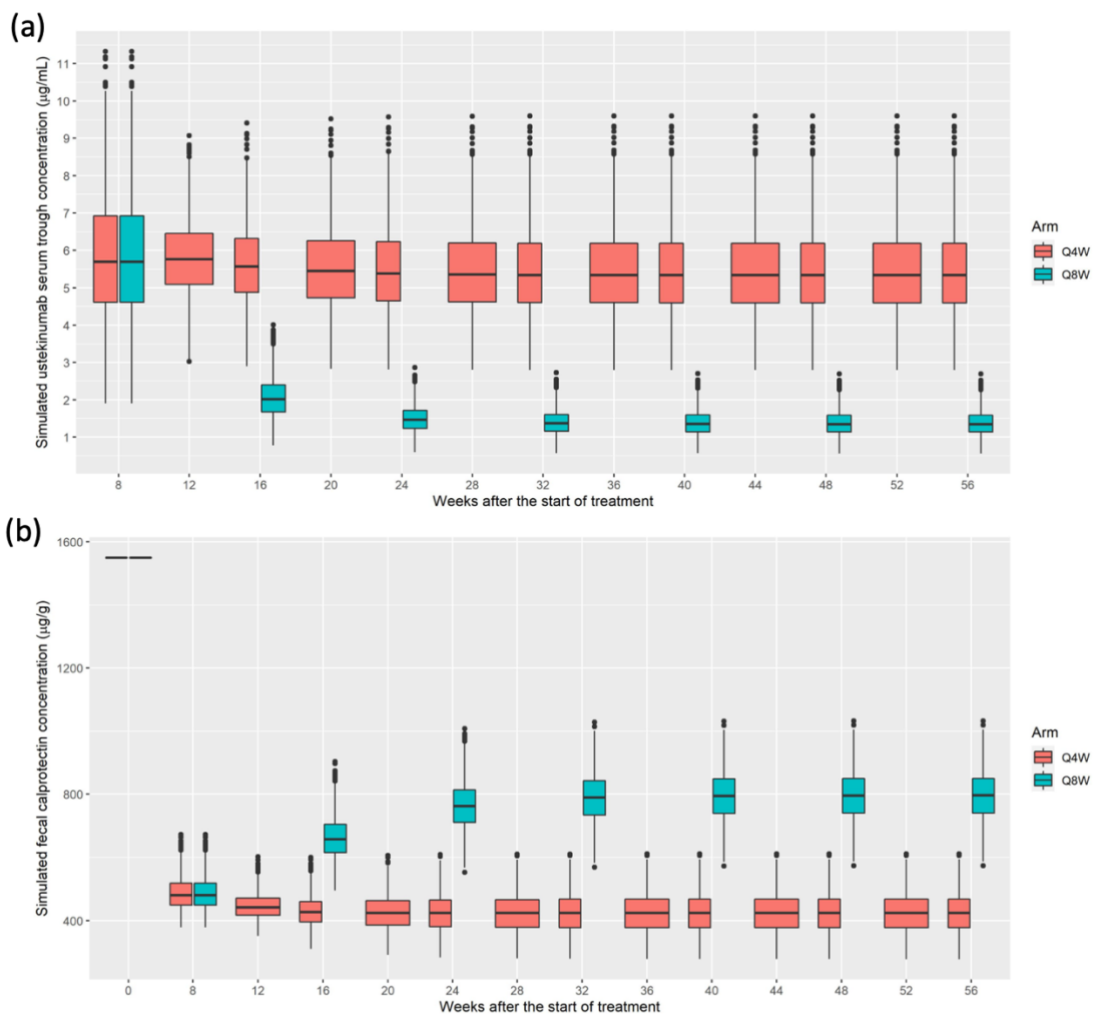


Figure 12. Simulated ustekinumab trough concentration and corresponding fecal calprotectin concentration. **(a)** Simulated ustekinumab serum trough concentrations at different treatment weeks following maintenance dosing every 4 weeks (q4w) or q8w in patients with Crohn's disease. **(b)** Simulated fecal calprotectin concentrations at different treatment weeks following maintenance dosing every 4 weeks (q4w) or q8w in patients with Crohn's disease. Figure reproduced with permission from Wang et al.¹⁶⁸

Table 9. Simulation of q4w and q8w ustekinumab dosing

Simulated results at steady state	Q4W	Q8W
Median trough serum concentration of ustekinumab	5.3 $\mu\text{g/mL}$	1.3 $\mu\text{g/mL}$
Median trough concentration of fecal calprotectin	424 $\mu\text{g/g}$	796 $\mu\text{g/g}$
Endoscopic remission rate	6.7%	4.2%
Endoscopic response rate	22.2%	16.7%

Abbreviations: Q4W, maintenance dosing every 4 weeks; Q8W, maintenance dosing every 8 weeks. Table reproduced with permission from Wang et al.¹⁶⁸

4. Discussion

We report the development of three sequential population pharmacometric models that link ustekinumab dose, ustekinumab exposure, FC and endoscopic outcomes in patients with moderate-to-severe CD in a prospective cohort study.

A one-compartment popPK model with first-order absorption and first-order elimination was previously published to describe the PK of ustekinumab following SC administration.^{148,169} Due to the availability of an intermediate ustekinumab concentration after IV infusion, the PK of ustekinumab in this study was well-described by a two-compartment model. The model was able to estimate the bioavailability because the ustekinumab serum concentrations were collected following both IV and SC dosing in each patient. Our model was not able to estimate the absorption rate constant due to the lack of ustekinumab concentration measurements shortly after SC injection.¹⁷⁰ The typical value of CL was estimated to be 0.235 L/day with a high precision. The model-estimated elimination half-life in patients with CD was 20.4 days, which is consistent with the reported ustekinumab half-life of approximately 19-22 days in other indications, including psoriasis, psoriatic arthritis, and UC.^{167,171,172}

A good understanding of the factors that affect ustekinumab exposure can contribute to the optimization of treatments in patients with CD. Bodyweight and serum albumin have been reported to explain the variability of ustekinumab PK in patients with CD¹⁷³, UC¹⁶⁷ and plaque psoriasis¹⁷¹. Albumin was identified as a covariate in our study. However, the bodyweight effect was not identifiable based on our dataset. We decided using standard allometric scaling to allow realistic model simulations when extrapolating outside the bodyweight range in our dataset, while at the same time not impacting the goodness of fit of our model (a drop in OFV by 33.6 points as compared to the base model without allometric scaling). Only one patient developed antibodies toward ustekinumab, thereby making it impossible to identify the effect of immunogenicity on ustekinumab clearance. The low incidence of anti-ustekinumab antibodies is consistent with low immunogenicity of ustekinumab observed in the IM-UNITI studies (27 of 1154 patients [2.3%]).¹⁶⁹

The popPK-PD model linking ustekinumab and FC concentrations quantifies the anti-inflammatory effect of ustekinumab that promotes the suppression of neutrophil trafficking

through the inflamed bowel wall and the release of calprotectin upon neutrophil death.^{174–176} As opposed to an endoscopic assessment, FC is a relatively patient-friendly, continuously traceable, non-invasive, surrogate biomarker with proven high sensitivity and high specificity for endoscopic disease activity.^{177–179} The model was not able to estimate the maximal effect of ustekinumab on inhibiting the production of FC, thereby it was fixed to 1, assuming ustekinumab can inhibit FC release completely. The ustekinumab serum concentration for achieving half-maximal drug effect was estimated to be 2.7 µg/mL, implying a 50% decrease in FC release at this serum ustekinumab concentration. The high IIV in baseline FC indicates a wide variability in baseline disease activity. The high IIV on IC₅₀ suggests that different patients require a different ustekinumab C_{trough} when aiming for the same FC target concentration. Although a high PD variability between patients was observed, the pcVPC of the popPK-PD model shows promising predictive performance at the population level.

The logistic regression model linked FC to the probability of achieving endoscopic outcomes at w24 of therapy. In recent years, endoscopic outcomes have become important therapeutic targets.³⁵ However, endoscopic outcome rates of ustekinumab under standard Q8W maintenance dosing remain unpromising in the UNITI trial.¹⁵⁶ Our model predicted that, in highly refractory patients with CD, endoscopic response and remission rates can be maximized to 35.8% and 19.9%, respectively, by reducing FC to an undetectable level.

Although the mechanism of action is different (anti-TNF vs. anti-IL12/23), a similar dose-exposure-FC-endoscopic remission sequence has been modeled for infliximab in biological-naïve patients with active luminal CD.¹⁸⁰ The median FC observed at baseline was similar between our study (1398 µg/g) and the infliximab study (1462 µg/g). The maximum drug effect on the production rate constant of FC was fixed to 100% in both popPK-PD models. The estimated maximum probability of achieving endoscopic remission was 19.9% in our current ustekinumab outcome model, while being 76.9% in the infliximab outcome model. The much lower maximum probability estimated in our model reveals the difficulty for achieving endoscopic remission in biological-(multi)refractory patients.

Standard Q8W maintenance dosing was compared to a treat-to-target dose optimization algorithm based on FC, CRP, and clinical symptoms in an ongoing clinical trial (Study of Treat to Target Versus Routine Care Maintenance Strategies in Crohn's Disease Patients Treated

With Ustekinumab, STARDUST; NCT03107793; study completed after this popPK-PD analysis).¹⁸¹ Preliminary results only shown a small numerical improvement in endoscopic outcome rates based on this PD monitoring algorithm.¹⁸² In our study, we recommend an individualized, two-step monitoring approach, consisting of FC monitoring and, only when FC is/remains high, ustekinumab monitoring (**Figure 13**). High ustekinumab concentrations after dose escalation indicate a possible mechanistic/PD failure, while low ustekinumab C_{trough} imply a PK failure. Loss of response due to PK failure during maintenance therapy may require re-induction. The optimal concentration cut-offs of FC and ustekinumab are yet to be identified based on larger studies. Costs associated with dose adjustments should also be considered, demanding a future pharmacoeconomic evaluation using popPK-PD-PE modeling.¹⁸³

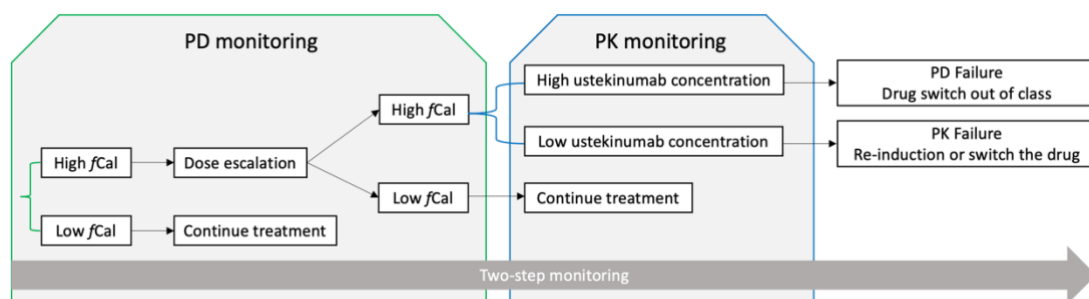


Figure 13. Recommended flowchart for therapeutic drug monitoring of ustekinumab in patients with Crohn's disease. fCal, fecal calprotectin; PD, pharmacodynamic; PK, pharmacokinetic; switch the drug, switch to a drug with another mechanism of action. Figure reproduced with permission from Wang et al.¹⁶⁸

The treatment costs are relatively the same between the two dose escalation approaches, doubling of dosing frequency (90 mg, Q4W) and doubling of the dose (180 mg, Q8W), since ustekinumab can be self-administered and does not require hospitalization. However, doubling of the dosing frequency during maintenance therapy was predicted to result in higher median C_{trough} than doubling of the dose (5.3 $\mu\text{g/mL}$ and 2.7 $\mu\text{g/mL}$). Therefore, dose escalation through doubling the dosing frequency is a more effective approach than doubling the dose.

Model-based simulation predicted improved endoscopic response and endoscopic remission rates after interval shortening to Q4W maintenance dosing in highly refractory patients. However, the absolute proportions of patients achieving endoscopic response (22.2%) or

endoscopic remission (6.7%) on Q4W dosing are still limited, partially because of the estimated low maximum endoscopic response and remission rates in highly refractory patients. The impact of interval shortening to Q4W maintenance dosing on endoscopic outcomes is also under investigation in an ongoing clinical trial (Loss of RESponse to Ustekinumab Treated by Dose Escalation, REScUE; NCT04245215).¹⁸⁴

Our analysis has several limitations. First, relatively sparse sampling and the lack of concentration measurements shortly after injection made it impossible to estimate k_a . Nevertheless, while partly relying on previous literature, we were able to develop a popPK model that adequately described our data. Still, rich ustekinumab (and FC) concentration data (i.e., intermediate concentrations, for instance through dry blood spot sampling) would allow establishing a more physiologically plausible popPK-PD model. Despite this limitation, the models described and predicted the data of matched ustekinumab exposure, FC concentration and endoscopic outcomes well. Second, SES-CD subscores was not available in our data. Future studies should collect these data to allow investigation of the relationship between disease extent and FC. Nevertheless, predictions of endoscopic outcome rates in the simulation should be interpreted with care. Our population outcome model was built using FC concentrations at w8 and endoscopic outcomes at w24, while simulations were performed assuming that the same relationship holds between maintenance FC and endoscopic outcomes at steady state. Still, our models provide insight into the ustekinumab dose-exposure-biomarker-response relationship, can be used to design future studies of ustekinumab in patients with CD, and may be used to guide dose optimization.

In conclusion, we developed a popPK-PD dose-exposure-biomarker-response model that links ustekinumab exposure and FC to endoscopic outcomes in patients with moderate-to-severe CD based on real-world data from a prospective cohort study. The developed model can guide clinical trial design and support model-informed dose optimization to improve endoscopic outcome rates. Our analyses showed that interval shortening from Q8W to Q4W maintenance dosing would result in lower FC concentrations, but the predicted gain in the proportion of patients achieving endoscopic response and remission was limited in highly refractory patient.

Appendix A

Supplementary tables

Table S1. *Univariate covariate analysis.*

Covariate effect on clearance	OFV	ΔOFV	df	P value
<i>Base model</i>	-305.8	-	-	-
Albumin	-344.4	-38.6	1	<0.001
Hemoglobin	-336.1	-30.3	1	<0.001
Fecal calprotectin	-321.9	-16.1	1	<0.001
Simple endoscopic score for Crohn's disease	-320.7	-14.9	1	<0.001
C-reactive protein	-319.8	-14.0	1	<0.001
Age	-316.7	-10.9	1	<0.001
Total number of biologicals used before	-316.2	-10.4	3	0.015
Previous exposure to anti-TNF agents	-314.1	-8.3	1	0.004
2-item patient-reported outcome	-312.7	-6.9	1	0.008
Smoking behaviour	-312.2	-6.4	1	0.040
Disease location	-309.8	-4.0	2	0.132
Corticosteroids at induction	-309.4	-3.6	2	0.167
Body mass index	-307.5	-1.7	1	0.186
Sex	-306.7	-0.9	1	0.335
Lean body mass	-305.9	-0.1	1	0.681
Immunomodulators	-305.8	0.0	1	0.861
Baseline disease activity	-305.8	0.0	1	0.884

Abbreviations: *Anti-TNF, anti-tumor necrosis factors; df, degree of freedom; OFV, objective function value; ΔOFV, change in objective function value as compared to the base model. Table reproduced with permission from Wang et al.¹⁶⁸*

Supplementary Figures

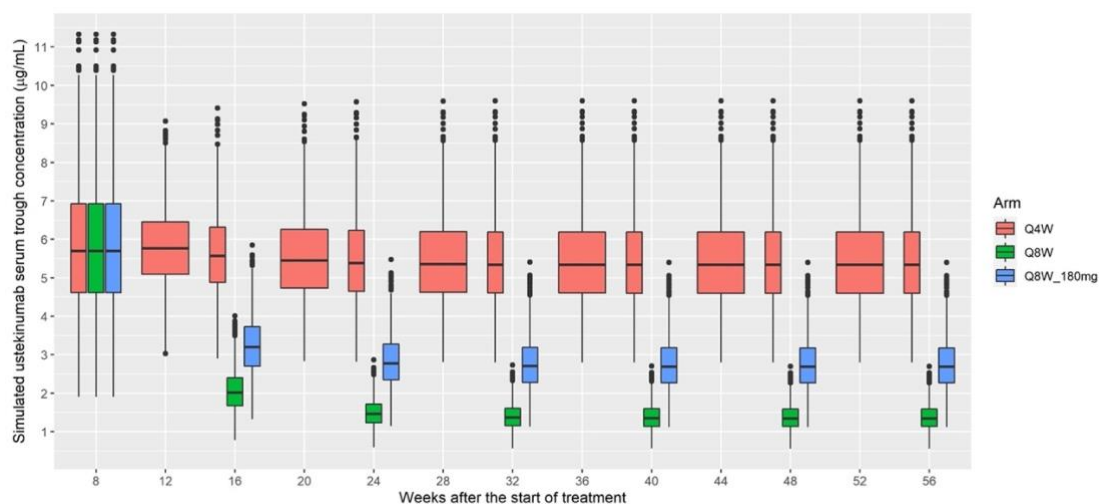
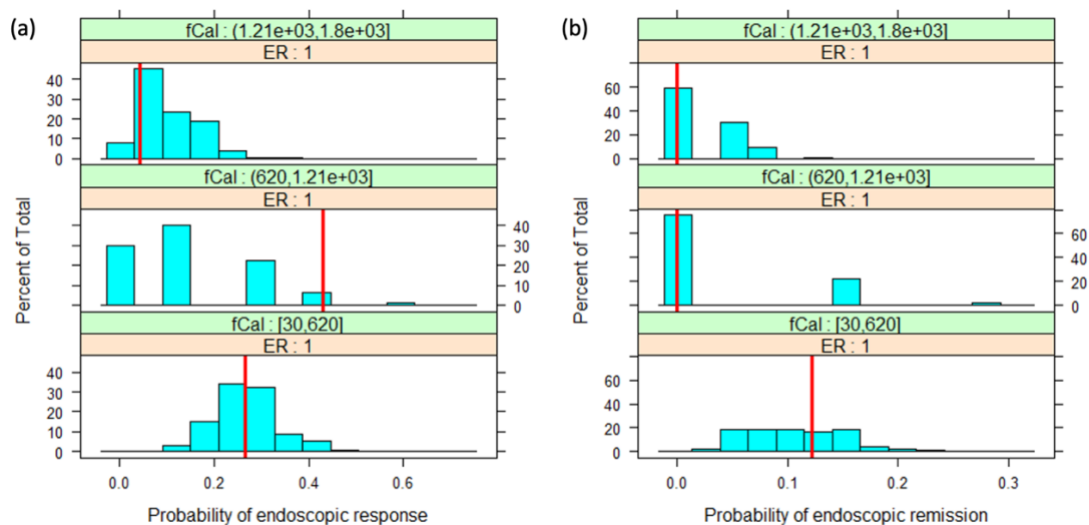


Figure S2. Simulated ustekinumab trough concentration following different dosage regimens of ustekinumab during maintenance. Ustekinumab serum trough concentrations at different treatment weeks following maintenance dosing of 90 mg every 4 weeks (Q4W), 90 mg Q8W, and 180 mg Q8W are presented. Figure reproduced with permission from Wang et al.¹⁶⁸

NONMEM control streams

Pharmacokinetic model

\$PROBLEM PopPK analysis of Ustekinumab in patients with Crohn's disease

\$INPUT ID EVID MDV TIME AMT CMT DV ALB

\$DATA Ustekinumab_PKPD_NONMEM_BLOQ.csv IGNORE=@

\$SUBROUTINE ADVAN6 TOL=3

\$MODEL COMP = (1) ; site of absorption

COMP = (2) ; central compartment

COMP = (3) ; peripheral compartment

\$PK

CLALB = ((ALB/42.65)**THETA(9))

CLCOV = CLALB

A_0(1)= 0

A_0(2)= 0

KA = THETA(1)

TVV1 = THETA(2)*((WEIGHT/65.27)**1)

TVCL = THETA(3)*((WEIGHT/65.27)**0.75)

TVF1 = THETA(4)

TVQ = THETA(5)*((WEIGHT/65.27)**0.75)

TVV2 = THETA(6)*((WEIGHT/65.27)**1)

TVCL = CLCOV*TVCL

F1 = TVF1*EXP(ETA(2))

V1 = TVV1

V2 = TVV2

Q = TVQ

CL = TVCL*EXP(ETA(1))

KE = CL/V1

K12 = Q/V1

K21 = Q/V2

\$DES

DADT(1)=-KA*A(1)

DADT(2)= KA*A(1) -KE*A(2) -K12*A(2) +K21*A(3)

DADT(3)= K12*A(2) -K21*A(3)

\$ERROR

CP = A(2)/V1

IPRED= LOG(A(2)/V1+0.00001)

IRES = DV-IPRED

W = SQRT(THETA(8)**2 + (THETA(7)/CP)**2)

IF (W.EQ.0) W=1

IWRES= IRES/W

LLOQ = LOG(0.25)

IF (DV.GE.LLOQ) THEN

```

F_FLAG = 0
Y      = IPRED+EPS(1)*W
ELSE
F_FLAG = 1
Y      = PHI((LLOQ-IPRED)/W)
ENDIF

```

```

$THETA (0.142) FIX    ; KA
      (0,5.05)      ; V1
      (0,0.235)     ; CL
      (0,0.624)     ; F1
      (0,0.0426)    ; Q
      (0,0.748)     ; V2
      (-0.272)      ; Additive error
      (0,0.136)     ; Proportional error
      (-1.21)       ; CLALB

```

```

$OMEGA 0.093      ; 1. omCL
      0.13        ; 2. omF

```

```

$SIGMA 1 FIX

```

```

$EST METHOD=1 INTER LAPLACIAN NUMERICAL MAXEVAL=99999 SADDLE_RESET=1
$COV Matrix=S PRINT=E
$TABLE ID EVID MDV TIME AMT CMT DV ALB IPRED F1 KA V1 V2 CL Q NOAPPEND NOPRINT
ONEHEADER FILE=data1.tab

```

Pharmacokinetic-pharmacodynamic model

\$PROBLEM PopPK-PD analysis of Ustekinumab in patients with Crohn's disease

\$INPUT ID EVID MDV TIME AMT IPREDUST DV IF1 IKA IV1 IV2 ICL IQ LOQ1

\$DATA data2.csv IGNORE=@

\$SUBROUTINE ADVAN13 TOL=9

\$MODEL COMP = (1) ; LINEAR ABSORPTION
COMP = (2) ; CENTRAL COMPARTMENT
COMP = (3) ; Peripheral COMPARTMENT
COMP = (4) ; CPT,DEFOBS

\$PK

A_0(1) = 0

A_0(2) = 0

A_0(3) = 0

F1 = IF1

KA = IKA

V1 = IV1

CL = ICL

KE = CL/IV1

Q = IQ

V2 = IV2

K12 = Q/V1

K21 = Q/V2

;;; baseline model ;;;

SHP = THETA(5)

TVBASE = THETA(1)

BASE = TVBASE*EXP((((EXP(ETA(1))))**SHP)-1)/SHP) ; B1 estimate baseline

A_0(4) = BASE

;;; drug effect ;;;

KOUT = THETA(2)*EXP(ETA(2))

KIN = BASE*KOUT

IMAX = THETA(3)

IC50 = THETA(4)*EXP(ETA(3))

\$DES

DADT(1)= -KA*A(1)

DADT(2)= KA*A(1) -KE*A(2) -K12*A(2) +K21*A(3)

DADT(3)= K12*A(2) - K21*A(3)

CONC = A(2)/V1 ; UST concentration

DADT(4)= KIN*(1-((IMAX*CONC)/(IC50+CONC)))-KOUT*(A(4))

\$ERROR CP = A(2)/V1

IPRED = A(4) ; fecal calprotectin

IRES = DV - IPRED

W = IPRED*THETA(6)

IF (W.EQ.0)

```

W      = 1
IWRES = IRES/W

IF (LOQ1.EQ.0) THEN
  F_FLAG = 0
  Y      = IPRED+EPS(1)*W
ENDIF

IF(LOQ1.EQ.1) THEN
  F_FLAG = 1
  Y      = PHI((30-IPRED)/W)
ENDIF

IF(LOQ1.EQ.-1) THEN
  F_FLAG = 1
  Y      = PHI((IPRED-1800)/W)
ENDIF

$THETA (0,1550,1800) ; BASE (µg/g fecal calprotectin in stool)
(0,0.0416)           ; KOUT (day-1)
1 FIX                ; IMAX
(0, 2.71)             ; IC50 (µg/mL ustekinumab)
0.133                ; SHP
(0,0.551,1)          ; Proportional error

$OMEGA 1.27           ; IIV_BASE
0.772               ; IIV_EC50
0.0933              ; IIV_KOUT

$SIGMA 1 FIX

$ESTIMATION MAXEVAL=9999 NOABORT PRINT=1 NSIG=3 SIGL=9 METHOD=1 INTER SLOW LAPLACE
NUMERICAL SADDLE_RESET=1
$COV      Matrix=S PRINT=E
$TABLE    ID EVID MDV TIME AMT IPREDUST IF1 IKA IV1 IV2 ICL IQ DV LOQ1 BASE KOUT IMAX IC50
SHP FILE=data2.dta

```

Pharmacodynamic model

\$PROBLEM PopPD analysis of endoscopic remission in patients with Crohn's disease

\$INPUT ID WEEK FCAL DV=Endoscopic_Remission_Week_24

\$DATA data3.csv IGNORE=@

\$PRED

; ----- transitions from score no Endoscopic Remission to Endoscopic Remission -----

; 0 --> 1

IMAX = THETA(1) +ETA(1)

IC50 = THETA(2)

P01 = P01MAX-P01MAX*IMAX*(FCAL/(IC50+FCAL))

; ----- transition fractions -----

IF (DV.EQ.1) Y = P01

IF (DV.EQ.0) Y = 1 - P01

\$THETA (0,0.199,1) ; P01MAX

1 FIX ; IMAX

(0,214) ; IC50 (µg/g fecal calprotectin in stool)

\$OMEGA 0 FIX ; ETA1

\$COV PRINT=E

\$TABLE ID WEEK FCAL DV IMAX IC50 P01MAX P01 NOPRINT ONEHEADER FILE=data3_1.dta

\$PROBLEM PopPD analysis of endoscopic response in patients with Crohn's disease

\$INPUT ID WEEK FCAL DV=Endoscopic_Response_Week_24

\$DATA data3.csv IGNORE=@

\$PRED

; ----- transitions from score no Endoscopic Response to Endoscopic Response -----

; 0 --> 1

IMAX = THETA(1) +ETA(1)

IC50 = THETA(2)

P01 = P01MAX-P01MAX*IMAX*(FCAL/(IC50+FCAL))

; ----- transition fractions -----

IF (DV.EQ.1) Y = P

IF (DV.EQ.0) Y = 1 - P

\$THETA (0,0.358,1) ; P01MAX

1 FIX ; IMAX

(0,694) ; IC50 (µg/g fecal calprotectin in stool)

\$OMEGA 0 FIX ; ETA1

\$COV PRINT=E

\$TABLE ID WEEK FCAL DV IMAX IC50 P01MAX P01 NOPRINT ONEHEADER FILE=data3_2.dta

Chapter II. PopPK simulation of SC infliximab and vedolizumab

This chapter is based on

Wang Z, Verstockt B, Sabino J, Ferrante M, Vermeire S, Dreesen E. Therapeutic Drug Monitoring Can Guide the Intravenous-to-Subcutaneous Switch of Infliximab and Vedolizumab: A Simulation Study. *Clin Gastroenterol Hepatol* 2023; 21: 3188-3190.e2.

Research letter

SC formulations of infliximab and vedolizumab were recently added to the armamentarium for maintenance treatment of patients with moderate-to-severe Crohn's disease and ulcerative colitis. These formulations showed non-inferior efficacy and safety to the IV formulations in the pivotal clinical trials and offer the convenience of self-injecting at home to patients with IBD.^{92,185,186}

IBD patients and physicians have recognized the added value of SC infliximab and vedolizumab in clinical practice.¹⁸⁷ However, further investigation is required to identify the patients most likely to benefit from switching to SC dosing. Only switching from label IV dosing was evaluated in the pivotal trials, but in real-world practice, patients may also receive IV infliximab or vedolizumab with different doses and intervals. In patients who switched to label-recommended Q2W SC infliximab, loss of response was observed mainly in patients on escalated IV dosing.⁹³ Although an escalated IV dosing regimen could pose a risk of relapse upon SC switch, there may still be SC dosing options that result in similar exposure and can thus be considered adequate.

We hypothesize that TDM can guide the IV-to-SC switch. However, there are no clear recommendations for TDM due to the limited experience and lack of data in this new clinical context. Therefore, we performed popPK simulations to explore the exposure profiles following various label and off-label IV-to-SC dosing scenarios (**Appendix B Supplementary Methods**).

Following IV-to-SC switch according to label, the serum C_{trough} will gradually increase over eight weeks (for infliximab) and 16 weeks (for vedolizumab) before reaching the new steady state (SS; infliximab: median 12.8 [IQR 9.2-17.6] mg/L, vedolizumab: 31.5 [IQR 23.2-42.0] mg/L; **Figure 14A-B**, left). A linear relationship was observed between IV and SC C_{trough} at SS ($C_{trough,ss}$) for both drugs (**Figure 14A-B**, right). Therefore, under IV-to-SC switch according to label, we recommend taking one $C_{trough,ss}$ sample right before the first SC dose ($C_{trough,ss,IV}$) and comparing it to a $C_{trough,ss}$ sample taken right before the fifth SC infliximab dose or ninth SC vedolizumab dose ($C_{trough,ss,SC}$). Using the right panels of **Figure 14A-B** as clinical support charts, clinicians can easily compare their measured $C_{trough,ss,SC}$ with the expected $C_{trough,ss,SC}$.

range (Y-axis) that corresponds to the previously measured $C_{\text{trough,ss,IV}}$ (X-axis). For instance, a patient with a $C_{\text{trough,ss,IV}}$ of 5 mg/L infliximab is expected to achieve a median $C_{\text{trough,ss,SC}}$ of 16.2 (95% prediction interval [PI] 14.0-18.5) mg/L after switching to SC dosing. Observing a $C_{\text{trough,ss,SC}}$ below the 95% PI suggests that the infliximab or vedolizumab clearance has increased, which is an alarm signal for higher disease activity.⁶ We recommend monitoring these patients more closely.

Following switch according to label, TDM will only be informative when performed two (infliximab) or four (vedolizumab) months after the first SC dose, when SS is reached. The lack of TDM support in the first months upon switch may be problematic due to the risk of temporary underexposure, as the eight-week averaged area under the concentration-time curve (AUC_{8w}) decreases upon switch and only normalizes when SS is reached (**Figure 14C-D**). It is unknown whether this temporary decrease in AUC_{8w} poses a clinical risk. As a precaution, one may consider administering the first SC dose four instead of eight weeks after the last IV dose, thereby hitting SS much faster (immediately for infliximab and in two months for vedolizumab), allowing TDM earlier during the uncertain time upon switching.

The Q2W SC dosing regimen of infliximab has been selected with the purpose of (1) exceeding a $C_{\text{trough,ss}}$ of 5 mg/L and (2) generating a SS AUC_{8w} ($AUC_{8w,ss}$) that aligns closely to that achieved by Q8W 5 mg/kg IV dosing.⁷ Consequently, patients on an escalated IV regimen switching to standard Q2W SC dosing are likely to be underexposed. Depending on the exposure metric that is believed to be the main driver of response ($C_{\text{trough,ss}}$ or $AUC_{8w,ss}$), Q1W instead of Q2W SC dosing may be required to guarantee unchanged exposure before versus after switch (**Appendix B Table S1**). Patients on Q6W or Q8W IV infliximab can safely switch to Q2W SC infliximab without a meaningful decrease in $C_{\text{trough,ss}}$, while patients on Q4W IV infliximab need Q1W SC infliximab (**Figure 14E**). Vedolizumab $C_{\text{trough,ss}}$ will always be maintained after switching to Q2W dosing (**Figure 14F**). While the $C_{\text{trough,ss}}$ target is easily maintained, it is more challenging to guarantee stable $AUC_{8w,ss}$ upon switch. Surprisingly, switching to Q2W SC infliximab resulted in a lower $AUC_{8w,ss}$ than all possible IV infliximab dosing regimens used in clinical practice (**Appendix B Figure S1a**). As for the $C_{\text{trough,ss}}$ targets, it is easier for vedolizumab than for infliximab to maintain the $AUC_{8w,ss}$ target. Only patients on Q4W IV vedolizumab should switch to Q1W SC dosing (**Appendix B Figure S1b**).

In conclusion, we performed a popPK simulation study to inform clinicians how to integrate TDM of infliximab and vedolizumab into their clinical practice as a tool to monitor patients and guide decision-making on the optimal dosing strategy. First, clinical support charts were developed to help clinicians evaluate PK changes that may indicate changes in disease activity after switch.⁵ Second, switching four instead of eight weeks after the last IV dose can hit SS faster, thereby avoiding the risk of temporary underexposure. However, the clinical relevance of this theoretical risk is currently unknown. Third, we investigated SC dosing opportunities for patients on escalated IV dosing regimens. Our recommendations differ dependent on the belief whether it is $C_{\text{trough,ss}}$ or $AUC_{8w,ss}$ that drives response. Following the widely accepted assumption that $C_{\text{trough,ss}}$ is the relevant exposure metric, only patients on Q4W IV infliximab need Q1W SC dosing. An exposure-response model built on rich data of the clinical development program can also be used to investigate the relative importance of $C_{\text{trough,ss}}$ versus $AUC_{8w,ss}$ as a driver for clinical outcomes. Our popPK predictions align well with the limited real-world data available today^{5,8}, demonstrating the power of *in silico* research tools. Ideally, company data are modeled in conjunction with real-world data to increase insights into PK mechanisms, exposure-response relationships, the impact of immunogenicity, and the potential for dashboard-guided IV-to-SC switching.

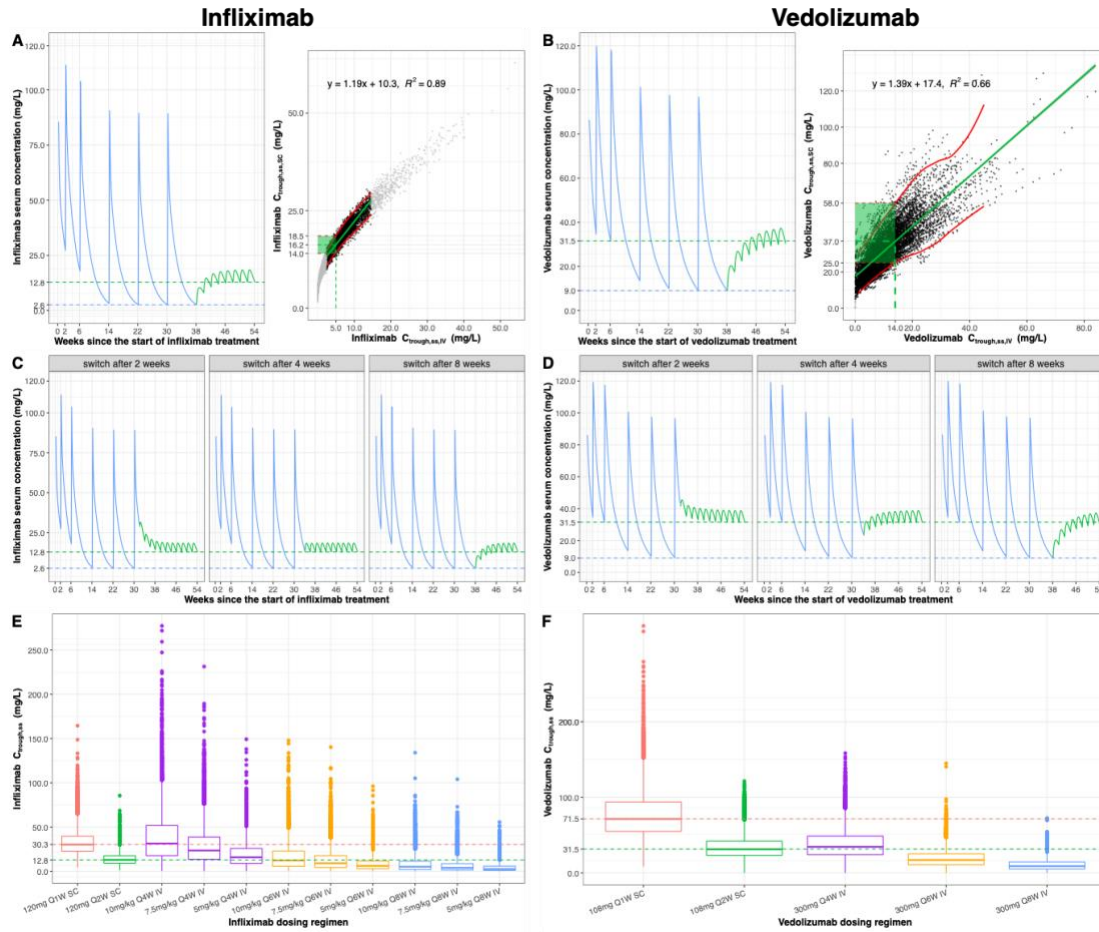


Figure 14. (A-B) Left: Simulated time course of median serum concentrations in the typical patient switching from 5 mg/kg Q8W IV to 120 mg Q2W SC infliximab and 300 mg Q8W IV to 108 mg Q2W SC vedolizumab. Horizontal dashed lines correspond to steady-state trough concentrations. Right: Linear regression between $C_{trough,ss,IV}$ and $C_{trough,ss,SC}$ on label dosing. The regression line (green) and 95% prediction interval (PI, red) of the simulated data are plotted. The green shaded area represents the 95% PI of $C_{trough,ss,SC}$ corresponding to a 5 mg/L (infliximab) and 14 mg/L (vedolizumab) $C_{trough,ss,IV}$. The linear relationship is only valid within the infliximab serum concentration range of 2.5 mg/L to 14.6 mg/L. (C-D) Simulated time course of median serum concentrations in the typical patient switching from 5 mg/kg Q8W IV to 120 mg Q2W SC (infliximab) and 300 mg Q8W IV to 108 mg Q2W SC (vedolizumab). Horizontal dashed lines correspond to steady-state serum trough concentrations. (E-F) Simulated $C_{trough,ss}$ following various maintenance dosing regimens. Horizontal dashed lines correspond to median $C_{trough,ss}$ following 120 mg (infliximab) and 108 mg (vedolizumab) Q2W or Q1W SC dosing. $C_{trough,ss}$, trough concentration at steady state; IV, intravenous; Q1W, weekly; Q2W, every two weeks; SC, subcutaneous. Figure reproduced with permission from Wang et al.⁷⁶

Appendix B

Supplementary Methods

The infliximab popPK model used in this study was recreated based on a published model that was developed to describe the PK of infliximab following IV and SC administration.⁴⁴ IIV was accounted for during the simulations. A total of 10,000 virtual patients were simulated for each dosing scenario. Covariates and residual errors were ignored during the simulation because they explain only a small part of the overall IIV. Three infliximab IV doses of 5 mg/kg, 7.5 mg/kg, and 10 mg/kg with dosing intervals of Q4W, Q6W, and Q8W, made up a total of nine IV dosing regimens that were simulated. Switching from Q8W 5 mg/kg IV to two SC dosing regimens, Q2W SC 120 mg and Q1W SC 120 mg, was also simulated.

The published popPK models of vedolizumab were developed based on either only IV or only SC data. Therefore, we combined a published popPK model based on IV data with the bioavailability and absorption parameters from a model based on SC data reported in a 2019 conference abstract.^{72,188} The IIV on bioavailability was not mentioned in the SC model and was set to a coefficient of variation of 20%. No covariates (except for disease type: CD and UC) and residual errors were included in the simulation. A vedolizumab IV dose of 300 mg with three dosing intervals (Q4W, Q6W, and Q8W) was simulated. Switching from Q8W 300 mg IV to two SC dosing regimens, Q2W SC 108 mg and Q1W SC 108 mg, was also simulated. For each dosing scenario, 10,000 virtual patients were equally divided into two disease groups (CD and UC; 5,000 patients each).

All models were coded in NONMEM (version 7.5; Icon Plc, Gaithersburg, MA, USA). Simulations were performed using NONMEM with a GNU Fortran 95 compiler. Data were analyzed in R (version 4.2.0; R Foundation for Statistical Computing, R Core Team, Vienna, Austria) with the RStudio integrated development environment (version 1.2.5001; RStudio, Inc., Boston, MA, USA).

Supplementary Tables

Table S1. Recommended subcutaneous dosing regimens for an unreduced exposure after intravenous-to-subcutaneous switch.

Infliximab IV dosing regimens	Infliximab SC dosing regimen required	
	$C_{trough,ss}$ target	$AUC_{8w,ss}$ target
Q8W 5 mg/kg	Q2W 120 mg	Q1W 120 mg SC
Q8W 7.5 mg/kg	Q2W 120 mg	Q1W 120 mg SC
Q8W 10 mg/kg	Q2W 120 mg	> Q1W 120 mg SC*
Q6W 5 mg/kg	Q2W 120 mg	Q1W 120 mg SC
Q6W 7.5 mg/kg	Q2W 120 mg	> Q1W 120 mg SC*
Q6W 10 mg/kg	Q2W 120 mg	> Q1W 120 mg SC*
Q4W 5 mg/kg	Q1W 120 mg	> Q1W 120 mg SC*
Q4W 7.5 mg/kg	Q1W 120 mg	> Q1W 120 mg SC*
Q4W 10 mg/kg	Q1W 120 mg	> Q1W 120 mg SC*
Vedolizumab IV dosing regimens	Vedolizumab SC dosing regimen required	
	TC_{ss} target	$AUC_{8w,ss}$ target
Q8W 300 mg	Q2W 108 mg	Q2W 108 mg
Q6W 300 mg	Q2W 108 mg	Q2W 108 mg
Q4W 300 mg	Q2W 108 mg	Q1W 108 mg

Abbreviations: $AUC_{8w,ss}$ eight-week averaged area under the concentration-time curve at steady state; IV, intravenous; QXW, every X weeks; SC, subcutaneous; $C_{trough,ss}$ trough concentration at steady state.

* If it is even considered appropriate. Table reproduced with permission from Wang et al.⁷⁶

Supplementary Figures

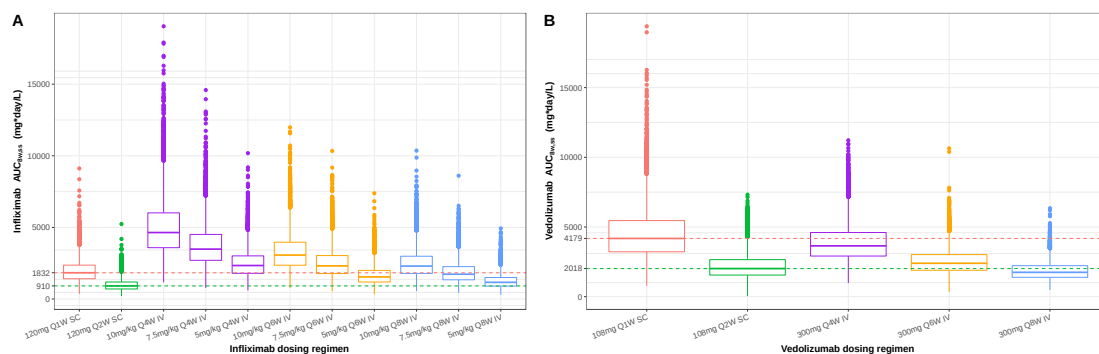


Figure S1. Simulated $AUC_{8W,SS}$ following various maintenance dosing regimens of infliximab (A) and vedolizumab (B). Horizontal dashed lines correspond to median $AUC_{8W,SS}$ following 120 mg (infliximab) and 108 mg (vedolizumab) Q2W or Q1W SC dosing. $AUC_{8W,SS}$ eight-week averaged area under the concentration-time curve; IV, intravenous; Q1W, weekly; Q2W, every two weeks; SC, subcutaneous. Figure reproduced with permission from Wang et al.⁷⁶

Chapter III. PopPK-PD modeling and simulation of SC infliximab

This chapter is based on

Wang Z, Ferrante M, Vermeire S, Dreesen E. Population pharmacokinetics–pharmacodynamics modeling and simulation of infliximab CT-P13 during intravenous-to-subcutaneous switching in patients with inflammatory bowel diseases. [Submission in preparation]

Abstract

Background: The SC formulation of infliximab CT-P13 has demonstrated non-inferior pharmacokinetics PK and comparable efficacy to the IV formulation. However, it remains unclear if patients on escalated IV dosing can switch to standard SC dosing without losing therapeutic response. This study aims to elucidate the dose–exposure–response relationship during the IV-to-SC switch in patients with CD and UC using popPK–PD modeling and simulation.

Methods: Data were collected from two single-dose Phase I studies in healthy volunteers and a Phase I study in patients with CD and UC. In patients, frequent PK sampling was conducted during the switch from 5 mg/kg IV infliximab to Q2W SC infliximab injections of 120/180/240 mg. Fecal calprotectin was measured in patients at baseline, w2, w6, and Q8W thereafter until w54. Endoscopic remission was assessed at screening, w22/30, and w54. We performed popPK–PD modeling and simulation (n=1000) using NONMEM 7.5 to evaluate FC time courses and probabilities of endoscopic remission post-switch.

Results: A total of 253 healthy volunteers and 179 patients (101 CD, 78 UC) contributed 7351 infliximab and 1201 FC concentrations. Infliximab clearance increased with decreasing serum albumin, increasing body weight, and presence of neutralizing antibodies. FC levels decreased when overall infliximab exposure (not just trough concentrations) increased. Reductions in FC were associated with increased probabilities of endoscopic remission in patients with UC, but not CD. Simulations showed that the standard IV-to-SC switch at w6 of infliximab therapy further decreases FC and results in a significantly higher probability of endoscopic remission at w22/30 (54% if no switch vs. 63% if switch). Patients on escalated Q6W/Q8W IV maintenance regimens can switch to 120 mg Q2W SC infliximab without an increase in FC. Patients on Q4W IV infliximab require higher SC dosing to avoid an FC increase.

Discussion: We quantitatively characterized the relationship between infliximab CT-P13 dosing, serum concentrations, FC levels, and endoscopic remission in patients with CD and UC switching to SC therapy. Our model serves as a powerful tool for investigating various IV-to-SC dosing scenarios of infliximab CT-P13.

1. Introduction

Inflammatory bowel diseases, with CD and UC as the two main subtypes, are lifelong, debilitating conditions characterized by chronic inflammation of the gastrointestinal tract.^{6,7} Infliximab, a chimeric immunoglobulin G1k monoclonal antibody targeting TNF- α , was the first biological drug approved for IBD treatment.⁵⁴ CT-P13 was the first infliximab biosimilar and demonstrated biosimilarity in RCTs and post-marketing observational studies across multiple indications, including IBD.^{189,190}

Recently, a SC formulation of CT-P13 has been approved for maintenance therapy in IBD after demonstrating non-inferior PK compared to IV infliximab.⁹² The label-recommended dosing for SC CT-P13 involves a fixed dose of 120 mg Q2W, following the standard weight-based IV induction of 5 mg/kg at w0 and w2 of induction, or starting from 4 weeks after the last IV infusion during maintenance.¹⁹¹ The introduction of SC CT-P13 is expected to offer operational and cost advantages over IV infliximab therapy for both clinicians and patients.¹⁹²

The SC administration of CT-P13, similar to other SC administered biologicals, is characterized by slower absorption, incomplete bioavailability, and lower peak concentrations compared to its IV formulation.¹⁹³ Clinical trials and simulation studies have shown that the label-recommended IV-to-SC switch results in significantly higher infliximab trough concentrations yet similar cumulative drug exposure.^{92,93,194}

Due to loss of response over time, up to 50% of patients on IV infliximab require dose escalation, with doses up to 10 mg/kg or intervals as short as four weeks.¹⁹⁵ For these patients, switching to 120 mg SC infliximab Q2W may pose a risk of a decrease in cumulative exposure, and they may benefit from an escalated SC dosing regimen.¹⁹⁴ This was observed in the REMSWITCH study, where patients receiving 10 mg/kg infliximab IV needed to double the SC dose to avoid treatment relapse after switching.⁹³ Further research is needed to identify the right patients and timing for switching to SC formulations and to explore optimal strategies in case of loss of response.¹⁸⁷ Therefore, this study aims to elucidate the dose–exposure–response relationship during the infliximab IV-to-SC switch in IBD patients through popPK–PD modeling and simulation.

2. Methods

2.1. *Patients and study design*

We utilized data from healthy volunteers in two single-dose Phase I studies (CT-P13 1.5 and 1.9; NCT03446976) and patients with IBD in a two-part Phase I study (CT-P13 1.6; NCT02883452).

The CT-P13 1.5 study was an open-label, dose-escalation, single-dose Phase I trial assessing the safety and PK of SC CT-P13 in 38 healthy volunteers. Participants received either a single SC injection of 120 mg (n=6), 180 mg (n=7), or 240 mg (n=7), or a single IV infusion of 3 mg/kg (n=10) or 5 mg/kg (n=8).⁴⁴

The CT-P13 1.9 study was an open-label, randomized, single-dose, parallel-group Phase I trial comparing PK and safety of SC CT-P13 in 220 healthy volunteers, all receiving a single SC injection of 120 mg.⁴⁴

The CT-P13 1.6 study was an open-label, randomized, two-part Phase I trial evaluating the PK, efficacy, and safety of SC and IV CT-P13 in patients with active CD and UC. Part 1 aimed to identify the optimal SC dose and enrolled 45 biologic-naïve patients with moderate-to-severe CD. After initial IV dosing (5 mg/kg at w0 and w2), patients were randomized to IV (5 mg/kg Q8W) or SC (120 mg, 180 mg, or 240 mg Q2W) regimens from w6 to w54. Part 2 aimed to demonstrate PK non-inferiority between SC and IV formulations in 136 patients (57 CD, 79 UC). After initial IV dosing (5 mg/kg), patients in the SC arm received weight-based doses (120 mg for ≤80 kg; 240 mg for >80 kg) Q2W from w6 to w54. Detailed study design and conduct have been previously reported.^{92,196}

2.2. *PK and PD data*

In the CT-P13 1.5 and 1.9 studies, serum samples were collected pre-dose and at 2 (0.3 in the IV arm), 3, 6, 12, and 24 hours, and at 2, 3, 7, 14, 28, 42, 56, and 84 days after the injection or the start of infusion. In the CT-P13 1.6 study, pre-dose serum samples for PK assessment were obtained at all study visits from week 0 to week 54, including a PK monitoring visit between w22 and w30. The detailed PK sampling scheme for the CT-P13 1.6 study has been reported previously.^{92,196}

Serum infliximab concentrations were measured using the Gyrolab xP method (BioAgilytix, Durham, NC, USA) in the CT-P13 1.5 study, and the Meso Scale Discovery electrochemiluminescence method (Meso Scale Diagnostics, Rockville, MD, USA) in the CT-P13 1.9 and 1.6 study. Cross-validation studies demonstrated that the performance of the two methods has acceptable comparability.⁴⁴ The LLOQ was 0.1 mg/L for both methods.

Antibodies towards infliximab (ATI) and neutralizing antibodies (NAb) were analyzed using an electrochemiluminescence platform with an affinity capture elution step (Celltrion, Inc, Incheon, Republic of Korea). These drug-tolerant assays could detect anti-drug antibody concentrations of ≥ 25 ng/mL in the presence of 120 mg/L infliximab in serum.

In the CT-P13 1.6 study, biomarker responses (C-reactive protein; CRP, and fecal calprotectin; FC) were evaluated pre-dose at baseline and weeks 2, 6, 14, 22, 30, 38, 46, and 54. Endoscopy was performed at screening, weeks 30, and 54 in Part 1, and at screening, weeks 22, and 54 in Part 2. Endoscopic remission was defined as a SES-CD ≤ 2 for patients with CD and a Mayo endoscopic subscore ≤ 1 for patients with UC.

2.3. *Model development*

2.3.1. PopPK model

Infliximab concentrations from the three studies were pooled to develop a popPK model. First-order conditional estimation with interaction Laplacian and the ADVAN6 differential equation solver were used for parameter estimation. Concentrations below the LLOQ were handled using the M3 method.¹⁹⁷ When consecutive samples were below the LLOQ in between two doses, only the first sample was retained in the database.

A base popPK model was fitted to the infliximab concentration–time data. One-, two- and three-compartment models with linear elimination were explored. Interindividual variability was tested for different PK parameters. Additive, proportional, and combined error models were explored to describe residual unexplained variability. Individual PK parameters were assumed to be log-normally distributed. Covariate model building was performed to yield the final popPK model (see Section 2.3.4).

2.3.2. Fecal calprotectin model

A sequential popPK–PD modeling analysis method was used to fit PK and biomarker data (*i.e.*, FC concentrations). PK parameters estimated from the final popPK model were fixed and were used to generate individually predicted infliximab concentrations, which were subsequently linked to the observed FC concentration–time courses in an indirect response model in which infliximab concentrations were assumed to either inhibit FC release or promote FC degradation. Covariate model building was performed to yield the final popPK–PD model (see Section 2.3.4).

2.3.3. Endoscopic remission model

A sequential popPD analysis method was applied to fit endoscopic remission data. Empirical Bayes estimates of individual parameters from the FC model were used to predict individual FC concentrations. These predicted FC concentrations were then linked to the observed endoscopic outcomes (active endoscopic disease [state=0], endoscopic remission [state=1], or missing endoscopic data [state=2]) in a logistic regression Markov model.

The individual predicted FC concentration at weeks 2, 6, and 14 of therapy was evaluated as an explanatory variable for the exposure–response relationship for achieving endoscopic remission at week 22 in the CT-P13 1.6 study Part 2 or week 30 in the CT-P13 1.6 study Part 2. The individual predicted FC concentrations at weeks 2, 6, 14, 22, 30, 38, and 46 were evaluated as an explanatory variable for maintaining endoscopic remission at week 54 of therapy. More specifically, the individual predicted FC concentrations were used to predict transition probabilities between consecutive endoscopic evaluations (active endoscopic disease [state=0], endoscopic remission [state=1], and missing endoscopic data [state=2]), with the contribution of time effects to transition probabilities also assessed. The Laplacian method was used to approximate the marginal likelihood.

2.3.4. Covariate analysis

The continuous, time-varying covariates included age, serum albumin, ATI titer, bodyweight, CRP, hemoglobin, platelet count, SC dose amount, and white blood cell count. The categorical, time-varying covariates included concomitant steroids, concomitant

immunomodulators, route of administration, presence of ATI, and presence of NAb. Baseline covariates included sex assigned at birth and concomitant immunomodulators and steroids.

Covariates were tested in the base model using stepwise forward addition followed by backward deletion. In the forward step, a covariate was included if it decreased the OFV by at least 6.63 points ($\alpha=0.01$, degree of freedom [df]=1). During backward deletion, a covariate was retained if its removal increased the OFV by more than 10.83 points ($\alpha=0.001$, df=1). The final model resulted after backward deletion. All covariates were tested in both the popPK and popPK-PD models. For continuous covariates, power and exponential parameterization were evaluated, while linear parameterization was tested for categorical covariates. Continuous covariates were centered around their median values. Missing time-varying covariates were imputed using the last observation carried forward, while missing baseline covariates were imputed with the median value from the population.

2.4. Model evaluation

The most parsimonious model was selected based on comparisons using standard procedures, including the precision and plausibility of parameter estimates, the OFV, goodness-of-fit plots, and prediction-corrected visual predictive check (pcVPC) plots (n=1000 simulated replicates of the original dataset). Sampling importance resampling (SIR) with samples and resamples equal to 5000 and 1000, respectively, was performed to obtain 95% confidence intervals of parameter estimates.

2.5. Simulations

Monte Carlo simulations were conducted using the final popPK and FC models to simulate infliximab and FC concentrations over time for various IV-to-SC switch scenarios. Each scenario was represented by a virtual patient with CD and UC with typical covariate values (bodyweight of 75 kg, serum albumin of 45 g/L, and absence of NAb). For each IV-to-SC switch scenario, 1000 simulations were performed using the \$SIM function in NONMEM. Probabilities of achieving endoscopic remission at week 22/30 or week 54 were calculated from the endoscopic remission model based on the simulated FC concentrations.

First, infliximab and FC concentrations over time were simulated for the IV-to-SC switch scenarios in the CT-P13 1.6 study: after initial IV dosing (5 mg/kg at weeks 0 and 2), treatment

continued with IV (5 mg/kg Q8W) or SC (120 mg, 180 mg, or 240 mg Q2W) from week 6 to week 54.

Second, infliximab and FC concentrations over time were simulated for switching from both standard and escalated IV maintenance regimens (all combinations of 5 mg/kg, 7.5 mg/kg, or 10 mg/kg and Q4W, Q6W, or Q8W infusion intervals) to standard and escalated SC maintenance regimens (120 mg Q2W, 120 mg QW, and 240 mg Q2W). Corresponding rates of endoscopic remission were calculated.

2.6. *Software*

Data were imported in R (version 4.4.1; R Foundation for Statistical Computing, R Core Team, Vienna, Austria) for data wrangling (including R packages *tidyverse* [version 2.0.0]) and visualization (including R packages *ggplot2* [version 3.5.1], *nonmem2R* [version 0.2.5]) in the RStudio integrated development environment (version 2024.04.2; RStudio, Inc., Boston, MA, USA). All nonlinear mixed-effects modeling and simulation were performed using NONMEM (version 7.5; Icon Development Solutions, Gaithersburg, MD, USA) with a GNU Fortran 95 compiler and the Perl-speaks-NONMEM (PsN; version 5.3.0) toolkit on the interface software Pirana (version 23.1.2; Certara, Inc., Princeton, NJ, USA). The NONMEM control streams generated during the current study are available in the **Appendix C**.

3. Results

3.1 Study populations

Participants without any quantifiable infliximab concentration were excluded. The analysis included 253 out of 258 (98%) healthy volunteers enrolled in the CT-P13 1.5 and 1.9 studies and 179 out of 181 (99%) patients with IBD enrolled in the CT-P13 1.6 study (101 CD, 78 UC). Baseline demographics and clinical characteristics of the trial participants are summarized in **Table 10**. Median FC level at baseline was numerically higher in patients with UC than in patients with CD (893 µg/g vs. 589 µg/g, **Table 10**).

Table 10. Baseline characteristics of study participants included in the model development population

Characteristic	Healthy volunteers	Patients with CD	Patients with UC
Number of participants	253	101	78
<i>Demographics</i>			
Sex, women, n (%)	30 (12)	47 (47)	34 (44)
Age, years, median [IQR]	33 [26–41]	36 [28–47]	36 [29–53]
Bodyweight, kg, median [IQR]	76.5 [68.2–83.7]	66 [58–79]	70 [61–80]
<i>Serology</i>			
C-reactive protein, mg/L, median [IQR]	0.9 [0.5–2.1]	4.0 [1.8–16.1]	2.2 [0.8–4.8]
Fecal calprotectin, µg/g, median [IQR]	–	589 [158–1483]	893 [480–1639]
Hemoglobin, g/L, median [IQR]	141 [130–150]	126 [115–138]	130 [118–142]
Platelet count, ×10 ⁹ /L, median [IQR]	243 [205–277]	294 [236–370]	297 [256–344]
Serum albumin, g/L, median [IQR]	45 [43–47]	43 [39–46]	45 [42–46]
White blood cell count, ×10 ⁹ /L, median [IQR]	5.7 [4.8–6.9]	7.0 [5.3–8.9]	7.2 [6.0–9.3]
<i>Disease activity</i>			
Full Mayo Clinic Score*, median [IQR]	–	–	8 [7–9]
Mayo Endoscopic Subscore†, 2 : 3, n (%)	–	–	42 (55) : 34 (45)
CDAI, median [IQR]	–	283 [250–338]	–
SESCD‡, median [IQR]	–	10 [7–16]	–
<i>Concomitant medication</i>			
Concomitant corticosteroids, n (%)	–	58 (57)	40 (51)
Concomitant immunomodulators, n (%)	–	51 (50)	31 (40)

Abbreviations: CD, Crohn's disease; CDAI, Crohn's disease activity index; IBD, inflammatory bowel disease; IQR, interquartile range; SESC, simple endoscopic score for Crohn's disease.

* Full Mayo Clinic Score was missing for 2 patients with ulcerative colitis.

† Mayo Endoscopic Subscore was missing for 2 patients with ulcerative colitis.

‡ Simple Endoscopic Score for Crohn's disease was missing for 42 patients with Crohn's disease.

3.2 PopPK model

In total, 253 healthy volunteers and 179 patients with IBD contributed 7351 infliximab serum concentrations to the popPK analysis. A schematic representation of the base model is shown in **Figure 15**. A two-compartment model with first-order absorption and linear elimination fitted the data well (**Figure 16A**, **Appendix C Figure S1**). Allometric scaling was included on the

central volume of distribution V_C and V_P (exponents 1), and CL and Q (exponents 0.75). The popPK model also includes the effect of serum albumin and the presence of NAb on CL. These two covariates explained 4% and 6% of the IIV on CL and F, respectively. **Equation 16** shows how infliximab clearance of patient i at time j ($CL_{i,j}$) increases with higher bodyweight, lower serum albumin, and the presence of NAb:

$$CL_{i,j} = \theta_{CL} \times \left(\frac{\text{serum albumin}_{i,j}}{45}\right)^{-1.18} \times \left(\frac{\text{bodyweight}_{i,j}}{75}\right)^{0.75} \times (1 + 0.565 * NAb) \times e^{\eta_i} \quad (16)$$

θ_{CL} represents the typical population clearance and η_i represents the deviation of patient i from the typical clearance. The PK parameter estimates of the base and final models, as well as SIR results, are listed in **Table 11**. The typical distribution half-life was approximately 1.4 days and the elimination half-life was 9.4 days.

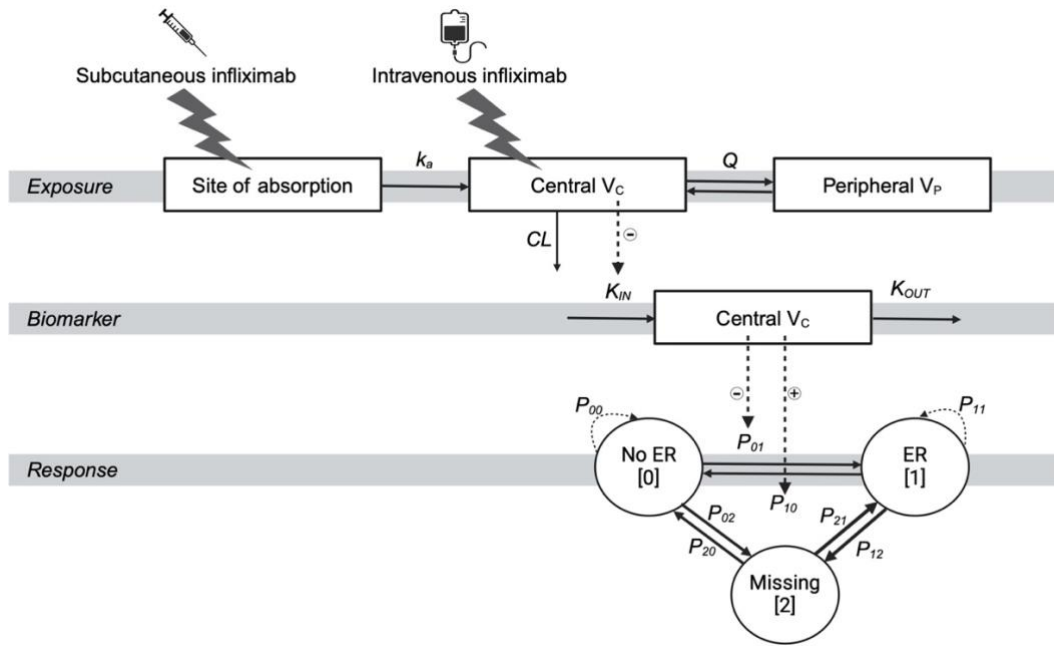


Figure 15. Schematic of the population pharmacokinetics–pharmacodynamics (popPK–PD) model of infliximab CT-P13 in healthy volunteers and patients with inflammatory bowel disease. The schematic includes the two-compartment infliximab popPK model, the indirect response fecal calprotectin model, and the logistic regression endoscopic remission model with transitions between active endoscopic disease [state=0], endoscopic remission [state=1], and missing endoscopic assessment [state=2]. CL, clearance; ER, endoscopic remission; F, absolute bioavailability; k_a , absorption rate constant; k_{IN} , production rate constant; k_{OUT} , degradation rate constant; P_{xy} , probability of going from endoscopic disease activity state x to state y , with x and y being either 0, 1 or 2; Q , intercompartmental clearance; V_C , volume of distribution in the central compartment; V_P , volume of distribution in the peripheral compartment.

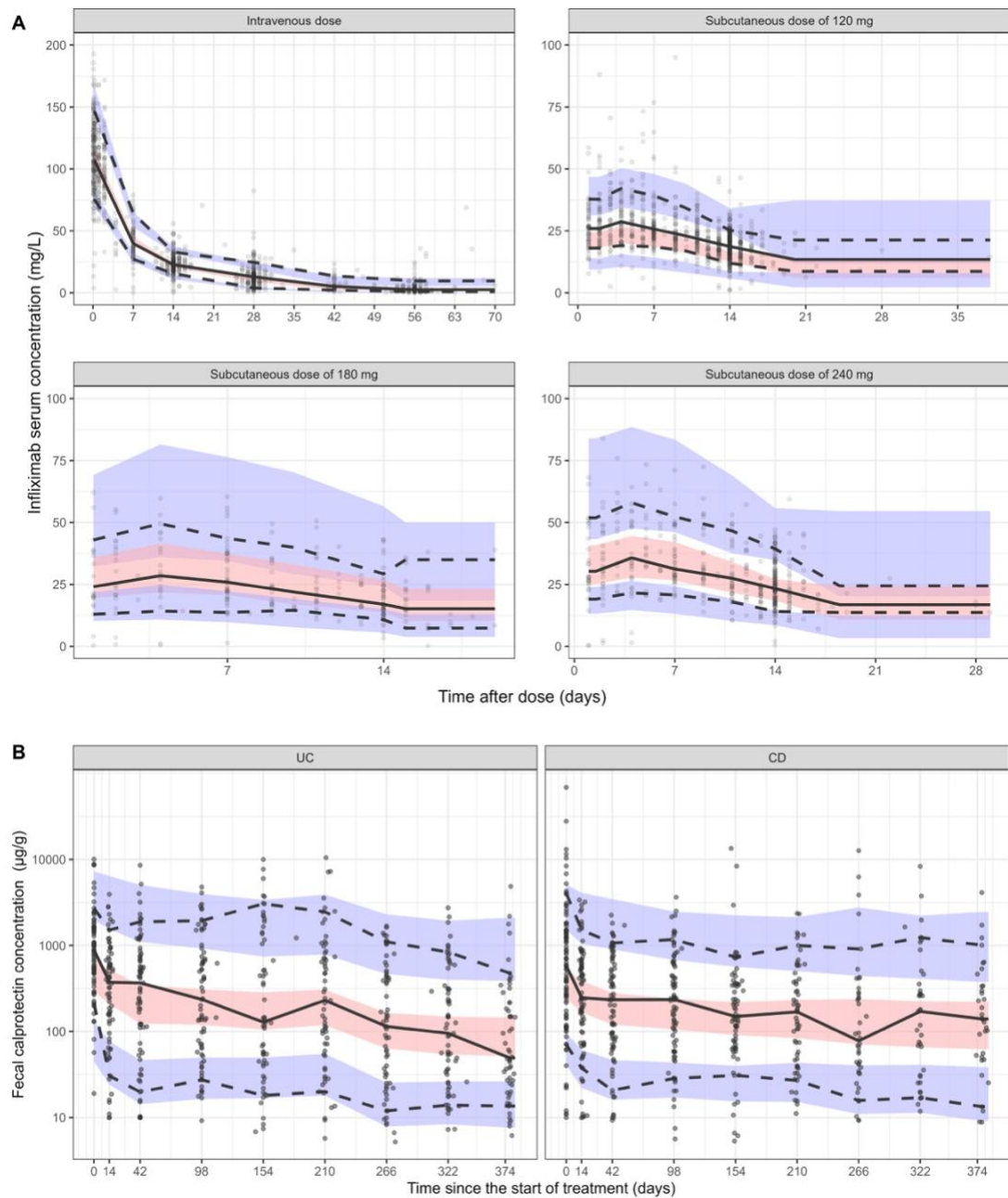


Figure 16. The prediction-corrected visual predictive check (pcVPC) of the final infliximab population pharmacokinetics model (A) and the final fecal calprotectin indirect response model stratified by Crohn's disease (CD) and ulcerative colitis (UC) (B). The pcVPC of the final pharmacokinetics model is stratified by formulation and dose. Gray dots are the prediction-corrected observed concentrations in patients with inflammatory bowel diseases. The black solid and dash lines represent median and 5th and 95th percentile of the prediction-corrected observations, respectively. Simulation-based 95% confidence intervals around the median and 5th and 95th percentiles of the observed data are shown in the form of red and blue shaded areas, respectively.

Table 11. Parameter estimates of the models describing infliximab, fecal calprotectin, and the probability of endoscopic remission

Parameter	Estimate (%RSE) [%Shrinkage]	Estimate (%RSE) [%Shrinkage]	Median (95% CI)
Population PK model			
	Base model (OFV=24211)	Final model (OFV=23430)	SIR
Absorption rate constant k_a (day ⁻¹)	0.200 (1.4)	0.198 (3.3)	0.199 (0.193, 0.204)
Bioavailability F (%)	0.696 (2.6)	0.735 (2.3)	0.735 (0.729, 0.741)
Central volume of distribution V_c (L)	2.99 (2.4)	3.20 (2.3)	3.20 (3.07, 3.32)
Bodyweight on V_c	1 FIX	1 FIX	—
Peripheral volume of distribution V_p (L)	1.15 (4.5)	1.23 (3.9)	1.22 (1.12, 1.30)
Bodyweight on V_p	1 FIX	1 FIX	—
Inter-compartmental clearance Q (L/day)	0.395 (6.5)	0.396 (1.5)	0.395 (0.379, 0.412)
Bodyweight on Q	0.75 FIX	0.75 FIX	—
Clearance CL (L/day)	0.341 (1.9)	0.352 (1.8)	0.349 (0.340, 0.360)
Serum albumin on CL	—	-1.18 (3.9)	-1.17 (-1.28, -1.06)
Bodyweight on CL	0.75 FIX	0.75 FIX	—
Presence of neutralizing antibody on CL	—	0.565 (2.6)	0.567 (0.535, 0.598)
IIV on k_a (%)	45.2 (4.8) [30]	47.3 (4.9) [28]	47.4 (42.6, 52.0)
IIV on F (%)	43.1 (2.0) [13]	36.9 (2.0) [16]	36.8 (36.4, 37.3)
IIV on V_c (%)	25.5 (6.2) [39]	25.5 (6.7) [39]	26.0 (23.0, 29.2)
IIV on V_p (%)	54.6 (7.8) [30]	51.3 (7.6) [32]	50.0 (44.4, 54.1)
IIV on CL (%)	29.0 (3.9) [17]	24.8 (4.1) [22]	24.7 (22.4, 27.4)
IIV on Q (%)	42.4 (14.0) [64]	49.7 (7.9) [60]	49.9 (44.4, 54.1)
Additive residual error (mg/L)	0.786 (1.1) [14]	0.715 (1.3) [14]	0.717 (0.699, 0.731)
Proportional residual error (%)	17.8 (0.4) [14]	17.4 (0.4) [14]	17.4 (17.2, 17.6)
Fecal calprotectin model			
	Base model (OFV=25666)	Final model (OFV=25343)	SIR
BASE ($\mu\text{g/g}$)	214 (9.1)	387 (6.6)	412 (349, 494)
Disease type UC on BASE	—	0.315 (14.5)	0.315 (0.187, 0.429)
k_{in} (day ⁻¹)	2.84 (168)	11.7 (1.7)	13.1 (9.7, 16.7)
I_{max} (%)	—	1 FIX	—
IC ₅₀ in patients with CD (mg/L)	—	11.8 (11.5)	12.5 (8.7, 16.7)
IC ₅₀ in patients with UC (mg/L)	—	4.86 (12.7)	4.96 (3.47, 6.70)
IIV on BASE (%)	100.5 (14.2) [41]	85.1 (2.8) [42]	87.6 (76.3, 99.3)
IIV on IC ₅₀ (%)	—	108 (9.2) [55]	107 (89, 126)
IIV on k_{in} (%)	—	—	—
Proportional residual error (%)	25.4 (2.7) [5]	21.0 (1.2) [8]	21.1 (20.0, 22.2)
Endoscopic remission model			
	Base model (OFV=1250466)	Final model (OFV=1250427)	SIR
$P_{max,01}$	43.2 (8.6)	—	—
$P_{max,01}$ in patients with CD (%)	—	32.6 (15.9)	33.1 (25.5, 41.2)
$P_{max,01}$ in patients with UC (%)	—	100 FIX	—
$P_{max,02}$ (%)	28.0 (16.9)	28.0 (16.9)	27.9 (21.3, 35.8)
$P_{max,10}$ (%)	12.1 (35.1)	12.1 (35.5)	12.9 (6.7, 20.7)
$P_{max,12}$ (%)	15.7 (32.9)	15.7 (32.9)	16.7 (9.6, 25.7)
$P_{max,20}$ (%)	31.2 (37.2)	31.2 (37.2)	33.3 (15.7, 52.4)
EC _{50,01} in patients with UC ($\mu\text{g/g}$)	—	287 (27.3)	296 (207, 432)

Abbreviations: BASE, baseline fecal calprotectin (FC); CI, confidence interval; EC₅₀, FC concentration at half-maximal effect; IIV, interindividual variability; k_{in} , production rate of FC; PK, pharmacokinetic; $P_{max,xy}$, maximal probability of transition from state x to state y; SIR, sampling importance resampling; %CV, percentage coefficient of variation calculated as $\sqrt{e^{variance^2}} \times 100\%$; %RSE, percentage relative standard error.

3.3 Fecal calprotectin model

In total, 179 patients with IBD contributed 1201 FC concentrations to the FC modeling analysis. A Ln-transform both sides approach was used for developing the FC model. PK parameters estimated from the final popPK model were fixed during FC modeling. The indirect response FC model, consisting of a baseline model and E_{max} drug model, described the data well. **Equation 17** shows the differential equation of the infliximab effect on FC:

$$\frac{dFC(t)}{dt} = k_{IN} \times \left[1 - I_{max} \times \frac{CONC(t)}{IC_{50} + CONC(t)} \right] - k_{OUT} \times FC(t) \quad (17)$$

The drug effect model was constructed to inhibit the production rate constant (k_{IN}) of FC through an inhibitory E_{max} model equation driven by infliximab CONC (**Figure 15**). The drug effect is governed by three drug-specific parameters: I_{max} , IC_{50} , and k_{IN} . I_{max} is the maximal drug effect. IC_{50} is the infliximab serum concentration required to achieve a half-maximal drug effect. k_{OUT} is the degradation rate of FC and is equal to k_{IN} divided by baseline FC (BASE).

Equation 18 shows the initial condition of the drug model:

$$FC(0) = BASE = \frac{k_{IN}}{k_{OUT}} \quad (18)$$

An I_{max} value of 0.996 (relative standard error of 0.2%) was estimated but was not significantly different from 1. Therefore, the parameter was fixed to 1. Compared to patients with CD, patients with UC were estimated to have 32% higher baseline FC and 58% lower IC_{50} . The pcVPC plot showed a good agreement between observed and model-predicted FC concentrations (**Figure 16B**). The parameter estimates of the final FC model as well as the results obtained from SIR are presented in **Table 11**.

3.4 Endoscopic remission model

We built an endoscopic remission model based on data from 125 patients with IBD (58 CD, 67 UC). We identified the individual predicted FC as a predictor of endoscopic remission in patients with UC, but not in patients with CD (**Figure 17, Appendix C Figure S2**). In patients with UC, FC at week 14 predicted the transition probability from active endoscopic disease to endoscopic remission at week 22/30 and at week 54 (**Figure 17A&B**). Furthermore, FC at week 30 was a predictor of the probability of maintaining endoscopic remission from week 22/30 to week 54, but its effect could not be precisely estimated (**Figure 17C**). The

parameter estimates from the endoscopic remission model, along with the results obtained from SIR, are presented in **Table 11**. **Equation 19–21** illustrates the effect of FC on transition probabilities in patients with UC.

$$P01 = P_{max,01} - P_{max,01} * \left(\frac{FC}{EC_{50,01} + FC} \right) \quad (19)$$

$$P02 = P_{max,02} \quad (20)$$

$$P00 = 1 - P01 - P02 \quad (21)$$

P_{max} represents the maximum transition probability and EC_{50} represents the half-maximal effective FC. The value of $P_{max,01}$ was estimated at 0.99 and was therefore fixed to 1. The maximum probabilities of achieving ($P_{max,01}$) and losing ($P_{max,10}$) endoscopic remission were constant over time. A $P_{max,01}$ of 100% indicates an approximately 100% chance of achieving endoscopic remission if FC is undetectable. A patient with UC with an FC concentration of 288 µg/g ($\sim EC_{50,01}$) at week 14, has a 50% chance of achieving endoscopic remission at week 22/30 and at week 54.

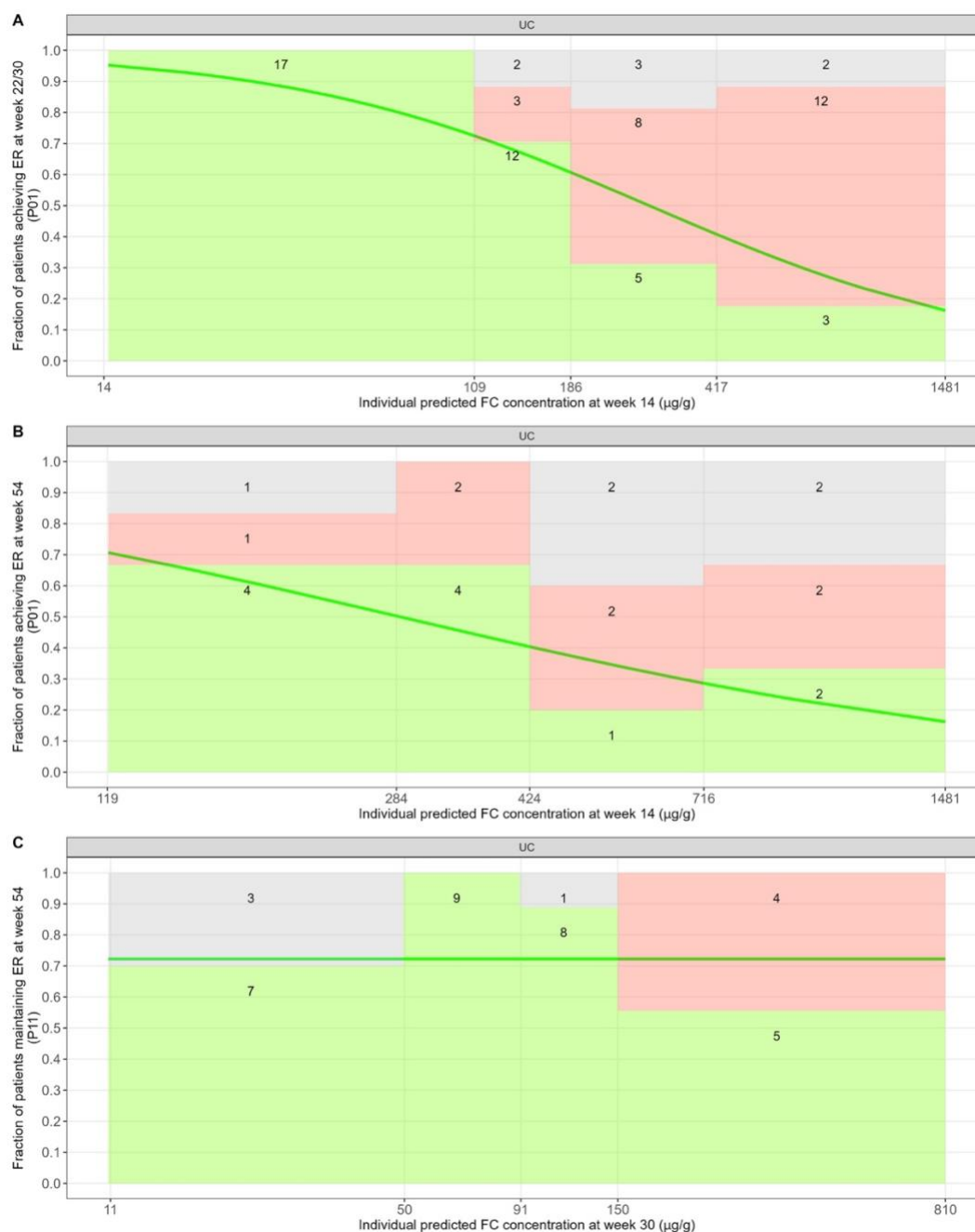


Figure 17. Goodness-of-fit plot of the endoscopic remission model for patients with ulcerative colitis (UC). Observed (green tiles) and predicted (green line) fractions of patients achieving endoscopic remission (ER) at week 22/30 (A) and at week 54 (B), and fractions of patients maintaining ER at week 54 (C) are presented. Red tiles represent the observed fractions of patients not attaining or maintaining ER, respectively. Grey tiles represent the observed fractions of patients without endoscopic assessment. Numbers of patients per state are indicated in each quartile of fecal calprotectin (FC) concentration. P_{01} , the probability of achieving endoscopic remission. P_{11} , the probability of maintaining endoscopic remission.

3.5 Simulations

We simulated time courses of infliximab and FC concentrations for patients making the IV-to-SC switch at week 6 (per the CT-P13 1.6 study). Serum trough concentrations of infliximab almost immediately reached steady state following the label-recommended switch to SC 120 mg Q2W (**Figure 18A**). It took approximately 10 and 14 weeks to reach steady state after switching to SC 180 mg and 240 mg Q2W, respectively.

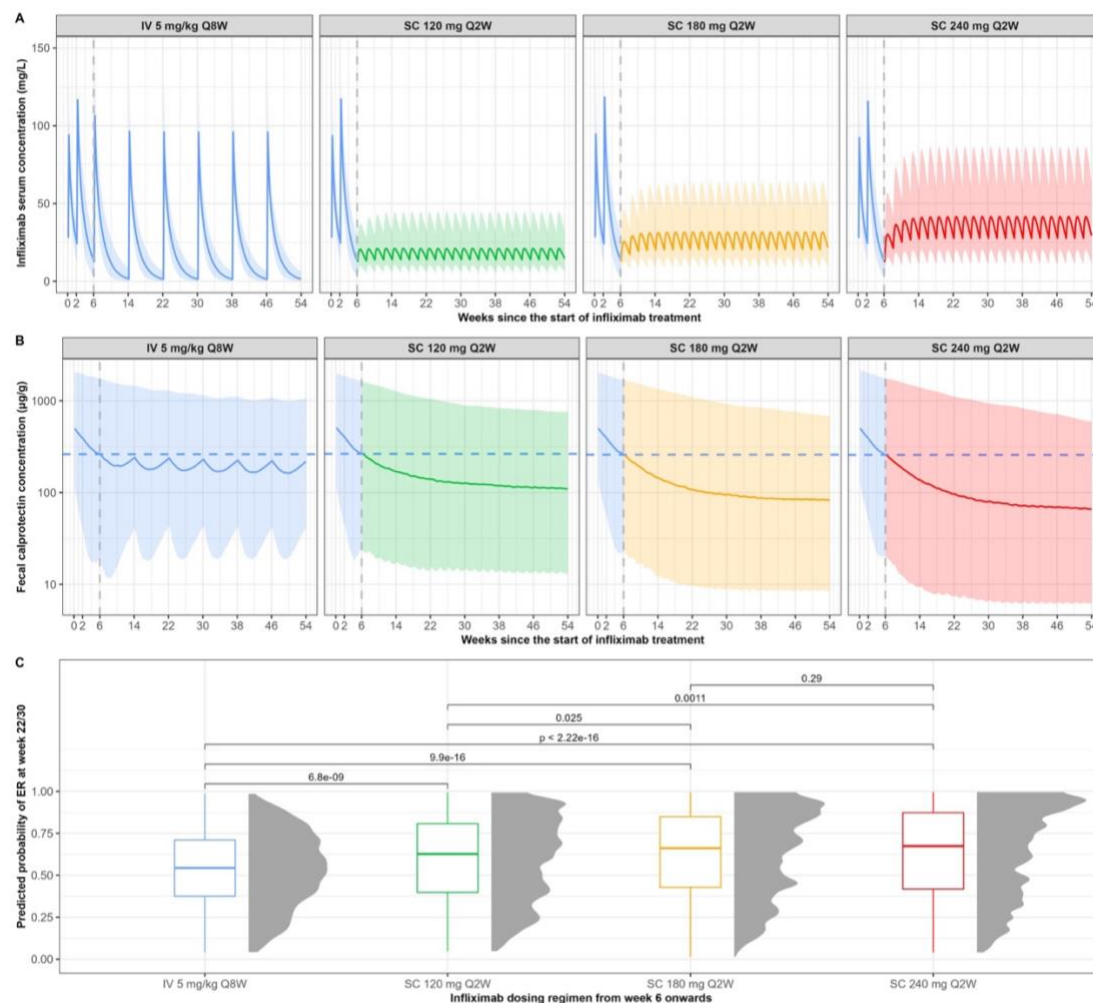


Figure 18. Simulated infliximab and fecal calprotectin concentration profile over time in patients with ulcerative colitis following the intravenous-to-subcutaneous (IV-to-SC) switch at week 6 in the CT-P13 1.6 study. **(A)** The simulated median (solid) and 90% prediction interval (shaded area) of infliximab concentration over time. **(B)** The simulated median (solid) and 90% prediction interval (shaded area) of fecal calprotectin concentration over time. **(C)** The predicted probability of endoscopic remission at week 22/30 based on individual predicted FC concentration at week 14 of therapy. The shaded area adjacent to the box plot illustrates the density distribution of the data. The significance was calculated based on a t-test.

Switching from standard 5 mg/kg IV induction to Q2W SC dosing resulted in a continued decrease in FC levels for all three dose levels (**Figure 18B**). For example, in patients with UC making the standard IV-to-SC switch, the predicted pre-dose FC level dropped from 263 µg/g (90% PI, 21–1695) at week 6 to 111 µg/g (90% PI, 13–757) at week 54 ($p<0.0001$, **Appendix C Figure S3A**). Furthermore, the predicted pre-dose FC level at steady state was significantly lower with standard SC dosing compared to standard IV dosing, at 111 µg/g (90% PI, 13–757) vs. 218 µg/g (IQR, 41–1067), respectively ($p<0.0001$).

Interestingly, despite infliximab trough concentrations decreasing from week 6 to week 14 following standard 5 mg/kg Q8W IV maintenance dosing, FC levels at trough remain stable in patients with UC (**Figure 18B**). Similar findings were observed in patients with CD (**Appendix C Figure S3B & Figure S4**).

The predicted probability of achieving endoscopic remission at week 22/30 increased from 54% (90% PI, 16–88) to 63% (90% PI, 18–94) in patients with UC when switching to label-recommended 120 mg Q2W SC maintenance dosing instead of continuing standard IV maintenance therapy ($p<0.0001$, **Figure 18C**). However, the predicted probability of achieving endoscopic remission was not impacted by FC levels in patients with CD.

We also simulated time courses of FC levels for patients who switched formulations at week 30 of maintenance therapy (**Figure 19A**). For patients on Q6W or Q8W IV regimens, switching to 120 mg Q2W SC infliximab is possible without FC levels increasing (**Figure 19A**). However, for patients receiving Q4W IV infliximab, switching to the standard 120 mg Q2W SC poses a risk of FC elevation after switch. To mitigate this risk, SC dose escalation to either 120 mg QW or 240 mg Q2W is required to control FC in these patients (**Figure 19B**). These trends were further confirmed by comparing the FC levels at week 30 (before SC switch) and at week 54 (at SC steady state) (**Appendix C Figure S5**). The same findings were observed in patients with CD, although our model does not predict any impact on endoscopic remission in these patients (**Appendix C Figure S6 & S7**).

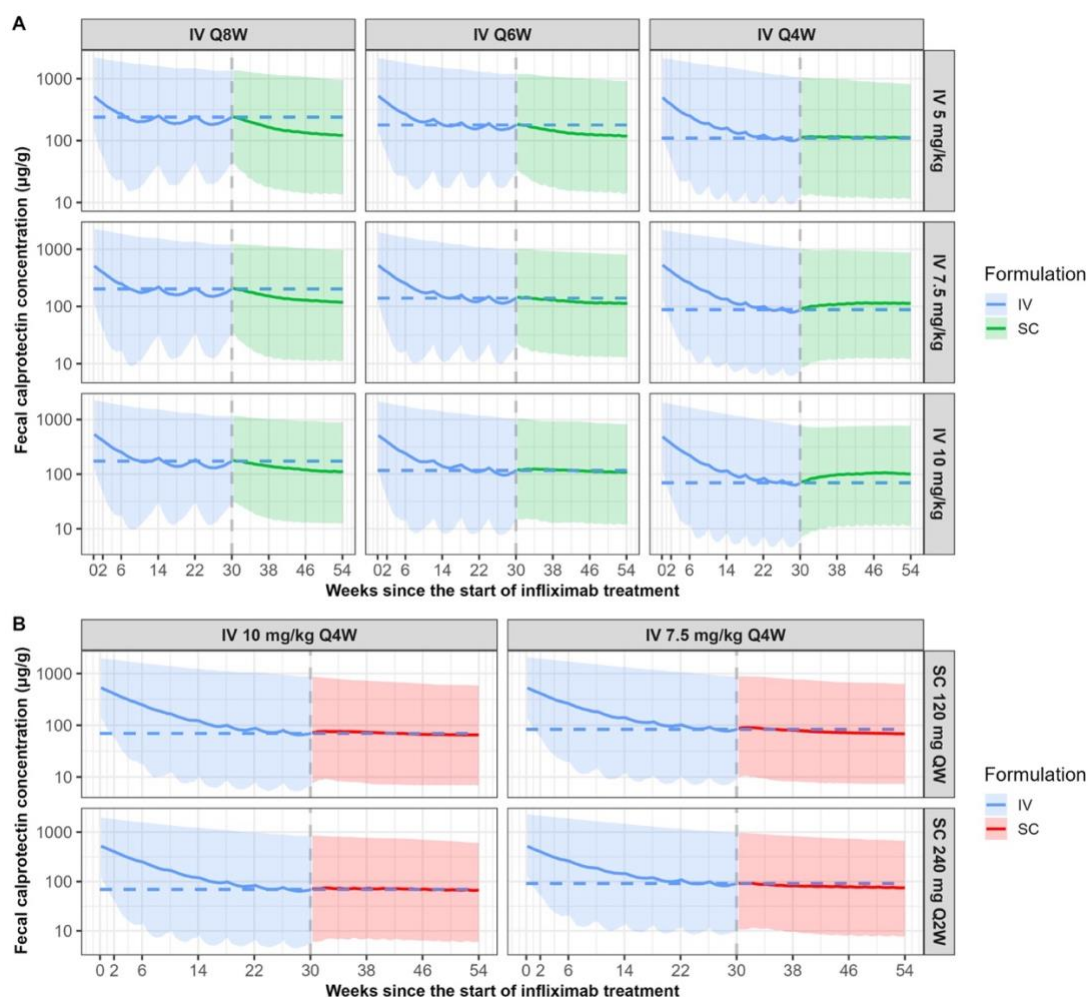


Figure 19. Fecal calprotectin over time following the switch from various escalated intravenous (IV) dosing regimens to subcutaneous (SC) dosing regimens of infliximab maintenance therapy in patients with ulcerative colitis. **(A)** Switching from both standard and escalated IV maintenance therapy to the standard bi-weekly 120 mg SC dosing of infliximab CT-P13. **(B)** Switching from both every four weeks (Q4W) IV maintenance therapy to the escalated SC dosing of 120 mg QW and 240 mg Q2W. The solid and dashed lines represent fecal calprotectin over time. The blue solid lines and shaded areas represent the simulated median and 90% prediction interval of fecal calprotectin over time following IV infliximab CT-P13. The green and red solid lines and shaded areas represent the simulated median and 90% prediction interval of fecal calprotectin over time following SC infliximab CT-P13. The blue horizontal dashed lines indicate the fecal calprotectin levels at the IV-to-SC switch. The grey vertical dashed lines indicate the timepoint when the SC dose starts.

4. Discussion

We established the foundations of an infliximab platform model by integrating PK data from healthy volunteers together with PK, biomarker, and endoscopy data from patients with CD and UC. Our model was well-calibrated and serves as a powerful tool for investigating various IV-to-SC switch scenarios. Our model may also be applied for personalized dosing through *Bayesian forecasting*, targeting both infliximab exposure and clinical outcomes like FC response and endoscopic remission.^{3,4}

We developed a popPK model using the largest dataset of SC infliximab available to date, incorporating data from both patients with IBD and healthy volunteers. Hanzěl and colleagues estimated the effect of bodyweight on infliximab PK using data from the CT-P13 1.6 study in their previously reported popPK study.¹⁹⁶ In contrast, we relied on allometric scaling, a more generalized approach. This approach allows for model extrapolations to predict dosing for pediatric patients based on their bodyweight differences from adults. Clinical validation of these extrapolations is awaited. Furthermore, infliximab clearance increased with decreasing serum albumin, increasing body weight, and presence of NAb. However, these covariates explained only 4% of the PK variability among patients, leaving 25% unexplained, underscoring the need for *Bayesian forecasting* when considering model-informed precision dosing.³

The FC model quantified the relationship between infliximab exposure and FC concentrations, reflecting infliximab's anti-inflammatory action, which suppresses neutrophil trafficking and calprotectin release.^{174–176} The maximal effect of infliximab on inhibiting FC release was estimated at 100%. However, the infliximab concentration required to achieve this complete inhibition is unrealistically high – over 485 mg/L for patients with UC and over 1168 mg/L for patients with CD – making it practically impossible. Infliximab appeared to be more effective for suppressing FC release in patients with UC, as reflected by lower serum infliximab concentration needed to achieve half-maximal inhibition of FC release in patients with UC than in patients with CD (4.9 mg/L vs. 11.8 mg/L). Significant variability was observed in the FC model parameters, indicating that patients respond differently to infliximab and require different infliximab trough concentrations to achieve the same FC level. The developed FC

model accounts for individual sensitivity to infliximab and can be used for personalized dose optimization, targeting both sufficient infliximab exposure and an adequate FC response.

In recent years, therapeutic goals in IBD have shifted toward endoscopic remission.³⁵ In our endoscopic remission model, reductions in FC were found to be associated with increased probabilities of endoscopic remission in patients with UC, but not CD. Reducing FC to undetectable levels by week 14 of infliximab therapy maximized the probability of achieving endoscopic remission to 100% for patients with UC. Additionally, FC measurements at weeks 2, 6, and 22 predicted endoscopic remission at week 22/30 as accurately as FC at week 14, indicating the generalizability of the model.

We showed that the standard IV-to-SC switch of infliximab at week 6 further decreases FC and results in a significantly higher probability of endoscopic remission at week 22/30. However, the significance was derived from 1000 simulations and may not be evident in clinical trials with smaller sample sizes. We did observe the trend in the CT-P13 1.6 study, where the patients who switched to the standard SC 120 mg Q2W at week 6 had a numerically higher rate of endoscopic remission at week 22/30 compared to those who continued on standard IV 5 mg/kg Q8W (59% [n=23/39] vs. 50% [n=21/42]). However, a sample size calculation indicates that 480 patients per arm would be necessary to confirm biochemical superiority of standard SC infliximab maintenance therapy over standard IV infliximab maintenance therapy.

When making the IV-to-SC switch beyond week 6, during maintenance therapy, standard SC infliximab dosing also provided superior or equal FC responses compared to standard and escalated IV dosing up to Q6W. However, for patients receiving Q4W IV infliximab during maintenance, especially when dosing higher than 5 mg/kg, SC dose escalation to 120 mg QW or 240 mg Q2W would be necessary to maintain adequate FC suppression after switching from to SC. These findings are consistent with both *in silico* simulations and real-world evidence indicating that patients on Q4W IV infliximab require escalated SC doses to ensure adequate infliximab exposure and clinical response.^{93,194}

The FC model describes an inverse relationship between infliximab concentrations and FC levels. However, while infliximab trough concentrations decrease from week 6 to week 14 following standard 5 mg/kg Q8W IV maintenance dosing, FC levels did not increase. Although

seemingly contradictory, this is explained by the contributions of the high peak and intermediate infliximab concentrations, which are much higher than during SC dosing. This suggests that (i) trough concentrations are only a surrogate for FC response, with overall infliximab exposure being the true driver, and (ii) patients on standard 120 mg Q2W SC dosing are not "overdosed" as they require higher trough concentrations to achieve similar overall exposure to standard IV dosing. This also means that dosage de-escalation of SC infliximab to the established 5 mg/L trough concentration target encompasses the risk of therapy failure. Even though overall exposure may be the true driver of a biomarker – and thus a therapeutic – response to infliximab therapy, this does not necessarily call for rich PK sampling. Infliximab trough concentrations can still serve as an acceptable surrogate for overall exposure and therapeutic response. Importantly, because infliximab concentration–time curves differ significantly between IV and SC dosing, trough concentrations should not be interpreted on the same scale when assessing their predictive ability for therapeutic outcomes.

Our study has limitations. First, extrapolation of the model to inform the switch from escalated IV to SC dosing should be performed with caution. Results from ongoing trials such as DISCUS-IBD (Switching from Dose-Intensified intravenous to Subcutaneous infliximab in Inflammatory Bowel Disease; ACTRN12622001458729) and AMERETTO (Subcutaneous Infliximab After a Previous Intravenous Dose Optimization; NCT06113913) are anticipated to clinically confirm our findings.¹⁹⁸ Second, our study only included biologic-naïve patients, so extrapolating to those with prior biologic exposure should be done with care. Nevertheless, our model serves as a solid foundation for a continued model-informed development plan of SC infliximab. We hope to update the model by including the emerging real-world data of SC therapy in IBD.

In conclusion, we established the foundation of an innovative clinical platform model that quantitatively captures the relationships between infliximab dosing, exposure, biomarkers including serum albumin and FC, and endoscopic outcomes in patients with IBD. The developed model is a powerful tool for exploring various IV-to-SC dosing scenarios. This paper demonstrates that pharmacometrics modeling and simulation can extend the utility of clinical trial data to answer additional clinically relevant questions. In the future, we will expand our platform model to include the emerging real-world data and so, our model can accelerate and

de-risk clinical trial design and, ultimately, inform personalized model-based dose optimization of SC infliximab CT-P13 to account for the heterogeneity in the IBD patient population.

Appendix C

Supplementary Figures

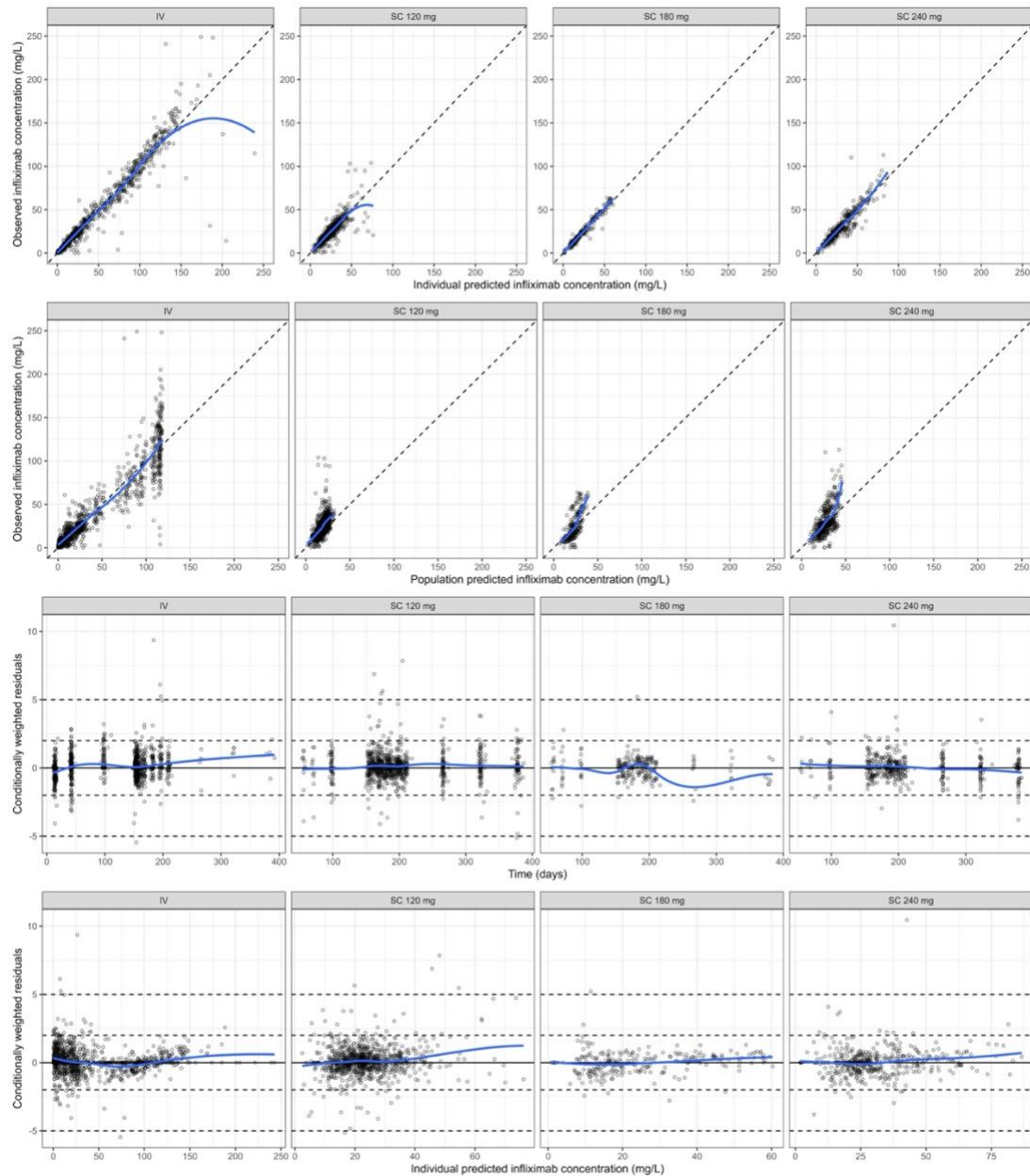


Figure S1. Goodness-of-fit plot of the final population pharmacokinetic model. IV, intravenous; SC, subcutaneous.

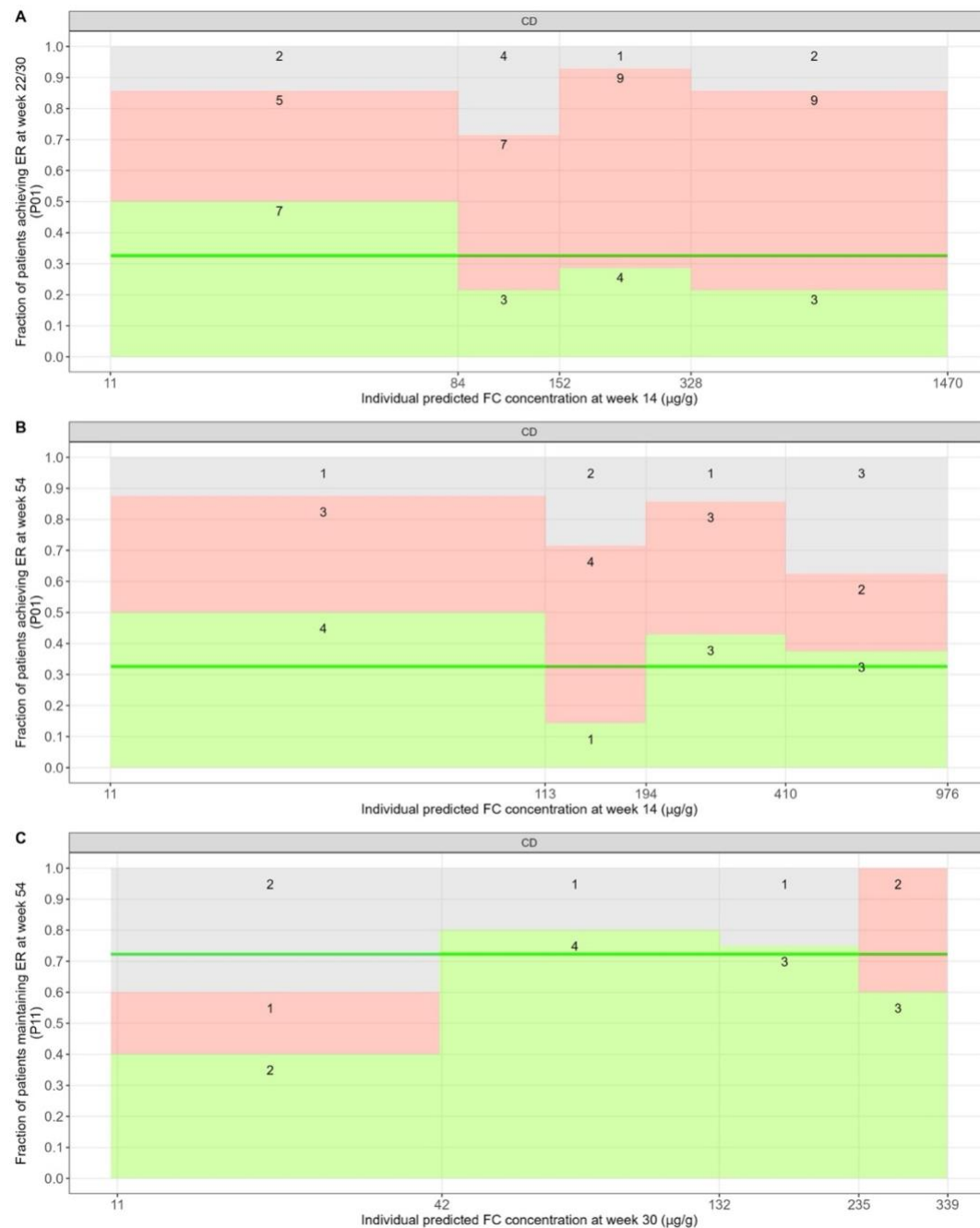


Figure S2. Goodness-of-fit plot of the endoscopic remission model for patients with Crohn's Disease (CD). Observed (green tiles) and predicted (green line) fractions of patients achieving endoscopic remission (ER) at week 22/30 (**A**) and at week 54 (**B**), and fractions of patients maintaining ER at week 54 (**C**) are presented. Red tiles represent the observed fractions of patients not attaining or maintaining ER, respectively. Grey tiles represent the observed fractions of patients without endoscopic assessment. Numbers of patients per state are indicated in each quartile of fecal calprotectin (FC) concentration. P_{01} , the probability of achieving endoscopic remission. P_{11} , the probability of maintaining endoscopic remission.

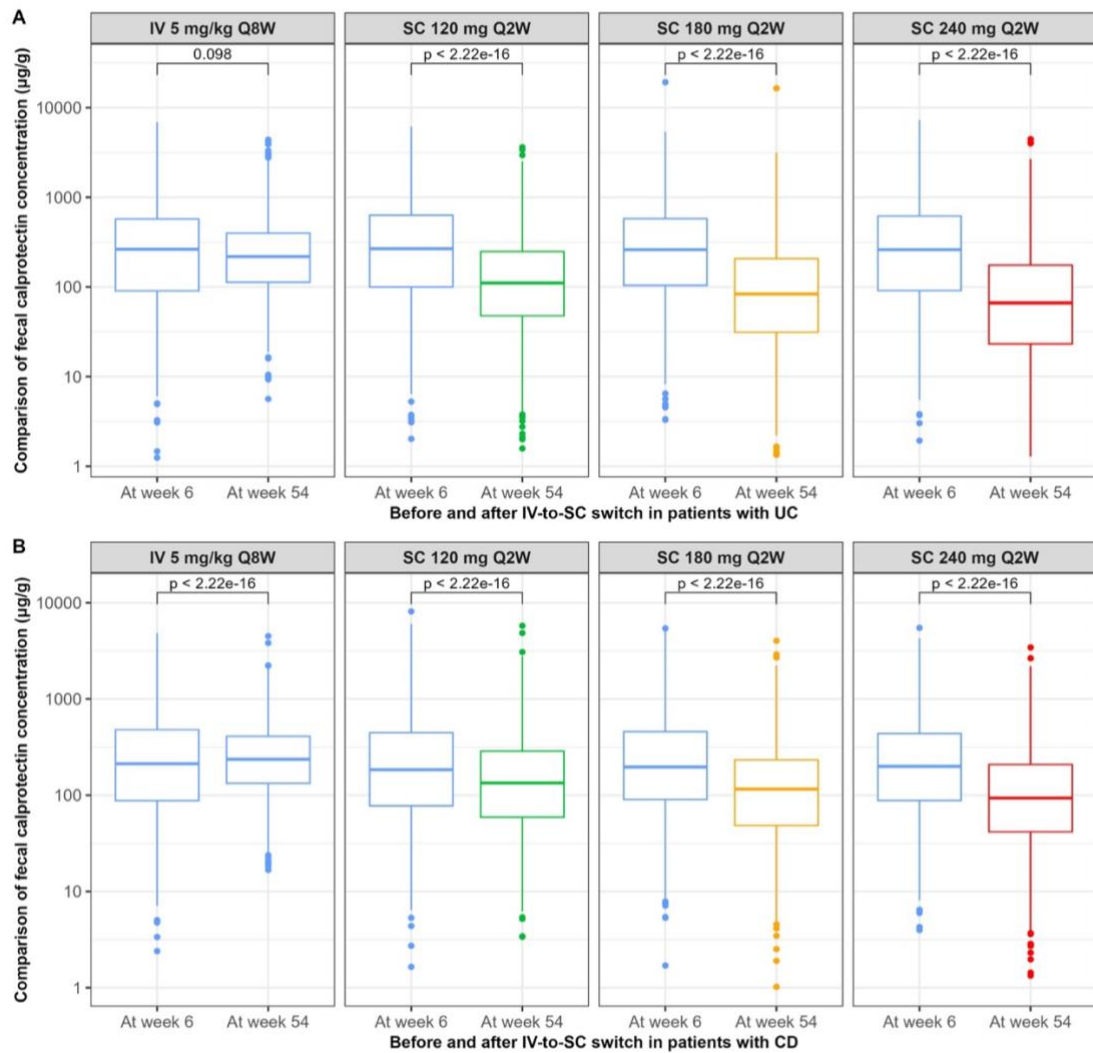


Figure S3. The comparison of simulated fecal calprotectin concentration before switch at week 6 and after switch at week 54 in patients with ulcerative colitis (UC) and Crohn's disease (CD). The significance was calculated based on paired t-test. IV, intravenous; QxW, every x weeks; SC, subcutaneous.

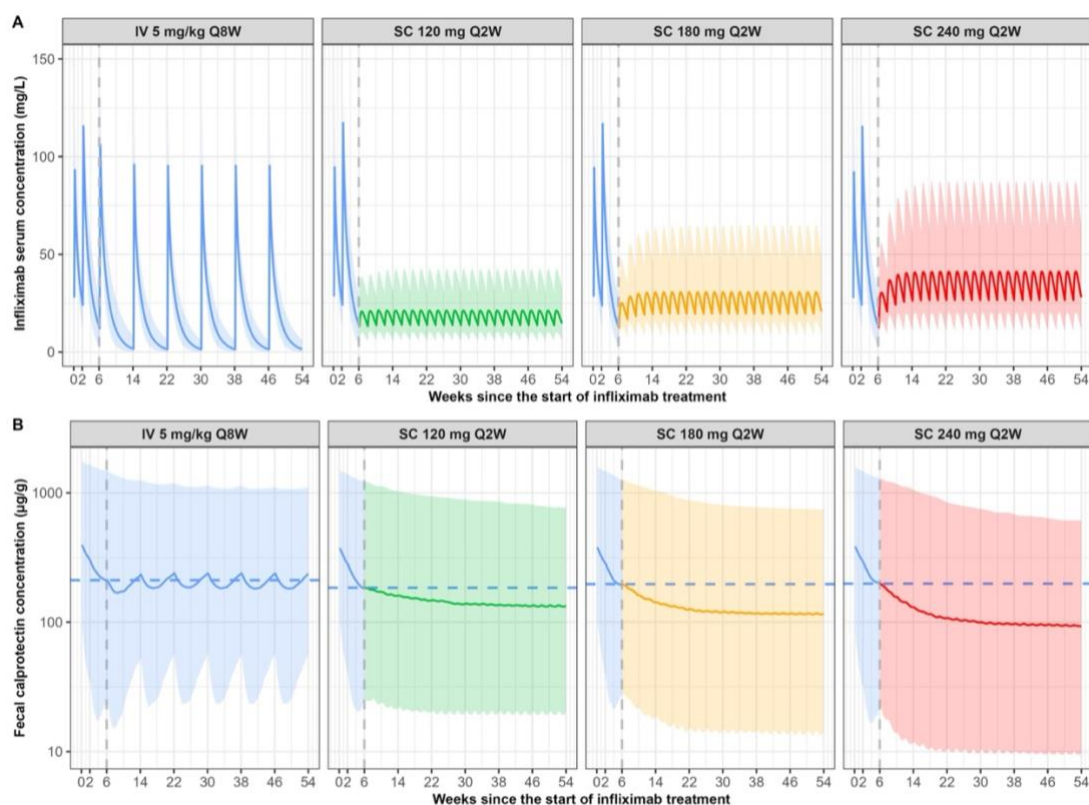


Figure S4. Simulated infliximab and fecal calprotectin concentration profile over time in patients with Crohn's disease following the intravenous-to-subcutaneous (IV-to-SC) at week 6 in the CT-P13 1.6 study. **(A)** The simulated median (solid) and 90% prediction interval (shaded area) of infliximab concentration over time. **(B)** The simulated median (solid) and 90% prediction interval (shaded area) of fecal calprotectin concentration over time. IV, intravenous; QxW, every x weeks; SC, subcutaneous.

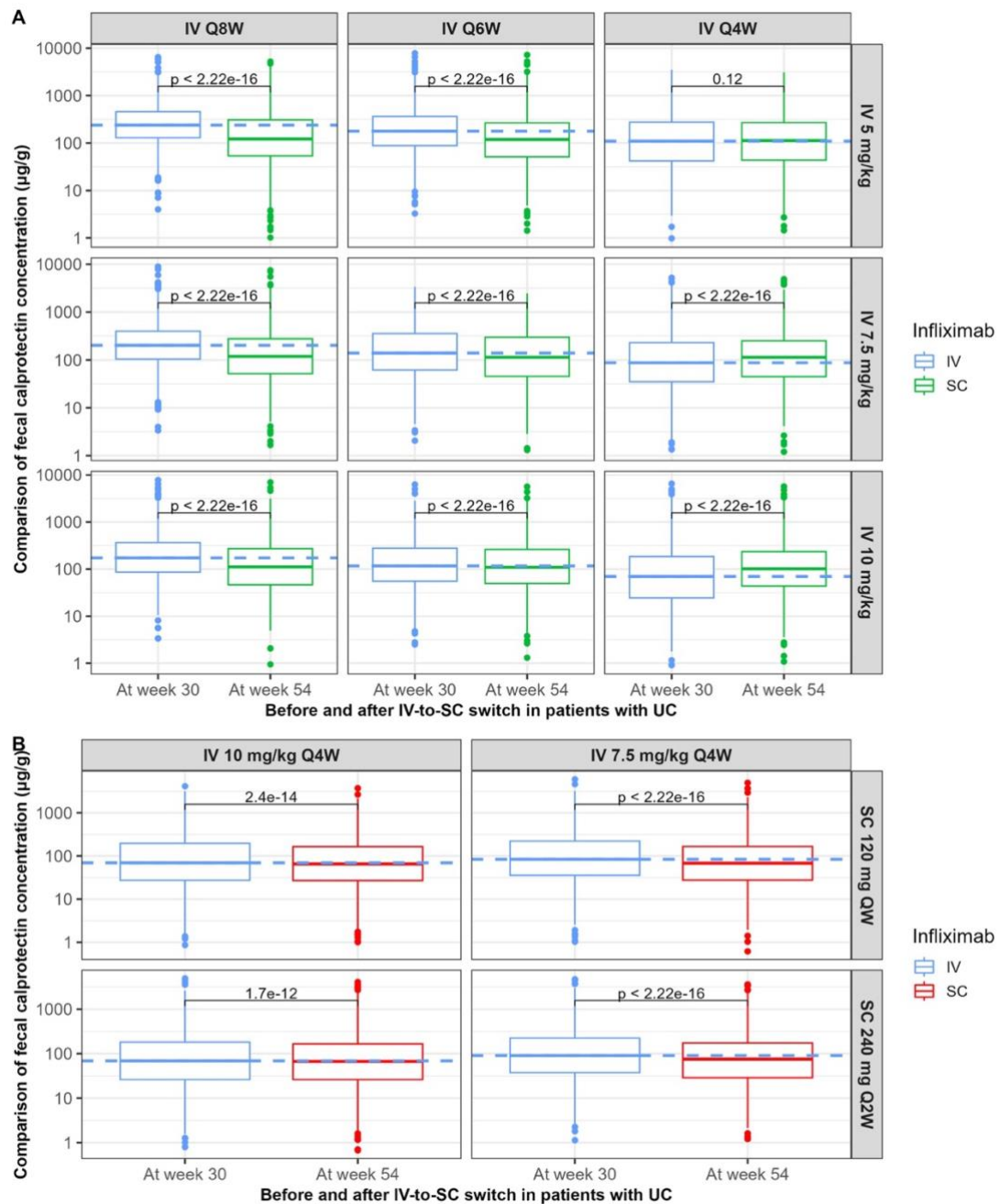


Figure S5. Comparison of simulated fecal calprotectin concentration before SC switch at week 30 and after SC at week 54 in patients with ulcerative colitis (UC). The significance was calculated based on paired t-test. IV, intravenous; QxW, every x weeks; SC, subcutaneous. The blue horizontal dashed lines indicate the fecal calprotectin concentration at the IV-to-SC switch.

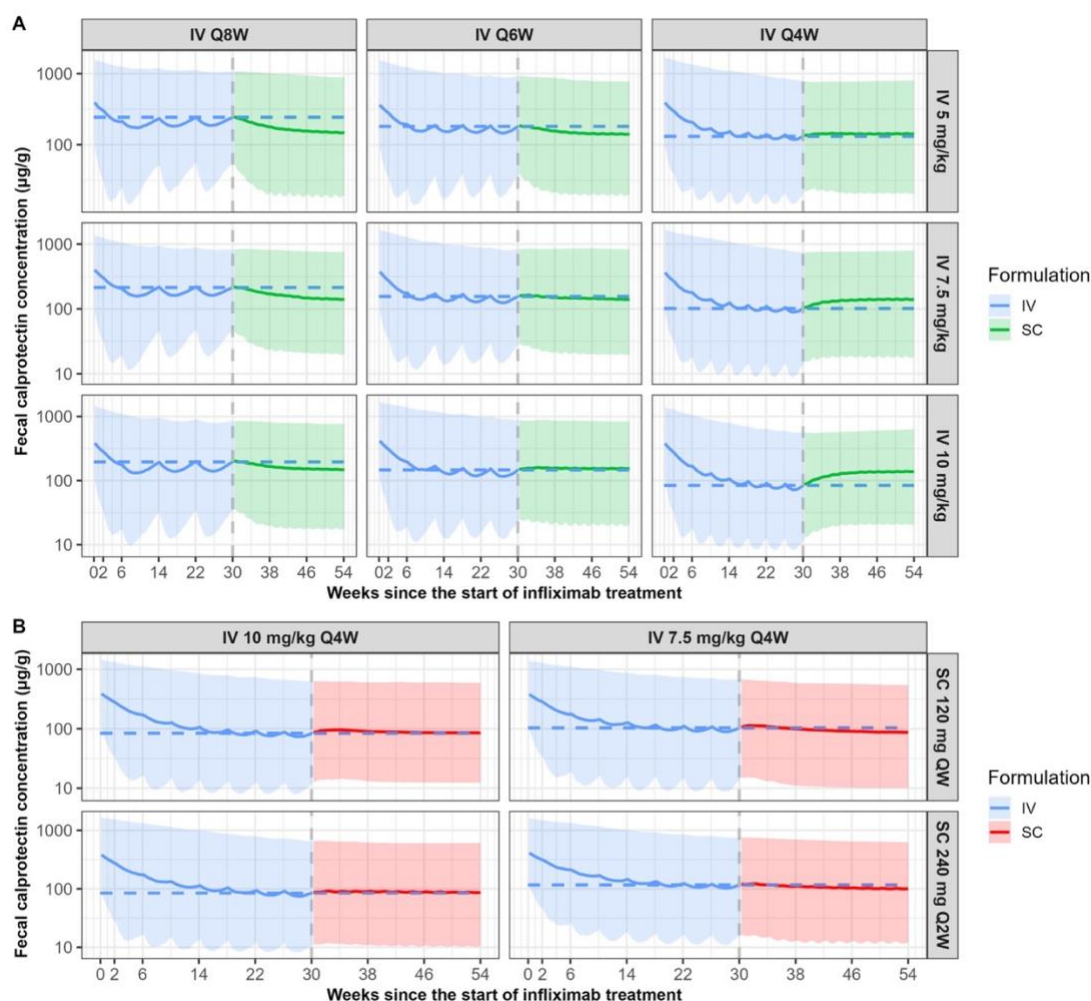


Figure S6. Fecal calprotectin over time following the switch from various escalated intravenous (IV) dosing regimens to subcutaneous (SC) dosing of infliximab CT-P13 in patients with Crohn's disease. **(A)** Switch from both standard and escalated IV maintenance to the standard bi-weekly 120 mg SC dosing of infliximab CT-P13. **(B)** Switch from both every four weeks (Q4W) IV maintenance to the escalated SC dosing of 120 mg QW and 240 mg Q2W. The solid and dashed lines represent fecal calprotectin over time. The blue solid lines and shaded areas represent the simulated median and 90% prediction interval of fecal calprotectin over time following IV infliximab CT-P13. The green and red solid lines and shaded areas represent the simulated median and 90% prediction interval of fecal calprotectin over time following SC infliximab CT-P13. The blue horizontal dashed lines indicate the fecal calprotectin concentration at the IV-to-SC switch. The grey vertical dashed lines indicate the timepoint when the SC dose starts.

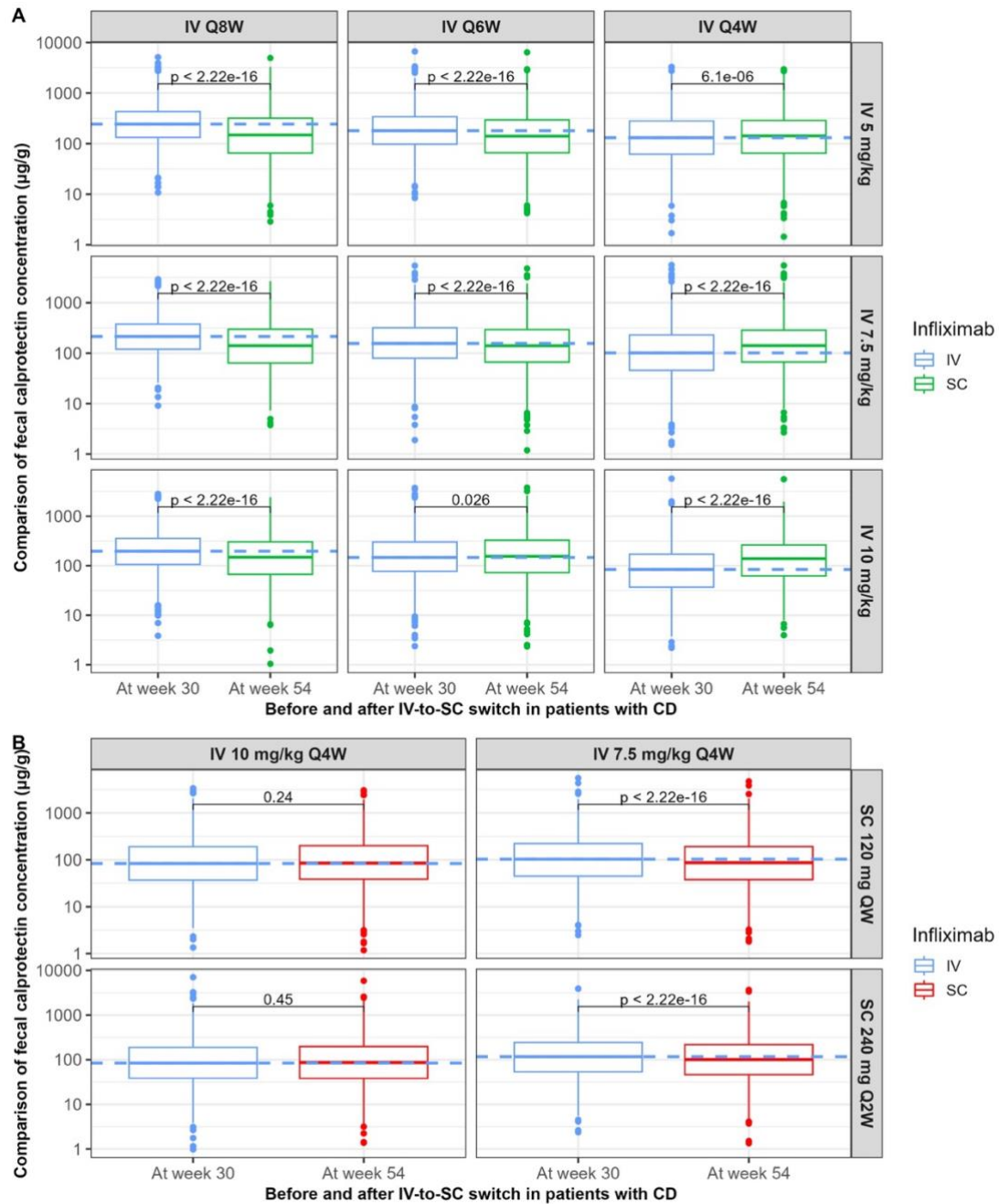


Figure S7. Comparison of simulated fecal calprotectin concentration before SC switch at week 30 and after SC at week 54 in patients with Crohn's disease (CD). The significance was calculated based on a paired t-test. IV, intravenous; QxW, every x weeks; SC, subcutaneous. The blue horizontal dashed lines indicate the fecal calprotectin concentration at the IV-to-SC switch.

NONMEM control streams

Subcutaneous infliximab CT-P13 Population Pharmacokinetics model

\$PROBLEM PopPK analysis of SC infliximab CT-P13

\$INPUT ID TIME TAD EVID MDV AMT RATE CMT DV NAB_CAT BWT ALB

\$DATA IFX_SC_PK_NONMEM_reduced_BLOQ.csv IGNORE=@

\$SUBROUTINE ADVAN6 TOL=5

\$MODEL NCOMP=4

COMP=(DEPOT) ; Absorption compartment

COMP=(CENTRAL) ; Central compartment

COMP=(PERIPH1) ; Peripheral compartment

COMP=(AUC) ; For calculating AUC

\$PK

;;; CLCOV-DEFINITION START

IF(NAB_CAT.EQ.0) CLNAB_CAT = 1 ; Most common

IF(NAB_CAT.EQ.1) CLNAB_CAT = (1 + THETA(7))

COVALB= (ALB/45)**THETA(8)

;;; CLCOV-DEFINITION END

;;; CL-RELATION START

CLCOV=CLNAB_CAT*COVALB

;;; CL-RELATION END

A_0(1)= 0

TVF1 = THETA(1)

F1 = TVF1* EXP(ETA(1))

;absorption

TVKA = THETA(2)

KA = TVKA* EXP(ETA(2))

;Elimination

TVCL = THETA(3)

CL = TVCL*EXP(ETA(3))*((BWT/75)**0.75)*CLCOV

;distribution

TVV1 = THETA(4)

V1 = TVV1*EXP(ETA(4))*((BWT/75)**1)

;inter-compartment parameters

TVQ = THETA(5)

Q = TVQ*EXP(ETA(5))*((BWT/75)**0.75) ; Inter compartmental clearance

TVV2 = THETA(6)

V2 = TVV2*EXP(ETA(6))*((BWT/75)**1) ; Volume of distribution in Peripheral COMPARTMENT

```

S1 = V1
KE = CL/V1
K23 = Q/V1
K32 = Q/V2

$DES
DADT(1) = -KA*A(1)
DADT(2) = KA*A(1) -K23*A(2) +K32*A(3) -KE*A(2)
DADT(3) =      K23*A(2) -K32*A(3)
DADT(4) = A(2)/V1

$ERROR
AUC = A(4)
IPRED = A(2)/V1
IRES = DV - IPRED
W=SQRT(SIGMA(1,1)*IPRED**2 + SIGMA(2,2))
IF (W.EQ.0) W = 1
IWRES = IRES/W

LLOQ=0.1
IF (DV.GE.LLOQ) THEN
  F_FLAG = 0
  Y=IPRED * (1+EPS(1))+ EPS(2)
ELSE
  F_FLAG = 1
  Y = PHI((LLOQ-IPRED)/W)
ENDIF

$THETA
(0,0.710448976413465,1) ; 1 F
(0,0.182051705731063) ; 2 KA
(0,0.31780853877249) ; 3 CL
(0,2.88612641947434) ; 4 V1
(0,0.400857505279394) ; 5 Q
(0,1.22013524068482) ; 6 V2
0.580370532599085 ; 7 CL_NAB
-1.04312417779917 ; 8 CL_ALB

$OMEGA
0.184415382197782 ; 1. omF1
0.213900623039383 ; 2. omKA
0.0817951904548301 ; 3. omCL
0.0594414109627844 ; 4. omV1
0.205062730763017 ; 5. omQ
0.28068517305762 ; 6. omV2

$SIGMA
0.0294396584380003 ; PROP ERROR
0.589506863886974 ; ADDI ERROR

```

```
$EST METHOD = 1 INTER LAPLACIAN NUMERICAL MAXEVAL=9999 SADDLE_RESET=1
$COV Matrix=S PRINT=E
$TABLE ID TIME IPRED PRED TIME DV IWRES CWRES NOPRINT ONEHEADER FILE=sdtab10_1
$TABLE ID F1 KA CL V1 Q V2 ETA(1) ETA(2) ETA(3) ETA(4) NOPRINT NOAPPEND ONEHEADER
FILE=patab10_1
```

Subcutaneous infliximab CT-P13 Population Pharmacokinetic-Pharmacodynamic model

\$PROBLEM PopPK-PD analysis of SC infliximab CT-P13

```
$INPUT ID TIME TAD EVID MDV AMT RATE CMT DV NAB_CAT BWT ALB CRP
      DISEASE ; DISEASE=1 UC, DISEASE=2 CD
      OBS_LABEL ; 1=IFX conc above LOQ; 2=IFX conc below LOQ; 3=FC conc
$DATA IFX_SC_PKPD_NONMEM_In.csv IGNORE=@
```

```
$SUBROUTINE ADVAN6 TOL=5
```

```
$MODEL NCOMP=5
```

```
COMP=(DEPOT) ; Absorption compartment
COMP=(CENTRAL) ; Central compartment
COMP=(PERIPH1) ; Peripheral compartment
COMP=(AUC) ; For calculating AUC
COMP=(FC) ; For fecal calprotectin
```

```
$PK
```

```
;;; CLCOV-DEFINITION START
```

```
IF(NAB_CAT.EQ.0) CLNAB_CAT = 1 ; Most common
```

```
IF(NAB_CAT.EQ.1) CLNAB_CAT = ( 1 + THETA(7))
```

```
COVALB= (ALB/45)**THETA(8)
```

```
;;; CLCOV-DEFINITION END
```

```
;;; CL-RELATION START
```

```
CLCOV = CLNAB_CAT*COVALB
```

```
;;; CL-RELATION END
```

```
TVF1 = THETA(1)
```

```
F1 = TVF1* EXP(ETA(1))
```

```
;absorption
```

```
TVKA = THETA(2)
```

```
KA = TVKA* EXP(ETA(2))
```

```
;Elimination
```

```
TVCL = THETA(3)
```

```
CL = TVCL*EXP(ETA(3))*((BWT/74.65)**0.75)*CLCOV
```

```
;distribution
```

```
TVV1 = THETA(4)
```

```
V1 = TVV1*EXP(ETA(4))*((BWT/74.65)**1)
```

```
;inter-compartment parameters
```

```
TVQ = THETA(5)
```

```
Q = TVQ*EXP(ETA(5))*((BWT/74.65)**0.75) ; Inter compartmental clearance
```

```
TVV2 = THETA(6)
```

```
V2 = TVV2*EXP(ETA(6))*((BWT/74.65)**1) ; Volume of distribution in Peripheral COMPARTMENT
```

```
S1 = V1
```

```

KE = CL/V1
K23 = Q/V1
K32 = Q/V2

;;; BASECOV-DEFINITION START
IF(DISEASE.EQ.2) BASE_DISEASE = 1 ; CD patients
IF(DISEASE.EQ.1) BASE_DISEASE = ( 1 + THETA(14)) ; UC patients
;;; BASECOV-DEFINITION END

;;; FC Baseline model;;
TVBASE = THETA(9)
SHP = THETA(10)
BASE = TVBASE*EXP(ETA(7))*BASE_DISEASE ; B1 estimate baseline

KIN = THETA(11)*EXP(ETA(8))
KOUT = KIN/BASE

;; Initial amount in compartment
A_0(1) = 0 ; absorption comp
A_0(2) = 0 ; central comp
A_0(3) = 0 ; peri comp
A_0(4) = 0 ; AUC comp
A_0(5) = BASE ; FC

EMAX = THETA(12)*EXP(ETA(9))
;;; BASECOV-DEFINITION START
IF(DISEASE.EQ.2) EC50 = THETA(13)*EXP(ETA(10)); CD patients
IF(DISEASE.EQ.1) EC50 = THETA(14)*EXP(ETA(10)); UC patients
;;; BASECOV-DEFINITION END

$DES
DADT(1) = -KA*A(1)
DADT(2) = KA*A(1) - K23*A(2) + K32*A(3) - KE*A(2)
DADT(3) = K23*A(2) - K32*A(3)
DADT(4) = A(2)/V1
CONC = A(2)/V1 ; IFX concentration
DADT(5) = KIN*(1-((EMAX*CONC)/(EC50+CONC)))-KOUT*(A(5))

$ERROR
AUC = A(4)
FC = A(5)
IFX = A(2)

LLOQ=0.1
;;; for above lloq ifx concentrations
IF (OBS_LABEL.EQ.1) THEN
  F_FLAG = 0
  IPRED = A(2)/V1
  IRES = DV - IPRED
  W=SQRT(SIGMA(1,1)*IPRED**2 + SIGMA(2,2))

```

```

Y=IPRED * (1+EPS(1))+ EPS(2)
ENDIF

```

```

;;; for below lloq ifx concentrations
IF (OBS_LABEL.EQ.2) THEN
  F_FLAG = 1
  IPRED = A(2)/V1
  IRES = DV - IPRED
  W = SQRT(SIGMA(1,1)*IPRED**2 + SIGMA(2,2))
  Y = PHI((LLOQ-IPRED)/W)
ENDIF

```

```

;;; for FC concentrations
IF (OBS_LABEL.EQ.3) THEN
  F_FLAG = 0
  IPRED = LOG(A(5)+0.001)
  IRES = DV - IPRED
  W = SQRT(SIGMA(3,3)**2 + (SIGMA(4,4)/(FC))**2)
  Y = IPRED * (1+EPS(3))+ EPS(4)
ENDIF

```

```

IF(W.EQ.0) W = 1
IWRES = IRES/W

```

```

$THETA
(0, 0.735,1) FIX ;1 F
(0, 0.198) FIX ;2 KA
(0, 0.352) FIX ;3 CL
(0, 3.2) FIX ;4 V1
(0, 0.396) FIX ;5 Q
(0, 1.23) FIX ;6 V2
(0.565) FIX ;7 CL_NAB
(-1.18) FIX ;8 CL_ALB
(0, 387,5000) ;9 BASE
(0) FIX ;10 SHP
(0, 11.7) ;11 KIN
(1) FIX ;12 EMAX
(0, 11.8) ;13 EC50 CD
(0, 0.315) ;14 BASE_DISEASE
(0, 4.86) ;15 EC50 UC

```

```

$OMEGA
0.136 FIX ; 1. omF1
0.224 FIX ; 2. omKA
0.0616 FIX ; 3. omCL
0.065 FIX ; 4. omV1
0.247 FIX ; 5. omQ
0.263 FIX ; 6. omV2
0.725 ; 7 IIV_BASE
0 FIX ; 8 IIV_KIN

```

0 FIX ; 9 EMAX
1.18 ; 10 EC50 CD

\$SIGMA

0.0303 FIX ; PROP ERROR IFX CONC
0.511 FIX ; ADDI ERROR IFX CONC
0.044 ; PROP FC
0 FIX ;ADD FC

\$EST METHOD = 1 INTER LAPLACIAN NUMERICAL MAXEVAL=9999 SADDLE_RESET=1
\$COV Matrix=S PRINT=E

\$TABLE ID TIME IPRED IWRES CWRES NOPRINT ONEHEADER FILE=sdtab4_1
\$TABLE ID F1 KA CL V1 Q V2 BASE KIN ETA(7) NOPRINT NOAPPEND ONEHEADER FILE=patab4_1
\$TABLE ID ATI_CAT ATI_OVERALL NAB_CAT SEX COIMMU CO_STEROID_BASELINE CO_STEROID
DISEASE NOPRINT NOAPPEND ONEHEADER FILE=catab4_1

Subcutaneous infliximab CT-P13 Population outcome model

\$PROBLEM Population outcome analysis of SC infliximab CT-P13

\$INPUT ID

VISIT ; weeks
DAYS ; TIME in days
ENDREM=DV ; 1= :) 0 = : (
PREV ; endoscopic remission at previous visit
DISEASE ; 1=UC, 2=CD
FC_w14 ; individual predicted FC at week 14[EXPOSURE]

\$DATA IFX_SC_ENDRM_NONMEM.csv IGNORE = @

;----- transitions from score 0 -----

; 0 --> 1 --- When no calpro: high chance! Baseline ~ 1

; from 0 to 1

;;; EMAX01-DEFINITION START

IF(DISEASE.EQ.2) EMAX01 = THETA(1) +ETA(1); CD patients

IF(DISEASE.EQ.1) EMAX01 = THETA(9) +ETA(1) ; UC patients

;;; EMAX01-DEFINITION END

EC5001 = THETA(2)

;;; FC effect only for UC patients

IF(DISEASE.EQ.2) P01 = EMAX01 ; cd patient

IF(DISEASE.EQ.1) P01 = EMAX01 - EMAX01 * (FC_w14/(EC5001+FC_w14)) ;BASELINE01 is
the baseline --- - calpro effect, because higher calpro results in lower chance of going from 0
to 1

; from 0 --> 2 [dropout]

EMAX02 = THETA(3)

ET5002 = THETA(4)

P02 = EMAX02

; from 0 to 0

P00 = 1-P01-P02

;----- transitions from score 1 -----

; 1 --> 0 --- When no calpro: low chance! Baseline ~ 0

; from 1 to 0

EMAX10 = THETA(5) ; population probability of going from 1 to 0

EC5010 = THETA(6)

P10 = EMAX10

; from 1 --> 2 [dropout]

EMAX12 = THETA(7)

P12 = EMAX12

; from 1 to 1

P11 = 1-P10-P12

```

;----- transitions from score 2 : missing data -----
; 2 --> 0 --- independent of calpro
; from 2 to 0
    EMAX20 = THETA(8) ; population probability of going from 2 to 0
    P20 = EMAX20

; from 2 --> 2 is not possible
    P22 = 0

; from 2 --> 1
    P21 = 1-P20

; not possible for 2 to 2

;----- transition's probabilities -----
IF (PREV.EQ.0.AND.DV.EQ.1) Y= P01
IF (PREV.EQ.0.AND.DV.EQ.2) Y= P02*(1-P01)
IF (PREV.EQ.0.AND.DV.EQ.0) Y= 1 - P01 - P02*(1-P01)

IF (PREV.EQ.1.AND.DV.EQ.0) Y= P10
IF (PREV.EQ.1.AND.DV.EQ.2) Y= P12*(1-P10)
IF (PREV.EQ.1.AND.DV.EQ.1) Y= 1 - P10 - P12*(1-P10)

IF (PREV.EQ.2.AND.DV.EQ.0) Y= P20
IF (PREV.EQ.2.AND.DV.EQ.1) Y= P21
IF (PREV.EQ.2.AND.DV.EQ.2) Y= 0

$THETA
(0, 0.523,1) ;TH1 EMAX01 CD
(0, 519) ;TH2 EC5001
(0, 0.28,1) ;TH3 EMAX02
(0) FIX ;TH4 ET5002
(0, 0.121,1) ;TH5 EMAX10
(0) FIX ;TH6 EC5010
(0, 0.157,1) ;TH7 EMAX12
(0, 0.312,1) ;TH8 EMAX20
(1) FIX ;TH9 EMAX01 UC

$OMEGA
0 FIX

$COV PRINT=E MATRIX=S;R
$ESTIMATION METHOD=COND LAPLACE LIKE MAX=9999 MSFO=MSFO8
$TABLE ID VISIT FC DV PREV DAYS EMAX01 EC5001 EMAX10 EC5010 Y P01 P00 P10 P11 ONEHEADER
NOPRINT FILE=sdtab100_4

```

Chapter IV. Clearance monitoring of mAbs

This chapter is based on

Wang Z, Hoffert Y, Zhang W, Kantasiripitak W, Verstockt B, Sabino J, Ferrante M, Declerck P, D'Haens G, Laharie D, Vermeire S, Dreesen E. Therapeutic antibody clearance better predicts endoscopic outcomes than trough concentrations in patients with Crohn's disease. [Submission in preparation]

Abstract

Background: Monitoring of mAb clearance has been suggested for predicting treatment outcomes in patients with inflammatory bowel diseases. We investigated clearance monitoring in patients with CD using data from clinical trials.

Methods: Data from patients with moderate-to-severe CD starting infliximab (n=108) and ustekinumab (n=80) therapy were repurposed. Endoscopic remission was assessed at week (w)12 and w54 of infliximab therapy. Endoscopic response was assessed at w24 of ustekinumab therapy. We performed Bayesian forecasting to estimate time-varying mAb clearance using measured drug concentrations, covariates, and population pharmacokinetics models (NONMEM7.5).

Results: Patients achieving endoscopic remission at w12 had significantly lower infliximab clearance and higher infliximab serum trough concentrations (C_{trough}) at w2 and w6 of standard 5 mg/kg infliximab induction therapy, patients achieving endoscopic remission at w12 had significantly lower infliximab clearance and higher infliximab serum C_{trough} than patients without endoscopic remission ($p < 0.05$). During infliximab maintenance therapy (with dose optimizations), patients achieving endoscopic remission at w54 had significantly lower infliximab clearance ($p < 0.001$) but not higher infliximab C_{trough} than patients without endoscopic remission. Moreover, infliximab clearance during maintenance had similar discriminative ability as FC (area under the receiver operating characteristics curve 0.63 [95% CI 0.57–0.69] vs. 0.67 [0.61–0.73], respectively). Ustekinumab clearance, but not C_{trough} , was significantly different between endoscopic responders and non-responders during standard ustekinumab induction and maintenance therapy ($p < 0.05$).

Conclusion: While C_{trough} loses its ability to predict treatment response when doses are optimized or not administered by bodyweight, mAb clearance remains a relevant predictor. Our interactive clearance monitoring tool can provide valuable information to forecast the chance of achieving endoscopic outcomes and distinguish between exposure-driven and mechanistic failure when patients with CD are dose-optimized.

1. Introduction

Since entering the market in 1999, mAbs have revolutionized the treatment of patients with IBD. Many studies have reported associations between lower C_{trough} of mAbs and worse clinical outcomes in patients with IBD.^{70,95,199–208} Therefore, following the well-established principle that drug exposure drives response, it stands to reason that dose escalations to elevate C_{trough} could potentially improve treatment outcomes.

A patient's exposure to mAbs is determined not only by the administered dose but also by the body's capacity to clear it. While dosing is an external and thus modifiable variable, the body's intrinsic mAb clearance capacity is not directly modifiable. Therefore, to overcome intrinsic variability in clearance, TDM-based dose optimizations can be performed to ensure adequate C_{trough} exposure.^{209,210} In other words, TDM is an individualized dosing strategy that accounts for PK variability, aiming at the same target C_{trough} – and presumably the same chances of treatment response – in all patients.

Despite increasing evidence for TDM over the last decade, not all patients benefit from dose optimizations. When a patient achieves the target C_{trough} but does not respond to treatment, mechanistic, PD-driven treatment failure is assumed. Moreover, when TDM-based dose optimizations are required for patients to achieve C_{trough} targets, the C_{trough} variability in the population reduces, blurring the exposure–response relationship and undermining the predictive ability of C_{trough} . This was observed in TDM studies such as TAILORIX, where C_{trough} lost its predictive ability of therapeutic outcomes.¹⁵ In the search for predictors of therapy success during TDM, we propose clearance monitoring.

Several factors can contribute to increased mAb clearance, including the presence of anti-drug antibodies, increased drug consumption due to a higher inflammatory load, and IBD conditions like leaky gut syndrome.^{50,211} We hypothesize that mAbs clearance, unlike C_{trough} , serves as a biomarker for IBD disease activity during TDM. To date, the benefits of clearance monitoring compared to C_{trough} monitoring remain unclear, and no tools exist to facilitate clearance monitoring.

In this study, we aimed to disentangle the bidirectional relationship between mAb PK and PD, and investigate the benefits of monitoring mAb clearance in patients with CD based on data

from clinical trials. Moreover, we aimed to develop an interactive software platform to facilitate clinical validation and showcase how our theoretical work can have a clinically meaningful impact on IBD management.

2. Methods

2.1. *Patients and study design*

Infliximab data from adult biologic-naïve patients with CD were obtained from a randomized, double-blind, controlled, multicenter study (“a randomized controlled trial investigating tailored treatment with infliximab for active luminal Crohn’s disease”, TAILORIX). The design and conduct of the study have been previously reported.⁸⁷ All patients received standard IV infliximab infusions of 5 mg/kg at w0, w2, and w6. At w14 of treatment, patients were randomized to one of three infliximab maintenance groups: dose increases in steps of 2.5 mg/kg based on clinical symptoms and biomarkers or serum infliximab concentrations, a dose increase from 5 mg/kg to 10 mg/kg based on the same criteria, or a dose increase to 10 mg/kg based on clinical symptoms alone. Additionally, a single additional infliximab infusion of 2.5 or 5 mg/kg at the 4-week interval was allowed when infliximab C_{trough} was below 1 mg/L. The study was approved by the local institutional review board of each participating center and by the French Health Authority (AFSSAPS) and was registered under EudraCT number 2011-003038-14. Written informed consent was obtained from all participants.

Ustekinumab data from adult CD patients, predominantly refractory or intolerant to anti-tumor necrosis factor agents, were obtained from a prospective, real-life cohort study at the UZ Leuven (Leuven, Belgium). The design and conduct of the study have been previously reported.⁷⁴ All included patients received a standard ustekinumab IV loading dose of approximately 6 mg per kilogram bodyweight (260 mg for <65 kg bodyweight, 390 mg for 65–85 kg, and 520 mg for >85 kg) at w0, followed by a fixed SC maintenance dose of 90 mg Q8W. No dose optimization was performed. Written informed consent was obtained from all participants (B322201213950/S53684).

2.2. *PK and PD data*

In the TAILORIX trial, serum and fecal samples were collected at w2 and w6 during infliximab induction therapy, then Q8W from w14 until w54, just before each dose. Patients’ infliximab C_{trough}, CRP, and FC were determined at all visits. Endoscopic remission (defined as CDEIS <3) was assessed at w12 and w54 of infliximab therapy. Levels of FC were measured using the

Quantum Blue assay, with a LLOQ of 100 µg/g and an ULOQ of 1800 µg/g (Bühlmann Laboratories, Schönenbuch, Switzerland). Serum infliximab concentrations were measured using a validated ELISA, with an LLOQ of 0.3 mg/L.

In the Leuven ustekinumab study, serum and fecal samples were collected at w4, w8, w16, and w24. Patients' ustekinumab serum concentration, CRP, and FC were determined at all visits. Endoscopic remission (defined as SES-CD ≤2) and response (≥50% decrease from baseline in SES-CD) were assessed at w24 of ustekinumab therapy. Levels of FC were measured using the Quantum Blue assay, with a LLOQ of 30 µg/g and an ULOQ of 1800 µg/g (Bühlmann Laboratories, Schönenbuch, Switzerland). Serum ustekinumab concentrations were measured using a validated ELISA, with an LLOQ of 0.25 mg/L.

2.3. Bayesian forecasting

Maximum *a posteriori* Bayesian forecasting was performed to estimate the antibody clearance at each sampling time point using the measured drug concentration at that time point and in-house built popPK models (NONMEM, version 7.5; Icon plc, Gaithersburg, MA, USA).^{168,212} The model codes are attached in the **Appendix D**. Covariates of FC, serum albumin, CDAI, and antibodies towards infliximab were included in the algorithm for estimating infliximab clearance. Covariates of serum albumin and bodyweight were used to estimate ustekinumab clearance.

2.4. Statistical analysis

Generalized linear mixed-effects models with binomial logistic regression were used to predict the probability of endoscopic outcomes based on single predictors: C_{trough}, mAb clearance, CRP, and FC (**Equation 22**).

$$\text{Probability of outcome} = 1 - \frac{1}{1 + e^{\text{intercept}_i + \theta_{\text{slope}} \times \text{predictor}}} \quad (22)$$

θ_{slope} represents the typical population slope factor describing the impact of different predictors on the probability of achieving the endoscopic outcome, intercept_i represents the parameter intercept of patient *i*. A non-parametric bootstrap method was used to calculate the 90% PI of the predicted probability of outcome.

Receiver operating characteristics (ROC) analysis was used to evaluate the discriminative ability of C_{trough} , mAb clearance, CRP, and FC for endoscopic outcomes.

Descriptive statistics were stated as frequencies and percentages for discrete variables and median and IQR for continuous variables. Wilcoxon signed-rank test was used to analyze paired measurements. Optimal thresholds of infliximab and ustekinumab clearance were selected using the Youden J statistic.²¹³ For all statistical analyses, a two-sided p -value of 0.05 or less was considered statistically significant.

Data were imported into R software (version 4.0.3; R Foundation for Statistical Computing, R Core Team, Vienna, Austria) for calculation of summary statistics (*tidyverse* package; version 2.0.0), generalized linear mixed-effects modeling of the endoscopic outcomes (*lme4* package; version 1.1-35.1), ROC analysis (*pROC* package; version 1.18.5), and graphical evaluation (*ggplot2* package; version 3.5.1).

2.5. Software tool development

A novel software tool for clearance monitoring of infliximab and ustekinumab was developed using the *Shiny* package (version 1.7.4) in R (version 4.0.3; R Foundation for Statistical Computing, R Core Team, Vienna, Austria), in the RStudio integrated development environment (version 1.3.1093; RStudio, Inc., Boston, MA, USA). The popPK model code of infliximab and ustekinumab was transposed in R using the *mrgsolve* package (version 1.0.9). Bayesian forecasting was implemented using the *mapbayr* package (version 0.10.0).²¹⁴

3. Results

3.1 *Study populations*

We repurposed data of 108 out of 122 (89%) patients with CD who enrolled in the TAILORIX trial. Eligible patients are those who contributed at least one detectable infliximab concentration, and had an endoscopic assessment at w12 or w54 (**Table 12**). We repurposed data of 80 out of 86 (93%) patients with CD who enrolled in the UZ Leuven ustekinumab trial. Eligible patients are those who contributed at least one detectable ustekinumab concentration, and had an endoscopic assessment at w24. Baseline demographics and clinical characteristics of the patients are summarized in **Table 12**.

Table 12. Patient characteristics at baseline

Parameter	Infliximab cohort	Ustekinumab cohort
Number of patients	108	80
<i>Demographics</i>		
Sex, women, n (%)	65 (60)	54 (68)
Age, years, median [IQR]	30.1 [22.4–44.7]	37.1 [27.9–49.1]
Bodyweight, kg, median [IQR]	63.3 [55.0–73.1]	64.5 [55.7–80.7]
Body mass index, kg/m ² , median [IQR]	22.1 [19.2–24.2]	23.3 [19.7–26.6]
<i>Serology</i>		
C-reactive protein, mg/L, median [IQR]	20.0 [8.0–36.3]	12.5 [4.6–19.8]
Fecal calprotectin, µg/g, median [IQR]	1462.5 [742.2–1800.0]	1398.0 [620.4–1800.0]
Serum albumin, g/L, median [IQR]	39.3 [34.3–42.5]	40.1 [37.9–42.5]
Hemoglobin, g/dL, median [IQR]	12.7 [11.8–13.8]	12.6 [11.9–13.3]
<i>Disease activity</i>		
Crohn's disease endoscopic index of severity, median [IQR]	11.0 [7.6–16.0]	NA
Simple Endoscopic Score for Crohn's disease, median [IQR]	NA	12 [8–18]
Disease location, ileal : Colonic : Ileocolonic, n (%)	25:19:64 (23:18:59)	23:10:47 (29:12:59)
Disease behavior, Inflammatory : Stricturing : Penetrating, n (%)	80:15:13 (74:14:12)	33:33:14 (41:41:18)
<i>Previous and concomitant medication</i>		
Previous exposure to anti-TNF, n (%)	0	75 (94)
Total number of biologicals previously exposed to, 0 : 1 : 2 : 3, n (%)	108:0:0:0 (100:0:0:0)	4:15:21:40 (5:19:26:50)
Concomitant corticosteroids, n (%)	35 (32)	28 (35)
Concomitant immunomodulators, n (%)	40 (37)	2 (2.5)
Smoking status, Active : Previous : Never, n (%)	31:30:47 (29:28:43)	40:19:21 (50:24:26)

Abbreviations: anti-TNF, anti-tumor necrosis factor; IQR, inter-quartile range; NA, not applicable

3.2 Clearance versus trough concentrations

During standard 5 mg/kg infliximab induction therapy, patients achieving endoscopic remission at w12 had significantly higher infliximab C_{trough} and lower infliximab clearance at w2 and w6 of treatment ($p < 0.01$, **Figure 20A–B**). During infliximab maintenance therapy with dose optimizations, infliximab C_{trough} was not significantly different between patients with and without endoscopic remission at w54 (median [IQR]: 5.7 [4.1–9.3] mg/L and 6.5 [3.9–8.5] mg/L, respectively) ($p = 0.92$, **Figure 20C**). However, infliximab clearance during maintenance therapy was significantly lower in patients with endoscopic remission than in patients without endoscopic remission at w54 (0.25 [0.22–0.29] L/day and 0.28 [0.25–0.33] L/day, respectively) ($p < 0.01$, **Figure 20D**).

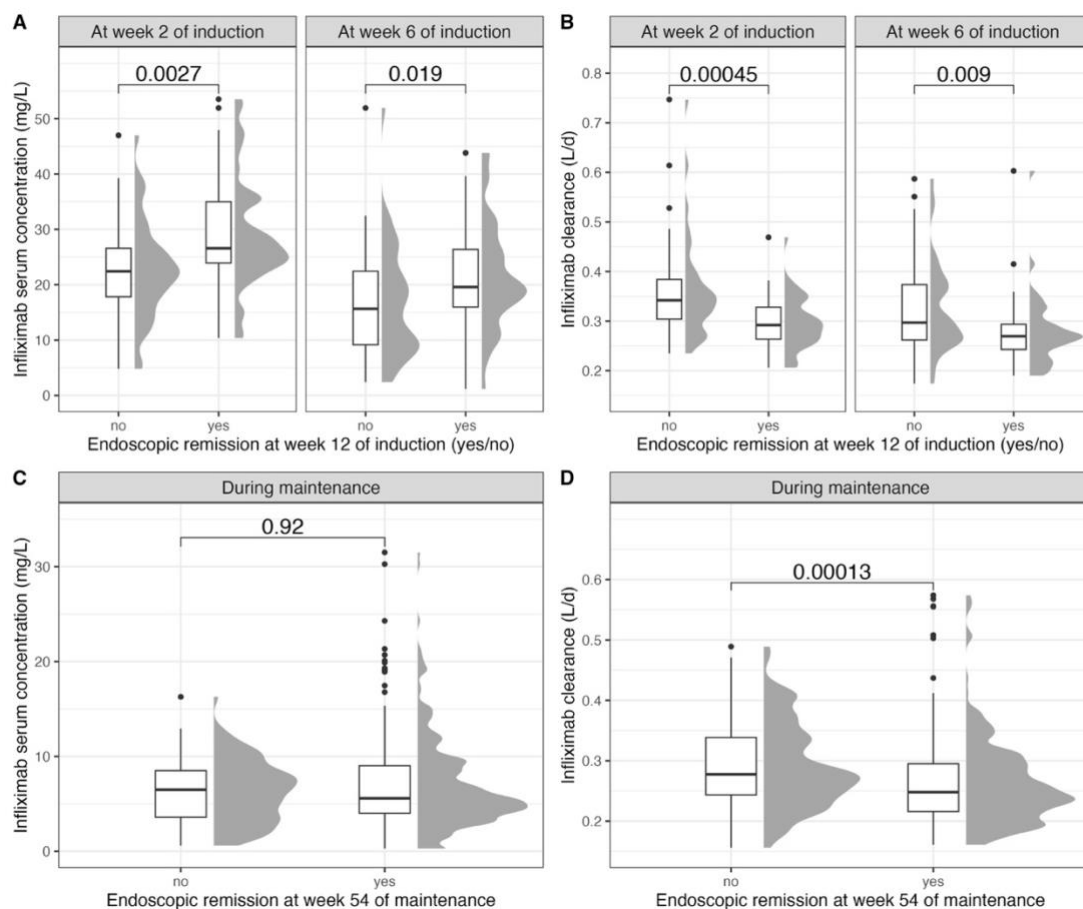


Figure 20. The comparison of infliximab serum trough concentration and infliximab clearance between patients who achieved and did not achieve endoscopic remission at week 12 (**A&B**) and at week 54 (**C&D**). The shaded area adjacent to each box plot illustrates the density distribution of the data. Two-sided p values were presented. No correction for multiple testing was performed.

During ustekinumab induction therapy, all patients received an IV loading dose stratified by bodyweight. Although dose stratification aimed to provide an approximate dose of 6 mg/kg, the actual dose per kilogram of bodyweight varied among patients (**Appendix D Figure S1**). Ustekinumab concentrations both during induction and maintenance therapy were not significantly different between endoscopic responders and non-responders at w24 (**Figure 21A**). Nevertheless, ustekinumab clearance at w4, w8 and w16 was significantly and consistently lower in patients who achieved endoscopic response at w24 ($p < 0.05$, **Figure 21B**).

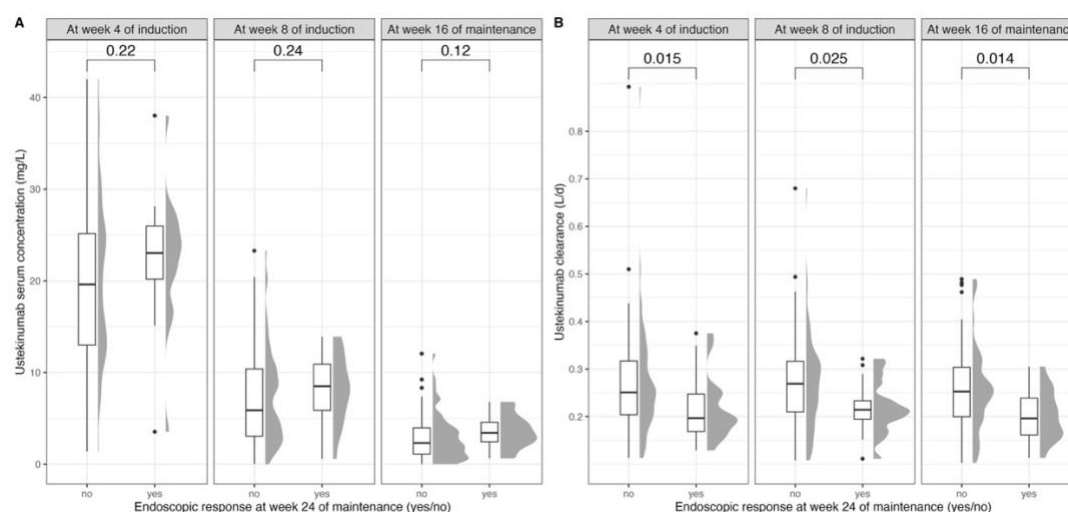


Figure 21. Ustekinumab serum concentrations (**A**) and clearance (**B**) between patients who achieved and did not achieve endoscopic response at week 24. The shaded area adjacent to the box plot illustrates the density distribution of the data. Two-sided p values are presented. No correction for multiple testing was performed.

3.3 Comparison of drug trough concentrations, clearance, and biomarkers for endoscopic outcomes

During infliximab induction therapy, infliximab C_{trough} , infliximab clearance, CRP, and FC had similar ability to discriminate between patients with and without endoscopic remission at w12 (**Figure 22A**). During infliximab maintenance therapy, neither infliximab C_{trough} nor CRP could discriminate between patients with and without endoscopic remission at w54 (**Figure 22B**). However, infliximab clearance could predict endoscopic remission at w54 with discriminative ability similar to that of FC (**Figure 22B**). An infliximab clearance of 0.25 L/day

during maintenance therapy was identified as an optimal cut-point to classify patients with and without endoscopic remission (sensitivity=0.48, specificity=0.75).

Generalized linear mixed-effects models were developed, linking infliximab clearance during induction to endoscopic remission at w12, and infliximab clearance during maintenance to endoscopic remission at w54 (**Equation 22**). VPCs for the outcome models are shown in **Figure 22C-D**, demonstrating a good model calibration. The probability of achieving endoscopic remission at w54 with the identified optimal threshold of infliximab clearance of 0.25 L/day during maintenance was 77% (90%PI 70–85%) (**Figure 22D**).

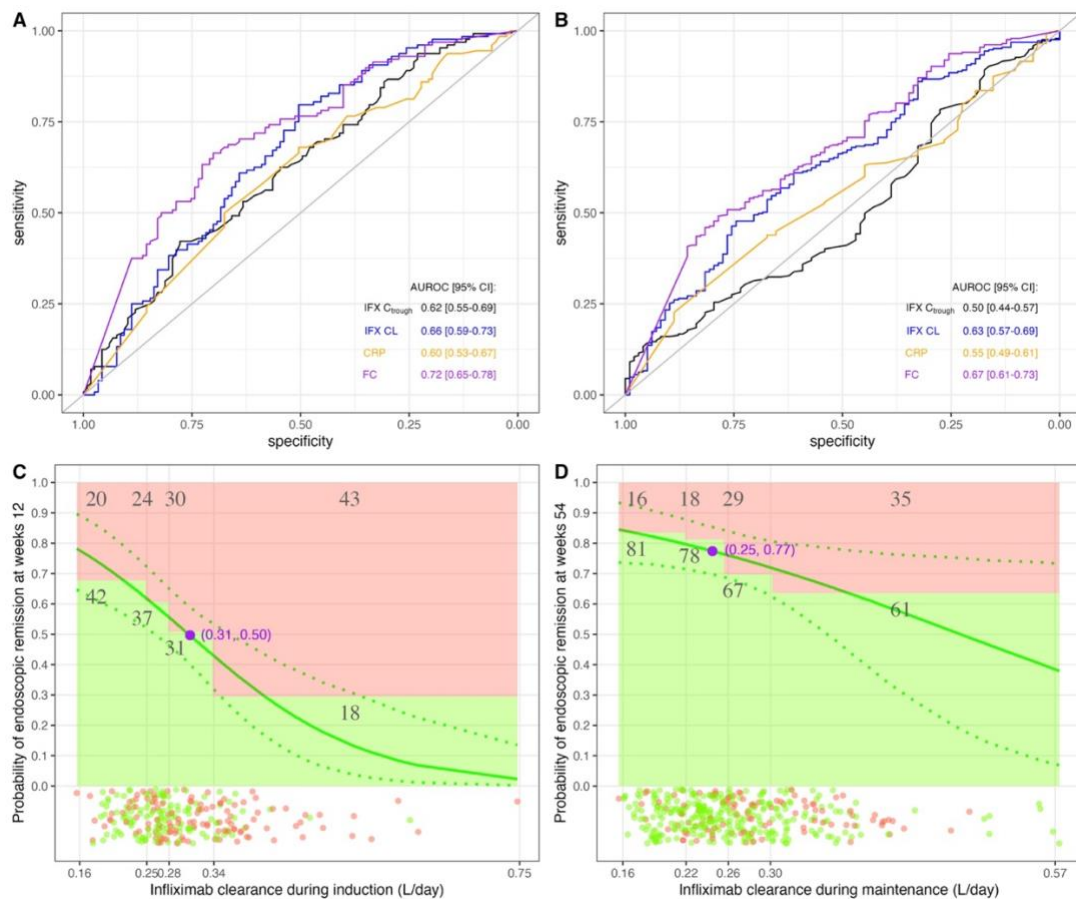


Figure 22. Receiver operating characteristic (ROC) analysis of different predictors (A–B) and visual predictive checks (VPC) of the infliximab clearance–endoscopic remission models (C–D) during induction and during maintenance for predicting the endoscopic remission at week 12 and week 54 of infliximab therapy. In Panel A&B, the ROC curves of infliximab trough concentrations (IFX C_{trough} ; black line), infliximab clearance (IFX CL; blue line), C-reactive protein (CRP; orange line), and fecal calprotectin (FC; purple line) are presented. The median and 95% confidence interval (CI) of the area under the ROC curve (AUROC) were annotated. In Panels C&D, the probability of achieving endoscopic remission based on infliximab clearance during induction (C) or maintenance (D) is plotted. Observed and predicted median fractions of patients achieving endoscopic remission are represented by green tiles and lines, respectively. Red tiles indicate the observed fractions of patients not achieving endoscopic remission. Green dashed lines represent the 90% prediction interval around the median prediction. Green and red dots under each plot are the estimated infliximab clearance from patients achieving and not achieving endoscopic remission, respectively. Purple dots and coordinates represent the identified optimal threshold of infliximab clearance and the associated probability of endoscopic remission. Grey numbers indicate the absolute number of patients in each quartile of infliximab clearance.

During ustekinumab induction therapy, neither ustekinumab C_{trough} nor CRP could discriminate between patients with and without endoscopic response at w24 (Figure 23A). Ustekinumab clearance and FC showed statistically comparable discriminative ability, but were not statistically better than ustekinumab C_{trough} (Figure 23A). During ustekinumab

maintenance therapy (**Figure 23B**), neither ustekinumab C_{trough} nor CRP could discriminate between patients with and without endoscopic response at w24. However, ustekinumab clearance during maintenance could predict endoscopic response at w24 with discriminative ability similar to FC (**Figure 23B**). A ustekinumab clearance of 0.22 L/day during maintenance therapy was identified as an optimal cut-point to classify patients with and without endoscopic response at w24 (sensitivity=0.71, specificity=0.70).

Generalized linear mixed-effects models were developed, linking ustekinumab clearance during induction and maintenance to endoscopic response at w24 (**Equation 22**). VPCs for the outcome models are shown in **Figure 23C-D**, demonstrating a good model fit to the data. The probability of achieving endoscopic response at w24 with the identified optimal threshold of ustekinumab clearance of 0.22 L/day during maintenance was 24% (95% CI 15–32%) (**Figure 23D**).

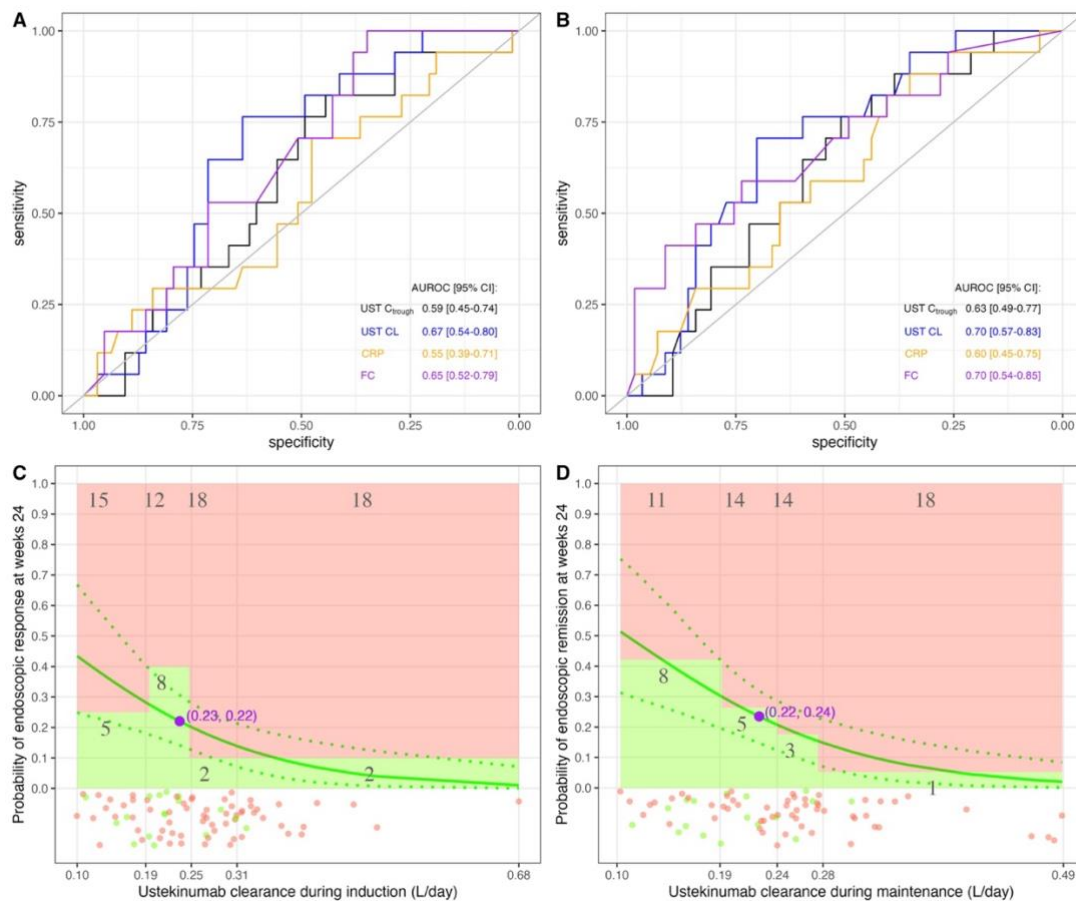


Figure 23. Receiver operating characteristic (ROC) analysis of different predictors (A–B) and visual predictive checks (VPC) of the ustekinumab clearance–endoscopic outcome models (C–D) during induction and during maintenance for predicting the endoscopic response post-maintenance at week 24 of ustekinumab therapy. In Panels A&B, the ROC curves of ustekinumab trough concentration (UST C_{trough}), UST clearance (UST CL; blue line), C-reactive protein (CRP; orange line), and fecal calprotectin (FC; purple line) are presented. The median and 95% confidence interval (CI) of area under ROC curve (AUROC) were annotated. In Panels C&D, the probability of achieving endoscopic response at week 24 based on ustekinumab clearance during induction (C) or maintenance (D) is plotted. Observed and predicted median fractions of patients achieving endoscopic remission are represented by green tiles and lines, respectively. Red tiles indicate the observed fractions of patients not achieving endoscopic remission. Green dashed lines represent the 90% prediction interval around the median prediction. Green and red dots under each plot are the estimated ustekinumab clearance from patients achieving and not achieving endoscopic response, respectively. Purple dots and coordinates represent the identified optimal threshold of ustekinumab clearance and the associated probability of endoscopic response. Grey numbers indicate the absolute number of patients in each quartile of ustekinumab clearance.

3.4 Software tool for clearance monitoring

We developed an evidence-based interactive software tool that can predict the probability of achieving endoscopic outcomes in a clinical setting based on the estimated clearance of

infiximab and ustekinumab. User input parameters include the infiximab/ustekinumab dosing information (*Dosing* tab) and information about patient characteristics (popPK model covariates; *Individual* tab) and an infiximab/ustekinumab concentration (*Individual* tab). Bayesian forecasting is performed in the background to obtain a clearance estimate. The concentration–time plot shows the individually predicted concentration–time profile based on patient covariates (“*Covariates-based prediction*”) and based on both drug concentration and covariates (“*Bayesian forecasting*”). If no drug concentration is provided in the *Individual* tab, the concentration–time profiles for both *Covariates-based prediction* and *Bayesian forecasting* will overlap.

Furthermore, the probability of achieving the endoscopic outcome associated with the individually predicted drug clearance is visualized in the tool. A third plot shows how the individually predicted clearance deviates from the typical clearance in the studied population. Finally, a summary table is generated, which includes administration time points, observations, predictions, covariates, and PK parameters. The infiximab and ustekinumab clearance estimates by the software tool align well with those estimated by NONMEM (**Appendix D Figure S2**). Additional details and instructions for the use of the software tool are available in the web application hosted at https://lpmx.shinyapps.io/clearance_monitoring/ (**Figure 24**).

4. Discussion

In this study, we demonstrated that mAb clearance is a better predictor than C_{trough} for endoscopic outcomes in patients with CD and provides a discriminative ability statistically similar to the current gold-standard biomarker, FC. Additionally, we provide an interactive software tool for clearance monitoring of infliximab and ustekinumab to facilitate clinical validation of our findings and, ultimately, support clinical decision-making in routine IBD management.

Monoclonal antibody clearance is an important predictor of disease activity of IBD. The mAb C_{trough} reflects a patient's intrinsic drug clearance rate, but this reflection can be easily confounded by two factors: the administered dose and the volume of distribution (generally related to bodyweight). Therefore, only when these two variables are controlled, we can interpret differences in C_{trough} as differences in mAb clearance. When standard dosing is based on bodyweight, as for infliximab, variations in the volume of distribution are accounted. This approach accounts for PK variability and thus maintains the association between C_{trough} and clearance. That is why during the TAILORIX induction phase, changes in clearance were immediately reflected in changes in infliximab concentrations, allowing C_{trough} to predict post-induction endoscopic remission. However, during the TAILORIX maintenance phase, where dosing was optimized to achieve similar infliximab C_{trough} in all patients, the exposure–response relationship inevitably disappeared. In all other scenarios where variability in the volume of distribution was either not accounted for (e.g., fixed SC dosing during ustekinumab maintenance therapy) or not adequately accounted for (e.g., insufficient dose stratification by bodyweight during ustekinumab induction), there is a risk that drug concentrations will not accurately reflect changes in drug clearance. This may render C_{trough} an unreliable predictor of therapeutic outcomes.

Variability in dosing causes noise that masks the relationship between C_{trough} and treatment outcomes. Bayesian forecasting considers dosing information when calculating mAb clearance, thereby reducing the noise and increasing the discriminative ability of mAb clearance as a predictor for treatment outcomes in IBD. Therefore, instead of solely relying on drug exposure to reflect clearance, we suggest to directly monitor clearance. However, unlike drug concentration, clearance cannot be directly measured. It needs to be estimated

using a popPK model, complicating its implementation in daily clinical practice. To overcome this, we developed an interactive software tool incorporating these model algorithms. By providing the patient's covariates and a drug concentration, the clearance monitoring software tool can simultaneously predict infliximab/ustekinumab clearance and the associated probability of achieving endoscopic outcomes for individual patients. The tool was developed based on the open-source R package *mapbayr*, which has been validated against NONMEM for producing reliable maximum *a posteriori* Bayesian estimation of PK parameters.²¹⁴ However, applicability of the tool to SC infliximab has not been investigated and needs further research.

Since clearance is an intrinsic patient characteristic, therapeutic decisions should not rely solely on clearance monitoring. Besides, the relatively poor discriminative ability of the infliximab and ustekinumab models (AUROCs and classification accuracy ratios below 0.7), though similar to fecal calprotectin, undermines confidence in the predicted probability of endoscopic outcomes. We propose a combined C_{trough} and clearance monitoring approach for infliximab, based on a predefined C_{trough} target and a clearance threshold of 0.25 L/day (**Figure 25**). In patients with an infliximab C_{trough} below the target, dose escalation can be performed without clearance monitoring to first raise C_{trough} above the target. For patients with an infliximab C_{trough} above the target, incorporating clearance monitoring alongside traditional TDM offers two advantages: (i) For patients not in clinical or biochemical remission, clearance monitoring can pre-screen those who may be experiencing mechanistic failure before additional endoscopic assessment, potentially reducing the need for endoscopy. (ii) For patients currently in clinical or biochemical remission, clearance monitoring can identify those at high risk of treatment relapse, allowing for more precise allocation of medical resources to patients who require close monitoring. We illustrated this approach using an infliximab C_{trough} target of 5.0 mg/L during maintenance therapy, as recommended by the AGA Institute Guideline.⁶⁵ However, the C_{trough} target and definitions of clinical and biochemical remission can be adjusted based on clinical practice, such as the C_{trough} target of 3–7 mg/L used in the TAXIT algorithm.⁸⁶ Further research is needed to demonstrate the superiority of combining model-informed TDM with clearance monitoring over traditional TDM.

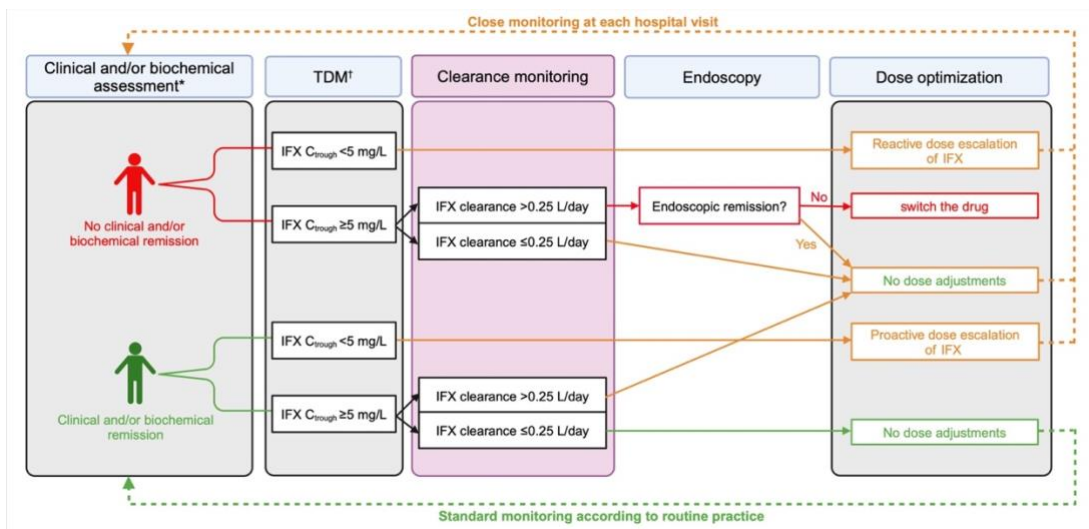


Figure 25. Recommended flowchart for a combined concentration and clearance monitoring approach of infliximab during maintenance therapy in patients with Crohn's disease. C_{trough} , trough concentration; IFX, infliximab; mAbs, monoclonal antibodies; switch the drug, switch to a drug with another mechanism of action; TDM, therapeutic drug monitoring. * The definition for clinical and biochemical remission can be adapted according to the clinical practice. † While the infliximab C_{trough} target of 5 mg/L is based on the AGA institute guideline, other C_{trough} targets can be chosen according to clinical practice, such as the 3–7 mg/L range used in the TAXIT algorithm.^{65,86}

Our study has several important strengths. Firstly, we evaluated the discriminative ability of infliximab and ustekinumab clearance for endoscopic outcomes, the current treatment goal in IBD.³⁵ Additionally, we compared the discriminative ability of clearance with not only C_{trough} , but also other biomarkers like CRP and FC. Furthermore, we proposed a combined C_{trough} and clearance monitoring approach for infliximab that can be integrated into the existing TDM workflow to reduce the need of endoscopy and identify patients at high risk of treatment relapse. Finally, we translated our theoretical findings into a clearance monitoring software tool to facilitate external validation of a stratified medicines approach to improve mAb therapy in IBD. The clearance monitoring software tool is highly flexible, allowing clearance calculation from concentration taken at any time, facilitating remote sampling.

Our study also has limitations. First, in the ustekinumab cohort, endoscopic response at week 24 was achieved in only 21.5% of patients, whereas endoscopic remission at week 24 in only 7.2%. It was not possible to identify predictors of endoscopic remission due to the low number of patients reaching that outcome, hence our focus on endoscopic response instead. Second, in the infliximab cohort, endoscopic remission was defined based on CDEIS, which is not commonly used in clinical practice, as compared to SES-CD. In addition, the impact of

different assays for mAb concentration measurement on clearance estimates remains uninvestigated. Last but not least, the clearance monitoring software tool was not externally validated due to the lack of suitable external data.

In conclusion, while C_{trough} loses the ability to predict endoscopic remission when doses are optimized or not administered by bodyweight, mAb clearance remains a relevant predictor, with similar discriminative ability as FC (but not demanding stool sampling). Our clearance monitoring software tool can provide valuable information to forecast the chance of achieving endoscopic outcomes and distinguish between exposure-driven and mechanistic failure through a combined concentration and clearance monitoring approach.

Appendix D

Supplementary Figures

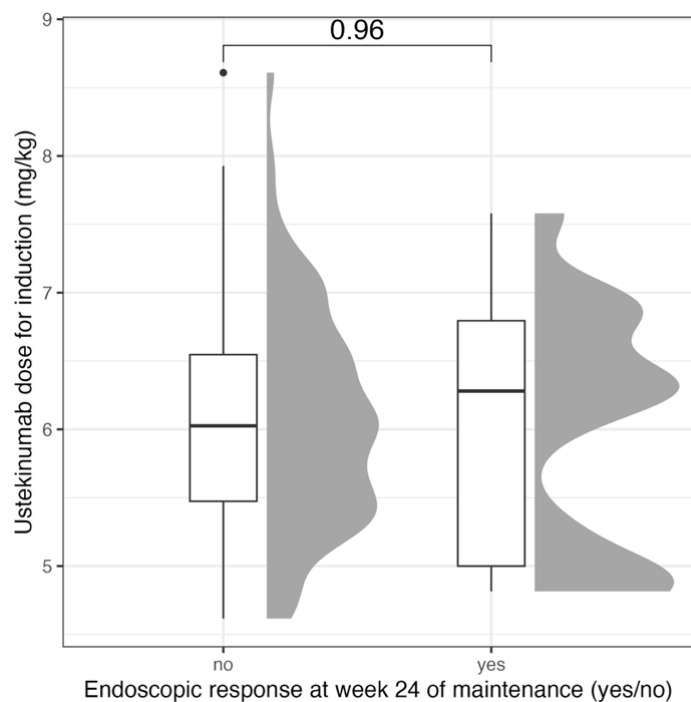


Figure S1. Ustekinumab induction dose between patients who achieved and did not achieve endoscopic response at week 24 of maintenance. The shaded area adjacent to the box plot illustrates the density distribution of the data. Two-sided p value is reported.

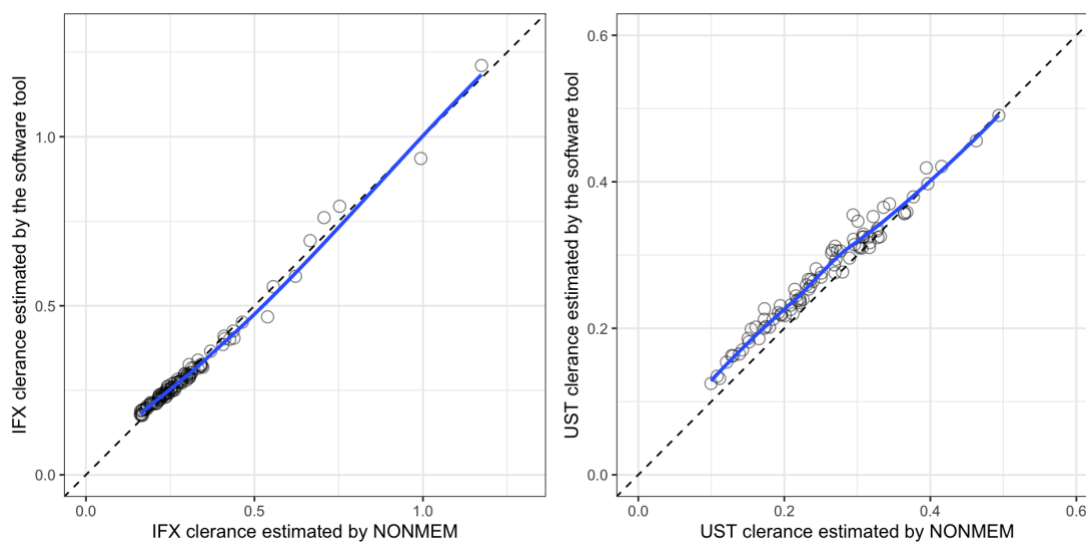


Figure S2. A comparison of the predicted infliximab and ustekinumab clearance between NONMEM and the developed clearance monitoring software tool. The blue line is a locally weighted smoother.

NONMEM control streams

Infliximab Pharmacokinetics model

```
$PROBLEM PopPK analysis of infliximab in TAILORIX
$INPUT ID TIME CENS AMT RATE EVID MDV LNDV=DV FLAG ATICAT FC ALB CDAI
$DATA data1.csv IGNORE=@
$SUBROUTINE ADVAN6 TRANS1 TOL=5
$MODEL NCOMP=2 COMP=(CENTRAL,DEFDOS,DEFOBS) COMP=(PERIPH)
$PK
CLFC = ((FC/273.68)**THETA(9))
CLCDAI = ((CDAI/115.58)**THETA(8))
IF(ATICAT.EQ.0) CLATICAT = 1 ; Most common
IF(ATICAT.EQ.1) CLATICAT = (1+THETA(7))
CLALB = ((ALB/42.95)**THETA(6))
CLCOV = CLALB*CLATICAT*CLCDAI*CLFC

TVCL = THETA(1)
TVCL = CLCOV*TVCL
TVV1 = THETA(2)
TVV2 = THETA(3)
TVQ = THETA(4)

CL = TVCL * EXP(ETA(1))
V1 = TVV1 * EXP(ETA(2))
V2 = TVV2 * EXP(ETA(3))
Q = TVQ * EXP(ETA(4))

K = CL/V1
K12 = Q/V1
K21 = Q/V2
S1 = V1
S2 = V2

IF(NEWIND.EQ.0) THEN
LLOQ = 0.3 ; mg/L
ENDIF

$DES DADT(1) = -K*A(1) - K12*A(1) + K21*A(2)
DADT(2) = K12*A(1) - K21*A(2)

$ERROR (ONLY OBSERVATIONS)
IPRED = LOG(0.10) ; value that is less than 1/2 lloq
W = THETA(5) ; residual error
IF(F.GT.0) IPRED = LOG(F)
IRES = IPRED-DV
SD = SQRT(W**2)
IWRES = IRES/SD

LLOQ=LOG(0.3)
```

```

IF (DV.GE.LLOQ) THEN
  F_FLAG = 0
  Y = IPRED+SD*ERR(1)
ELSE
  F_FLAG = 1
  Y = PHI((LLOQ-IPRED)/SD)
ENDIF

$THETA (0,0.277) ; TVCL
(0,4.9) ; TVV1
(0,0.844) ; TVV2
(0,0.0201) ; TVQ
-0.364 ; ADDS
(-1000,-1.13,100000) ; CLALB
(-1,0.754,5) ; CLATICAT
(-100,0.0609,100000) ; CLCDAI
(-100,0.0752,100000) ; CLFC

```

```

$OMEGA 0.0781 ; ETA_CL
0 FIX ; ETA_V1
0 FIX ; ETA_V2
0 FIX ; ETA_Q

```

```

$SIGMA 1 FIX
$ESTIMATION MAXEVAL=9999 NOABORT PRINT=1 NSIG=3 METHOD=1 INTER LAPLACE
$COVARIANCE UNCONDITIONAL PRINT=E
$TABLE ID TIME CENS AMT RATE EVID MDV LNDV=DV FLAG ATICAT FC ALB CDAI V1 K K12 K21 Q V2
CL PRED IPRED RES IRES SD IWRES CWRES ETA(1) NOAPPEND NOPRINT ONEHEADER FILE=data1.tab

```

Ustekinumab Pharmacokinetics model

\$PROBLEM PopPK analysis of Ustekinumab in patients with Crohn's disease

\$INPUT ID EVID MDV TIME AMT CMT DV ALB

\$DATA Ustekinumab_PKPD_NONMEM_BLOQ.csv IGNORE=@

\$SUBROUTINE ADVAN6 TOL=3

\$MODEL

COMP = (1) ; site of absorption

COMP = (2) ; central compartment

COMP = (3) ; peripheral compartment

\$PK

CLALB = ((ALB/42.65)**THETA(9))

CLCOV = CLALB

A_0(1)= 0

A_0(2)= 0

KA = THETA(1)

TVV1 = THETA(2)*((WEIGHT/65.27)**1)

TVCL = THETA(3)*((WEIGHT/65.27)**0.75)

TVF1 = THETA(4)

TVQ = THETA(5)*((WEIGHT/65.27)**0.75)

TVV2 = THETA(6)*((WEIGHT/65.27)**1)

TVCL = CLCOV*TVCL

F1 = TVF1*EXP(ETA(2))

V1 = TVV1

V2 = TVV2

Q = TVQ

CL = TVCL*EXP(ETA(1))

KE = CL/V1

K12 = Q/V1

K21 = Q/V2

\$DES

DADT(1)=-KA*A(1)

DADT(2)= KA*A(1) -KE*A(2) -K12*A(2) +K21*A(3)

DADT(3)= K12*A(2) -K21*A(3)

\$ERROR

CP = A(2)/V1

IPRED= LOG(A(2)/V1+0.00001)

IRES = DV-IPRED

W = SQRT(THETA(8)**2 + (THETA(7)/CP)**2)

IF (W.EQ.0) W=1

IWRES= IRES/W

LLOQ = LOG(0.25)

IF (DV.GE.LLOQ) THEN

```

F_FLAG = 0
Y      = IPRED+EPS(1)*W
ELSE
F_FLAG = 1
Y      = PHI((LLOQ-IPRED)/W)
ENDIF

$THETA
(0.142) FIX ; KA
(0,5.05)    ; V1
(0,0.235)   ; CL
(0,0.624)   ; F1
(0,0.0426)  ; Q
(0,0.748)   ; V2
(-0.272)    ; Additive error
(0,0.136)   ; Proportional error
(-1.21)     ; CLALB

$OMEGA
0.093       ; 1. omCL
0.13        ; 2. omF

$SIGMA 1 FIX

$EST METHOD=1 INTER LAPLACIAN NUMERICAL MAXEVAL=99999 SADDLE_RESET=1
$COV Matrix=S PRINT=E
$TABLE ID EVID MDV TIME AMT CMT DV ALB IPRED F1 KA V1 V2 CL Q NOAPPEND NOPRINT
ONEHEADER FILE=data2.tab

```

Chapter V. MIPD-guided infliximab dosage de-escalation

This chapter is based on

Wang Z*, De Greef I*, Kantasiripitak W, Wicha S, Tops S, Nigro M, Outtier A, Verstockt B, Sabino J, Thomas D, Vermeire S, Ferrante M*, Dreesen E*. Model-informed dosage de-escalation of infliximab in patients with inflammatory bowel diseases. [Submission in preparation]

* Designated shared co-first or co-senior authorship

Abstract

Background: Extending the infliximab infusion interval has been attempted in patients with CD and UC who sustained treatment response after an earlier interval shortening. We aimed to compare TDM and MIPD for successful infliximab dosage de-escalation.

Methods: In this prospective, historically controlled Phase 4 trial, patients with steroid-free clinical and biological remission on shortened (less than 8-weekly) infliximab intervals underwent interval extension guided either by TDM (historical control) or MIPD, targeting an infliximab C_{trough} of 5 mg/L (prospective intervention). The primary endpoint was sustained steroid-free clinical and biological remission over one year. Infliximab concentrations were measured before each infusion. Bayesian forecasting was performed using the *TDMx Infliximab* module.

Results: The study included 52 patients: 21 patients (11 CD, 10 UC) in the TDM arm and 31 patients (19 CD, 12 UC) in the MIPD arm. MIPD achieved a numerically higher rate of sustained steroid-free clinical and biological remission (52% vs. 29%, $p=0.17$) and a significantly higher rate of sustained steroid-free clinical remission (81% vs. 38%, $p<0.01$) over one year compared to TDM. The average dosing interval over one year was comparable between the TDM arm (8.0 [IQR 7.8–8.0] weeks) and the MIPD arm (8.0 [IQR 7.9–8.0] weeks). The average infliximab dose per infusion over one year was similar between the TDM arm (7.5 [IQR 7.4–8.2] mg/kg) and the MIPD arm (8.8 [IQR 6.6–10.9] mg/kg).

Conclusion: MIPD-guided dosage de-escalation of infliximab resulted in higher rates of relapse-free survival compared to TDM. Future studies with larger sample sizes are needed to validate these findings.

1. Introduction

Infliximab, an anti-TNF mAb, is commonly used for treating patients with CD and UC.^{215,216} Standard dosing involves a 5 mg/kg IV infusion at weeks 0, 2, and 6 during induction therapy, followed by Q8W maintenance infusions. Nearly a third of patients exhibit primary non-response, showing no clinical benefit after the induction phase.⁵⁷ Additionally, up to half of the patients may initially respond but experience secondary loss of response over time.⁵⁶ Low C_{trough} of infliximab and the presence of ATI have been linked to this loss of response in IBD.^{217–219}

For patients with low C_{trough} but no ATI, dose escalation has proven beneficial, by increasing the dose and/or shortening the interval between infusions.²²⁰ Shortening the infusion interval is often more effective and efficient in raising C_{trough} and may be preferred over dose increases.²²¹ However, interval shortening has disadvantages including increased patient inconvenience due to more frequent hospital visits, higher direct and indirect costs due to lost workdays, and an elevated workload for infusion services.²²² The financial impact is furthermore significant, especially when a shortened dosing interval is maintained over an extended period. Additionally, long-term use of escalated infliximab dosing raises concerns about safety risks, including a higher likelihood of malignancy and opportunistic infections.^{223,224} Therefore, dosage de-escalation is primarily recommended for patients who are considered to have a low risk of relapse.^{225,226}

In routine practice, dosage de-escalation is often attempted once patients stabilize on an increased dose or shortened dosing interval of infliximab. Dosage de-escalation was suggested for patients receiving infliximab or adalimumab whose C_{trough} consistently exceed 10–15 mg/L.²²⁷ However, data on the success of infliximab de-escalation following previous dose escalation are still limited. Recent evidence from the NOR-DRUM B and PRECISION trials supports the benefits of TDM and MIPD of infliximab during maintenance therapy, demonstrating improved treatment outcomes.^{89,98} The AGA Institute Guideline also recommends an infliximab C_{trough} target of 5.0 mg/L during maintenance therapy for maintaining disease remission.⁶⁵ These findings indicate that personalized dosing strategies targeting the infliximab C_{trough} , can enhance the therapeutic effectiveness. Additionally, studies have shown that TDM-guided dosage de-escalation of infliximab, which uses C_{trough} to

assess the feasibility of dose reduction, reduces the risk of relapse compared to dosage de-escalation strategies not informed by TDM.^{228,229}

A freely available software tool was previously developed and validated for guiding MIPD of infliximab based on C_{trough} in patients with IBD.⁸⁴ Nevertheless, the efficacy of TDM- or MIPD-informed infliximab dosage de-escalation, aiming at a target C_{trough} , remains uninvestigated. Therefore, we aimed to compare infliximab dosage de-escalation based on TDM with de-escalation guided by an MIPD software tool in patients with CD and UC who regained treatment response following an earlier dose escalation of infliximab maintenance therapy.

2. Methods

2.1. *Study design and participants*

We conducted a monocentric, historically controlled Phase 4 trial at UZ Leuven (Leuven, Belgium), that is, the Model-informed Dosage De-escalation of Infliximab in Patients With Inflammatory Bowel Diseases (MODIFI) trial. The trial was approved by the local Institutional Review Board (S64521) and was registered under EudraCT number 2020-004012-10. Informed consent was obtained from all participants. The prospective, open-label intervention study (MIPD arm) was conducted between February 2022 to September 2024. The historical control data (TDM arm) were retrospectively collected from the UZ Leuven electronic health records. Serum samples were available from the CCare Biobank (B322201213950/S53684).

Eligible patients were at least 18 years of age and a diagnosis of moderately to severely active CD or UC, confirmed by clinical, endoscopic, histological, or imaging criteria. At screening, patients had to be on escalated infliximab maintenance therapy (i.e., dose >5 mg/kg or dosing interval <8 weeks), and have regained steroid-free, combined clinical and biological remission following a previous loss of response to standard infliximab maintenance therapy. Patients had to have an infliximab serum C_{trough} of at least 5 mg/L at screening. Clinical remission was defined based on the PRO2: for CD, a mean daily abdominal pain score ≤ 1 and a liquid stool frequency score ≤ 1.5 ; for UC, a rectal bleeding score of 0 and a stool frequency score ≤ 1 .^{20,230} Biological remission was defined as CRP <5 mg/L and fecal calprotectin <250 mg/kg. Patients with unclassified IBD, with systemic or topical corticosteroid use, with an ostomy or an ileal anal pouch anastomosis, and patients who received infliximab prophylactically were excluded.

2.2. *Study procedures*

In this study, patients underwent infliximab dosage de-escalation either guided by TDM (the historical control arm) or by MIPD (the prospective intervention arm). In both TDM and MIPD arms, study visits were scheduled as follows: at screening (the infusion visit before the start of dosage de-escalation), at baseline (T0, the start of the infliximab dosage de-escalation), and approximately Q8W thereafter until one year, totaling six study visits after T0 (from T0+1

to T0+6). In the MIPD arm, two additional visits to the general practitioner (GP) were scheduled between screening and T0, and between T0 and the next infusion (T0+1).

Patients in the TDM arm underwent infliximab dosage de-escalation (by dose reduction and/or interval extension) according to the TAXIT (Trough Concentration Adapted Infliximab Treatment) algorithm, aiming to maintain an infliximab C_{trough} between 3–7 mg/L after the dosage de-escalation.⁸⁶ Patient's PRO2, infliximab C_{trough} , and CRP were determined right before infliximab infusion. Stool samples were not collected in these patients.

Patients in the MIPD arm underwent infliximab dosage de-escalation by interval extension following an MIPD dosing algorithm implemented in the stand-alone *TDMx* software tool.⁸⁴ Each patient received an individual infliximab dose predicted by the dosing algorithm and accepted by the treating physician. Venous blood sampling for serum albumin, CRP, and infliximab determinations was prospectively performed at trough (right before the next infusion) at every infusion visit and at two intermediate time points (by the GP). Patients collected fresh stool samples at screening, at six months (T0+3), and at one year (T0+6), stored them at 4°C in the home fridge, and transported them cooled within 24 h to the hospital for FC determination. Patients were asked to self-report PRO2 scores to the electronic healthcare system on a weekly basis for one year.

Infliximab C_{trough} were measured using the apDia Infliximab ELISA (apDia, Turnhout, Belgium), with a LLOQ of 0.3 mg/L.²³¹ ATI were only quantified in samples with an unquantifiable infliximab concentration using an in-house developed drug-sensitive anti-infliximab bridging assay (LLOQ at 5.0 mg/L MA-IFX10F9 equivalents).²³² FC measurements were performed with the BÜHLMANN fCal® ELISA kit (Bühlmann, Schönenbuch, Switzerland) with a LLOQ of 30 mg/kg. The LLOQ for CRP was 0.6 mg/L.

2.3. Precision dosing software

The *TDMx* MIPD software tool integrated with multi-model algorithms consisting of four popPK models of infliximab was used to guide dosage de-escalation.⁸⁴ The freely available MIPD software tool was jointly developed by KU Leuven and the University of Hamburg, and underwent validation in adult patients with CD and UC receiving infliximab treatment at UZ Leuven (<https://tdmx.shinyapps.io/infliximab/>).

In the prospective MIPD arm, we used an infliximab C_{trough} target of 5 mg/L, as suggested by the AGA guideline.⁶⁵ According to the trial protocol, a higher C_{trough} target can be set if a patient experiences a loss of clinical remission for at least three consecutive weeks, as indicated by weekly-reported PRO2 scores, or a loss of biological remission observed at the infusion visit. The algorithm for C_{trough} target escalation is illustrated in **Appendix E Figure S1**. The infliximab dose during the trial could go up to a maximum of 1800 mg, with dose banding in step sizes of 50 mg. The dosing interval for infliximab could be extended to at least 8 weeks and maximally 12 weeks.

The MIPD software tool considered the patient's most recent infliximab C_{trough} and (if available) an intermediate concentration (maximum *a posteriori Bayesian forecasting*), as well as IBD type (CD/UC), the most recent serum albumin level, fecal calprotectin level, absence/presence of ATI, bodyweight, and the prior infliximab dose (*a priori* prediction). Utilizing this data, the model selection algorithm of the MIPD software automatically chose the model with the highest maximum likelihood estimate. By specifying the desired infusion interval, the software predicted an individualized infliximab dose with at least an 80% probability of attaining the C_{trough} target for the specified infusion interval.

2.4. Outcomes and follow-up

The primary endpoint was the proportion of patients achieving sustained steroid-free, combined clinical and biological remission over one year after infliximab dosage de-escalation. Steroid-free indicates the absence of any dose of any oral or rectal steroid use over one year. The definitions of clinical and biological remission are provided above.

Secondary endpoints include the proportion of sustained steroid-free combined clinical and biological remission, sustained steroid-free clinical remission, and sustained steroid-free biological remission from T0 at each study visits. To avoid over-testing in the MIPD arm and ensure a fair comparison with the TDM arm for treatment outcomes, only PRO2 assessments at each infusion visit were used to evaluate clinical remission. Weekly PRO2 recordings between infusions were solely used for the evaluation of infliximab C_{trough} target escalation.

Additional secondary endpoints are the average infliximab C_{trough} , the total infliximab dose over the course of one year, and the proportion of C_{trough} target attainment at each study visit.

In the TDM arm, C_{trough} target attainment was defined as an observed infliximab C_{trough} of 3–7 mg/L. In the MIPD arm, C_{trough} target attainment was defined as an observed infliximab C_{trough} within the [100%–120% C_{trough} target] interval.

At each study visit, the treating physician evaluated data for adverse events (AEs), serious AEs (SAEs), and concomitant medication of immunomodulators or corticosteroids. For patients in the prospective MIPD arm, health-related quality of life and indirect cost (calculated based on cost related to ability to work, sick leave, and productivity) were measured before (at screening) and after dosage de-escalation (at T0+6) by using the 32-Item Inflammatory Bowel Disease Questionnaire (IBDQ-32) and the Institute for Medical Technology Assessment Productivity Cost Questionnaire (iPCQ), respectively.^{233,234}

2.5. Sample size calculation

This was an exploratory study due to the lack of published data on which to base assumptions. We performed a sample size calculation targeting 80% power at a 5% significance level. This calculation was based on the assumption that 37% of patients in the historical TDM control arm and 72% in the prospective MIPD intervention arm would achieve the primary endpoint over one year following dosage de-escalation. We estimated a requirement of 60 patients, with 30 in each study arm (1:1 ratio).

2.6. Statistical analysis

Data exploration, data visualization, and statistical analyses were performed in R (v4.2.0; R Foundation for Statistical Computing, R Core Team, Vienna, Austria) within the RStudio integrated development environment (version 2022.02.3; RStudio, Inc., Boston, MA, USA).

Subjects who withdrew early due to safety concerns, failure of infliximab therapy, and withdrawal of informed consent were categorized as failures. Missing values of CRP, fecal calprotectin and PRO2 were imputed using the last observation carried forward method.

The chi-squared test was used for testing the null hypothesis that proportions in both study arms were the same (two-sided alternative hypothesis). Unless stated otherwise, data are expressed as median [IQR]. For the survival or time-to-event analysis, a Kaplan-Meier curve was constructed. Wilcoxon Test was used to calculate p values of unpaired two-samples. All statistical tests were 2-sided, with p values of .05 or less denoting statistical significance.

All authors, with the exception of SGW, had access to the study data. All authors reviewed and approved the final manuscript.

3. Results

3.1 Patient population

A total of 52 patients underwent infliximab dosage de-escalation following earlier dose escalation (21 in the TDM arm (11 CD and 10 UC) and 31 in the MIPD arm (19 CD and 12 UC). At screening, median CRP was 0.9 [IQR 0.6–1.7] mg/L, median FC was 90.0 [IQR 32.1–140.8] mg/kg, and the median infliximab C_{trough} was 8.8 [IQR 6.2–10.8] mg/L in the MIPD arm. Patient disposition of the screening, inclusion, completion, and discontinuation in the MIPD arm is presented in **Figure 26**. Baseline characteristics of the included patients are presented in **Table 13**.

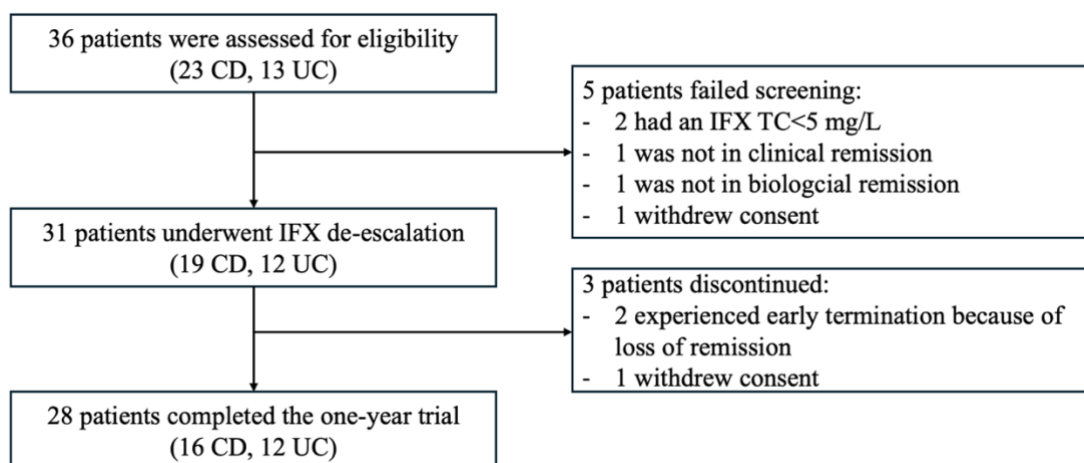


Figure 26. Patient disposition of the screening, inclusion, completion, and discontinuation in the prospective model-informed precision dosing arm. CD, Crohn’s disease; IFX, infliximab; TC, trough concentration; UC, ulcerative colitis.

Table 13. Patient characteristics at baseline

Characteristic	TDM arm (n=21)	MIPD arm (n=31)
Sex, female, n (%)	7 (33)	15 (48)
Age, years, median [IQR]	47 [36–56]	34 [23–47]
Bodyweight, kg, median [IQR]	85 [70–95]	71 [65–87]
Diagnosis, CD, n (%)	11 (52)	19 (61)
Disease duration, years, median [IQR]	13 [10–17]	11 [7–19]
Previous IBD surgery, yes, n (%)	7 (33)	4 (13)
Prior use of biologics other than infliximab, yes, n (%)	4 (19)	8 (26)
Serum albumin, g/L, median [IQR]	44.0 [42.6–45.0]	44.5 [42.4–46.1]
C-reactive protein, mg/L, median [IQR]	1.1 [0.7–2.1]	0.9 [0.6–1.4]
PRO2 of UC patients, 0:1, n (%)	10 : 0 (100:0)	11 : 1 (91:9)
PRO2 of CD patients, median [IQR]	2 [0–6]	2 [0–7]
Infliximab dose before dosage de-escalation, n		
20 mg/ kg Q4W	1	0
7.5 mg/kg Q4W : Q6W : Q7W	1 : 1 : 1	2 : 10 : 2
5.0 mg/kg Q4W : Q6W : Q7W	3 : 8 : 6	0 : 12 : 5
Infliximab trough concentration, mg/L, median [IQR]	5.7 [4.3–9.1]	9.2 [6.5–14.0]

Abbreviations: CD, Crohn's disease; IBD, inflammatory bowel disease; IQR, interquartile range; n, number; NA, data not available (If NA not mentioned in the table, there were no missing data); PRO2, two-items patient-reported outcome; UC, ulcerative colitis.

3.2 Infliximab dosing

In the TDM arm, 76% of patients (16/21) were on Q6W or Q7W escalated infliximab dosing at screening, with another 24% (5/21) on a Q4W schedule (**Figure 27**). All patients began de-escalation at T0, with 67% (14/21) extending their interval to Q8W or longer.

In the MIPD arm, 87% of patients (27/31) were on a Q6W or Q7W dosing schedule at screening. Most patients had their intervals extended to Q8W, with one patient extended to Q9W.

The standardized 8-weekly infliximab dose per kilogram bodyweight administered at screening did not differ significantly between the TDM arm (6.7 [IQR 5.9–8.2] mg/kg) and the MIPD arm (6.8 [IQR 6.7–10.0] mg/kg). Similarly, the average infliximab dose per infusion over the course of one year was comparable between the TDM arm (7.5 [IQR 7.4–8.2] mg/kg) and the MIPD arm (8.8 [IQR 6.6–10.9] mg/kg). Furthermore, the average dosing interval over one

year was comparable between the TDM arm (8.0 [IQR 7.8–8.0] weeks) and the MIPD arm (8.0 [IQR 7.9–8.0] weeks), as shown in **Appendix E Figure S2**.

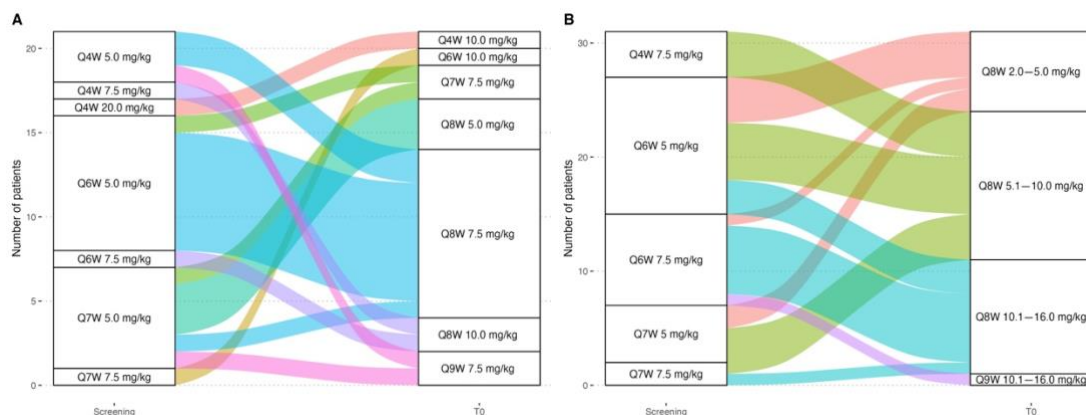


Figure 27. The transition from escalated infliximab dosing regimens at screening to de-escalated infliximab dosing regimens at baseline (T0) in the historical control TDM arm (A) and prospective MIPD arm (B). MIPD, model-informed precision dosing; QxW, every x weeks; TDM, therapeutic drug monitoring.

Overall, 29% of patients (n=9) had their C_{trough} target increased once, and 13% (n=4) had their C_{trough} target increased twice due to treatment relapse, defined as the loss of clinical or biological remission. Among these patients, one withdrew consent following a single relapse. Additionally, two patients were discontinued from the study because of persistent relapse even after setting a higher C_{trough} target twice.

3.3 Primary endpoint

The proportion of patients achieving the primary endpoint – sustained steroid-free combined clinical and biological remission over one year – was numerically higher in the MIPD arm (52%, 16 out of 31 patients) compared to the TDM arm (29%, 6 out of 21 patients) ($p=0.17$; **Figure 28A**). Among patients with UC, the rate of achieving the primary endpoint in the MIPD arm was threefold higher than in the TDM arm (67%, 8 out of 12 patients, vs. 20%, 2 out of 10 patients, $p=0.08$), as depicted in Figure 28C. However, no statistically significant differences in primary endpoint attainment were observed between the two treatment arms when considering all patients collectively or those with CD or UC, as shown in **Figure 28**.

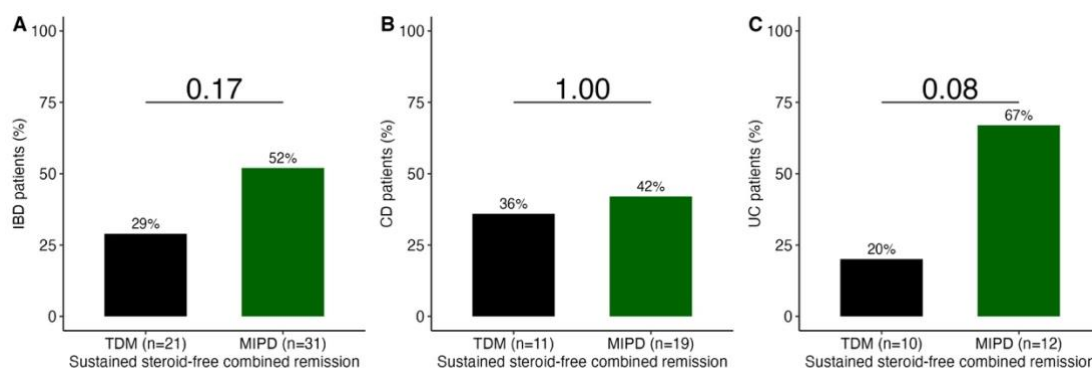


Figure 28. A comparison of sustained steroid-free combined clinical and biological remission over one year between the therapeutic drug monitoring (TDM) and model-informed precision (MIPD) arm. Two-sided *p* values were reported. CD, Crohn's disease; IBD, inflammatory bowel disease; UC, ulcerative colitis.

3.4 Secondary endpoints and exploratory analysis

3.4.1 Sustained remission

The proportion of patients in sustained steroid-free combined clinical and biological remission at each study visits was not different between the TDM and MIPD arms (**Figure 29**) and did not differ between CD and UC (**Appendix E Figure S3**).

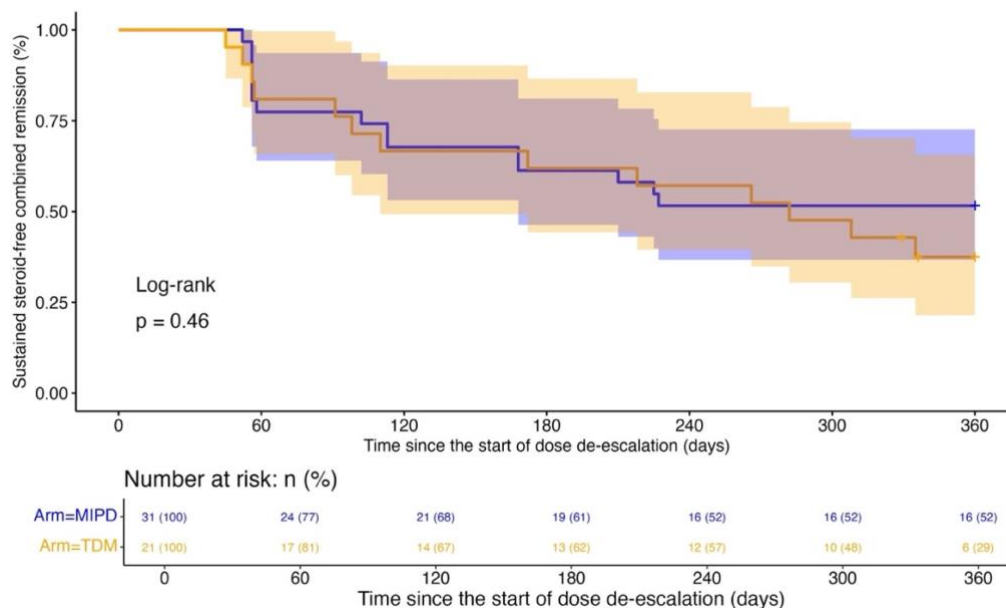


Figure 29. Kaplan-Meier curve and the number of patients at risk representing the survival of sustained steroid-free combined clinical and biological remission during one year after the start of infliximab dosage de-escalation between the MIPD and TDM arm. MIPD, model-informed precision dosing; TDM, therapeutic drug monitoring. The solid lines are the survival curves. The shaded areas are 95% confidence intervals around the survival curves.

The proportion of patients achieving sustained steroid-free clinical remission *over* one year was significantly higher in the MIPD arm compared to the TDM arm, both in the total patient population (81%, 25 out of 31 patients, vs. 38%, 8 out of 21 patients) and among the subpopulation of UC patients (100%, 12 out of 12 patients, vs. 30%, 3 out of 10 patients), as demonstrated in **Figure 30A&C** ($p<0.01$). However, the proportion of patients achieving sustained steroid-free biological remission at one year did not statistically differ between the MIPD and TDM arms, both in the total population and in the subpopulation analyses, as illustrated in **Figure 30D–F**. Similarly, survival analysis showed a statistically higher rate of sustained steroid-free clinical remission in the MIPD arm compared to the TDM arm, though no such difference was observed for biological remission, as depicted in **Appendix E Figures S4 and S5**.

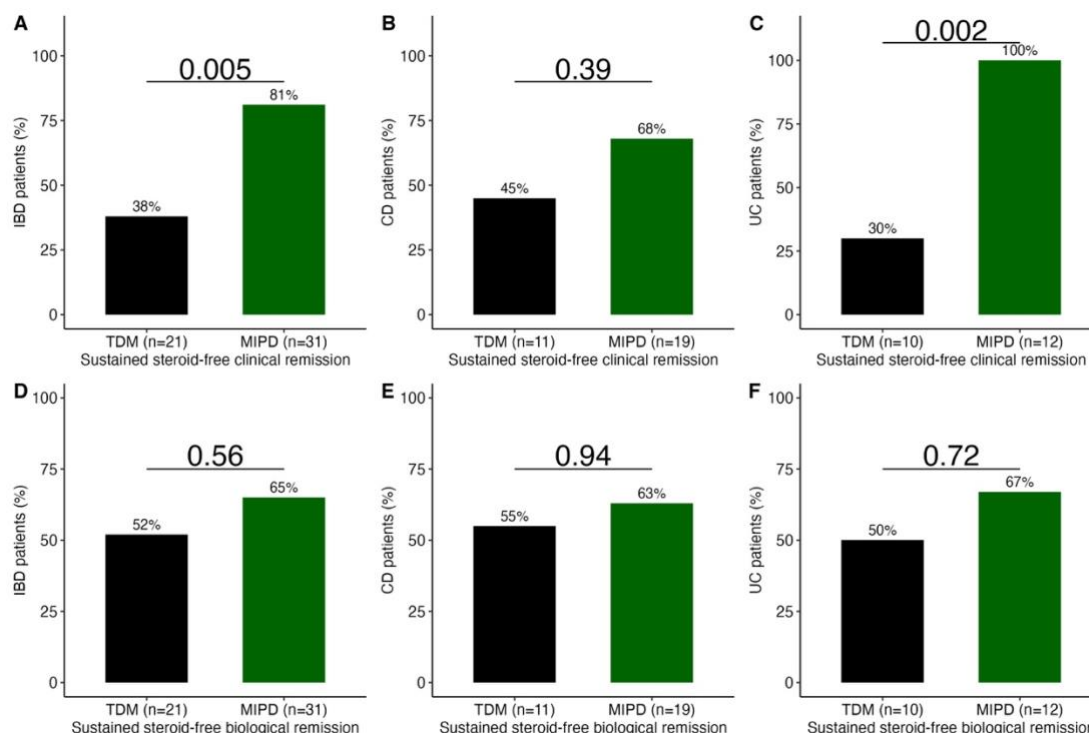


Figure 30. A comparison of sustained steroid-free clinical remission (A–C) and biological remission (D–F) over one year between the therapeutic drug monitoring (TDM) and model-informed precision (MIPD) arm. Two-sided p values were reported. CD, Crohn’s disease; IBD, inflammatory bowel disease; UC, ulcerative colitis.

The time course of CRP levels was similar between the TDM arm and the MIPD arm, with an average of 7% and 10% of patients, respectively, recording CRP levels above 5 mg/L at each infusion visit (**Figure 31A**).

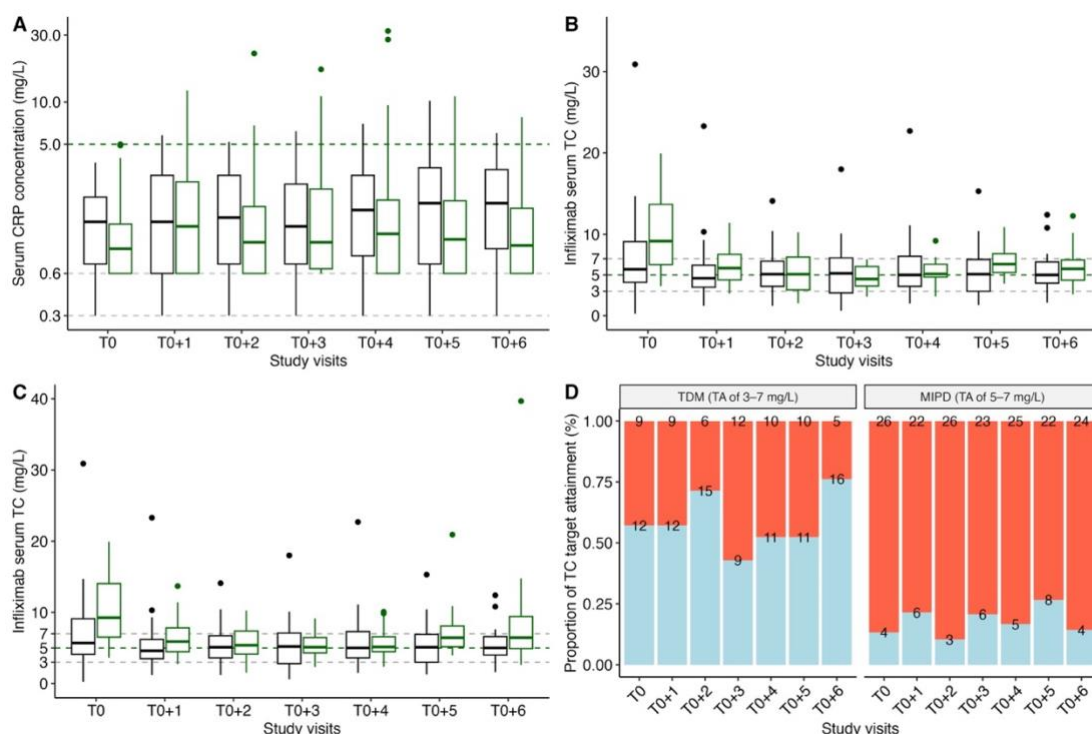


Figure 31. The time course of serum CRP (**A**), infliximab TC in patients targeting 5 mg/L (**B**), infliximab TC across all patients (**C**), and the proportion of infliximab TC target attainment (**D**) over study visits. In Panels A, B, and C, black and green boxes depict data from the TDM and MIPD arm, respectively. In Panel D, blue and orange bars represent the proportion of patients who did or did not achieve the TC target in the two treatment arms, respectively. Black numbers above each bar in Panel D indicate the number of patients who achieved or did not achieve the target. In the TDM arm, target attainment was defined as an observed infliximab TC of 3–7 mg/L. In the MIPD arm, target attainment was defined as an observed infliximab TC within the [100%–120% TC target] interval. CRP, C-reactive protein; MIPD, model-informed precision dosing; T0, baseline; T0+x, the x visits after baseline; TA, target attainment; TC, trough concentration; TDM, therapeutic drug monitoring.

3.4.2 Pharmacokinetics analysis

The median infliximab C_{trough} at baseline was significantly higher in the MIPD arm (9.2 [IQR 6.5–14.0] mg/L) than in the TDM arm (5.7 [IQR 4.3–9.1] mg/L) ($p < 0.01$). Following a single model-informed dose at baseline, the infliximab C_{trough} immediately decreased to a median of 5.9 [IQR 4.5–7.8] mg/L with 68% of patients >5 mg/L (**Figure 31B&C**). Furthermore, MIPD resulted in reduced variability in infliximab C_{trough} among patients at nearly all study visits. A higher proportion of target attainment was observed in the TDM arm, although the

proportions of target attainment are not directly comparable between both study arms (**Figure 31D**). None of the patients exhibited undetectable infliximab concentrations; consequently, measurements of ATI were not performed during the trial.

3.4.3 Safety, cost, and quality of life

None of the patients in the MIPD arm received co-medication with immunomodulators or corticosteroids during the trial. In the TDM arm, one patient received immunomodulators.

In the MIPD arm, one patient experienced two SAEs (severe abdominal pain at T0+2, which resolved after a three-day hospitalization, and appendicitis at the study discharge visit T0+6). Two other patients in the MIPD arm experienced one SAE (one patient had a COVID-19 infection and another patient experienced palpitations).

Total infliximab consumption over one year showed no significant differences between the two study arms, amounting to 3,732 [3,277–4,678] mg in the TDM arm and 3,700 [2,650–4,800] mg in the MIPD arm. The incidence of productivity losses in the MIPD arm was similar before and after dosage de-escalation (**Appendix E Table S1**). Quality of life remained stable, as indicated by similar IBDQ-32 scores between Screening and T0+6 (**Appendix E Figure S6**).

4. Discussion

In this historically controlled trial, infliximab dosage de-escalation guided by MIPD targeting a C_{trough} of 5.0 mg/L did not demonstrate statistical superiority over dosage de-escalation guided by TDM targeting a C_{trough} interval of 3–7 mg/L in maintaining sustained steroid-free combined clinical and biological remission in patients with CD and UC. However, the MIPD arm did show a numerically higher proportion of patients achieving sustained combined clinical and biological remission over one year, with a significantly higher proportion maintaining sustained steroid-free clinical remission over one year. This is the first trial to prospectively evaluate MIPD for guiding infliximab dosage de-escalation, as well as the first to directly compare the efficacy of MIPD and TDM in patients with IBD.

Prospective TDM studies have largely failed to demonstrate the efficacy of TDM in improving treatment outcomes for patients with IBD, thus hindering its widespread adoption in clinical practice.^{5,209} However, the success of the recent PRECISION and NOR-DRUM B trials has reignited interest in this area by supporting the benefits of proactive TDM and MIPD.^{89,98} Many ongoing prospective trials like TITRATE (Induction For Acute Ulcerative Colitis; NCT03937609), REMODEL-CD (Precise Infliximab Exposure and Pharmacodynamic Control; NCT05660746), and OPTIMIZE (Proactive Infliximab Optimization Using a PK Dashboard versus Standard of Care in Patients With CD; NCT04835506) are expected to further validate the benefits of MIPD in infliximab therapy.³ Nevertheless, these ongoing studies primarily compared TDM or MIPD with standard dosing, but did not explore direct comparisons between TDM and MIPD in achieving superior treatment outcomes in patients with IBD. This gap is significant, as transitioning from TDM to MIPD involves additional efforts, making MIPD appealing only if it demonstrates superior performance compared to TDM. On the other hand, TDM has become an umbrella term that covers the use of a diversity of assays, concentration targets, and dose optimization algorithms, while MIPD only aims to improve the efficiency and accuracy of dose optimization. Therefore, it would not be surprising that the superiority of MIPD over TDM is difficult to confirm. The main advantage of MIPD may eventually lie in the standardization of dose optimization practices.

In our study, the proportion of patients achieving the primary endpoint was substantially higher in the MIPD arm than in the TDM arm (52% vs. 29% in the total population and 67%

vs. 20% in patients with UC). However, these differences were not statistically significant, likely resulted from an insufficient sample size. A sample size calculation based on observed rates – which are considered clinically relevant – suggests that 70 patients per arm would be necessary to detect a statistically significant difference in the primary endpoint between the TDM and MIPD in a future, confirmatory trial. For UC patients, the required sample size could be reduced to 16 per arm to observe a statistical difference. Yet, we did observe a statistically significant improvement in sustained steroid-free clinical remission *over one year* in the MIPD arm compared to the TDM arm, with remission rates of 81% vs. 38% in the total population and 100% vs. 30% in patients with UC. This aligns with previous findings that an exposure-response relationship was observed only in patients with UC, not CD, who underwent infliximab dosage de-escalation after regaining response on escalated infliximab therapy.²³⁵

Approximately 74% of patients required an infliximab dose higher than 5 mg/kg Q8W at baseline. Furthermore, more than one-third of patients needed a baseline dose of 10 mg/kg or more. The high doses required to attain the C_{trough} target of 5 mg/L in these patients suggest that those who previously experienced a loss of response may exhibit more rapid drug elimination, indicative of higher infliximab clearance. In both adult and pediatric patients with IBD, increased clearance of infliximab, adalimumab, and ustekinumab has been associated with poorer clinical outcomes.^{236–239} Monitoring monoclonal antibody clearance has also been proposed for the early detection of asymptomatic disease flares and undetectable ATIs in patients with IBD.² Clearance monitoring may be a better tool in guiding dosage de-escalation in these "unstable" patients who have previously experienced a loss of response.

Overall, 42% of patients experienced a total of 17 C_{trough} target escalations, with 59% (n=10/17) of these escalations triggered by an increase in CRP levels to a value between 5 mg/L and 10 mg/L. Additionally, we observed more frequent CRP elevations during the flu season. These findings suggest that a CRP threshold of 5 mg/L may be too stringent for defining biological remission, particularly when CRP-based remission is used to determine target escalation rules. This stringency directly led to the early termination of one patient who was in stable clinical remission but had to discontinue the trial due to two consecutive CRP values of 7.5 mg/L and 6.8 mg/L, as per the dose escalation rules (see **Appendix E Figure S1**).

A low rate of C_{trough} target attainment was observed in the MIPD arm when strictly applying the target attainment definition according to the protocol. However, the definition of target attainment in the MIPD arm was more stringent than in the TDM arm, making direct comparisons between the two arms difficult. By filtering only patients targeting a C_{trough} of 5 mg/L and allowing a $\pm 40\%$ flexibility in C_{trough} attainment (corresponding to a target C_{trough} interval of 3–7 mg/L), the MIPD arm showed similar target attainment rates to the TDM arm.

This study has several limitations. First, the common drawback of using historical controls was the potential cause for unfair comparisons, as the two arms might not be similar.²⁴⁰ Second, the limited sample sizes and imbalance between both arms rendered the study underpowered to detect statistically significant differences in endpoints. Third, the historical control TDM arm was not directly comparable to the MIPD arm. The median infliximab C_{trough} at baseline was significantly higher in the MIPD arm, whereas patients in the TDM arm started with a baseline C_{trough} close to 5 mg/L. Additionally, patients in the TDM arm were mostly established on an escalated infliximab dose for an extended period and were presumably in more stable remission before dosage de-escalation. In contrast, patients in the MIPD arm underwent dosage de-escalation immediately upon achieving remission. Furthermore, PRO2-based clinical remission was assessed weekly in the MIPD arm, while it was only evaluated at each infusion visit in the TDM arm, potentially leading to a relative overestimation of treatment relapse in the MIPD arm.

In conclusion, this is the first study to prospectively evaluate and compare MIPD and TDM in guiding infliximab dosage de-escalation in patients who regained response on escalated infliximab maintenance therapy. The results from this study indicate that MIPD achieved a numerically higher rate of sustained steroid-free clinical and biological remission and statistically significant higher rates of sustained steroid-free clinical remission compared to TDM. These findings contribute to the comparison of TDM and MIPD and enhance our understanding of their roles in guiding the dosage de-escalation of infliximab. Future confirmatory studies with randomization and sufficient sample size are needed to validate our findings. Additionally, clearance monitoring may offer further insights into managing "unstable" patients who have previously experienced a loss of response.

Appendix E

Supplementary methods

Evaluation of predictive performance

The model-predicted versus measured infliximab C_{trough} were compared by calculating the relative bias (rBias, **Equation S1**) and the relative root mean square error (rRMSE, **Equation S2**) to determine accuracy and precision, respectively.

$$rBias = \frac{1}{n} \times \sum_1^i \left(\frac{TC_{predicted,i} - TC_{measured,i}}{TC_{measured,i}} \right) \times 100\% \quad \text{Equation S1}$$

$$rRMSE = \sqrt{\frac{1}{n} \times \sum_1^i \left(\frac{(TC_{predicted,i} - TC_{measured,i})^2}{TC_{measured,i}^2} \right)} \times 100\% \quad \text{Equation S2}$$

with n representing the total number of patients, and i for each patient. A rBias between $\pm 20\%$ with a 95% confidence interval including zero was deemed clinically acceptable.^{84,241} No rRMSE threshold for clinical acceptability was prespecified. Lower rRMSE values indicated more precise predictions.

The Bland–Altman plot was used to assess the agreement between the predicted and measured C_{trough} across the range of measured C_{trough} . The predicted and measured C_{trough} were classified at the prespecified target C_{trough} , with classification accuracy calculated as in **Equation S3**.

$$Classification\ accuracy = \frac{TN + TP}{n} \times 100\% \quad \text{Equation S3}$$

with TN and TP representing the numbers of true negative (predicted and measured $C_{trough} \leq$ target C_{trough}) and true positive (predicted and measured $C_{trough} \geq$ target C_{trough}) predictions, respectively, and n representing the total number of predictions.

Supplementary results

Prediction performance of MIPD

MIPD showed good predictive accuracy for predicting infliximab C_{trough} , with a median rBias of -4% [-17–16%] and 0 [-15–24%] in patients with CD and UC, respectively (**Figure S7A**). One patient with UC had a rBias of 228%, due to the discrepancy between the predicted and measured C_{trough} of 5.0 mg/L and 1.5 mg/L, respectively. MIPD prediction also had good precision, indicated by the acceptable rRMSE of 28% in patients with CD and 40% in patients with UC (**Figure S7B**). After excluding the patient with a rBias of 228%, the rRMSE in patients with UC reduced to 30%.

MIPD resulted in a high classification accuracy of 77% at the pre-defined C_{trough} target, with accuracies of 69% in patients with UC and 82% in patients with CD (**Figure S7C**). The Bland–Altman analysis reveals a mean bias of -0.26% with a 95% CI ranging from -0.60 to +0.08. Overall, the Bland–Altman plot shows no trend in the agreement of the two predicted and measured C_{trough} , with mean bias including 0 on a 5% significance level (**Figure S7D**).

Supplementary figures

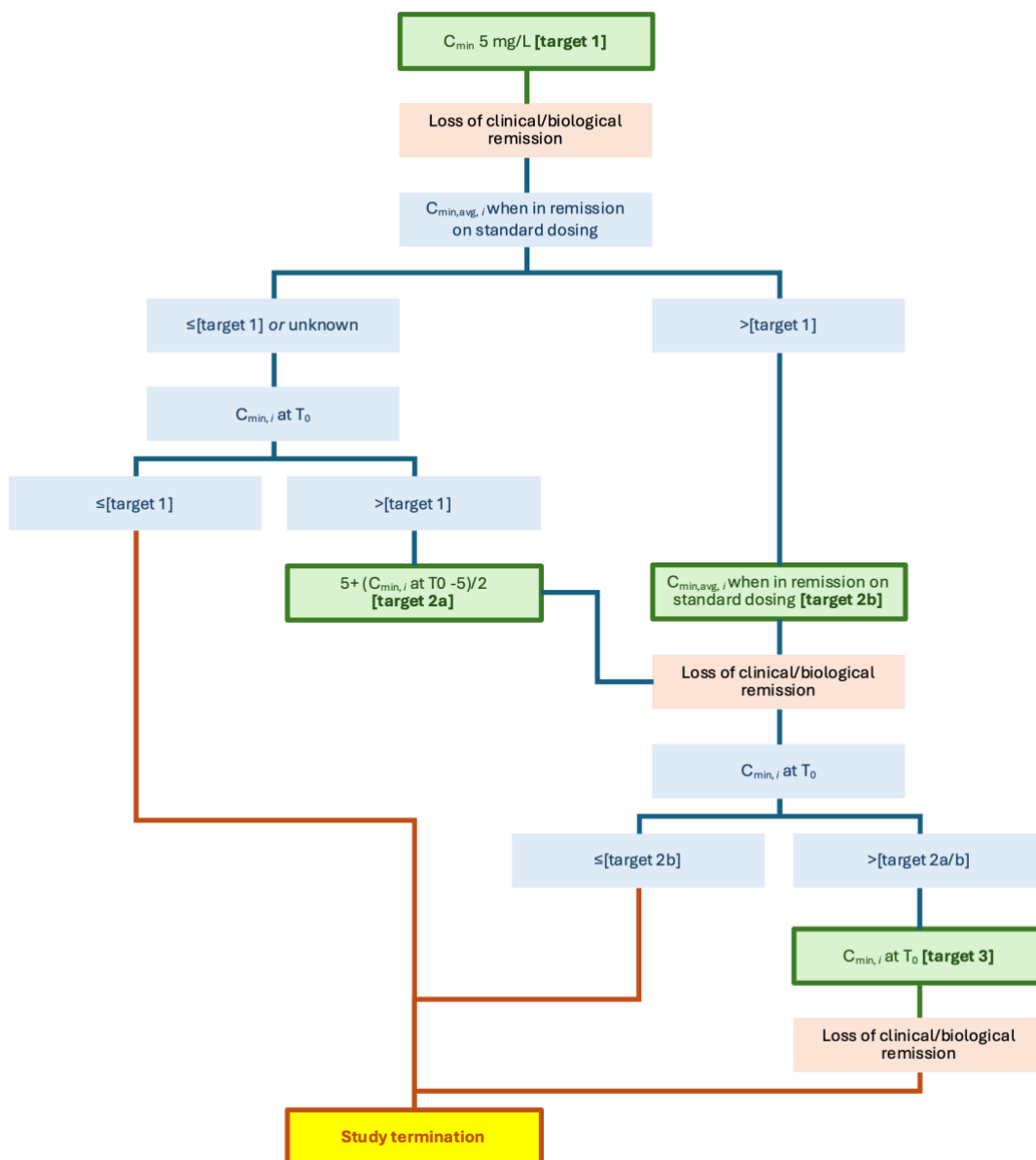


Figure S1. Trough concentration target used in the prospective intervention group. The 5 mg/L infliximab trough concentration (C_{min}) target (for all patients) is based on the recommendation by the American Gastroenterological Association Institute.⁶⁵ Loss of clinical response is defined based on patient-reported outcomes (rectal bleeding score > 0 + stool frequency score >1 [ulcerative colitis], mean daily abdominal pain score >1 + liquid stool frequency score >1.5 [Crohn's disease]). Loss of biological remission is defined as C-reactive protein ≥ 5 mg/L and fecal calprotectin ≥ 250 mg/kg. $C_{min(,avg),i}$ (average) infliximab trough concentration of patient i ; T_0 : time of study inclusion (baseline, day 0).

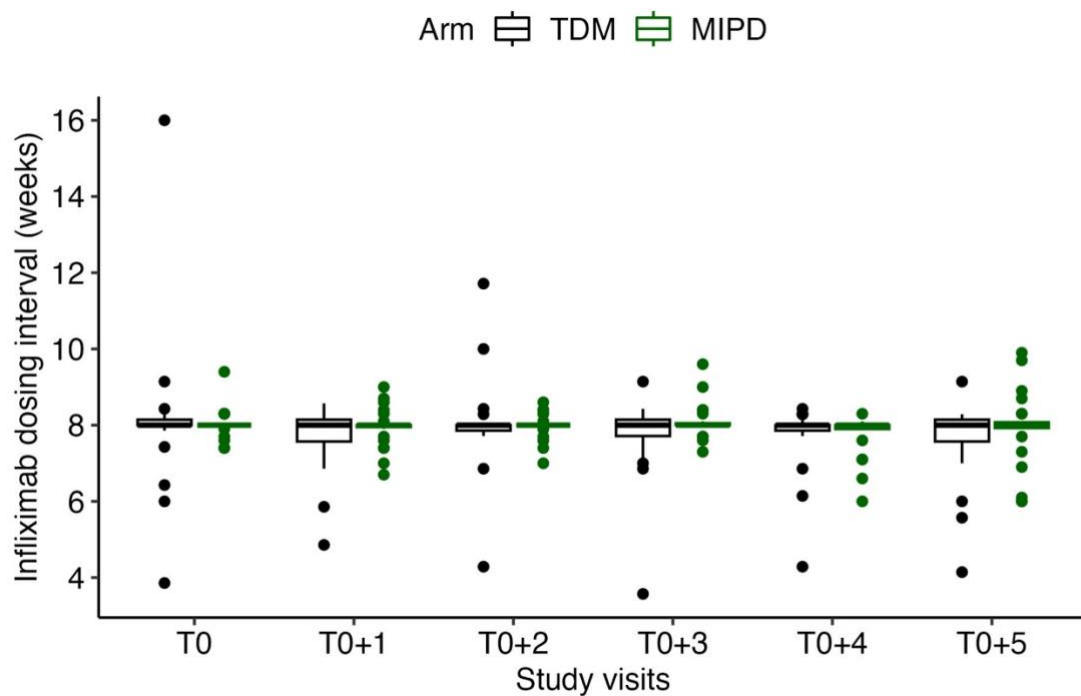


Figure S2. Comparison of infliximab dosing intervals as each study visit between the TDM and the MIPD arm. MIPD, model-informed precision dosing; T0, baseline; T0+x, the x visits after baseline; TDM, therapeutic drug monitoring.

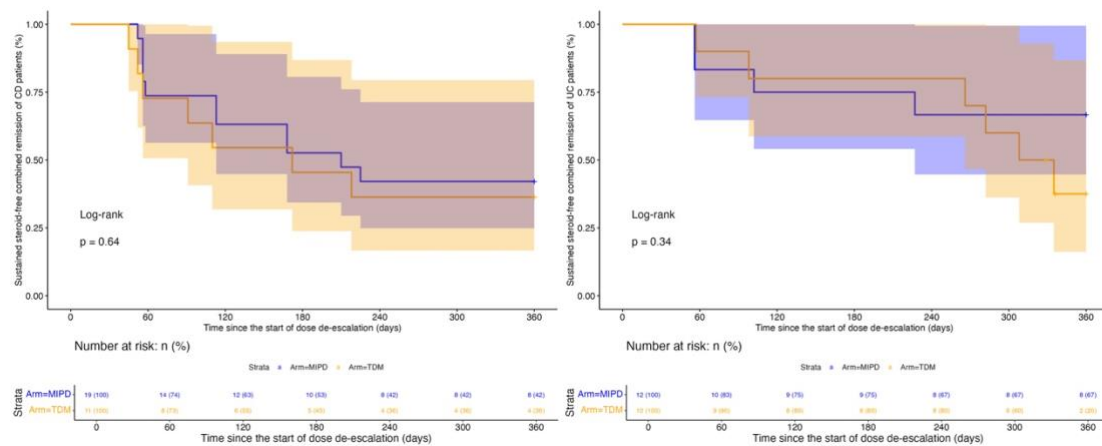


Figure S3. Kaplan-Meier curve and number of patients at risk representing survival of sustained steroid-free combined clinical and biological remission during one year after the start of infliximab dosage de-escalation between the MIPD and TDM arm in patients with CD (*left*) and UC (*right*). CD, Crohn's disease; MIPD, model-informed precision dosing; TDM, therapeutic drug monitoring; UC, ulcerative colitis. The solid lines are the survival curves. The shaded areas are confidence intervals around the survival curves. The risk tables below each plot show the number of patients at risk at different time points throughout the study period.

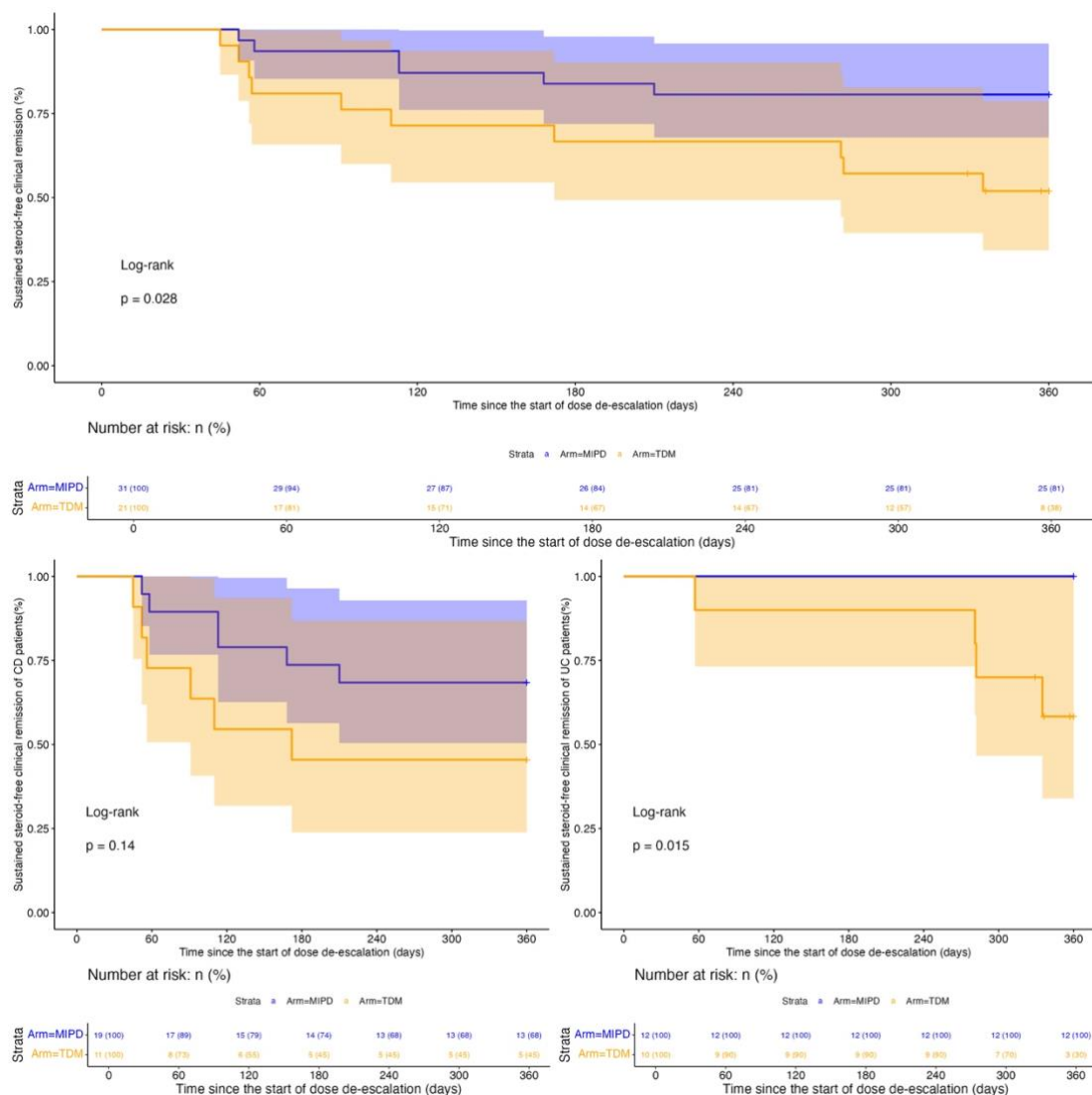


Figure S4. Kaplan-Meier curve and number of patients at risk representing the survival of sustained steroid-free clinical remission during one year after the start of infliximab dosage de-escalation between the MIPD and TDM arm in total patients with IBD (**top**), CD (**bottom left**) and UC (**bottom right**). CD, Crohn's disease; IBD, inflammatory bowel diseases; MIPD, model-informed precision dosing; TDM, therapeutic drug monitoring; UC, ulcerative colitis. The solid lines are the survival curves. The shaded areas are confidence intervals around the survival curves. The risk tables below each plot show the number of patients at risk at different time points throughout the study period.

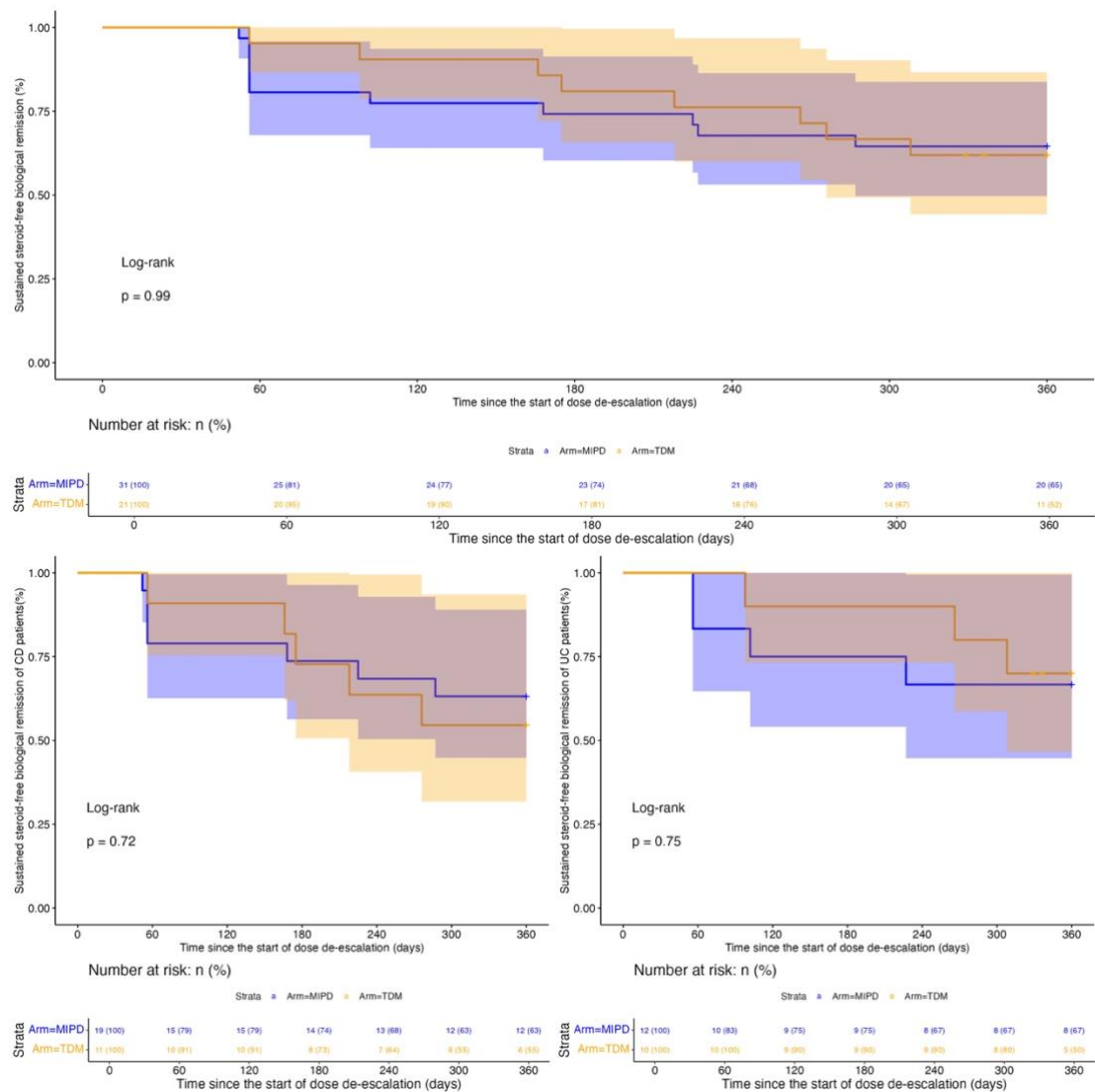


Figure S5. Kaplan-Meier curve and number of patients at risk representing the survival of sustained steroid-free biological remission during one year after the start of infliximab dosage de-escalation between the MIPD and TDM arm in total patients with IBD (**top**), CD (**bottom left**) and UC (**bottom right**). CD, Crohn's disease; IBD, inflammatory bowel diseases; MIPD, model-informed precision dosing; TDM, therapeutic drug monitoring; UC, ulcerative colitis. The solid lines are the survival curves. The shaded areas are confidence intervals around the survival curves. The risk tables below each plot show the number of patients at risk at different time points throughout the study period.

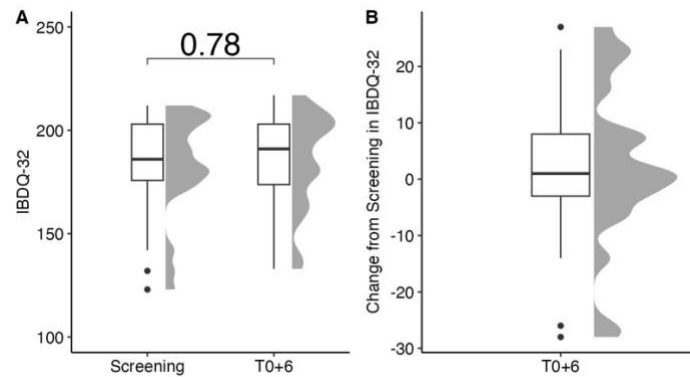


Figure S6. The Inflammatory Bowel Disease Questionnaire (IBDQ-32) score distribution and change in IBDQ-32 score from screening. The shaded area adjacent to each box plot illustrates the density distribution of the data. Two-sided p value is reported.

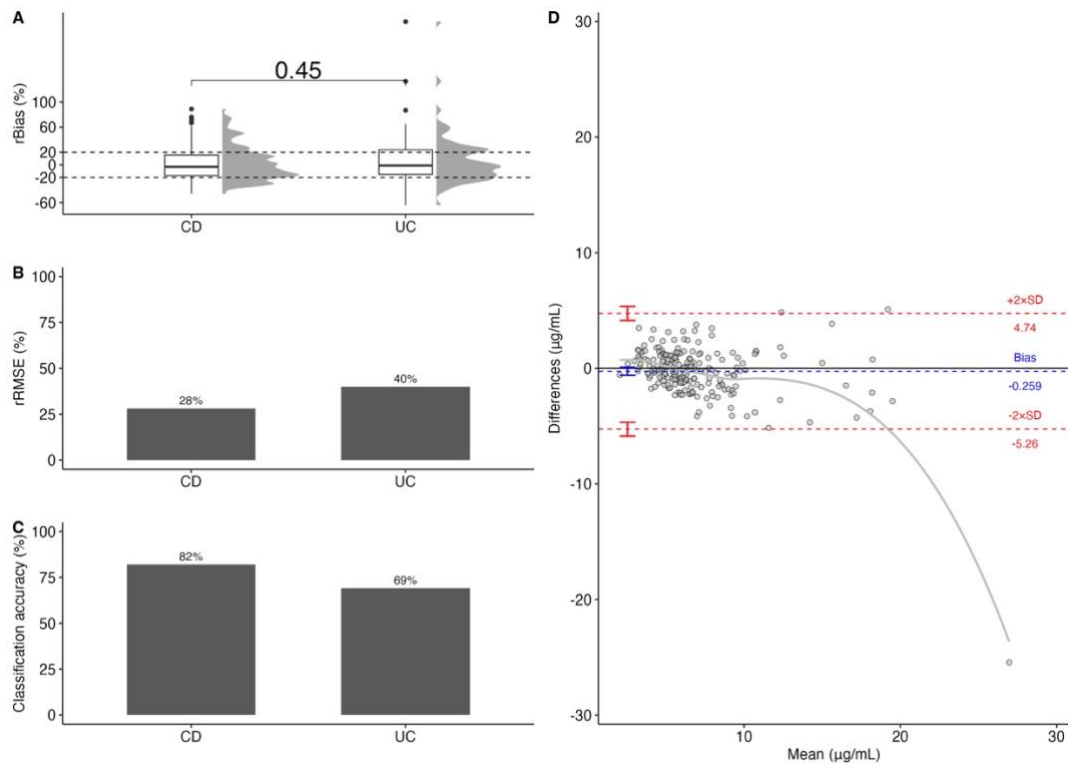


Figure S7. The rBias (A), rRMSE (B), classification accuracy (C) and the Bland–Altman plot (D) for assessing the prediction of infliximab trough concentrations. Panels A, B, and C are stratified based on patients with CD and UC. In Panel A, the shaded area adjacent to each box plot illustrates the density distribution of the data. In Panel D, the blue and red dashed line represents the mean including its standard deviation of difference between predicted and measured TC. The blue and red intervals represent the 95% confidence interval for the mean and the standard deviations. CD, Crohn’s disease; rBias, relative bias; rRMSE, relative root mean square error; UC, ulcerative colitis.

Supplementary Tables

Table S1. *The comparison of the Productivity Cost Questionnaire at screening and at study discharge*

Characteristic	At Screening (n=31)	At Discharge (n=28)
Female sex, n (%)	15 (48%)	12 (55%)
Age, years, median [IQR]	35.1 [23.8–47.2]	35.0 [25.7–47.7]
Level of education, n (%)		
- Low	3 (10%)	3 (11%)
- Secondary	13 (42%)	11 (39%)
- University or above	8 (26%)	6 (27%)
- Missing	7 (22%)	6 (21%)
With a paying job, n (%)	23 (74%)	18 (64%)
Experience of productivity losses during the past month, n (%)	4 (13%)	6 (21%)

IQR, interquartile range; n, number of patients.

General discussion and perspectives

Parts of the discussion are based on

Wang Z, Dreesen E. Therapeutic drug monitoring of anti-tumor necrosis factor agents: lessons learned and remaining issues. *Curr Opin Pharmacol*. 2020;55:53–9.

Kantasiripitak W*, **Wang Z***, Spriet I, Ferrante M, Dreesen E. Recent advances in clearance monitoring of monoclonal antibodies in patients with inflammatory bowel diseases. *Expert Rev Clin Pharmacol*. 2021;14(12):1455-1466.

Alsoud D*, Moes D*, **Wang Z***, Soenen R, Layegh Z, Barclay M, Mizuno T, Minichmayr I, Keizer R, Wicha S, Wolbink G, Lambert J, Vermeire S, De Vries A, Papamichael K, Padullés-Zamora N, Dreesen E. Best Practice for Therapeutic Drug Monitoring of Infliximab: Position Statement from the International Association of Therapeutic Drug Monitoring and Clinical Toxicology. *Ther Drug Monit*. 2024;46(3):291-308.

Wang Z, Dreesen E. Therapeutic Drug Monitoring 2.0: Time to Assemble the Puzzle. *Inflamm Bowel Dis*. 2024;30 (2):316-317.

* Designated shared co-first authorship

In clinical drug development, “one dose fits all” is a key objective. However, variability in drug exposure and response often results in suboptimal treatment of individual patients. The dictum of “one dose fits all” has been shown problematic in patients with IBD starting mAbs therapies.

Over a decade ago, personalized medicine was introduced to infliximab treatment, offering tailored therapeutic recommendations through TDM.²⁴² TDM-guided dosing for therapeutic mAbs typically relies on flowcharts or decision trees, as seen in trials like TAXIT, TAILORIX, and NOR-DRUM.^{86–89} These approaches are intuitive and straightforward for clinical use. However, dosing recommendations based on such analogical methods lack precision and specificity, leading to a slow, trial-and-error process for dose optimization. Multiple dose adjustments are often required before achieving the desired C_{trough} target. For instance, the TAXIT algorithm, which adjusts dosing intervals in two-week increments and doses in 5 mg/kg steps, has proven inefficient in precisely reaching target concentrations.⁸⁶

In contrast, popPK models, also known as nonlinear mixed-effects models, offer a more sophisticated approach. These models can quantify PK variability and predict future drug exposures, allowing for more accurate dose adjustments. When supported by a popPK model and therefore inevitably assisted by a software tool, TDM evolves into MIPD. Despite the recognized potential of MIPD, significant efforts are still needed to validate its effectiveness and integrate it into clinical practice.²⁴³ Several key challenges need to be addressed in the areas of definition and terminology, understanding the value of MIPD, scientific aspects, practical implementation, and drug development and regulatory issues (**Figure 32**).²⁴⁴ In this thesis, we focused on tackling several scientific and practical challenges associated with MIPD. These include the development of popPK(-PD) models for MIPD, identifying PK and biomarker targets that correlate with therapeutic outcomes, translating scientific insights into user-friendly MIPD software tools, validating MIPD in prospective clinical trials, and providing guidelines for effective implementation of MIPD in clinical practice.

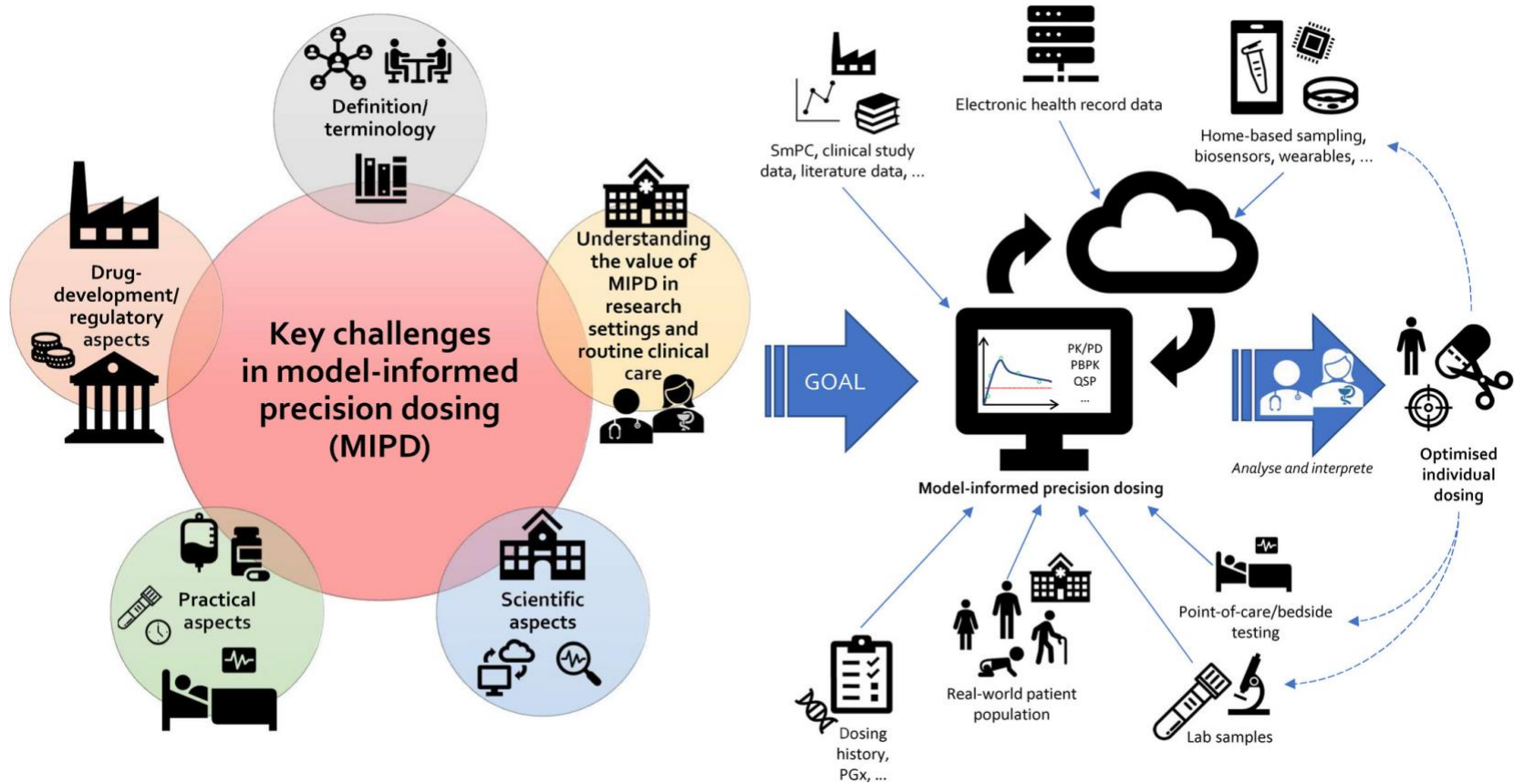


Figure 32. Key challenges and perspectives for model-informed precision dosing (MIPD). PBPK, physiologically-based pharmacokinetics; PD, pharmacodynamics; PGx, pharmacogenomics; PK, pharmacokinetics; QSP, quantitative systems pharmacology; SmPC, summary of product characteristics. Figure reproduced with permission from Kluwe et al.²⁴⁴

We identified several critical knowledge gaps that could hinder the implementation of MIPD for mAb treatment in patients with IBD. First, there is a lack of a popPK-PD model for ustekinumab in patients with CD. Second, there is insufficient guidance on TDM and MIPD for the recently approved SC formulations of infliximab and vedolizumab. Third, current predictors, such as C_{trough} , are inadequate for correlating with endoscopic outcomes in patients undergoing TDM. Lastly, a prospective study comparing the advantages of MIPD over traditional TDM is still needed.

In this doctoral research, we adhered to Sheiner's "learn and confirm" cycle (**Figure 33**).^{245,246} This approach begins with learning from available data, such as retrospective studies, and is followed by confirming these findings through prospective studies. We developed a popPK-PD model for ustekinumab in patients with CD, provided guidance for TDM and MIPD of SC infliximab and vedolizumab, identified mAb clearance as a superior predictor over C_{trough} for endoscopic outcomes, and prospectively evaluated the benefits of MIPD-guided dosing interval extension for infliximab compared to a TDM-guided approach. These contributions address three essential components of MIPD: the availability of popPK-PD models, the identification of PK-PD target, and the prospective validation of MIPD with a software tool in clinical trials.

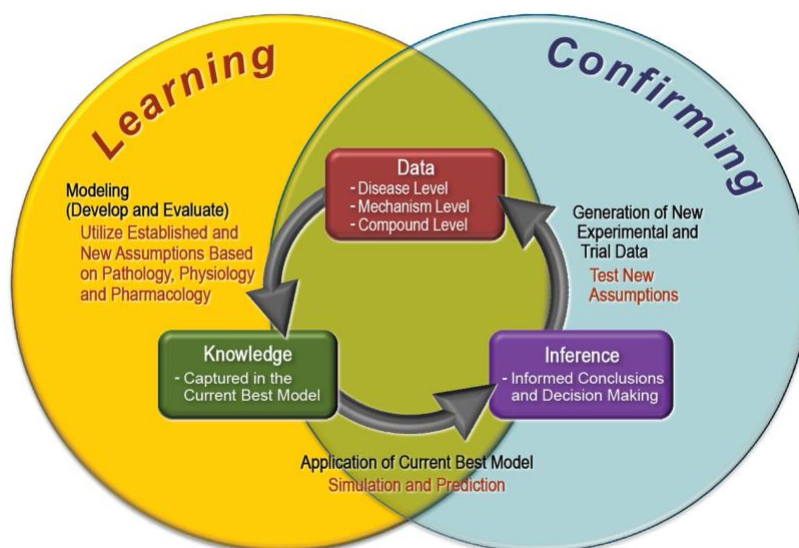


Figure 33. The "Learn and Confirm Cycle". The colored boxes represent essential components of the cycle. The arrows represent processes that link these components. This figure was reproduced with permission from EFPIA MID3 Workgroup et al.²⁴⁶

Dose optimization of ustekinumab

Pivotal trials for ustekinumab have revealed an exposure-response relationship, showing that higher serum levels of the drug are associated with clinical remission both after induction and during maintenance phases.^{201,247} Evidence indicates that a significant increase in FC levels after week 8 of induction is associated with suboptimal endoscopic outcomes at week 24.⁷⁴ Besides, there are several popPK models developed for ustekinumab in patients with IBD.^{95,248–250} A semi-mechanistic PK-PD model was reported for ustekinumab in patients with CD, described by a two-compartment TMDD model and an indirect response model of FC.²⁵⁰ Nevertheless, the quantitative relationship between ustekinumab exposure, FC levels, and endoscopic results in patients with CD remains to be defined. Employing a popPK-PD model could offer valuable insights into the factors influencing ustekinumab exposure and associated biomarker responses.

In the first chapter of this doctoral research, we developed a popPK-PD model consisting of three sequential components: a two-compartment popPK model that links the dose of ustekinumab to its exposure, an indirect response model that describes the inhibitory effect of ustekinumab exposure on the production rate constant of FC, and a logistic regression outcome model that correlates FC levels with endoscopic outcomes. The developed models can guide clinical trial design and support model-informed dose optimization—either stratified or individualized—to enhance endoscopic outcomes. Additionally, the developed popPK-PD model could potentially replace existing popPK models used for MIPD, facilitating integrated PK (*i.e.*, based on drug concentration) and biomarker (*i.e.*, based on FC) monitoring of ustekinumab in patients with IBD.

Over the past decade, the number of available biological therapies has doubled, introducing novel molecules like IL-23p19 inhibitors (risankizumab and mirikizumab). As the spectrum of treatment options widens with these new biologicals, there is a growing necessity for differentiating between these therapies. This highlights the importance of conducting head-to-head clinical trials. Notably, in recent head-to-head trials, ustekinumab showed similar efficacy to adalimumab in achieving clinical remission at one year but was less effective than risankizumab in achieving endoscopic remission at the same time point.^{251,252} There is also a

growing interest in dual biologic therapies. Dual biologic therapy of ustekinumab and vedolizumab appeared to be a promising and safe option for patients with refractory CD, particularly in those who demonstrate a partial response to monotherapy of either agent.²⁵³

Perspectives

In summary, results from prospective trials revealed that higher ustekinumab concentrations correlated with normalized CRP and FC levels, alongside endoscopic improvements.^{96,251} Based on popPK-PD modeling and simulation, we proposed an individualized, two-step monitoring approach, consisting of FC monitoring and, only when FC is/remains high, ustekinumab C_{trough} monitoring. Additionally, results from the ongoing RESCUE (Loss of Response to Ustekinumab Treated by Dose Escalation) study, which investigates ustekinumab dose intensification in patients with CD experiencing secondary loss of response, are expected to enhance our understanding of ustekinumab dose optimization.

Therapeutic drug monitoring in a new era of subcutaneous biologicals

In 2020, the EMA approved the SC formulation of infliximab (CT-P13) and vedolizumab for maintenance treatment of patients with IBD, following clinical trials that demonstrated comparable PK, efficacy, and safety to their IV formulation.^{92,185,186} Despite the superior adherence often associated with IV administration, many patients and clinicians prefer SC therapy because of its convenience.^{254–256} In addition, preliminary results suggest a potential for lower immunogenicity with the SC formulation.^{92,256} The introduction of SC formulations for infliximab and vedolizumab marks a significant milestone in the era of SC therapies for IBD, with nearly all mAbs used in IBD now available in SC form (except Natalizumab).

TDM practices in IBD have traditionally focused on trough sampling. However, the emergence of SC infliximab has challenged this belief. The European Public Assessment Report (EPAR) of SC infliximab indicates that dosing is aimed not only at targeting C_{trough} but also overall drug exposure (*i.e.*, AUC).⁴⁴ Although C_{trough} is consistently higher in patients receiving SC infliximab and vedolizumab compared to those on IV dosing, simply comparing drug levels over time does not fully reflect AUC of the two formulations.^{92,185,186} Yet, it is still unclear whether it is C_{trough} or AUC that drives the treatment response of SC infliximab therapy.

The DIAMOND consensus preliminarily evaluated the role of TDM for SC infliximab and vedolizumab, underscoring the need for more scientific data to select the right patients and timing for switching to these newer formulations, and to explore the optimal strategy in case of loss of response.¹⁸⁷ A summary of statements agreed by the DIAMOND expert panel on TDM of SC infliximab is presented in **Table 14**.

Table 14. A summary of statements agreed by the DIAMOND expert panel

Statement	Agree*
TDM should be performed before IV-to-SC infliximab switch	84%
There is a role for reactive TDM in patients losing response under maintenance therapy with SC infliximab	89%
The optimal infliximab trough concentration for patients under maintenance therapy with SC infliximab should still be determined	88%
It is not the trough concentration that drives the therapeutic response to SC infliximab, but the overall exposure	73%
Based on the scientific data currently available, one cannot conclude that SC infliximab is less immunogenic than IV infliximab	84%

*The 'agree' percentage represents the percentage of the expert panelists scoring the statement 7–10 on a 10-point Likert. AUC, area under the concentration–time curve; IV, intravenous; SC, subcutaneous; TDM, therapeutic drug monitoring. Table adapted with permission from Fierens et al.¹⁸⁷

In **Chapter II** of this doctoral research, we conducted a popPK simulation study to assess the PK non-inferiority of switching from both standard and escalated IV infliximab and vedolizumab to SC dosing. The results indicated that switching from IV to SC infliximab 120 mg Q2W maintains non-decreased $C_{trough,ss}$ in patients with IBD, except for those previously receiving IV Q4W, who may need SC 120 mg QW.⁹³ Additionally, the results indicate that patients on either intensified or standard IV therapy can transition to SC vedolizumab at 108 mg Q2W without a reduction in $C_{trough,ss}$. However, transitioning from an escalated IV dosing regimen to the standard SC dosing regimen for both infliximab and vedolizumab may lead to a decreased AUC post-switch. Based on our results, Celltrion, the manufacturer of SC CT-P13, developed a clinical support tool for TDM of infliximab during the IV-to-SC switch (**Figure 34**).

In **Chapter III**, we introduced the first popPK-PD model for SC infliximab. Three sequential models were developed: a 2-compartment popPK model linking infliximab dose to exposure; an indirect response FC model describing the inhibitory effect of infliximab exposure on FC; and an endoscopic remission model linking FC to transitions between endoscopic remission and active endoscopic disease. Simulations from this model demonstrated that standard dosing of SC infliximab provides an enhanced FC response compared to both standard and escalated IV dosing. This model could support personalized dosing strategies based on combined monitoring of infliximab and FC levels. Additionally, we identified the effect of bodyweight on the clearance of SC infliximab, which enables the simulation and prediction of SC dosing for pediatric patients. Moreover, the modeling was based on data from pivotal trials shared by Celltrion, exemplifying effective collaboration between academia and industry.

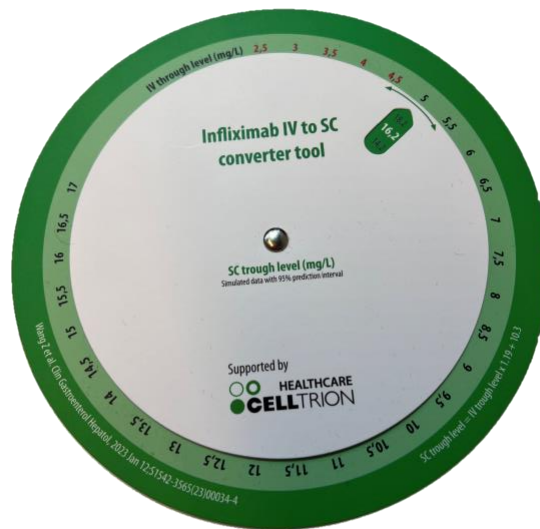


Figure 34. An intravenous (IV)-to-subcutaneous (SC) converter tool for therapeutic drug monitoring of SC infliximab.⁷⁶ The tool is only valid for switching from the standard IV infliximab 5 mg/kg every eight weeks to the standard SC infliximab 120 mg every other week. By pointing the arrow to the trough level at IV steady-state before the switch, the tool can predict the median and 95% prediction interval (PI) of the trough level at SC steady-state post-switch. Observing a trough level at SC steady-state below the 95% PI is an alarm signal for increased infliximab clearance and higher disease activity.

Perspectives

In summary, the rollout of SC infliximab and vedolizumab, together with other SC-administered mAbs, is expected to facilitate decentralized IBD care, offering practical treatment options for both clinicians and patients without sacrificing effectiveness. However, there is still a need to validate the effectiveness of TDM for SC infliximab and vedolizumab, understand the implications of elevated C_{trough} , and establish concentration thresholds that are associated with remission. Additionally, the potential drawbacks of lower peak concentrations (C_{peak}) with SC therapy need to be assessed, since C_{peak} are crucial for inducing remission in severe cases, like acute severe UC or perianal CD.²⁵⁷

The effectiveness and safety of transitioning from escalated IV dosing to SC dosing remain unclear. Results from ongoing trials such as DISCUS-IBD (“Switching from Dose-Intensified intravenous to Subcutaneous infliximab in Inflammatory Bowel Disease”) and AMERETTO (“Subcutaneous Infliximab After a Previous Intravenous Dose Optimization”) are anticipated to clarify this and may also delineate the impact of C_{peak} , C_{trough} , and AUC on treatment response in IBD.¹⁹⁸

The investigation of SC infliximab for induction therapy is gaining interest, particularly through the DIRECT-CD trial, which examines its efficacy in both induction and maintenance phases for CD patients. Early results suggest that SC infliximab can provide rapid symptomatic and biochemical improvements from the first day of induction treatment.²⁵⁸

The IV-to-SC shift also supports TDM through home sampling techniques like patient-led intracapillary blood sampling, dried blood sampling (DBS), and volumetric absorptive microsampling (VAMS). These methods facilitate the collection of comprehensive PK and biomarker data in both clinical practice and research settings. Such comprehensive sampling is crucial for AUC calculation, determining PK-PD targets for TDM and enhancing our understanding of TDM's role in IBD management.

Clearance monitoring: one step further of therapeutic drug monitoring

The mAb clearance is a PK parameter that not only affects drug exposure but is also closely related to IBD disease activity (**Figure 35**). A bidirectional relationship between PK and PD is common for mAbs used in IBD.²

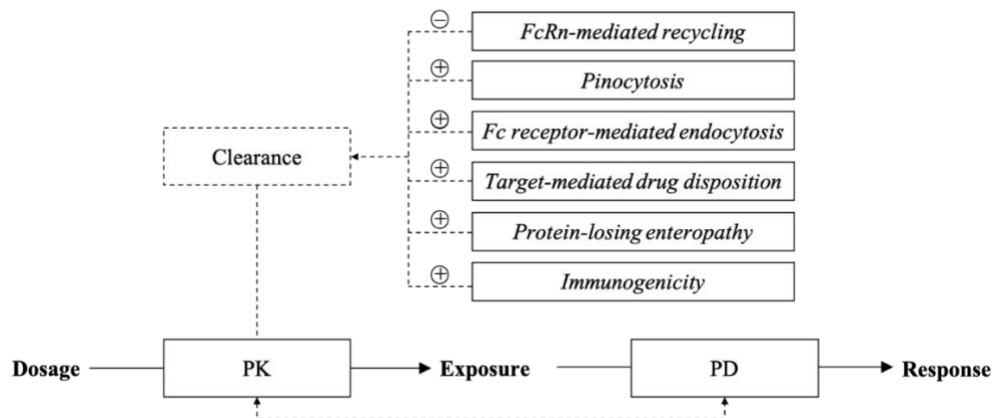


Figure 35. The dose-exposure-response relationship. Underlying mechanisms of the clearance of monoclonal antibodies are indicated. The dashed arrow between pharmacokinetics (PK) and pharmacodynamics (PD) denotes the bidirectional relation between both. FcRn, neonatal Fc receptor. Figure reproduced with permission from Kantasiripitak and Wang et al.²

The concept of clearance monitoring was recently introduced into the field of IBD.^{2,259,260} A comparison of the workflows of TDM, clearance monitoring, and MIPD is shown in **Figure 36**. Like TDM, clearance monitoring focuses on the “passive” monitoring, while not providing any recommendations for “active” dose optimization.² Adding dosage simulations transforms clearance monitoring into MIPD, offering concrete dosing support to physicians.

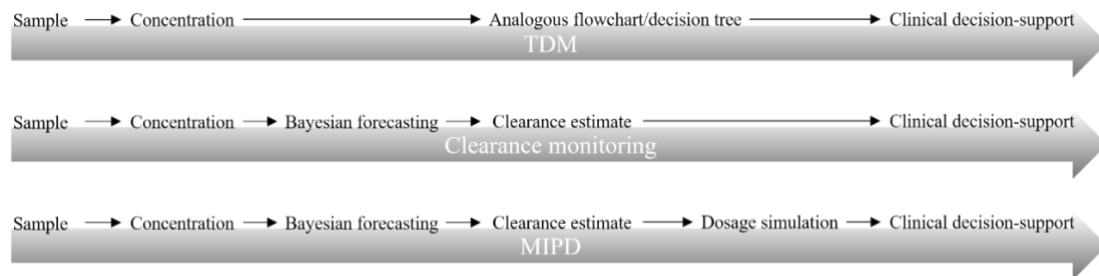


Figure 36. A comparison of the workflows of therapeutic drug monitoring (TDM), clearance monitoring, and model-informed precision dosing (MIPD). Figure reproduced with permission from Kantasiripitak and Wang et al.²

Monitoring mAb clearance over time allows a better insight into immunogenicity and disease status of patients with IBD. These insightful understandings could improve (cost-)efficacy and safety of mAbs treatments. In addition, clearance monitoring may outperform exposure-guided monitoring in identifying changes in PK and PD during dose adaptations (**Figure 37**). Therefore, clearance monitoring has the potential to improve mAbs treatment of patients with IBD in terms of effectiveness, drug utilization, and economics.

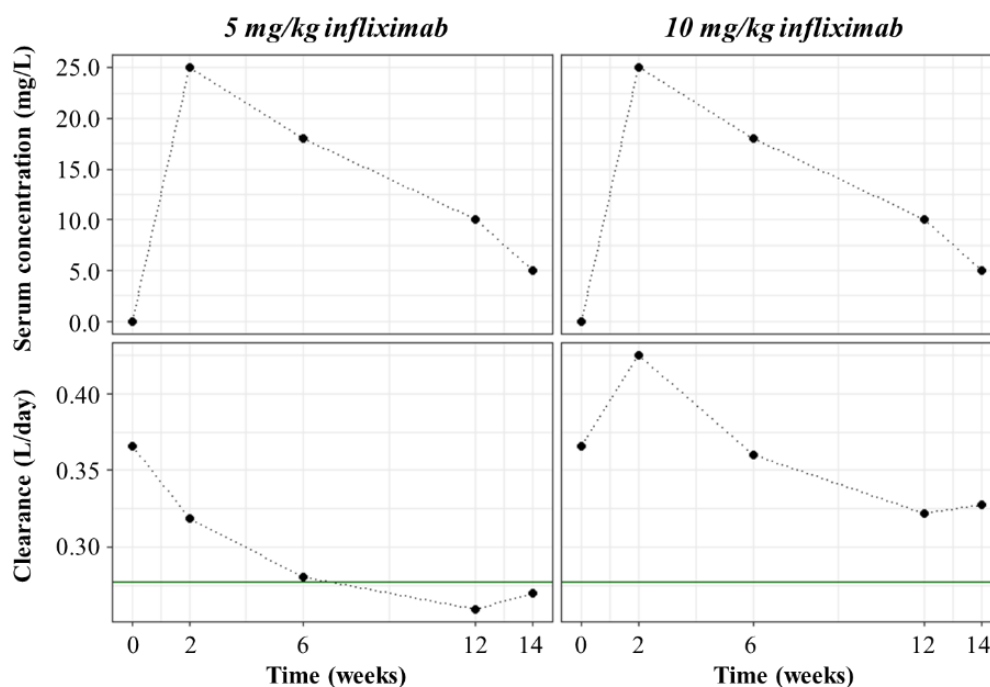


Figure 37. The time courses of observed serum concentrations and estimated clearances of infliximab in two virtual patients with identical covariate constellations, one receiving 5 mg/kg infliximab, the other receiving 10 mg/kg infliximab. The solid line in the clearance panels indicates the typical infliximab clearance in the population (0.277 L/day). NONMEM (version 7.5; Icon plc, Gaithersburg, MD, USA) was used for Bayesian forecasting. Figure reproduced with permission from Kantasiripitak and Wang et al.²

Although the benefits of clearance monitoring are theoretically obvious, its clinical implication still requires effort. Since drug clearance is also estimated during MIPD, limitations of integrating MIPD in routine clinical practice also hold for clearance monitoring. One of the main limitations is the lack of evidence for its clinical and economic benefits generated from prospective RCTs.² To facilitate a broad integration of clearance monitoring into clinical practice, its clinical utility and cost-effectiveness need investigation. Other limitations include the complexity of its implementation into the clinical workflows

considering the need for frequent blood sampling, availability of rapid concentration measurements, fit-for-purpose of the BF software tools, and training of health care personnel.¹¹⁸ Since measuring concentrations is the cornerstone of TDM, clearance monitoring, and MIPD, standardization of the bioanalytical assays is crucial. Also, interdisciplinary collaborations between academic researchers, the private sector, clinical teams, and regulators could accelerate clinical implementation.

In **chapter IV** of this doctoral thesis, we demonstrated that mAb clearance is a better predictor than C_{trough} for endoscopic outcomes in patients with CD and provides a discriminatory ability statistically similar to the current gold-standard biomarker FC. Additionally, we developed an open-access software tool for clearance monitoring of infliximab and ustekinumab to facilitate clinical validation of our finding and support clinical decision-making in routine IBD management.

Perspectives

To date, the value of clearance monitoring has been identified in two IBD settings; (i) as a predictor for the relapse risk in stable patients who undergo treatment de-escalation, and (ii) as a predictor for the need for colectomy during induction treatment in patients with acute severe UC.^{260,261} We believe that clearance monitoring will be utilized for broader purposes: (i) as an indicator for the (yet undetectable) onset of immunogenicity, (ii) as a marker for (yet asymptomatic) disease activity, (iii) for dose optimization, and (iv) as a predictor for treatment outcome. In both adult and pediatric patients with IBD, increases in infliximab, adalimumab, and ustekinumab clearance were reported to be associated with worse clinical outcomes.^{236–239} However, before becoming a tool for precision medicine in the era of immunotherapy, the clinical value of clearance monitoring is awaited to be confirmed.

Model-informed precision dosing: time to assemble the puzzle

Apart from the NOR-DRUM B, PAILOT and the PRECISION study, prospective TDM studies largely failed to show the benefit of TDM for improving treatment outcomes in patients with IBD, discouraging the widespread implementation of TDM in clinical practice.³ The failure of these TDM trials can be attributed to several factors: long turnaround times for sample measurements, poorly defined exposure targets, inefficient algorithms of TDM-based dose adjustments, and the enormous PK variability among patients with IBD.¹ Over the years, concerted efforts have been made to address and resolve these issues.

Point-of-care testing (POCT) involves conducting medical diagnostic tests at the time and location of patient care, allowing healthcare professionals to perform tests on-site without the need for extensive sample preparation. This approach significantly reduces the turnaround time to approximately 15-20 minutes, enabling immediate dose optimization based on real-time PK information. This is valuable especially in acute settings such as severe UC where high within-patient PK variability commonly exists.²⁶² However, to date, the application of POCT devices in TDM of mAbs remains very limited. Hurdles of cost, connectivity to the laboratory information system, quality control, and external quality assurance procedures, still exist and need to be overcome to allow a wide clinical integration.

Traditionally, the exposure target for TDM was established based on ROC analysis, which has been criticized for high uncertainty and not accounting for IIV. PopPK-PD modeling and simulation are more appropriate and powerful alternatives for describing the exposure-response relationship.¹ Through PK-PD simulations, the popPK-PD model converts different PD targets (*e.g.*, clinical remission, biological remission, endoscopic remission, histological remission) into corresponding PK targets. Emerging home sampling techniques, like patient-led intracapillary blood sampling, DBS, and VAMS, offer high flexibility and patient acceptance, facilitating the collection of rich drug exposure (*i.e.*, PK) and biomarker (*i.e.*, PD) data in clinical practice and future studies.⁵ The reduced blood volume requirement of capillary blood sampling methods also mitigates challenges associated with pediatric patient sampling. The rich data collected will power the selection of suitable therapeutic targets for biological treatments based on popPK-PD analysis.

Inefficient TDM algorithms and substantial PK variability among patients can diminish the effectiveness of TDM. MIPD offers a solution to this dilemma. MIPD considers patient-specific covariates, measured drug concentrations, and IIV, enabling dose adaptations that precisely and swiftly achieve the PK target. For clinical implementation, MIPD necessitates the integration of the popPK(-PD) model into an MIPD software tool, for which many options are available.¹¹⁸ Results from the PRECISION trial indicated that MIPD of infliximab was more effective than standard dosing in maintaining remission during maintenance therapy.⁹⁸ However, it is still unclear whether MIPD leads to better outcomes compared to TDM, whether reactive or proactive. Further studies are needed to clarify this aspect.

In **Chapter V** of this doctoral research, we conducted the first prospective study comparing infliximab interval extension based on TDM with interval extension guided by a MIPD software tool for maintaining steroid-free, combined clinical and biological remission (*i.e.*, treatment outcome) during one year in patients with IBD who sustained treatment response following an earlier interval shortening. The MODIFI study is a monocentric, open-label, historically controlled Phase 4 trial at UZ Leuven beginning in February 2022. It consisted of two arms:

- (i) a historically controlled TDM arm (S63206; N=21);
- (ii) a prospective MIPD arm (S64521, NCT04982172; N=31).

In the MODIFI trial, MIPD achieved a numerically higher rate of sustained steroid-free clinical and biological remission (52% vs. 29%) and a significantly higher rate of sustained steroid-free clinical remission (81% vs. 38%, $p<0.01$) compared to TDM.

Perspectives

We believe the time is now to join forces and combine all innovative technologies to design the next prospective MIPD clinical trial. To maximize the power of MIPD, we recommend MIPD in combination with home sampling, POCT and integration with the electronic health record (EHR) system (**Figure 38**). First, to enhance MIPD's clinical utility, integrating MIPD software with EHR systems is essential to automate data extraction.^{141,263} Second, home sampling tools like capillary blood devices can efficiently monitor mAbs exposure and CRP levels, facilitating over-time PK and PD assessment.²⁶⁴ Third, abrupt changes in infliximab clearance can reflect changes in disease activity.² By using POCT, MIPD can utilize the latest PK and PD data, providing real-time insight into PK and disease activity and supporting dose optimization directly at the point of care.

Currently, the abundance of therapeutic options for IBD, including mAbs, JAK inhibitors, and combination therapies, enables clinicians to easily switch medications for both primary and secondary non-responders, reducing the reliance on TDM/MIPD-based dose optimization. Its time to realize that TDM/MIPD is not universally necessary and selecting the right target population is paramount. Recent attention has been directed towards vulnerable populations, such as those with acute severe UC, postoperative Crohn's disease, and pediatric patients.^{265–267} Thus, we propose to prioritize these patients who are most likely to benefit from TDM/MIPD in future precision dosing studies. In the future, MIPD with a simultaneous popPK-PD model may also be interesting for simultaneously predicting treatment outcomes that are associated with the predicted drug exposure.

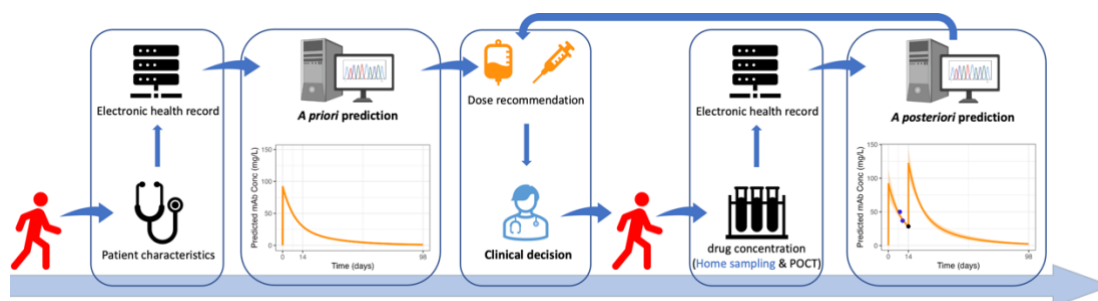


Figure 38. The suggested workflow of model-informed precision dosing of infliximab in clinical practice with illustration in a virtual patient. Conc, concentration; mAb, monoclonal antibody; POCT, point-of-care testing. Figure adapted with permission from Alsoud, Moes, and Wang et al.³

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PUBLICATIONS

Included as part of the doctoral thesis:

Wang Z, Dreesen E. Therapeutic drug monitoring of anti-tumor necrosis factor agents: lessons learned and remaining issues. *Curr Opin Pharmacol*. 2020;55:53–9. (PMID: 33128934)

Wang Z, Verstockt B, Sabino J, Vermeire S, Ferrante F, Declerck P, Dreesen E. Population pharmacokinetic-pharmacodynamic model-based exploration of alternative ustekinumab dosage regimens for patients with Crohn's disease. *Br J Clin Pharmacol*. 2022; 88(1): 323-335. (PMID: 34197653)

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Wang Z*, De Greef I*, Kantasiripitak W, Wicha S, Tops S, Nigro M, Outtier A, Verstockt B, Sabino J, Thomas D, Vermeire S, Ferrante M*, Dreesen E*. Model-informed dosage de-escalation of infliximab in patients with inflammatory bowel diseases. [Submission in preparation]

* Designated shared co-first or co-senior authorship

Not included in the thesis:

Faelens R, **Wang Z**, Bouillon T, Declerck P, Ferrante M, Vermeire S, Dreesen E. Model-Informed Precision Dosing during Infliximab Induction Therapy Reduces Variability in Exposure and Endoscopic Improvement between Patients with Ulcerative Colitis. *Pharmaceutics*. 2021;13(10):1623. (PMID: 34683916)

Soenen R, **Wang Z**, Grine L, Dreesen E, Schots L, Brouwers E, Declerck P, Thomas D, Lambert J. Therapeutic drug monitoring in dermatology: the way towards dose optimization of secukinumab in chronic plaque psoriasis. *Clin Exp Dermatol*. 2022;47(7):1324-1336. (PMID: 35245966)

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Yarur A*, Dervieux T*, Ungaro R, Spencer E, Bruss A, Berens L, Vermeire S, **Wang Z**, Panetta J, Dreesen E, Dubinsky M. Ustekinumab drug clearance is better associated with disease control than serum trough concentrations in prospective cohort of inflammatory bowel disease. [Submitted]

* Designated shared co-first or co-senior authorship

Oral presentations at international conferences

Wang Z, Verstockt B, Sabino J, Vermeire S, Ferrante F, Declerck P, Dreesen E. Modeling of the relationship between ustekinumab exposure, fecal calprotectin and endoscopic outcomes in patients with Crohn's disease. The Population Approach Group of Australia & New Zealand (PAGANZ) meeting 2021, online.

Wang Z, Kantasiripitak W, Verstockt B, João S, Ferrante M, Declerck P, D'Haens D, Laharie D, Vermeire S, Dreesen E. Infliximab and ustekinumab clearance during induction predicts post-induction endoscopic outcomes in patients with Crohn's Disease. The European Crohn's and Colitis Organization (ECCO) Annual Conference 2022, online.

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Ustekinumab clearance during induction predicts post-induction endoscopic response in patients with Crohn's disease.
The World Conference on Pharmacometrics (WCoP) 2022, online.

Wang Z, Vande Casteele N, Ferrante M, Vermeire S, Dreesen E.
A repeated time-to-event model linking infliximab clearance to the risk of relapse during maintenance treatment of patients with inflammatory bowel disease.
The International Association of Therapeutic Drug Monitoring and Clinical Toxicology (IATDMCT) Annual Congress 2023, Oslo, Norway.

Poster presentations at international conferences

Wang Z, Verstockt B, Sabino J, Vermeire S, Ferrante F, Declerck P, Dreesen E.
Model-based identification of an optimized ustekinumab dosage regimen for patients with crohn's disease.
The Digestive Disease Week (DDW) 2021, online.

Wang Z, Verstockt B, Sabino J, Vermeire S, Ferrante F, Declerck P, Dreesen E.
Modelling of the relationship between ustekinumab exposure, faecal calprotectin and endoscopic outcomes in patients with Crohn's disease.
The European Crohn's and Colitis Organization (ECCO) Annual Conference 2021, online.

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Pharmacokinetic-Pharmacodynamic Model-Based Exploration of Alternative Ustekinumab Dosage Regimens for Patients with Crohn's Disease.
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Wang Z, Kantasiripitak W, Verstockt B, João S, Ferrante M, Declerck P, D'Haens D, Laharie D, Vermeire S, Dreesen E.
Higher clearance of therapeutic antibodies is associated with worse endoscopic outcomes in patients with Crohn's disease.
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Wang Z, Afif W, Hanžel J, Kobayashi T, Papamichail K, Roblin X, Peyrin-Biroulet L, Vermeire S, Dreesen E.
Infliximab and prevention of colectomy in acute severe ulcerative colitis: an individual patient data meta-analysis.
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Wang Z, Verstockt B, Sabino J, Ferrante M, Vermeire S, Dreesen E.
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Wang Z, Verstockt B, Sabino J, Ferrante M, Vermeire S, Dreesen E.
Therapeutic drug monitoring can guide the intravenous-to-subcutaneous switch of infliximab and vedolizumab: a pharmacokinetic simulation study.
The American Society of Clinical Pharmacology and Therapeutics (ASCPT) Annual Meeting 2023, Atlanta, USA.

Wang Z, Vande Casteele N, Ferrante M, Vermeire S, Dreesen E.

A repeated time-to-event model linking infliximab clearance to the risk of relapse during maintenance treatment of patients with inflammatory bowel disease.

The Population Approach Group of Europe (PAGE) meeting 2023, A Coruña, Spain

Wang Z, Kantasiripitak W, Outtier A, Thomas D, Sabino J, Vermeire S, Ferrante M, Dreesen E.

Time-to-event modelling and simulation of the risk of relapse after infusion interval extension of infliximab in patients with inflammatory bowel diseases.

The Population Approach Group of Europe (PAGE) meeting 2024, Rome, Italy

Wang Z, Ferrante M, Vermeire S, Dreesen E.

Fecal Calprotectin Decreases Following Transition to Subcutaneous CT-P13 in Patients With Inflammatory Bowel Diseases on Escalated Intravenous Infliximab Maintenance Therapy.

The American College of Gastroenterology (ACG) Annual Congress 2024, Philadelphia, USA.

Wang Z, De Greef I, Kantasiripitak W, Wicha S, Tops S, Nigro M, Outtier A, Verstockt B, Sabino J, Thomas D, Vermeire S, Ferrante M, Dreesen E.

Model-Informed Dosing Interval Extension of Infliximab in Patients With Inflammatory Bowel Diseases: Six Months Interim Analysis of the MODIFI Trial.

The American College of Gastroenterology (ACG) Annual Congress 2024, Philadelphia, USA.

GRANTS

FWO PhD fellowship for strategic basic research 2021, 1SF2922N

FWO PhD fellowship for strategic basic research 2023, 1SF2924N

FWO travel grant for participation in a congress abroad 2023, K107623N

The American Society of Clinical Pharmacology and Therapeutics (ASCPT) Annual Meeting
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ASCPT Student/Trainee grant

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Grant for Short-Term Scientific Mission, COST Action ENOTTA

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**Acknowledgments,
personal contribution,
conflict of interest, and use
of generative AI**

Acknowledgments

Chapter I

Zhigang Wang wrote the manuscript; Erwin Dreesen and Zhigang Wang designed the research; Zhigang Wang performed the research; Bram Verstockt, João Sabino, Séverine Vermeire, Marc Ferrante, and Paul Declerck provided the data. All authors contributed to the review of the manuscript and approved the final version for submission.

Chapter II

Zhigang Wang wrote the manuscript; Erwin Dreesen and Zhigang Wang designed the research; Zhigang Wang performed the research. All authors contributed to the review of the manuscript and approved the final version for submission.

Chapter III

Zhigang Wang wrote the manuscript; Erwin Dreesen and Zhigang Wang designed the research; Zhigang Wang performed the research. Celltrion provided the data. All authors contributed to the review of the manuscript and approved the final version for submission. The authors thank Celltrion for generously providing the clinical data. The study results and discussion presented in this article are solely the personal views of the respective authors.

Chapter IV

Zhigang Wang wrote the manuscript; Erwin Dreesen and Zhigang Wang designed the research; Bram Verstockt, David Laharie, Geert D’Haens, and Séverine Vermeire provided the data; Zhigang Wang performed the research. All authors contributed to the review of the manuscript and approved the final version for submission.

Chapter V

Zhigang Wang wrote the manuscript; Erwin Dreesen, Marc Ferrante, Debby Thomas and Wannee Kantasiripitak designed the research; Erwin Dreesen, Marc Ferrante, Zhigang Wang, Ine De Greef, Wannee Kantasiripitak and Sophie Tops conducted the investigation; Zhigang Wang performed the analysis. All authors contributed to the review of the manuscript and approved the final version for submission.

Personal contribution

For **Chapter I, II, and IV**, the author of this thesis manuscript fully contributed to the design of the studies, the generation and assembly of data, the analysis and interpretation of the data, and wrote the manuscripts.

For **Chapter III**, the author of this thesis manuscript fully contributed to the design of the study, the analysis and interpretation of the data, and wrote the manuscripts.

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Conflict of interest

Zhigang Wang is supported by a doctoral research grant from the Research Foundation – Flanders (FWO), Belgium (grant number:1SF2922N, 1SF2924N).

Erwin Dreesen received consultancy fees from Alimentiv and argenx, lecture fees from Celltrion and Galapagos, and financial support from Celltrion, Janssen, Pfizer, Prometheus, R-Biopharm, and Sandoz, outside the submitted work, with all honoraria/fees being paid to KU Leuven and not to any personal account of Erwin Dreesen.

Séverine Vermeire received research support from AbbVie, J&J, Pfizer, Galapagos and Takeda; consulting and/or speaking fees from AbbVie, AbolerIS Pharma, AgomAb, Alimentiv, Arena Pharmaceuticals, AstraZeneca, Avaxia, BMS, Boehringer Ingelheim, Celgene, CVasThera, Cytoki Pharma, Dr Falk Pharma, Ferring, Galapagos, Genentech-Roche, Gilead, GSK, Hospira, Imidomics, Janssen, J&J, Lilly, Materia Prima, MiroBio, Morphic, MrMHealth, Mundipharma, MSD, Pfizer, Prodigest, Progenity, Prometheus, Robarts Clinical Trials, Second Genome, Shire, Surrozen, Takeda, Theravance, Tillots Pharma AG, Zealand Pharma.

Marc Ferrante received research support from AbbVie, Amgen, Biogen, Janssen, Pfizer, Takeda and Viatris; speaker's fees from AbbVie, Amgen, Biogen, Boehringer Ingelheim, Falk, Ferring, Janssen, Lamepro, MSD, Mylan, Pfizer, Sandoz, Takeda and Truvion Healthcare, consultancy fees from AbbVie, Boehringer Ingelheim, Celltrion, Janssen, Lilly, Medtronic, MSD, Pfizer, Sandoz, Takeda and Thermo Fisher.

Bram Verstockt received financial support for Research from AbbVie, Biora Therapeutics, Landos, Pfizer, Sosei Heptares and Takeda; Speaker's fees from Abbvie, Biogen, Bristol Myers Squibb, Celltrion, Chiesi, Falk, Ferring, Galapagos, Janssen, Lily, MSD, Pfizer, R-Biopharm, Sandoz, Takeda, Tillots Pharma, Truvion and Viatris; Consultancy fees from Abbvie, Alfasigma, Alimentiv, Applied Strategic, Astrazeneca, Atheneum, BenevolentAI, Biora Therapeutics, Boxer Capital, Bristol Myers Squibb, Galapagos, Guidepont, Landos, Lily, Merck, Mylan, Nxera, Inotrem, Ipsos, Janssen, Pfizer, Progenity, Sandoz, Sanofi, Santa Ana Bio, Sapphire Therapeutics, Sosei Heptares, Takeda, Tillots Pharma and Viatris; Stock options Vagustim.

João Sabino received research support from Galapagos and Viatris; speaker's fees from Abbvie, Falk, Takeda, Janssen, and Fresenius, consultancy fees from Janssen, Ferring, Fresenius, Abbvie, Galapagos, Pharmacosmos.

Isabel Spriet is supported by the Clinical Research Fund of UZ Leuven and reports consulting fees from Pfizer and Cidara; speakers' fees from Pfizer; travel support from Pfizer.

Yannick Hoffert is supported by a doctoral research grant from the Research Foundation – Flanders (FWO), Belgium (grant number: 1SHAA24N).

All other authors declare no conflict of interest.

Use of generative AI

Generative AI assistance tools were not used during the research/writing process of my thesis, except for mere language assistance.